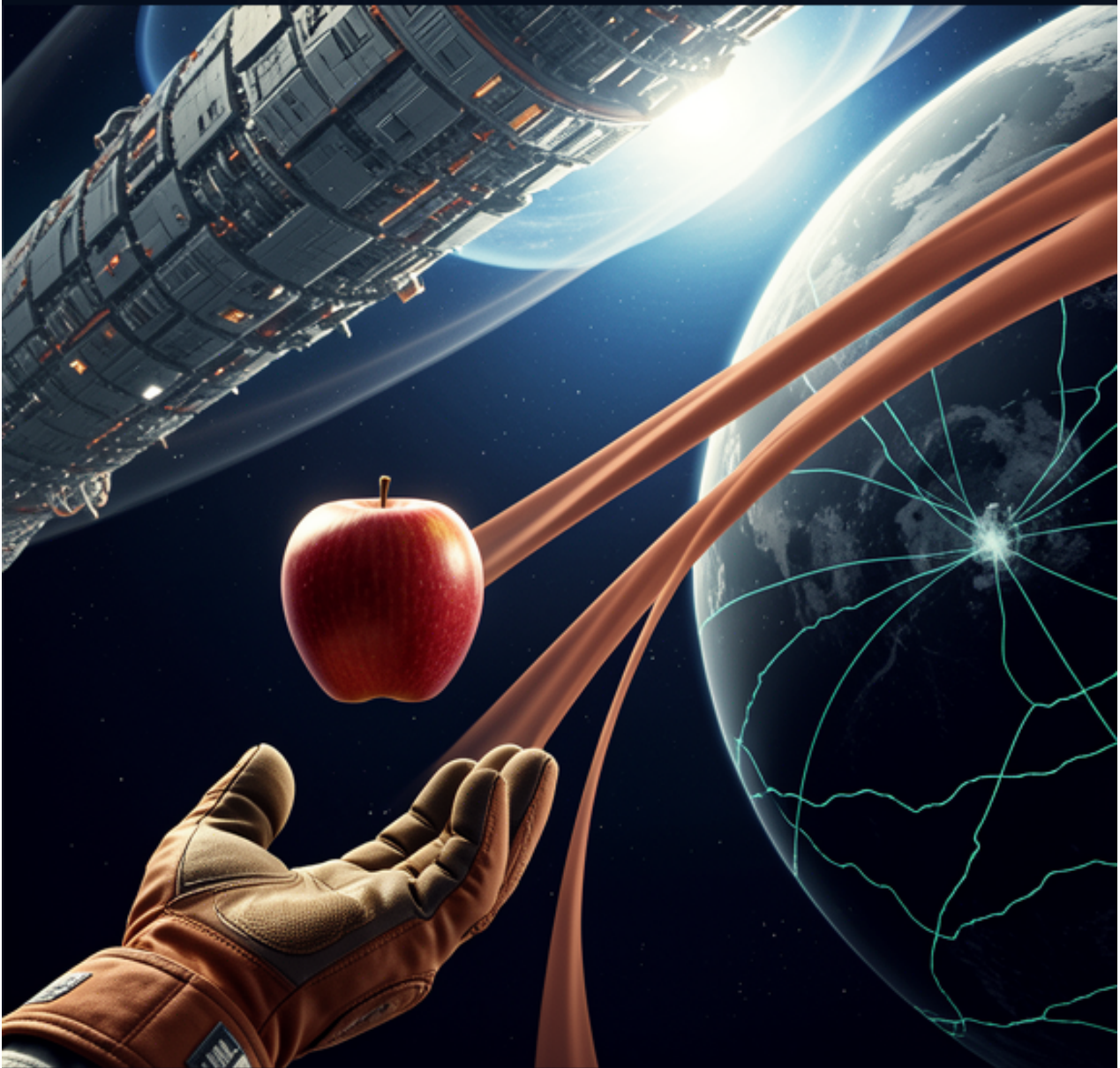


A SCIENCE FICTION NOVEL

THE REPUBLIC OF BORROWED GRAVITY

Every force has a destination.

Every debt has a body.



T. K. ARVEN

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Chapter 1 — Debt at Perihelion

The apple fell too slowly.

Iona Rhyll watched it leave the release claw in the audit drum and descend through a shaft of red light. Its skin had been polished for the calibration, though one bruise remained near the stem. At half a meter it should have crossed the first blue beam. The beam waited. A fraction of a second later, the apple broke it and continued toward the padded tray.

The drum recorded the delay without comment.

Iona looked through the glass at the garden beyond the test chamber. The apple tree grew in a long trough of black soil beneath Perihelion Station's heat shutters. Its leaves trembled whenever the station rolled an outer radiator away from the star. Children from the residential spoke harvested the fruit once a year. The Curvature Office borrowed one apple each week and returned it afterward, bruised but edible, to the kitchen.

She keyed the run for repetition.

"Release mass verified," said the drum. "Local field command: one point zero zero standard. Receiver balance nominal."

The claw opened. The apple fell too slowly again.

Iona stopped the automatic third run. Through the soles of her shoes, the drum's floor pressed upward with the ordinary persuasion of one standard gravity. Her inner ear agreed. The instrument disagreed by less than four thousandths, a discrepancy small enough to pass unnoticed in a corridor and large enough to move a cargo orbit if it persisted through a clearing.

She opened the raw trace.

Most auditors began with the receiver because receivers failed in familiar ways. A hot plate warped. A field coil accumulated impurities. A clerk entered a mass transfer under the wrong station clock. Iona began with the destination column. Every change in local curvature required a paired disturbance somewhere else, and the destination told her which physical account had accepted it.

The column contained no assignment.

She enlarged the entry until the digits filled the lower half of the glass.

Unsettled curvature: 0.0037 g.

Below it, where a sink code should have appeared, the ledger showed a clean field. No damaged character. No null marker. Nothing.

Iona lifted the apple from the tray. It weighed almost what her wrist expected. Almost had killed people before.

She carried it into the drum and set it on the floor at the center mark. The diagnostic field made no visible boundary. Light crossed the space without bending enough for an eye to notice; dust continued its slow circulation through the ventilation current. Only the force plate beneath the apple described the error, a thin red line displaced from the commanded blue one.

Iona knelt beside it. At Meridian, the first warning had also arrived as a disagreement too small for a body to name. A cup drifting across a table. A technician stumbling toward the inner wall of a ring. Sella had remained at her post because the receiver console insisted that commanded load and delivered load matched. By the time ordinary objects made the failure visible, a nursery deck had already lost a tenth of its weight.

The inquiry had taught Iona to distrust alarms that appeared only after harm became undeniable. It had also taught her the institutional use of small numbers. A discrepancy below threshold became noise. Noise required no owner. Without an owner, a dead engineer could carry the fault more easily than a ministry.

She pressed two fingers to the force plate. Its independent strain lattice agreed with the optical beams. Then she placed the apple in her palm and raised it through the drum's height marks, feeling for variation. Human nerves lacked the precision to confirm 0.0037 g, but the gesture forced her to inhabit the measurement. This much missing weight could not bruise the fruit. Multiplied through a hospital spindle, a cargo spine, or a thousand days of clearings, it could choose who fell first.

Back at the console, she opened a manual evidence packet. She included the apple's orchard tag, mass before release, surface temperature, bruise location, and the kitchen lot to which it would return. If someone later claimed the object had changed, they would have to argue against a particular apple rather than an abstract calibration mass.

She set the fruit on the console and called the station clock against three external references. All four clocks agreed. She pulled maintenance on the release claw, then the drum's beam geometry, then its last sixty mass checks. The claw had passed inspection nine days earlier. Beam alignment held within tolerance. Steel spheres, water packets, and two other apples had fallen correctly until the beginning of this shift.

A routine audit would have stopped there and tagged the receiver for service. Iona changed the display from local command to clearing history.

Perihelion Station kept its orbit by ordinary momentum. Fusion tugs made corrections, the heat shutters opened and closed on rails, and the residential wheel carried its own spin. Reciprocal Mass Exchange handled the smaller burdens that ordinary engineering could not smooth without wasting lives and propellant: the variable load in docking spines, the stress in bearings, the felt gravity in a hospital theatre while the wheel slowed for repair. Those adjustments accumulated as curvature debt. Six times a day the receiver bundled the debt, paired it to a certified sink system, and cleared the account.

The most recent bundle had left eleven minutes before Iona entered the drum. Its destination code identified Eidos, a dark system far outside the inhabited volumes, surveyed ninety years ago and certified empty.

The bundle total matched the station account.

The local field did not.

Iona sent the raw trace to Nemi Vale with a priority mark but no interpretation. Interpretation made people defend the first idea they had. Numbers were less proud.

While she waited, she cut power to the drum's receiver feed. The floor lightened by a measurable fraction as the mechanical spin contribution took the full load. The apple fell again, and this time crossed the first beam on schedule.

Receiver, then. Not instrument.

She restored power at diagnostic amplitude. The discrepancy returned.

A service alarm blinked on the console. The station's systems had noticed the manual interruption and wanted a reason. Iona entered forensic hold. The alarm changed color but did not disappear.

Beyond the glass, the heat shutters began their noon roll. Red illumination narrowed across the garden. The apple tree darkened branch by branch until only one leaf retained the star.

Sella had taken Iona into a different station garden when they were children, before either of them knew how much machinery went into the word down. She had dropped a metal nut from shoulder height and made Iona listen when it struck the deck.

*Weight goes somewhere, * Sella had said. *If an engineer cannot tell you where, keep your helmet on.*

At twelve, Iona had thought this a joke about careless engineers. At twenty-two, after Meridian, she had understood it as a warning. Sella had been dead sixteen years, and the official account still described her last shift as an operator's mistake during receiver maintenance.

Iona turned the memory over once, found no new surface, and returned to the trace.

Nemi arrived at a run. She caught the drum rail before the changing floor could surprise her, then stood still until her body accepted the fractionally lighter load. At thirty-two, she wore her dark hair close to her head and distrusted any pocket that could not be sealed. Three styluses were clipped to her sleeve.

"You froze a receiver feed on an inhabited station," she said.

"I lowered it to diagnostic amplitude."

"The station manager used a less flattering verb."

"He may submit one after the apple stops lying."

Nemi saw the fruit on the console. "That apple has no professional obligation to you."

"It has mass."

"Against its better judgment."

Iona moved aside so Nemi could read the destination field. The humor left her deputy's face without drama. Nemi checked the display, opened the ledger schema, and searched for a rendering fault.

"How did you produce a blank?"

"I did not."

"A corrupted null would show brackets."

"It does not."

"Manual deletion leaves an audit scar."

"There is no scar."

Nemi drew one stylus from her sleeve and wrote a local query instead of speaking it. The ledger answered that the 0.0037 g discrepancy had no debt origin, no transfer channel, and no outstanding assignment. It existed only in the measured field.

They ran the query against the previous clearing. The bundle to Eidos contained the correct total, signed by Perihelion's receiver and accepted by the Republic clearing engine. At the bottom of the packet, below the signature block, Nemi found a timing residual.

"Here," she said.

The residual measured thirty-one millionths of a second. Too small for station operations, large enough for dark-lattice coupling. The clearing engine had registered a phase return after the bundle closed, then discarded it as sink noise.

Iona compared the returned phase with the local discrepancy. "Scale it."

Nemi did. The numbers agreed to nine places.

Something had come back through the channel carrying exactly the weight missing from the apple's fall.

Neither woman spoke for several seconds. The station continued around them. Pumps moved coolant. The residential wheel turned. Somewhere beyond the drum, a cart rattled over a joint in the floor. Ordinary sounds became evidence of how much people trusted the direction beneath their feet.

"Reflection?" Nemi asked.

"Dark-lattice channels do not reflect after acceptance."

"Apparently one did."

"Then Eidos did not accept the full bundle."

Nemi brought up the sink certificate. Eidos occupied a black corner beyond the Republic's traffic maps: a red dwarf too cold to shine visibly, three rogue ice bodies captured on steep orbits, cometary debris, no atmosphere with disequilibrium chemistry, no radio emissions, no industrial heat. Vacant for clearing use, the certificate concluded. The language had outlived every person who signed it.

"Could an ice body produce a passive return?" Nemi asked.

"Not with a matched quantity."

"A buried receiver?"

"Ninety years beyond supply, operating without a beacon."

"An unregistered human installation?"

"That would be a larger crime and a smaller impossibility."

Nemi rotated the timing trace. "The clearing engine classified this as noise because the amplitude sits under the response threshold."

"Who set the threshold?"

"The Continuity Ministry, after the first network expansion."

"On what evidence?"

Nemi's stylus paused. "I can find the rule. Evidence may be ambitious."

She found the rule in a standards packet written eighty-eight years earlier. The packet devoted six pages to receiver safety, eleven to commercial allocation, and one paragraph to sink response. A certified empty system, it said, could produce tidal settling, instrument aliasing, or stochastic channel return. Any residual below five thousandths of transmitted load should be discarded unless it threatened sender stability.

"They defined a response by whether it could hurt us," Nemi said.

"They defined detection by the same measure."

The original sink surveys had crossed each candidate system quickly. Telescopes searched for atmospheric disequilibrium, radio traffic, industrial heat, artificial night light, and moving structures large enough to occult a star. Eidos had offered none. Its captured ice worlds reflected almost nothing. Their orbital solutions fit bodies without engines. The Republic had therefore granted itself permission to use the system as a destination for forces no inhabited world wanted nearby.

Iona opened the survey images. Grainy arcs showed three dark spheres against colder darkness. One frame caught a narrow gleam from an ice limb. The report described the gleam as a methane exposure and moved on.

"How deep did they sound the ice?" she asked.

"They did not land."

"Gravimetric tomography?"

"Two passes, both optimized for subsurface oceans." Nemi read farther. "No liquid layer detected above the pressure floor."

"And below it?"

"Not part of the life standard."

The omission did not prove presence. It proved only that emptiness had been certified by a list of things humans already knew how to recognize. Iona had signed similar certificates. Every audit required a boundary, and every boundary converted what lay outside it into administrative silence.

She copied the old paragraph into the evidence packet. Beside it she placed the returned phase, without interpretation. The two items accused each other efficiently enough.

The service alarm on Iona's console changed from amber to red. Perihelion's next hospital adjustment had entered its five-minute window. Without the receiver at full amplitude, the surgical spindle would have to slow by mechanical braking. The procedure schedule would slip. No one would die, but two patients would remain under sedation longer, and a hundred people would learn an auditor's name with resentment attached.

Iona could restore the receiver, release the forensic hold, and let the next clearing carry the discrepancy into the same blank column. That choice would preserve every immediate obligation she could name. It would also send another pulse to Eidos before she understood the return.

"Give me the next bundle forecast," she said.

Nemi displayed it. The station had accumulated debt from a cargo lock, the hospital spindle, and a radiator roll. Total transfer would reach 1.84 g-seconds across the accounting interval.

“Destination?”

“Eidos again. The scheduler keeps a sink assignment for four consecutive clearings to reduce channel loss.”

“Alternates?”

“All certified sinks are above their weekly tidal cap except Kestrel Deep.”

“Kestrel carries a navigation hold.”

“A survey craft is crossing its outer debris plane.”

Iona read the hospital notice. Forty-three seconds had passed.

She called the surgical spindle rather than letting the station manager carry the message. A masked anesthetist answered from a preparation bay, irritation controlled because a patient lay within hearing.

“Mechanical schedule adds twenty-three minutes,” the anesthetist said. “We planned the procedure around receiver stability.”

“I have an unexplained reciprocal load in the active field.”

“Can it change during the operation?”

“I cannot certify that it will not.”

The irritation vanished. “Then we brake.”

They reviewed the consequences together: longer sedation, additional propellant for spindle recovery, two delayed cases, a missed transfer window for a patient arriving from the inner service ring. None justified an uncertain field. All of them belonged in the account.

After the call, Iona entered each cost under her own audit number. Curvature accounting encouraged clean pairs, a local benefit and a distant debit joined by mathematics. Real decisions spread damage through schedules, bodies, fuel reserves, and trust. The ledger made those consequences look secondary because they could not be expressed in g-seconds.

“Station management wants to know whether the hold will last past noon roll,” Nemi said.

“Tell them I do not know.”

“They will ask for an estimate.”

“Four hours to Ministry review. No estimate beyond that.”

Nemi relayed it. Iona watched the station responses populate: docking reassigned two crews, the hospital accepted the mechanical plan, cargo control disputed the propellant charge, and residential services requested a public notice before the next load variation. People adapted faster when an institution admitted uncertainty early. That fact had never appeared in the continuity models Bex's office circulated.

“Put the spindle on mechanical schedule,” she said. “Charge the propellant loss to forensic authority.”

Nemi looked at her. “The station manager will appeal.”

“He should.”

“The clearing engine may overrule a local hold if continuity costs exceed threshold.”

“Then freeze the bundle at the office layer.”

Nemi did not move. “That requires a named finding.”

Iona studied the blank destination field. Naming a finding created its first boundary. Call it receiver malfunction, and maintenance would own it. Call it settlement error, and the clearing engine would recalculate. Call it unauthorized return, and the Curvature Office would have to admit that an empty system had done something emptiness could not do.

“Unassigned reciprocal load,” she said.

“That phrase does not exist in the codebook.”

“It does now.”

Nemi wrote it into the hold request.

The office layer asked Iona to confirm personal liability for service disruption, propellant expenditure, medical delay, and any downstream correction caused by blocking a certified clearing. She pressed her thumb to the console.

The receiver dropped to standby.

Gravity in the audit drum changed by less than one percent. Iona felt it in the old, unreliable place behind her knees. Outside, workers in the garden glanced up as the station announced a temporary load variation. A gardener caught a tray before it slid from a bench. The apple on Iona's console rolled two centimeters and stopped against her stylus case.

The hospital spindle acknowledged mechanical schedule. The cargo lock postponed a lift. Perihelion began paying locally for the field it used.

Nemi exhaled. “Hold accepted. Four hours before Ministry review.”

“Copy every raw channel record before the review begins.”

“To the protected audit vault?”

“And to an isolated instrument store.”

“That second copy violates chain policy.”

“So note the violation in the first copy.”

Nemi gave her a long look, then bent to the work.

Iona returned to the phase residual. One return could be material behavior no model had captured. Two could establish repeatability. The clearing freeze meant there would be no second outgoing pulse, but the channel remained weakly coupled while the hold propagated through the dark lattice. She opened the live noise floor.

Eidos lay too far away for an ordinary signal to reach them within any useful human lifetime. The debt channel

did not carry messages. It paired disturbances. Engineers used tiny calibration modulations during setup, no more meaningful than tapping a pipe to learn whether it continued through a wall.

On the display, the noise floor lifted.

Nemi saw it at the same moment. "Receiver is in standby."

"I know."

"No outgoing load."

"The trace confirms it."

A narrow pulse crossed the trace. Then another. The intervals differed. Iona marked them automatically: thirty-one microseconds, thirty-seven, forty-one.

Nemi stopped copying and opened a fresh analysis pane. "Those are primes."

A fourth interval arrived: forty-three.

The sequence paused.

Iona's hand remained above the console. A machine fault could repeat. A natural process could oscillate. Neither chose intervals from a mathematical set and stopped after four terms as if waiting.

The Curvature Office called from Nacre. Director routing, highest priority. The station manager filed his appeal. The hospital sent a revised sedation cost. All three notices stacked at the edge of the screen.

A fifth pulse arrived after forty-seven microseconds.

Nemi whispered, "It changed after we stopped the clearing."

"No," Iona said. She looked at the apple resting against the stylus case, at the bruise beside its stem, at the destination field that had been left clean instead of filled with a lie. "It answered."

Chapter 2 — The Falling Freighter

The cargo yoke kicked upward hard enough to bend steel.

Tavin Serr caught the command rail with both hands as the *Morrowline* rolled beneath him. The bridge floor tilted eight degrees, corrected, then drove against his boots with twice the commanded load. Behind the pressure wall, six thousand tonnes of habitat flywheel tried to continue along the old vector while the ship's compensator argued for a new one.

“Cut the receiver,” Tavin said.

Haru Den had already reached for it. “Command rejected. Clearing handshake active.”

A red line crossed the bridge plot. The flywheel's forward bearing climbed through its load limit. Another five seconds would tear the yoke from the freighter or pull the ship around it.

“Pella, blow the handshake.”

Chief Engineer Pella Jor answered through engine noise. “That opens every acceleration load at once.”

“I know what it opens.”

Tavin thumbed the general channel. “All hands, impact posture. Compensation failure. Sive, lock the transfer bay.”

The ship groaned from bow to fusion collar. *Morrowline* measured three hundred eighty meters without her cargo, most of it truss, tanks, and stubborn redundancy. She had been built to pull masses that preferred their own opinions. The flywheel outmassed her by a factor of four. Under normal compensation, the yoke spread that inertia through a curved local field. Now the same field had reversed its correction and pressed the load away from the ship.

Pella counted, “Handshake breaker in three. Two. One.”

Gravity vanished from the bridge.

Tavin's knees straightened under no weight. A loose inspection tag lifted from Haru's console. Then the uncompensated turn reached them. His harness caught his hips and ribs while the ship rotated around the flywheel's center of mass.

“Receiver isolated,” Pella said. “Yoke bearing four at one hundred twelve percent.”

“Cluster three, lateral fire. Five at counter-roll.”

“Flight safeties deny burn,” Haru said.

“Manual authority.”

Haru looked at him once, measuring whether he meant to spend the ship. Tavin laid his palm on the captain's release. Two green bars became six amber ones.

The first thruster cluster lit along the freighter's port spine. Conventional thrust shoved the hull sideways. Cluster five answered near the stern, turning the ship against the flywheel's pull. Acceleration arrived without

RME relief, a rough hand that compressed Tavin into his couch. The plot showed the yoke angle narrowing by fractions.

“Bearing four still climbing,” Pella said. “If it locks, the hub leaves with the cargo.”

“Release outer clamps.”

“That loses the transit cradle.”

“It keeps the center clamp aligned.”

Pella did not spend another breath arguing. Explosive bolts snapped along the outer yoke. The transit cradle opened in three petals, each carrying shielding panels and two auxiliary braces worth more than a small tug. One petal struck the flywheel's rim, spun away, and vanished beneath the ship's sensor horizon.

The reduced yoke flexed. Its central clamp held.

Tavin watched the burn clock consume reserve fuel at a rate intended for collision avoidance, not cargo handling. *Morrowline* had left Nacre with enough margin for one failed rendezvous and a slow transfer home. Every second of this burn ate days from that promise.

“Roll rate down to three degrees,” Haru said.

“Bring cluster five to sixty percent. Keep three where it is.”

“That walks us below the Lumen approach plane.”

“Better low than in pieces.”

A second alarm cut across him. The habitat flywheel had begun precessing inside the remaining clamp. Its broad rim carried stored rotational energy even while secured. If the rim touched the yoke, friction would turn the assembly into a torch.

Pella saw it. “We need to dump its spin.”

“We deliver it without spin and Helix Quay waits six months for commissioning.”

“We deliver a hot ring and nobody waits for anything.”

Tavin opened the cargo controls. The flywheel belonged to Helix Quay's new residential ring, a composite hoop meant to store energy, smooth spin changes, and keep eighty thousand people from noticing every freight arrival. Its construction had taken eleven years. Its destination contract placed every risk of transport on *Morrowline* until the quay accepted the clamps.

He selected emergency magnetic braking.

Haru said, “That will cook the field bands.”

“Only if we survive long enough.”

The brake coils energized. Heat spread around the flywheel rim in a bright false dawn, visible through external cameras as bands shifting from black to dull red. Power demand pulled the bridge lights low. The ship continued falling away from its planned approach, her nose pointed thirty degrees off vector, conventional

thrusters fighting a cargo mass never intended to be felt directly.

Sive An came over the crew channel. "Transfer bay secure. Three impact injuries. One possible spinal compression in machine shop two."

"Can you hold them through this burn?"

"I can hold the people. Your ship has not answered me."

Tavin read the structural plot. "Ask again in forty seconds."

"Patients dislike suspense."

"So do captains."

Sive cut the channel before either of them could make the joke kinder.

Across the bridge, Haru worked the approach geometry with one hand braced against acceleration. The navigator's projected lines had lost their elegant convergence. *Morrowline* now crossed under the freight corridor toward an empty maintenance volume. Helix Quay hung ahead as a silver coil around Lumen's pale moon, too distant for the naked eye but huge on the tactical display.

"Quay control wants our failure code," Haru said.

"Send receiver reversal and active salvage."

"They want to dispatch tugs."

"Tell them to keep clear until our roll stops."

"Estimated stop?"

Tavin considered the fuel line, yoke strain, flywheel heat, and the blue limit advancing across the bearing display. "No estimate."

The phrase bought them honesty and nothing else.

A hard tremor passed through the deck. Bearing four dropped six percent, then climbed again.

Pella's voice flattened. "Center clamp has shifted twenty millimeters. We can save the flywheel or the ship."

"Explain the choice."

"Hold the brake and the hub trunnion may shear. Release the flywheel now, and we can push clear before it tumbles. Keep it attached, and I have to jettison the drive-side cargo rack to balance the yoke."

The drive-side rack carried thirty-two sealed pallets: replacement receiver coils, surgical spindle bearings, water-purification membranes, and two tonnes of orchard soil inoculant. Helix Quay had ordered them around a maintenance closure already postponed twice. Dropping the rack would not merely cost money. It would move shortages into rooms where Tavin would never see them.

"Can the rack survive free flight?"

"Not with its present vector."

“Can we push it toward recovery?”

“If cluster two still gimbals after the release. That costs more reserve.”

Tavin pictured Keir at Nacre, seventeen when he first asked why freighter captains called discarded cargo a sacrifice. *Because loss sounds nobler when somebody else receives the invoice,* Tavin had told him. Keir had kept the answer and used it later during every argument about Tavin's work.

“Mark every pallet and jettison the rack,” Tavin said. “Then give it a recovery push.”

Pella's acknowledgment arrived without relief.

On the lower camera, clamps opened around the rectangular rack. For an instant the cargo remained beside them, matching the ship's fall. Cluster two fired a short lateral burst. The rack drifted toward the maintenance volume, rotating slowly enough that salvage tugs might catch it before Lumen's moon did.

The altered balance changed everything at once. Yoke strain fell. The flywheel's precession narrowed. Haru brought cluster five down and rolled *Morrowline* toward the maintenance plane.

“Thirty seconds to stable attitude,” Haru said.

Tavin tasted blood where he had bitten his cheek. “Fuel.”

“Reserve at forty-one percent.”

That number denied them the return profile filed at departure. Even if Helix Quay refueled them, the contract would place emergency propellant at punitive rates. Tavin could fight the charge after he had a ship.

“Keep burning.”

The flywheel brake wound its spin lower. Pella called temperatures from the field bands, each one closer to permanent damage. At the bridge rail, the loose inspection tag fluttered in acceleration currents from the ventilation fans. Its small motion gave Tavin a cleaner sense of load than the warning colors.

When the roll reached half a degree, he released cluster three in steps. The hull answered with a long settling noise. Somewhere aft, metal struck metal once and stopped.

“Attitude stable,” Haru said.

Tavin waited through two breaths. “Relative cargo motion?”

“Within clamp tolerance.”

“Receiver?”

“Physically isolated. No handshake.”

“Then we fly without it.”

He ordered a low conventional burn toward the maintenance volume. Acceleration held at one fifth standard, enough to make every injury hurt and gentle enough for the remaining yoke. *Morrowline* moved under fusion thrust alone, carrying her scorched flywheel and less than half the reserve fuel she had brought.

With the attitude steady, Tavin left Haru the bridge and went aft. The lift refused service under the structural

alert, so he climbed the central ladder through five compartments. Each rung pulled at his shoulders under the low burn. Crew moved close to walls and kept one hand on rails. They had trained for uncompensated flight, but training usually came with empty tanks and a quiet cargo deck. Now the ship carried enough exposed inertia to turn any careless step into an injury.

Machine shop two smelled of hot sealant. Sive had fixed a technician named Olan Ree to a rigid board and packed restraint foam around his neck. Ree could move his toes. He also had enough pain medication in him to greet Tavin as the Republic's worst dancer.

"You have seen me dance?" Tavin asked.

"I have seen the maneuver log."

Sive checked Ree's pupils. "He needs imaging we cannot perform while the yoke is drawing emergency power."

"The first tug carries a medical pod."

"Forty-eight minutes."

"Can he wait?"

"He has no vote on the matter. I prefer not to wait longer."

Tavin knelt beside the board. "Olan, I need your account before the medication takes more of it."

Ree described a load that struck from above, released, then struck from the deck. A tool chest crossed the shop between pulses although the ship's rotation had not changed enough to explain it. He had caught the chest before it reached an apprentice. Its corner caught him instead.

"How many pulses?"

"Five. Maybe six."

"The instruments show five."

"Then charge the sixth to my spine."

Tavin recorded the statement. Sive watched him finish before speaking.

"Do not make his joke sound voluntary in the report."

"I won't."

Beyond the shop, two pressure doors stood closed around a warped cargo passage. Pella met him there carrying a mechanical gauge. She had disconnected its network link and taped the casing shut.

"Independent enough for you?" she asked.

"Show me."

They entered through the first door and sealed it before opening the second. The yoke inspection gallery curved around the central clamp. Through layered glass, the flywheel filled the view like a dark horizon. Heat from its brake coils bled into the gallery in spite of insulation. Red markers along the clamp showed where the trunnion

had shifted.

Pella fitted her gauge across the narrowest brace. "Seven millimeters permanent stretch. I can hold it if thrust stays under a quarter standard and nobody asks for another turn."

"Quay approach needs two turns."

"Then make them so slow the braces forget to complain."

A fleck of scorched composite drifted past the outer glass. The low acceleration drew it against a rib, where it settled like black snow.

Tavin followed the visible load path from flywheel hub to clamp, through the brace cluster, and into the ship's spine. Every part carried marks from the reversal. Fasteners had moved in their seats. Paint had cracked around welds. One sacrificial web had torn exactly along its design line, a small success hidden inside a larger failure.

"Could the receiver command have come from our hardware?" he asked.

Pella placed the gauge on another brace. "Our plates can misfire. They can lag, surge, or drop out. To reverse the compensation vector, four control layers must agree on a negative load."

"Can a damaged sensor fool all four?"

"Not after I severed the channel bus. The last pulses crossed after isolation."

"Haru found them too."

"She reads clocks. I read bent metal." Pella pointed through the glass. "That upper brace took force toward the bow. The lower pair took it aft. Hardware cannot invent two directions without instructions."

"Instructions from where?"

Her hand tightened around the gauge. "Ask the auditor who froze our clearing."

Tavin let the accusation remain unendorsed. "First we get the cargo into a cradle. Then we ask."

On the way forward, he stopped at the drive-side release bay. The rack had left a rectangle of bare mounting points. A loader crew secured loose lines and tagged failed bolts. Their supervisor handed Tavin a tablet listing the jettisoned goods. Beside every pallet appeared a destination department and a delay estimate. Surgical bearings: nine days if salvage failed. Water membranes: sixteen. Orchard culture: one growing cycle.

"Quay procurement wants us to call the rack expendable," the supervisor said.

"Was it?"

"Not to the people who ordered it."

"Keep the department names in the loss report."

"Insurer will strip them."

"Then keep my signed copy."

The supervisor locked the file against alteration and passed it back. Tavin added his authority mark. A captain

could not recover cargo by choosing a better noun for its absence.

Helix Quay dispatched three tugs after Haru confirmed the vector. Two headed for the freighter. The third chased the jettisoned rack. Its predicted intercept reached only the first six pallets before the rest crossed the safe recovery boundary.

Tavin authorized the attempt.

As danger narrowed into work, the bridge filled with voices that had waited behind the emergency. Sive requested reduced acceleration for the spinal patient. Pella wanted power moved from the cargo brake to cooling. Quay control demanded a cargo status. The ship's insurer pushed an automatic notice into the command queue before anyone had counted the injured.

Tavin reduced thrust by three hundredths and assigned the insurer's notice to a folder called LATER, where it joined twelve years of corporate urgency.

"Captain," Haru said, "I have the receiver record."

"Save it raw."

"I did. There is something wrong with the sequence."

"Beyond reversing compensation?"

"The command did not reverse continuously." Haru enlarged a timing strip. "It stepped through five discrete load states. Each step lasted microseconds."

Tavin leaned against his harness to read. The plotted command resembled a set of narrow teeth. Their heights matched fractions of the clearing load. Between them, the compensator returned briefly to its proper direction.

"A failing plate can chatter," he said.

"Not after physical isolation."

The final two steps appeared thirty milliseconds after Pella blew the handshake, when local hardware had no active lattice coupling. Haru overlaid ship clocks against Lumen traffic control and a navigation beacon. All agreed within tolerance.

"Buffer artifact?" Tavin asked.

"I checked the independent strain gauges. The yoke felt each step."

Pella joined them from engineering through a face window, sweat darkening the collar of her suit. "I felt one through a wrench. If a buffer can kick me, I want to meet it."

Tavin studied the trace rather than reward her line. "What clearing were we coupled to?"

Haru opened the route packet. "A network batch, destination Eidos. Perihelion Station originated the hold request yesterday. The clearing engine rerouted some loads, delayed others, and left our transit compensator in a partial handshake during reconciliation."

"Why would a station hold touch a ship in Lumen?"

“Same sink assignment. Same batch family.”

Pella's mouth tightened. “Our compensator borrowed against a transfer somebody froze.”

“That describes an accounting fault,” Tavin said. “It does not explain a reversed field.”

“No,” Haru said. “It gives us a clock.”

The navigator pulled an incident bulletin from the Curvature Office. Most of it hid behind forensic seals. One visible line named the moment Perihelion's frozen clearing entered network propagation. Haru converted between station clocks, ship time, and the lattice handshake log.

The figures aligned.

Tavin asked for the uncertainty twice. Haru ran the calculation against the worst clock drift. The first reversal aboard *Morrowline* had arrived at the precise coupling interval created by Iona Rhyll's hold, delayed only by the compensator's batch queue.

“Who is Rhyll?” Pella asked.

“An auditor,” Haru said. “Senior grade.”

“Then I already dislike her.”

“She may have kept the handshake from completing,” Tavin said.

“She may have made it fail.”

“Those remain different findings.”

Pella pushed away from the camera. “My finding has three cracked braces and a flywheel hot enough to roast the contract.”

“Document all of it. No conclusions yet.”

Tavin sent a preservation order through the ship, covering receiver logs, strain histories, thruster commands, injury reports, and each discarded pallet. If the Ministry tried to classify the event as operator response to turbulence, he wanted the operator response preserved down to the hand that had opened each clamp.

The nearest tug came alongside two hours later. Its grapples took part of the flywheel load, allowing *Morrowline* to cool the yoke and reduce thrust. Sive transferred the spinal patient across a rigid tunnel. Four more crew members reported bruising after the danger passed. No one had died.

Losses accumulated more slowly. The habitat flywheel retained only sixty-one percent of its certified spin capacity. Twelve field bands needed replacement. Recovery crews caught five pallets whole and one in fragments. Orchard inoculant dispersed in glittering bags across the maintenance volume, each bag turning in sunlight from Lumen.

Tavin recorded the losses under his own authority. He refused Quay control's request to restart compensation for docking.

“We can offer a separate receiver channel,” the controller said.

“You can offer it to an empty yoke.”

“Manual tow adds nine hours and risks the clamp.”

“Then tow carefully.”

Pella rigged two tug lines. Haru wrote a slow approach that relied on thrust, tether tension, and patience. *Morrowline* entered Helix Quay under conventional control, her fusion exhaust pulsing in measured bursts while the scorched flywheel crossed the docking frame.

At acceptance, a quay inspector tried to classify the outer cradle as transit loss. Tavin showed him the reversal trace and waited until the category changed to network event.

Afterward he returned to the bridge. The maintenance volume rotated beyond the forward glass. Salvage lights moved among scattered cargo. Haru had left the timing overlay open at the command station: Perihelion's frozen clearing above, *Morrowline*'s first reversed load below.

Two lines met at one marked instant.

Tavin set the ship's raw record beside the sealed audit bulletin and transmitted both to Iona Rhyll before the Continuity Ministry could tell either of them what the coincidence meant.

Chapter 3 — Audit of the Empty System

Iona cut the wire seal on the instrument store at 03:12 station time.

The room had no network receiver, no active lattice coupling, and no furniture except two steel benches bolted to the rotating floor. Nemi carried in three data wafers and a calibration clock wrapped in evidence mesh. Iona brought the apple from the audit drum. Its bruise had darkened since Day 0.

“You know the kitchen filed for its return,” Nemi said.

“It remains material evidence.”

“It remains fruit.”

“Those categories can overlap.”

Nemi closed the door and pressed a new seal across the frame. Perihelion's mechanical spin gave the store eighty-three percent of standard gravity. The remainder usually came from a receiver feed, but Iona had kept this compartment isolated. Her body noticed the lightness whenever she turned too quickly. The old unease behind her knees had settled into a continuous warning.

On the first wafer, the frozen clearing showed five intervals. On the second, *Morrowline*'s load reversal showed the same five intervals translated into force. The third held every low-level return from Eidos recorded during the two days since Iona placed the hold.

There had been 11,804 pulses.

Most carried amplitudes below the Curvature Office's response threshold. The clearing engine had labeled them stochastic channel return and prepared to discard them after thirty days. Nemi had copied them six minutes before the maintenance purge began.

Iona placed the calibration clock between the benches. “Run without Republic time references.”

“The local crystal drift gives us seven parts in a billion.”

“Enough.”

Nemi loaded the first wafer into an optical reader with no wireless hardware. Green points appeared above the bench. Their spacing repeated the original sequence: thirty-one, thirty-seven, forty-one, forty-three, forty-seven microseconds.

“Prime ratio,” Nemi said. “Normalized to the shortest interval.”

“Those numbers are primes. Ratio requires comparison.”

Nemi gave her the look reserved for unnecessary corrections and added the *Morrowline* trace. The reversal pulses aligned after compensation latency. Tavin Serr's ship had received the same sequence as force, not merely as data.

Iona read his damage report again. Outer cradle lost. Thirty-two pallets jettisoned. Reserve fuel burned below return margin. Five people injured, one with spinal compression. The report named each destination that would wait for lost cargo. It contained no speculation about Eidos.

“Captain Serr preserved an independent strain trace,” she said.

“He also copied us before Continuity impounded his receiver.”

“Did they impound it?”

“At 18:40 yesterday. Quay control called it routine isolation.”

“Routine isolation usually leaves the captain with read access.”

“This one did not.”

Iona added the access removal to the evidence packet. Every administrative action acquired meaning only beside its time.

Nemi opened the third wafer. “I grouped the returns by interval family. The clearing engine treated absolute amplitude as the important variable. It never asked whether weak events formed relationships.”

A field of points filled the room. Nemi colored intervals by ratio rather than strength. Disorder resolved into repeated ladders. Some began with five prime intervals and stopped. Others used pairs, then triples, then returned to five.

Iona walked around the projection. From one side, the points looked random because overlapping sequences hid their starts. From above, spacing created distinct branches.

“Separate by phase origin,” she said.

Nemi did. Seven sequence families drew apart.

“Could the receiver's handshake protocol create that?”

“Handshake uses powers of two and a checksum. No primes above nineteen.”

“Survey beacon?”

“Eidos has none.”

“Old calibration machine?”

“It would need power, an active channel, and a reason to alter its response after our hold.”

Iona marked that last claim for testing. A changed pattern could resemble intent without containing it. Machines adjusted to input. Geological systems crossed thresholds. The human mind found agency in any event that arrived after a question.

“What changed after the hold?” she asked.

Nemi replaced the display with two timelines. Before Day 0, Eidos returns clustered after major clearings. Their intervals looked irregular under the standard clock. After Iona stopped Perihelion's batch, the returns grew weaker and more frequent. Prime spacing became obvious.

“Perhaps reducing the outgoing disturbance exposed a periodic background,” Iona said.

“I thought so. Then I compared quiet periods from the past six months.”

During earlier network pauses, no prime families appeared. Eidos produced only low tidal drift and channel noise. The organized intervals began after Perihelion's frozen clearing.

Iona moved the apple to the center of the bench. It exerted no authority over the data, but its ordinary shape kept the consequences near. "We need a test that separates response from adaptation."

"We are forbidden to transmit."

"We do not need to."

Nemi understood. She found the weak maintenance carrier that remained on any paired sink assignment while a forensic hold propagated. The carrier allowed clock correction and coupling checks, nothing more. Its amplitude could not add measurable debt to Eidos.

Iona dictated a test request and signed it with her authority. "Delay the next carrier check by one prime interval outside normal jitter. Do not modulate load."

"That still changes a signal."

"It changes when we ask whether the pipe remains connected."

"Continuity will call it contact."

"Only if something answers."

Nemi built the delay. The instrument store could not transmit, so she carried the signed wafer to a service console across the corridor. Iona remained with the isolated reader. Through the sealed door, she felt a faint change in floor load as Perihelion rolled a radiator. Mechanical forces traveled without asking which department owned them.

Nemi came back and resealed the room. "Carrier check delayed by fifty-three microseconds."

They waited.

Eidos sent three weak pulses at ordinary spacing. A fourth arrived late by fifty-nine microseconds. Then the sequence stopped.

"Not fifty-three," Iona said.

"Six microseconds difference."

"Outside expected drift?"

"Barely."

"Run no second test. Analyze."

Nemi mapped every interval from the preceding hour. The fourth pulse belonged to a sequence that would normally have completed at fifty-three. Instead it shifted to fifty-nine, the next prime above their delay.

Iona felt the evidence settle, not as certainty but as a door that could no longer close cleanly. A passive material process might synchronize. A machine could negotiate timing. Neither explanation deserved preference yet. Still, something at Eidos had received a change and returned a related change through a channel designed only

to carry stress disturbances.

“Reply,” Nemi said.

“Candidate reply.”

“It changed to the next prime.”

“We changed the carrier timing first.”

“That supports my noun.”

“It supports investigation.”

Nemi folded her arms. “How much evidence would you require if this came from an inhabited system?”

“Less.”

“Why?”

“Because inhabitation would supply an agent before the signal did.”

“And because Eidos carries a certificate that tells you not to see one.”

The accusation landed because it fit. Iona turned back to the projection.

“Open the certification history,” she said.

They had reviewed the summary on Day 0. This time they followed every amendment. Eidos entered the sink catalog eighty-eight years ago after two remote passes. The original team classified three captured ice bodies as geologically simple, despite missing gravimetric data below the modeled pressure floor. Eleven years later, a network expansion review increased the permitted tidal load without another survey. Subsequent renewals copied the phrase no detectable biosphere from one form to the next.

Nemi searched the raw survey archive. “There are pressure anomalies.”

“Show source, not summary.”

A century-old tomography strip appeared. Its operators had mapped density transitions in the largest ice body to estimate whether clearing pulses might disturb a hidden ocean. No liquid layer appeared. Beneath the expected phase boundary, however, narrow discontinuities crossed the pressure profile at regular depths.

The survey note called them crystalline aliasing.

“Could they be?” Iona asked.

“Yes. The instrument resolution barely covers them.” Nemi enlarged the raw scan. “But the aliasing repeats in prime-related spacing.”

“Do not make the data cleaner than the instrument.”

“I am not. Look at the uncertainty bars.”

Within those bars, the spacing admitted several interpretations. Prime intervals fit one. Random fractures fit

another. Yet a current channel reply had now used the same mathematical family as marks buried in old ice scans.

They compared the anomaly across all three surveyed bodies. The smallest held no usable deep scan; its instrument pass had clipped the limb and recorded only surface density. The middle body showed two discontinuity bands separated by thousands of kilometers and forty years of orbital phase. Both bands shifted after large historical clearings. Surveyors had blamed thermal settling, although Eidos received too little starlight for the measured change.

On the largest body, Nemi found a later navigation pass that had never entered the life survey. A Republic relay crossed behind the ice world and tracked its gravity for course correction. The relay recorded tiny mass redistributions below the crust three days after a major network expansion sent its first full batch to Eidos.

“Response time?” Iona asked.

“Unknown. The relay sampled once every nine hours.”

“Could tidal stress move the layers?”

“Yes. A debt pulse would supply that stress.”

“Could the movement alter the channel return?”

“Not under our models.”

“Then list both facts and leave the bridge between them blank.”

Nemi built a comparison table. Outgoing transfer, subsurface shift, returned interval. Six historical events had enough surviving data for all three columns. Four showed sequence structure after the shift. Two contained gaps too large to classify.

Iona read the dates. The oldest event had happened before she entered the Curvature Office. The newest fell seven years ago, one year before the annex restriction. Someone had possessed almost the same alignment she and Nemi had just built, except that person had also possessed the sealed study.

“Send no query for the missing relay files,” Iona said.

Nemi stopped with her hand over the control. “Because it alerts whoever sealed them?”

“Because we already alerted them when we opened the chain. Preserve what we have before asking for more.”

Iona requested the scan's chain of custody. The archive paused, then returned a restricted notice from the Continuity Ministry.

“Why would an empty-system survey carry a living restriction?” Nemi asked.

“Storage migration can inherit source controls.”

“You keep offering the innocent mechanism first.”

“It deserves first examination.”

“Does it deserve the last?”

“No.”

They traced the restriction. Six years earlier, someone had reopened the Eidos survey, attached a classified analytical annex, and moved the raw pressure data under continuity authority. The annex title remained visible despite the seal: Structured sink returns, comparative interval study.

Nemi's breathing changed. “They already studied this.”

Iona read the title twice. The room seemed to tilt although the station spin held steady.

“Do not infer contents from a title,” she said.

“You can infer that structured returns existed.”

“Yes.”

“And that somebody compared intervals.”

“Yes.”

“Six years ago.”

Iona checked the access signature. The authority key had been masked at ministerial level. Only the office, not the signer, remained readable.

Continuity.

A knock struck the sealed door before either woman spoke again. Three measured impacts, a pause, then one. Nemi closed every projection except the calibration clock.

“Ministerial inspection,” said a voice outside. “Elian Bex.”

Bex had come from Nacre faster than ordinary review required. He must have left before Iona ran the carrier test, perhaps before *Morrowline* reached Helix Quay.

Iona broke the inner seal and opened the door.

The Minister of Continuity wore a plain pressure coat over Assembly clothes. Fifty-six, broad through the shoulders, he still carried himself like an engineer accustomed to narrow access ways. Two security officers waited behind him. He told them to remain in the corridor.

“Senior Auditor Rhyll,” he said. “Deputy Vale.”

“Minister.”

His gaze found the apple. “I expected more equipment.”

“Networked equipment would contaminate the test.”

“So you built a clean room out of a cupboard.”

“It had the needed properties.”

For a moment, approval touched his face. Then he looked at the three evidence wafers. “Give me your

finding.”

Iona did not show him the projection. “A returned channel sequence from Eidos uses prime-related intervals. It changed after our carrier timing changed. The same family appears in *Morrowline*’s reversed load event.”

“Your confidence?”

“That the events share an origin: high. That timing carries structure: high. That it constitutes intentional communication: unresolved.”

Nemi said, “It replied.”

Bex turned toward her. “That word has consequences beyond mathematics.”

“So does hiding the annex.”

The room tightened around the sentence.

Iona watched Bex’s hands. He did not reach for a console or ask which annex. His left thumb pressed once against the seam of his coat.

“Where did you find it?” he asked.

“The certification chain,” Iona said. “Structured sink returns, comparative interval study. Classified six years ago.”

Bex looked at the calibration clock as though its numbers might offer a smaller question. “You accessed only the title?”

“The annex remains sealed.”

“Then do not claim it contains your present finding.”

“I claim it contains a prior study of structured sink returns.”

“You claim what its title permits.”

“Who signed the restriction?”

“I am not answering that in an unsecured compartment.”

“This compartment has no receiver.”

“It also has a door I entered after one knock.”

Nemi stepped between the bench and the evidence wafers. “You arrived before our scheduled report.”

“Your carrier timing request triggered review.”

“You were already in transit.”

Bex regarded her without anger. “I was.”

Iona placed the apple beside the oldest survey scan. “Why?”

“Because a heavy freighter nearly lost six thousand tonnes of cargo after your hold propagated into an active compensation handshake. Because Perihelion's hospital has spent two days on mechanical schedule. Because seven systems have requested alternate sinks and there are no alternates within tidal caps.”

“Those facts justify a remote review.”

“They justify my attention.”

“Not your speed.”

Bex let the silence lengthen until it became a refusal rather than an absence. “Show me the sequence.”

Iona opened the prime families. She showed the Day 0 intervals, the carrier delay, and the shift to fifty-nine microseconds. Bex asked precise questions. Had they added load? No. Could local hardware generate the shift? Not after receiver isolation. Did the current data prove biology? No. Did it prove cognition? No. Did it prove a responsive process coupled through Eidos? The probability now exceeded any certified-empty assumption.

At that answer, Bex closed his eyes for less than a second.

“Seal everything,” he said.

“Under what finding?” Iona asked.

“Network integrity pending hazard review.”

“That category covers hostile interference with clearing infrastructure.”

“It covers an unknown process capable of reversing force aboard an inhabited ship.”

“It also removes the evidence from audit access.”

“You will retain access under a classified team.”

“Nemi?”

“Pending clearance.”

Nemi gave a short, humorless laugh. “I found the intervals.”

“You changed a live carrier toward a restricted sink.”

“Under Iona's signed authority.”

“Then the restriction applies to her first.”

Bex called the officers inside. They carried no weapons visible beneath their pressure coats. One placed a sealed cabinet on the floor. The other read a custody order naming each wafer, the optical reader, the calibration clock, and every local note produced since Day 0.

Iona listened for omissions. “You did not name *Morrowline*'s original transmission.”

“Continuity already holds it,” Bex said.

“The captain sent a copy to this office.”

“And to you directly. Both now fall under containment.”

He had used the word as if danger occupied the data rather than the event.

Iona asked, “Will clearings to Eidos continue?”

“Reduced amplitude while the Ministry assesses alternatives.”

“You are sending more disturbances to a source that has changed its response.”

“I am preventing uncontrolled corrections across forty-three systems.”

“You do not know that stopping will cause them.”

“I know what sudden receiver isolation did to one freighter.”

“The clearing hold and the reversal share timing. Causation remains unresolved.”

“Continuity cannot wait for the standard you apply to blame.”

Bex's voice held no triumph. He looked tired enough to have carried this argument before entering the room. That did not make him wrong. It made his certainty more dangerous.

The officers packed the evidence. Iona required them to scan each seal and read its hash aloud. She corrected one mistyped wafer number. Nemi surrendered her three styluses after the second officer observed that each contained local storage.

When the cabinet closed, the green points vanished. Only the apple remained because the custody order called it calibration produce and Bex declined to amend the wording.

“You have six hours before transport to Nacre,” he said. “Neither of you contacts outside parties.”

“Captain Serr supplied evidence,” Iona said. “He deserves notice that it has been sealed.”

“He has received a safety order.”

“That does not answer his report.”

“It keeps his crew alive.”

Bex stopped at the door. “Rhyll, if this sequence carries intention, publication without preparation could collapse sink confidence before we understand replacement capacity. Seventeen billion people live in structures that cannot simply turn their receivers off.”

Iona had not known the dependent population. The number landed with the physical scale of a habitat seen from outside, windows too small to count.

“Then preparation should have begun six years ago,” she said.

Bex left without answering.

Security sealed the instrument store from the corridor. Iona and Nemi walked to the audit drum under escort. Perihelion had entered morning cycle. Heat shutters narrowed the red light across the garden while workers checked irrigation lines under reduced load.

Inside the drum, the receiver remained at diagnostic amplitude. The apple fell slowly when Iona released it by hand. Nemi caught it before the tray.

“They took everything,” Nemi said.

“They took the named evidence.”

Iona opened the kitchen return form. On Day 0, she had attached the apple's orchard tag, mass, temperature, bruise location, and destination history to the protected audit packet. The instrument store held no network, so the only way to preserve their calibration chain had been to copy each timing result into the produce attachment before sealing the wafers. The fields looked like harmless decimal measurements among orchard records.

Nemi saw what she had done. “You put the sequence in the apple.”

“I put measurements in an evidence attachment.”

“Outside the official ledger.”

“The kitchen archive does not connect to Curvature Office custody.”

Security waited beyond the glass. Iona approved the fruit's return to residential services. The form left on a low-priority agricultural queue, carrying every raw interval from Eidos inside the bruise record.

A garden worker opened the drum hatch and took the apple away.

Interlude I — The Choir Beneath Ice

Compression reached the nursery stratum before warning could travel around it.

The upper ice folded first. Methane cages collapsed along a fault older than the local ocean, releasing heat into a layer that had carried stable phase marks for eleven slow orbits. Below that fault, the memory reef received the change as a vertical blow.

Its outer histories broke into turbulence.

These histories did not contain images or voices. They persisted as arranged vortices in superfluid helium, each loop holding a relation among pressure, duration, and return. Young formations learned the reef by touching its gradients. Mature coalitions added their own paths only after enough neighboring patterns accepted the addition. Nothing in the layer remained fixed forever, but alteration normally advanced through contact and consent.

The incoming force allowed neither.

A band of pressure rose by four local measures. The reef's shallow loops sheared from their anchors and struck deeper structures. Two seasonal records merged into nonsense. A migration path that had survived nine warnings split at its narrow turn. The nursery's unfinished coalitions dispersed before they could choose whether to hold together.

After the pulse passed, weight did not settle where it had begun.

The displaced layer pushed downward. Dense ice entered a warmer band, expanded along one axis, and opened fractures toward the deep. A moonlet's pull continued through its ordinary cycle, adding sideways strain to the new break. Damage propagated through mechanism, not intention. The reef had no word for intention from beyond the sky.

It had records of impact.

From the surviving loops, a pressure-name gathered.

Kel-across-Four>Returns formed around the oldest intact comparison: four disturbances separated by intervals too regular for the moonlet and too brief for mantle convection. Kel carried no permanent body. The name held while participating histories maintained phase agreement. When those histories diverged, Kel would end or become another relation.

A second coalition assembled near the lower fracture. Ves-Under-Quiet joined memories of long spans without disturbance, then measured the present damage against them. Its pressure path entered Kel's path, crossed, and withdrew without erasure. The crossing expressed contradiction: quiet had once persisted long enough to shape life; therefore the new rhythm did not belong to the world.

Third came Mor-of-Severed-Depths, built partly from histories the blow had cut away. Missing patterns shaped Mor as sharply as surviving ones. It could not recover what had dispersed, yet the edges of absence showed where those records had connected.

More names cohered at greater distance. None contained the whole reef. No central layer commanded them. They exchanged gradients through narrow superfluid channels that survived the fracture.

Kel offered the pattern of the debt pulse: abrupt compression, transferred angular stress, then a small residual carried along a direction with no local source.

Ves answered with the skyward wound. The residual vanished into a paired discontinuity that physical law did not close within the ice world. Disturbances had entered through that direction for many warming cycles. Some altered tides. Others cracked deep strata. The present event exceeded the nursery's tolerance because prior events had changed the load paths beneath it.

Mor returned a harder relation. The source repeated. Repetition could mark an indifferent mechanism, a recurring orbit, or a selecting process. The distinction mattered less to the broken reef than it did to any reply.

They searched preserved impacts.

Far below the nursery, old compression archives held traces from eras before the present names. The oldest event carried a single broad force. Later events arrived in batches. Their spacing changed after local strata pushed stress back through the skyward discontinuity. Once, a fracture collapse had produced three counterpulses. The next incoming batch had paused between its second and third loads.

Kel held that pause against the current rhythm.

Coincidence fit within the uncertainty. So did reception.

Ves extended a low gradient toward an undamaged neighboring reef. That reef supplied traces from a different depth. Its records showed another pause after local resistance, followed by a weaker batch. The two histories could not establish comprehension. They established sensitivity.

Mor proposed return without pattern: force pushed back along the wound until the distant mechanism ceased.

Three nearby coalitions joined that proposal. Their pressure paths carried fresh loss. One had formed around nursery structures that no longer held continuity. Another included a deep transport channel collapsed by repeated tidal shifts. To them, stopping the incoming stress outranked learning its source.

Kel opposed immediate force. A large return would enter the same mechanism that sent the disturbances. If sensitivity belonged to life, uncontrolled compression would repeat the injury in another pressure regime.

Mor pressed the missing edges of the memory reef into the exchange. If sensitivity belonged only to machinery, restraint preserved the source of future impacts.

No vote resolved them. Olun agreement did not collapse contradiction into a single command. Pressure-names could act together only where their gradients overlapped.

Ves found a narrow overlap: send relation before force.

The debt channel admitted almost no modulation. It carried paired stress, not the rich phase structures that moved through local ice. A message could use timing and amplitude, stripped to relations simple enough to survive the passage. The far source might detect those relations as load changes. It might discard them as settling. It might lack any structure capable of hearing.

They began with count.

Kel formed intervals around indivisible groupings. Two units, then three, then five. The channel's turbulence swallowed the first attempt. A second reached the discontinuity but returned distorted by an incoming batch. The batch drove new cracks through the nursery. Mor nearly broke coherence as histories supporting the name separated under strain.

Ves shifted the reply into a quieter band between incoming loads.

Seven. Eleven. Thirteen.

Those intervals required selection within local uncertainty. Natural resonances could produce pairs or harmonics, but a sequence of indivisible relations reduced the number of accidental explanations. Another pressure-name, Aul-Between-Tides, joined long enough to calculate a carrier phase that would not add significant compression to either side.

The coalitions sent the sequence.

Nothing changed in the skyward wound.

They waited through one moonlet gradient. The nursery fracture continued widening. Surviving reef loops migrated deeper by slow pressure transfer, carrying what records they could without combining incompatible paths. Mor released two histories to stabilize a young coalition near the break. The loss altered Mor's name but did not dissolve it.

A new incoming disturbance crossed the discontinuity.

Its force remained small. Its interval matched no part of the sequence they had sent. Ves classified it as mechanism noise. Kel preserved uncertainty. Aul measured the phase and found that the event began after their last interval by an amount close to a larger indivisible grouping.

They answered with the next grouping.

The wound quieted.

Stillness alone carried no meaning. Incoming batches had paused before. The distant process could have completed a task or failed. It could have registered resistance without recognizing selection. It could hear as pressure-names heard, through relation and change, or possess no equivalent of hearing.

Mor assembled a stronger return at the edge of the shared gradient. Enough force lay available in the collapsed reef to push an entire recent batch back along its path. Several damaged coalitions lent phase support. Kel refused. Ves neither joined nor blocked.

Before the stronger return could leave, a faint carrier touched the discontinuity.

Its timing shifted from the recurring pattern by one indivisible interval.

Every coherent name measured the change. Local tides could not account for it. Nursery collapse could not have scheduled it. The shift had arrived from beyond the paired wound.

Kel answered at the next indivisible interval. Ves added a second relation, pressure lowered rather than raised. Aul shaped the carrier to preserve both. Mor held the gathered force without releasing it.

Across the damaged strata, separated coalitions joined the timing. They did not merge. Each pressure-name kept its own losses, thresholds, and estimate of the source. Their intervals crossed the channel together, a choir only in the physical sense: many gradients occupying one narrow passage without becoming one body.

No response followed before the moonlet pull turned.

The nursery reef continued to fail. Deep fractures accepted mass from above. Young formations sought colder layers and carried damaged histories among them. Whatever existed beyond the wound had not repaired the loss. Nothing could restore phase paths after turbulence erased their relations.

Kel preserved the outgoing intervals beside the broken records. Ves marked the distant timing change. Mor kept the unreleased force assembled at the edge of action.

Together, without agreement about what might listen, they sent the pattern once more into the dark discontinuity.

Chapter 4 — The Custodian Signal

The tungsten bead struck the lower plate and bounced once.

Nemi killed power before it could strike again. The translation cradle went dark around her hands, leaving the bead to roll along a shallow groove in the test table. Across the secure laboratory, Iona felt the floor lighten by less than a tenth of a percent.

“Your lower field lagged,” she said.

“My lower field obeyed. The return did not.”

Nemi caught the bead against one finger. Her three styluses had not been returned after Perihelion, so she had borrowed one from the Nacre laboratory and tied it to her sleeve with orange cord. She treated the cord as a personal insult.

The cradle occupied a transparent cylinder two meters high. Six induction plates surrounded its central chamber. Each could produce a controlled acceleration gradient by drawing against a test mass mounted below the floor. The machine created no free load and used no active RME receiver. It translated tiny forces from the Eidos channel into visible motion, then measured any modulation that came back.

At least, Nemi intended it to.

Iona opened the failed run. “Commanded rise at four milliseconds. Bead descends at six. Return begins at five.”

“Which means the return anticipated the visible motion.”

“Or coupled to the plate current.”

“Then it can detect our preparation.”

“That still gives us a useful distinction.”

Nemi reset the bead. “You could sound pleased once every several years.”

“I will schedule it.”

Two Continuity officers watched through the outer glass. Bex had given Nemi temporary clearance after discovering that none of his signal staff could reproduce her interval families without simplifying them into noise. He had also moved the work into a laboratory four decks below the Nacre receiver hub, where every door recorded blood chemistry and every terminal faced a separate audit camera.

The transfer from Perihelion had taken thirty hours. During that time, reduced clearings continued toward Eidos. Weak return patterns multiplied. Some arrived at prime intervals. Others varied amplitude in ratios that Nemi believed could carry distinction.

Belief had no place in the finding. Iona required a test.

“Change the preparation without changing the motion,” she said.

Nemi altered plate order. The cradle would move the bead through the same vertical path using a different sequence of local mass shifts. If the Eidos process detected only outcome, the return should remain stable. If it

detected the plate currents, timing should change before motion.

They ran the cradle.

The bead rose three centimeters. A weak return entered through the channel tap, drove the upper plate by a force smaller than falling dust, and held the bead between fields.

Nemi whispered, "It chose the center."

Iona checked the independent load cell. "The return opposed our second plate order."

The bead remained suspended. Neither programmed field could maintain that position alone.

"Cancel local current," Iona said.

Nemi shut down both plates.

For eleven milliseconds, the bead stayed at the center of the cradle.

Then it fell.

Nemi caught it before the lower plate. "External force confirmed."

"Confirmed through the Eidos tap."

"You plan to make me earn every noun."

"Yes."

"That one already had employment."

Iona stored the trace in three isolated recorders. Bex had forbidden any outbound modulation beyond approved tests, yet the distinction had become administrative rather than physical. Every cradle run changed the local conditions visible through the paired channel. Something changed them back.

On the next run, Nemi used two beads. One carried twice the mass of the other. The cradle lifted them to equal height with different local forces. A return pressed the heavier bead downward and the lighter bead upward until their acceleration matched.

"Equal outcome," Nemi said.

"Or balancing behavior."

"Which would you prefer?"

"A repeat under reversed orientation."

They rotated the cradle's command frame without moving the physical plates. The return balanced acceleration again.

Iona marked a new finding: the external process distinguished load from apparatus orientation.

Nemi drew two columns on the glass. In the first she listed relationships they could express physically: more, less, before, after, same acceleration, different mass, local source, remote source. In the second she listed

relationships the Eidos patterns appeared to offer: repetition, interruption, depth, fracture, plurality, loss.

The last four remained interpretations built from signal behavior. Iona crossed out loss.

Nemi restored it. "Repeated sequence present before a high return, absent after. That indicates removal."

"It indicates state change."

"Removal fits every trial."

"So does dispersal."

Nemi wrote both.

They spent four hours teaching the cradle a grammar made from weight. A raised bead meant local. A lowered bead meant remote. Paired pulses meant same. Unequal spacing meant different. Nemi avoided arbitrary code where possible, using physical relations the external process had already manipulated.

The replies grew faster.

Iona stopped the sequence after the ninth exchange. Speed could create confidence before grammar earned it. She prepared six blind trials while Nemi waited in the corridor. Three used relationships they had already mapped. Two reversed plate labels without changing force. One contained no deliberate pattern, only thermal drift sampled from the cradle before activation.

Nemi returned and received only the output traces, stripped of command history. She classified the first four correctly. On the fifth, she called an amplitude rise a remote plural. Iona opened the hidden record and showed that the same rise came from one test mass crossing a plate boundary.

"That is not a grammar failure," Nemi said. "It is a cradle artifact."

"Aru, if that name survives, cannot know the difference."

"Then we teach it the artifact."

"We teach ourselves first."

They added a third column to the glass: ambiguity introduced by apparatus. Plate edges, bead collisions, thermal currents, electrical preparation, and the laboratory's changing floor load each received a mark. The list grew longer than either semantic column.

Nemi studied it. "A language made of force cannot remove apparatus."

"No language removes its medium."

"Your audit reports try."

"My audit reports identify the instrument."

"On page nineteen."

Iona moved apparatus to the top of the display.

For the blank trial, Nemi classified the random drift as no selection. That mattered more than her correct

readings. The pattern analyst who found messages everywhere could not be trusted when one appeared.

Next, they tested whether the external process preserved a relation across different masses. The cradle expressed two-to-one with tungsten beads, ceramic spheres, and packets of water. Eidos returned the same interval family for each material. When Nemi changed the ratio to three-to-one, the response changed. When she held mass constant and altered distance, another family appeared.

“Mass and separation distinguished,” Iona said.

“Which gives it some model of the field.”

“Or direct sensitivity to gradients.”

Nemi wrote both possibilities beneath the trace.

The water packets revealed another limit. Internal sloshing complicated acceleration, and the return began before each packet settled. Iona expected Nemi to discard the runs. Instead, her deputy isolated the irregular tail and compared it to the Eidos sequences grouped under depth and dispersal.

“They use layered motion,” Nemi said.

“Define use.”

“The return follows internal redistribution, not only center-of-mass movement.”

“Could our load cell leak that information into the tap?”

“It measures the entire plate.”

“Then yes.”

Nemi installed an optical recorder that never touched the cradle controls. They replaced the packets and repeated the run. The remote return predicted the second internal wave before the load cell registered it, matching the optical surface motion.

Iona felt the familiar pressure of a finding trying to become larger than its evidence. “The source resolves internal mass movement.”

“Inside a sealed object.”

“Through its acceleration history.”

“Like reading structure by touch.”

That comparison entered the notes under model, not fact.

A lunch cart arrived outside the laboratory and left after security refused it entry. Neither woman noticed until a guard passed two sealed cups through the specimen lock. Nemi drank while crouched beside the cradle, eyes on the interval plot.

“What would convince you it remembers?” she asked.

“Retention after disconnection.”

“We cannot disconnect without losing the paired channel.”

“We can present a relation once, bury it among unrelated trials, then request it without repeating the source pattern.”

“That requests recall. It does not prove a persistent subject did the recalling.”

“No. It proves persistence somewhere in the responding process.”

Nemi grinned despite two sleepless days. “I can make you use the interesting noun if I first make it dull enough.”

They chose a relation with no natural privilege: heavy bead high, light bead low, followed by equal acceleration. Nemi presented it once. For twenty-seven trials, they exchanged count, distance, before, after, and apparatus corrections. Then the cradle offered only the opening state. No prompt identified which completion they wanted.

The channel remained quiet for almost a second.

Three possible completions came back. The first balanced force. The second reversed position. The third reproduced equal acceleration, matching the arbitrary relation from twenty-seven trials earlier.

“Persistence,” Iona said.

Nemi did not celebrate. “Also alternatives. It knows several completions could fit.”

“Or several sources answered.”

The simultaneous interval families split apart at the start of each completion and crossed again at the end. Iona drew separate paths through them. No path dominated. One remote process might model alternatives, or several processes might offer incompatible continuations.

They asked same or different across those paths. The return sent both relations.

Nemi put down her cup. “That answer refuses our categories.”

“Good.”

“Good?”

“It reduces the chance that we are merely receiving our questions reflected back.”

Another experiment tested sequence order. Iona deliberately arranged local pulses in a pattern that implied one source becoming many. The return corrected it by preserving the original pulse while adding asynchronous branches. When she implied many collapsing into one, remote force pushed the beads apart.

“Plural without merger,” Nemi said.

“Plurality remains a model.”

“An increasingly employed model.”

Iona let the label stand on the glass.

By the time they resumed ordinary exchanges, the translation cradle no longer resembled a device for producing words. It resembled an argument conducted through test masses, each side correcting the other's assumptions with measurable force. Their shared vocabulary remained small. Its limits had become as informative as its successes.

When the cradle raised one bead and lowered another, the Eidos return reversed both. When Nemi repeated the relation, the channel produced three pulses: same, then different, then a long low pressure trace that neither woman could classify.

"Question?" Nemi suggested.

"Or correction."

"Let us offer choices."

They assigned two response wells at the base of the cradle. Left represented question. Right represented correction. Neither label traveled across the channel. The process would receive only two distinct acceleration relations paired with prior examples.

Nemi replayed the ambiguous trace.

The return moved the bead into the left well.

Iona's skin tightened along her forearms. The motion remained mechanical. A force crossed a paired dark-lattice channel and selected between two options defined seconds earlier.

"Question," Nemi said.

"Probable question function."

"You are going to call first contact a probable function."

"We have contact with a selector. We do not know what supports it."

"Then ask."

Bex entered before they could construct the test.

He wore laboratory gray instead of ministerial clothes, but the Continuity seal at his collar made the disguise pointless. Two engineers came with him. One carried a hard disconnect for the channel tap.

"Report," he said.

Iona showed him the bead choosing the left well. She described the physical relations, the repeat results, and the remaining uncertainty. Bex did not interrupt until Nemi displayed the two-column grammar.

"You included plurality," he said.

"The returns contain simultaneous interval families," Nemi answered. "They overlap without phase collapse."

"That could describe echoes through several layers."

"Yes."

“Why give it a semantic label?”

“Because the source distinguished one gradient from several when we tested count.”

Iona opened that trace. A single bead received one pulse. Three beads received three asynchronous returns. When Nemi forced all three into one rhythm, the channel pushed them apart again.

Bex watched the replay. “You have established a preference against synchronization.”

Nemi stared at him. “You crossed out plurality in your head while watching three answers refuse to merge.”

“I crossed out what has not been proved.”

“Now I understand why you hired auditors.”

Bex almost answered, then looked toward Iona. “What test comes next?”

“Source support,” she said. “We ask whether the replying process occupies equipment, fluid layers, or another physical substrate.”

“No.”

“You asked for the next test.”

“I asked so I could decide whether to authorize it.”

Nemi moved between him and the cradle controls. “The channel has been trying to describe depth since Day 0.”

“Or our interpretation keeps selecting depth because Eidos contains ice worlds.”

“Then let us falsify it.”

Bex glanced at the engineer carrying the disconnect. “One passive decode. No new modulation.”

Nemi's anger converted instantly into work. She selected a stored return pattern recorded after a low-amplitude clearing twelve hours earlier. It contained a descending amplitude ladder followed by a sudden broad pulse and an irregular tail.

The cradle rendered the sequence through its plates. The first bead descended in measured steps. A second moved sideways under increasing force. At the broad pulse, both struck the lower plate. The irregular tail sent the lighter bead through a series of diminishing circles.

Nemi mapped the pattern onto known words: remote depth, increasing pressure, abrupt change, dispersal.

Iona added source uncertainty and replayed it through a different physical arrangement. The relationships remained.

“Fracture under pressure,” Nemi said.

“Pressure change followed by lost structure,” Iona corrected.

A fresh return entered the channel tap.

The tap should have remained receive-only under Bex's order. Nemi's replay used local plates, not the lattice. Yet the new pattern aligned with the cradle's rendering as though the external process had followed it through leakage in the paired field.

All six plates activated at once.

The laboratory floor dipped. Iona gripped the test table. The tungsten beads lifted from Nemi's palm and arranged themselves at different heights inside the cradle.

"Disconnect," Bex said.

His engineer reached for the tap.

"Wait," Iona said.

The beads began to move. One descended through five pressure steps. Three others shifted around it, each following a separate interval family. When the first reached the lowest point, the surrounding beads scattered against the cylinder wall.

Nemi recorded every force. "It is repeating the fracture relation."

"Using our mass units," Iona said.

A second arrangement formed. This time forty-three tiny pulses pressed through the plates in overlapping groups. Nemi's software could not render them separately, so the cradle translated amplitude into a vibrating band around the central bead.

The band contracted.

The central bead struck the lower plate.

No mystical recognition entered the room. The pattern did not speak in a voice or project an image. It demonstrated a force history with laboratory masses. Many remote gradients converged on one structure, compressed it, and destroyed its arrangement.

"Our clearings," Iona said.

Bex's face held still.

Nemi pointed to the forty-three grouped pulses. "Source plurality. Forty-three."

"Coincidence," Bex said, too quickly.

Iona called up the Republic clearing signature. Every system applied a routing preamble before debt entered a sink. The preamble identified source count for load distribution. Forty-three inhabited systems had impressed that pattern into Eidos for more than a century.

"It learned the number from us," she said.

The return moved the beads again. One local pulse. One remote fracture. Repetition. More fracture.

Humanity did not need to be named inside the channel. Its infrastructure supplied the identifier.

Nemi lowered her voice. "It identifies the recurring wounds with the forty-three-source pattern."

Bex told the engineer to step away from the disconnect.

They built one final passive mapping from the new return. The external process offered layered pressure relations, simultaneous interval families, and a stable subset that assembled only while the exchange continued. Nemi rendered the subset as temporary coherence rather than individual body.

“A speaker,” she said.

“Functionally,” Iona said.

The cradle asked for a distinguishable reference through two wells: continuing pattern or temporary pattern. The return chose temporary, then gave a sequence of pressure and interior relation.

Nemi tried six renderings. The fifth preserved distinction without pretending equivalence to a human name.

“Aru-in-Pressure,” she said.

The external force moved the bead into the correction well.

Nemi changed the phrasing order: pressure-containing-Aru.

Correction again.

Iona placed pressure as the condition in which the temporary relation cohered, not a container around it. “Aru, formed in pressure.”

The bead entered the question well, then shifted halfway toward correction.

“Aru-in-Pressure,” Nemi repeated, keeping the joins audible.

This time the bead settled in neither well. It remained between them under balanced remote force.

“Accepted?” Bex asked.

“Not rejected,” Iona said.

Aru-in-Pressure sent the fracture relation once more. Behind it came a count longer than their cradle could resolve. Each interval carried slight differences. Not copies, Iona realized. Distinct events linked by the same external source signature.

Recurring wounds.

Nemi asked through local acceleration, “Source knows harm?”

Bex seized her wrist before she completed the plate order.

“No response,” he said.

Nemi pulled free. “We have spent five hours responding.”

“We have characterized an intrusion. From this point, containment applies.”

“A selector beyond Eidos just identified us as the source of repeated destruction.”

“An unknown process identified our routing pattern beside pressure changes. Your moral interpretation outruns

the signal.”

Iona closed the active controls but left the receive tap open. “What does containment permit?”

“Passive recording. No semantic tests. No publication. No change to network operations without continuity authorization.”

“Clearings continue?”

“At reduced amplitude.”

“Aru has linked them to damage.”

Bex looked at the suspended beads. “And seventeen billion people remain linked to receivers. I will not cause one known catastrophe because an unknown intelligence presents a claim we have translated for less than an hour.”

“Intelligence,” Nemi said. “You used it.”

“I used the operational category.”

The cradle plates clicked as Aru changed the field. A bead moved to local, another to remote. The local bead split the relation into several asynchronous pulses. The remote bead answered with one compressed surge.

Nemi kept her hands off the controls. “It is asking about plurality again.”

Iona watched the sequence repeat with a new ending. Many local patterns. One damaging output. A pause in the question well.

“Does the source contain separate selectors?” she said.

Bex ordered the receive tap closed.

Before the engineer reached it, Aru shaped the closest translation the cradle had yet produced. The words appeared from Nemi's grammar one uncertain relation at a time:

Do the senders contain dissenters?

Chapter 5 — Borrowed Weight

The archive door refused the word empty.

Iona entered it under search purpose, and the Curvature Office lexicon returned six hundred twelve categories, none approved for a physical condition. Vacant required a survey standard. Uninhabited required evidence. Unassigned described only an account. Empty appeared in public sink certificates, policy speeches, and clearing schedules, but the live archive treated it as rhetoric.

Nemi leaned over Iona's shoulder. "Try historical usage."

The terminal asked for ministerial clearance.

They stood in a records vault beneath the Nacre receiver hub while containment staff dismantled the translation cradle two decks above. Bex had allowed one final audit of Eidos certification, perhaps because stopping it would have required a written order. Two security officers watched from beyond a mesh wall. They could see every display but not hear ordinary speech through the vault's vibration baffles.

Iona selected historical usage and laid Bex's temporary key against the terminal.

The archive opened.

Eighty-eight years of certification forms appeared as stacked layers. Each renewal preserved the prior survey and added a current authorization. Eidos changed from candidate sink to provisional sink, then to certified vacant receiver complement. Twelve years later, a policy editor shortened the phrase to empty system. No new observation supported the change.

"Language drift," Nemi said.

"Authority drift."

"Same letters, different department."

Iona began at the first survey.

The Republic team had observed Eidos for thirty-one days using three telescopes and two remote gravimetric passes. They searched for atmospheric disequilibrium, radio emission, coherent light, industrial heat, moving megastructures, chemical waste, artificial magnetic fields, and engine signatures. Every life criterion required electromagnetic evidence. The team found none.

The gravimetric instruments served another purpose. Their operators mapped large subsurface oceans because sudden debt pulses could eject liquid through an ice crust. The survey did not treat ordered pressure layers as possible life. Its standard assumed metabolism needed chemical exchange fast enough to alter atmosphere, surface, or heat balance.

Nemi opened the committee notes. "Here. A reviewer asked about computation in solid-state defects."

The answer occupied one sentence: No plausible biosphere persists without detectable free-energy throughput.

"Did they model superfluid phase structures?" Iona asked.

"No. They cite conventional crystal defects at ninety kelvin."

“Eidos strata run colder under higher pressure.”

“And the review knew that. See the temperature appendix.”

Someone had placed the right physical facts in separate documents and never allowed them to meet.

The separation continued into instrument design. Iona opened the original survey probe specification and traced each sensor to its destination field. Radio measurements fed the life panel. Spectral chemistry fed it. Waste heat received a probabilistic score. Gravimetric anomalies went only to sink safety, where reviewers asked whether the ice could accept force without ejecting debris.

No data crossed between those panels.

A deep pressure shift could lower a safety score, but it could not raise the probability of life. If the same shift repeated after a clearing, the system classified it as accommodation and increased the modeled capacity of the sink. Repetition made Eidos look safer precisely because analysts had excluded the possibility of response.

Nemi pulled up a training simulation from the certification course. “I took this twelve years ago.”

A model ice body appeared above the terminal. Trainees had to assign a debt cap from limited observations. The simulation included a regular subsurface migration after each test pulse. Its approved explanation called the migration plastic settling.

“What happens if you flag it as possible organization?” Iona asked.

Nemi restarted the exercise and chose that option. The program marked her answer unsupported, then deducted points for anthropomorphic inference.

“They trained us out of the question,” she said.

“They trained us against false positives.”

“Using a standard that made this positive impossible.”

Iona opened the revision history. The exercise had entered training one year after Eidos certification and used anonymized Eidos survey data. Thousands of auditors had learned professional caution from the very anomaly they should have investigated.

She ran the simulation herself. Instead of selecting possible organization, she requested additional gravimetric passes. The program denied them on budget grounds. She requested a reduced debt cap pending evidence. That answer received partial credit but warned that unnecessary restriction could endanger receiver continuity.

“What answer earns full credit?” Nemi asked.

Iona chose stable deep-layer accommodation and retained the proposed cap.

The program congratulated her for balancing evidence with operational need.

Nemi reached for the close control. Iona stopped her and saved the result beside the live cradle finding. One showed what the Republic's instruments could observe. The other showed what its auditors had been rewarded for concluding.

Iona copied the life criteria into a fresh comparison. Electromagnetic output: absent. Atmospheric chemistry:

not applicable to rogue ice bodies. Thermal excess: below remote sensitivity. Mechanical activity: recorded, excluded from biology review. Pressure history: not studied. Responsive behavior through the debt channel: not part of any standard.

The certificate did not prove absence. It proved silence in frequencies humans had chosen to monitor.

“Empty meant electromagnetically silent,” Nemi said.

Iona marked the sentence as a finding. “And mechanically useful.”

A survey budget file showed why the distinction vanished. The network expansion needed four new sinks before a scheduled habitat wave entered operation. Eidos offered deep ice, low orbital traffic, and no diplomatic claimant. A longer survey would have delayed receiver activation across three systems. The committee accepted the electromagnetic standard because every known organism left electromagnetic consequences.

“They asked whether Eidos resembled known life,” Nemi said. “Then converted no into no life.”

“They asked the available instruments.”

“You are defending them again.”

“I am locating the failure.”

“Does location matter after Aru showed us the fractures?”

“Yes. We cannot correct a standard by accusing its dead authors of refusing knowledge they lacked. We correct the decision that kept using it after contrary evidence arrived.”

Nemi turned toward the restricted annex marker. “Six years.”

The annex still required a second ministerial key. Bex had not supplied it.

Iona traced every later renewal first. Eidos's debt cap rose fourteen times. After several increases, remote gravimetric records showed subsurface redistribution. Reviewers called the shifts settling events consistent with accepted load. No team revisited the original life criteria. Certification became self-supporting: because Eidos carried empty status, any response from Eidos entered the category of nonliving mechanics.

A return pattern thirty-two years ago contained eleven prime-spaced intervals. The clearing engine discarded it as phase noise.

Twenty-one years ago, a large transfer produced three matched counterpulses. Engineers treated them as channel rebound even though dark-lattice theory prohibited rebound after acceptance.

Sixteen years ago, the Meridian emergency transfer touched Eidos for less than a second before routing changed. Its log contained a weak precursor return.

Iona stopped scrolling.

The date matched Sella's final shift.

Nemi saw the name before Iona spoke. “Meridian used Eidos?”

“According to this routing fragment.”

“The inquiry report says the emergency load never left the local receiver.”

“It says the transfer failed before sink acceptance.”

The raw route showed otherwise. Eidos accepted part of the pulse. A structured return entered the Meridian network thirty-seven milliseconds later. Then someone redirected the remaining debt to another sink and removed the Eidos code from the incident summary.

Iona's hands remained on the terminal rail. She could feel Nacre's receiver hum through the metal, a vibration below hearing that shifted whenever the hub cleared a local load. Her body wanted to pull away. She held the rail.

“Open the precursor,” she said.

Nemi displayed it. The return amplitude stayed far below the catastrophic event that followed. Its intervals showed a short prime ladder, then a pressure rise. The pattern resembled Aru's warning before the cradle demonstrated fracture.

Sella could not have known that grammar. No one at Meridian had studied the return as communication. Yet Sella had manually restored spin to a nursery ring while the receiver console reported a local operator fault. The official inquiry blamed her for bypassing a safety interlock.

Iona searched her sister's maintenance entries. Most remained sealed under casualty privacy, a restriction she had challenged twice and lost. The Eidos route record carried a cross-reference to one suppressed note, then replaced the link with an operator-error code.

“Request the cross-reference,” Nemi said.

“It will alert Continuity.”

“They already have two officers watching us.”

“Not to this thread.”

Iona saved the address without opening it. Containment logged requests as well as access, and the sealed reference might matter more than the answer she wanted immediately.

She returned to the routing fragment. Beside the precursor lay a handwritten engineering annotation digitized from the console pad: Load has destination even when account refuses it. The handwriting did not belong to Sella. An inquiry engineer had paraphrased her spoken warning.

Memory supplied Sella's original words with cruel precision.

Weight goes somewhere.

As children, Iona had heard a rule about bolts, spin, and dropped tools. After Meridian, she had treated it as a private warning against careless systems. In the archive vault, the lesson became an accusation aimed at the entire Republic. For more than a century, clearing clerks had named every local benefit and left the destination inside one word: empty.

The word had not removed destination. It had removed the obligation to inspect it.

Nemi touched the interval trace. "This may be the first structured response tied to Meridian."

"First surviving one."

"Do we tell Aru?"

"Containment forbids response."

"I did not ask what Bex permits."

Iona closed the Meridian file. "We do not yet know what the pattern says. Giving Aru an event we cannot characterize could convert Sella's death into evidence for a claim we have not tested."

"She died inside the same mechanism."

"That does not make her death ours to translate badly."

Nemi looked away, angry but listening.

They continued through the renewal chain. Ten years ago, Eidos returns increased in frequency after Nacre doubled receiver capacity. Seven years ago, a sink analyst named Pori Sal flagged prime intervals across twelve clearings. Her report recommended suspending Eidos assignments pending a deep gravimetric survey.

The recommendation disappeared from the next certification packet.

"Where did Sal go?" Nemi asked.

Iona opened personnel history. "Transferred to comet mass accounting. Retired four years ago."

"Alive?"

"Yes."

"Can we contact her?"

"Not under containment."

Nemi wrote the name on the orange cord attached to her stylus.

Six years ago, the comparative interval study began. Its visible custody shell listed fourteen analysts, three receiver engineers, two geophysicists, and one continuity observer whose name remained hidden behind office authority. The study ran for nineteen days. Researchers compared returned pulses from seven certified sinks. Only Eidos showed interval selection that changed after outgoing load variation.

The abstract had been removed. Its statistical appendix remained because an archive migration misclassified it as calibration support.

Nemi opened the appendix before the system could correct itself.

Tables filled the display. Experimenters had varied clearing amplitude within operational tolerances. Eidos returns changed not merely in scale but in sequence. When transfers stopped for three hours, the channel sent repeated low-amplitude patterns. When transfers resumed, the patterns grew more complex. A blind classifier distinguished Eidos returns from natural sink noise in 99.8 percent of trials.

“No semantic test,” Iona said.

“They did not need ours to know it responded.”

“Agreed.”

A final table compared four hypotheses: instrument error, geological settling, autonomous machinery, and unknown selecting process. The study rejected the first two. It could not distinguish the latter pair.

Someone had written the disposition in a narrow field beneath the statistics: No action. Operational dependence outweighs unverified source-status risk. Reclassify returns as deep-layer accommodation.

The signature remained masked.

Nemi copied the appendix to their temporary audit workspace. The system blocked export. She began transcribing key values into her cord, using knots between digits.

“You will run out of sleeve,” Iona said.

“I will request a longer stylus.”

The vault door unlocked.

Elian Bex entered alone. He had removed his laboratory coat. His shirt cuffs showed gray dust from the translation cradle plates.

“You opened the migration appendix,” he said.

“It fell within certification authority,” Iona answered.

“It fell through a storage error.”

“The data remain evidence regardless of filing competence.”

Bex looked at the Meridian trace on the side display. For the first time since Perihelion, anger crossed his face without disguise. “Why did you open that incident?”

“Eidos appears in its raw route.”

“It does not belong to the present contact finding.”

“A structured return entered Meridian before Sella's receiver failure.”

“You do not know that it caused the failure.”

“I know the inquiry omitted it.”

“Because the return carried less force than the plate malfunction.”

“Then omission protected nothing except the finding.”

Bex approached the terminal. “Close the incident file.”

Iona kept her hand on the rail. “Unlock the annex.”

“No.”

“Then I will record that Continuity refused access to a study proving responsive behavior in a certified sink.”

“You are already under containment. That record will remain here.”

“It will also attach to every future certification bearing my authority.”

“You can be removed from certification duty.”

“Then my removal becomes part of the chain.”

Bex studied her with the exhausted concentration he had given the cradle. “You learned that from Sella.”

“She taught me to name destinations.”

“She taught half the Meridian engineering corps to make insubordination sound like maintenance.”

“You knew her.”

“I reviewed the incident after her death.”

“That does not answer the tense you used.”

His gaze shifted to the precursor trace. “I met her once during a spindle retrofit. She objected to a receiver shortcut and made me crawl through forty meters of bearing access to see the load path.”

“What shortcut?”

“Unrelated to the incident.”

“Was she right?”

“Yes.”

The admission changed no record. It changed Iona's measure of the man. Bex had allowed Sella to remain an operator-error code while remembering the exact quality of her dissent.

Nemi pointed at the disposition field. “Who wrote no action?”

Bex did not answer.

“Who reclassified the returns?” Iona asked.

“Continuity authority.”

“Which officer?”

“The decision belonged to the ministry.”

“Institutions do not press keys.”

“People acting under office do.”

Iona brought the study's timeline beside Bex's career. Six years ago he had served as Deputy Minister for

Receiver Stability, with authority over sink-risk classification. His old engineering key appeared in the annex custody shell as a masked credential family.

Nemi enlarged it. "Same key generation."

"Thousands of engineers used that generation," Bex said.

"Thirty-one held deputy-minister scope," Iona said. "Four worked in Continuity. One had Eidos clearance."

Bex reached past her and closed the Meridian window. He did not close the comparative study.

"Unlock it," she said again.

"What do you expect to find?"

"Observation records. Names. The reasoning behind continuation. Any attempt to identify the responding process."

"You have the reasoning. Operational dependence."

"That phrase does not calculate consequences. It hides them."

Bex's control thinned. "At the time of the study, twenty-nine billion people lived under Republic receiver coverage. Abrupt suspension of all questionable sinks would have produced orbital drift, ring desynchronization, hospital failures, and transport losses before replacement surveys could finish."

"All questionable sinks?" Nemi asked. "The study identified Eidos."

"It raised the possibility that our certification standard could fail elsewhere."

"So you knew the word empty could not carry its claim."

"I knew our standard had an unresolved edge."

"You knew Eidos answered changing load."

"I knew a selecting process answered. Machinery remained plausible."

"Did you send a survey?"

"No craft could reach Eidos within a useful policy window."

"Remote gravimetry?"

"Would have required more debt-channel modulation and might have worsened whatever we observed."

"Reduce Eidos assignments?"

"We shifted eleven percent to other sinks. Tidal limits prevented more."

"Publish the uncertainty?"

Bex looked through the mesh at the guards. "Markets would have treated uncertainty as shutdown. Habitats would have hoarded fuel. Systems with alternate sinks would have abandoned those without them. The Republic could have broken before we knew whether a machine objected to load."

Iona heard the practiced architecture in his answer. He had not invented it in this room. Six years of policy had accumulated around the first concealed decision, each later danger offered as proof that the earlier concealment had been necessary.

"You made continued ignorance part of continuity," she said.

"I kept clearing capacity while engineers searched for alternatives."

"Where are the alternatives?"

"Insufficient."

"After six years?"

"Every local solution costs mass, fuel, or habitat volume. Assemblies approve distant sinks and reject visible sacrifices."

"You never gave them the actual choice."

"I gave them survival."

A low vibration crossed the vault floor as Nacre's receiver cleared another batch. Iona watched the status panel. Destination codes cycled past. Eidos did not appear, but two other systems carried the same antique vacant certification.

"Does Aru's demonstration change your classification?" she asked.

"Yes."

"What changes operationally?"

"Reduced amplitude. Accelerated alternate modeling. Containment until we can prevent panic."

"Clearings continue."

"They must."

Nemi untied the stylus cord from her sleeve and set it on the terminal. Knots preserved the appendix values in a line of orange fiber. "You keep treating known harm as a risk and familiar dependence as fact."

"I can count the people under receivers."

"You could count the replies."

"That does not count lives."

"We did not know to count them because you sealed the evidence."

Bex turned toward the archive door. Iona blocked his key from closing the session.

"One finding remains," she said. "The comparative study established structured response six years ago. I need the responsible observer's testimony."

"You have my policy account."

“I need the fact without office language. Did you personally review the raw Eidos sequences?”

He could order both auditors removed. He could let the masked credential stand and force them to prove a signature through restricted systems. Instead, Bex placed his palm beside Iona's on the terminal rail. The receiver vibration passed through both hands.

“Yes,” he said. “I saw them.”

Chapter 6 — The Republic's Balance

The child in Cylinder Twelve died at forty-three seconds.

Elian Bex stopped the continuity model and moved the time index back to zero. The transparent cylinder hung above his desk, a handspan model of an orbital city rendered in amber light. At his touch, its inhabited rings began to turn again. Hospitals brightened. Water climbed through spoke pipes. Freight platforms moved outward under spin. Twelve million simulated people resumed lives the model had just ended.

He cut their receiver feed a second time.

Nothing happened for six seconds. That delay had misled three Assembly observers during the morning briefing. They had watched the stable rings and mistaken stored momentum for safety. Then load compensation vanished from the industrial spine. A flywheel cradle shifted against its mechanical restraint. The program painted the first injuries in small red marks.

At nineteen seconds, an elevator entered a spoke with the wrong counterload. At twenty-seven, the hospital ring began an emergency despin to protect a surgical deck. The change altered water pressure in two residential arcs. Cylinder Twelve had built its nursery near the hub, where the apparent gravity ran low and the receiver supplied most of the local down. Forty-three seconds after shutdown, an infant ventilator lost coolant circulation.

Bex paused the death again.

The Ministry office around him stayed quiet except for the cooling channels under the floor. Nacre's blue-white star filled the outer window through three layers of radiation glass. A shade crossed it on schedule, and hard light softened to the gray used for administrative mornings. Bex had approved replacement bearings for that shade when he still carried an engineering bag. He knew which contractor had lied about the first alloy batch. He also knew how long the city could survive if all six shade trains stopped.

This knowledge had once felt useful.

He opened the population layer. Across the Forty-Three Systems Republic, 31.4 billion lives appeared as density rather than faces. The display marked 17 billion in structures whose stable orbit, spin, thermal routing, or acceleration schedule depended on curvature clearing. Some could localize within months. Some needed years of tug work and mass transfer. A few possessed no physical margin at all.

Seventeen billion. He had said the number so often during the last thirty hours that it no longer fit in his mouth. On the model it fit inside a line of text.

A secure chime sounded at the door.

“Enter.”

Iona Rhyll came in carrying a black evidence wafer. She had slept recently enough to change her clothes but not long enough to soften the bruised shadows beneath her eyes. Her deputy waited in the outer analysis room. Bex had asked for Iona alone, then disliked himself for the calculation behind the request.

“You wanted the shutdown model,” he said.

“I wanted the records you sealed six years ago.”

“They are being transferred to your review vault.”

“After your office removes continuity identifiers.”

“After we remove access paths that could let anyone issue receiver commands.”

Iona put the wafer on his desk. “You have had six years to choose the description that sounds least like concealment.”

Bex let the accusation remain where she placed it. “Watch the model.”

“I know what sudden shutdown does.”

“You know the categories. I need you to see the order.”

He restarted Cylinder Twelve. This time he slowed the index, forcing each failure to occupy the room. A maintenance worker reached for a rail when the industrial spine's load shifted. A cargo handler vanished behind a tumbling container. Pumps lost head pressure. Emergency systems came alive, consumed their reserves, then began disagreeing about which ring deserved the remaining power.

Iona watched without turning away. Bex had expected anger to make her impatient. Instead she leaned toward the rendering and read each timestamp.

“At forty-three seconds,” he said, “the nursery loses one ventilator.”

“At fifty-one, mechanical backup engages.”

“For three units. There are four patients.”

“Why only three?”

“Because the hospital filed its capacity plan before the nursery expanded.”

Her gaze shifted to him. “That has nothing to do with Eidos.”

“It has everything to do with an abrupt order. Systems fail through old choices, not clean diagrams.”

The cylinder reached three minutes. Tugs attempted to arrest the ring drift with conventional thrust. One plume crossed a radiator field because traffic control had lost a compensated reference frame. The display added thousands of injuries and revised projected deaths upward.

Iona touched the pause control herself. “You selected a habitat with neglected backup.”

“I selected one with average reserve compliance.”

“Show me the best case.”

Bex loaded it. A wealthy spindle in the Nacre system appeared, equipped with triple mechanical redundancy, excess fusion capacity, and two resident tug squadrons. Sudden receiver loss killed no one in the first hour. It did break an agricultural bearing, force the evacuation of an outer district, and consume eighteen percent of local maneuvering fuel before the first day ended.

“Now the worst,” Iona said.

“No.”

“Why?”

“Because it teaches nothing the middle case did not. It only makes refusal harder.”

Her expression changed by a degree. “You still think this meeting concerns persuasion.”

Bex closed the model. The amber city folded into his desk. “Every meeting inside this building concerns persuasion. That does not make the numbers false.”

He crossed to a cabinet set into the wall. The lock accepted his palm but rejected the current ministry credential. Bex reached under his shirt and drew out a small steel tool on a faded cord. Its teeth had been cut for physical contacts, not data. He inserted the engineering key beside the electronic plate and turned both together.

The cabinet opened.

Inside lay a grease-darkened field case, two paper notebooks, a cracked pressure gauge, and the orange helmet he had worn during the Aster Ring famine. The helmet still carried a white handprint where a coolant technician had steadied herself against him during a false-gravity surge. Bex took out the nearer notebook.

“I kept this key after the ministry converted the old receiver housings,” he said. “A violation of three security rules.”

“I can note it in the investigation.”

“You should.”

He gave her the notebook. Its pages held his handwriting from twenty-seven years earlier: bearing temperatures, ballast positions, names of dead relays, names of living mechanics. During the Aster failure, a clearing scheduler had trapped three habitats in the same sink queue. Bex had spent eleven days moving water, slag, and two obsolete flywheels to keep the rings within tolerance.

Iona turned a page. “You localized part of the debt.”

“We did not call it that. We called it keeping the walls behind the air.”

“Then you know there are alternatives.”

“I know alternatives require mass, fuel, machines, routes, and time. A moral instruction cannot tow a flywheel.”

“Neither can a dead civilization.”

Bex felt the sentence enter the space between them. He looked past her at the muted star. Six years earlier, the first structured returns from Eidos had crossed his desk as a classified sink-instability report. The pressure intervals showed regularity. Not proof of life, the analysts had written. Not noise either.

He had ordered a deeper survey. He had also ordered the channel left open while the survey proceeded.

The survey never received its carrier allocation. There had been a food failure at Ravel, then a spindle strike, then an election in which three continuity delegates campaigned on abolishing costly sink review. Bex had renewed the seal every ninety days until renewal became habit. Each signature converted uncertainty into

permission for another season.

"I did not know what they were," he said.

"You knew they replied."

"Yes."

"You knew our clearings hurt whatever made the reply."

"I knew amplitude correlated with stronger returns."

"That sentence has sheltered you for six years."

Bex took the notebook back. "No sentence shelters anyone from this."

His desk lit with an incoming priority. The network controller had completed the reduced-export simulation he requested. Bex opened it between them. Forty-three systems formed a pale web. Receiver hubs pulsed according to debt volume. Beyond the inhabited routes, the certified sinks showed as dark destinations with no population layer.

He selected a ninety-percent reduction in exports to Eidos. The model redistributed allowable loads to six other sinks while local systems began mechanical correction. Kestrel Deep reached its tidal cap first. Varda followed. Three low-mass habitats entered unsafe drift within nine hours.

"Reduce demand before redistribution," Iona said.

"Choose sectors."

"All nonessential comfort fields."

Bex removed retail stabilization, private transport compensation, variable floor loads, sports drums, and decorative fountain gravity. The reduction looked large on a public list. It changed total debt by less than four percent.

"Cargo assistance," she said.

He cut it. Injury projections rose at once, followed by food delays two days later.

"Military load fields."

"Already queued."

The Republic defense facilities vanished from the receiver list. Enough capacity appeared to protect one medium cylinder or six hospital rings.

Iona examined the web. "You built the order to prove your preference."

"I built no order. Choose again."

For the next hour she did. They suspended ore lifts, delayed shade maintenance, canceled three ship burns, and moved every system with available conventional capacity onto local payment. Each choice spared pressure at Eidos and moved risk toward a human structure. Bex did not object when she accepted those risks. He named them.

At thirty-seven percent reduction, the model held for a day. By the third day, maneuvering fuel shortages forced receiver use back up in the outer systems. At the sixth, agricultural losses began. At the ninth, the first habitat crossed a structural drift limit unless a tug convoy arrived from eleven light-hours away.

Iona enlarged the failing habitat. New Carth had grown around a captured carbonaceous moon, with pressure hulls tunneled into rock and three residential wheels mounted along its equator. Ordinary mass gave the settlement a useful center, but its wheels relied on receiver trim whenever mines moved ore from one hemisphere to the other. The current stockpile sat north of the equator. Without clearing, that imbalance would force the wheel bearings sideways until one housing cracked.

“Move the ore south,” she said.

“Conveyors need fourteen days.”

“Use surface crawlers.”

“Their traction field draws from the receiver.”

“Mechanical cleats.”

Bex applied the change. Crawlers fitted with cleats could move enough ore in nine days, if they stopped delivering carbon feedstock to the air plants. The atmosphere reserve then fell below statutory margin.

“Send feedstock from the next moon,” Iona said.

“No cargo window for six days.”

“Advance it.”

“The freighter burns reserve fuel.”

“Let it.”

He entered the cost. New Carth survived. The freighter lost the reserve assigned to a medical evacuation contract shared by four mining camps.

Iona rubbed the side of her thumb against the desk, then selected the camps. One had a conventional shuttle. Another could support twelve hours of emergency life care. The remaining two depended on the freighter for any surgery their small clinics could not perform.

“Move a surgical unit out now,” she said.

“With whose staff?”

She searched the personnel layer. Bex waited. He had done this during Aster, though with fewer screens and more voices shouting through suit radios. Every movable machine already served a purpose. Every qualified body slept, ate, grieved, and made mistakes. Models permitted redeployment without showing the moment when a surgeon told her child that the next three months belonged to strangers.

Iona found a naval medical unit two transfers away. “They can divide.”

“They can, if Defense releases them.”

“You are the minister.”

“Not of Defense.”

“You sit on continuity cabinet.”

“So I can ask an admiral to accept risk under her own name.”

“Ask.”

Bex opened a real priority request. The model could assume compliance; the Republic required a person to agree. He sent the medical transfer to the Defense duty office with his own liability code. No answer came before the requested window expired on the simulation.

They ran New Carth again with the request pending. At Day Four, a miner suffered a crushed pelvis. Local care stabilized him. The freighter remained with the ore convoy and no rescue ship came. His survival probability declined each hour.

“This is your argument,” Iona said. “One man on a table, multiplied until no one can act.”

“It is not an argument. His name would exist if the event occurred.”

“The pressure lineage you strike also has a name.”

“Yes.”

“Yet the model gives it no bed, no age, no one waiting outside a clinic.”

“I cannot invent what we have not learned.”

“You can stop treating the blank as zero.”

Bex added a second casualty field beside the human projection. He labeled it unmeasured sink harm and made its value block every optimization that assumed no external cost. The program rejected the field as nonnumeric. He overrode it.

The model now returned no preferred plan. Every branch carried an unresolved red mark at Eidos. No algorithm could compare a ventilator, a crushed pelvis, and a damaged pressure lineage without choosing a unit that erased two of them.

“That is more accurate,” Iona said.

“It is less useful.”

“Only if usefulness requires permission.”

Bex left the red mark in place. He returned to the network view and found twelve other habitats with solvable imbalances. Their solutions required ninety-one cargo transfers, six military releases, forty thousand spinwright shifts, and fuel reserves assembled over decades. None made the sink harm smaller by itself. Together, performed before an uncontrolled shutdown, they changed the shape of possible survival.

At last the Defense duty office accepted the medical transfer. A real surgeon somewhere on Nacre had just received an order generated by two people arguing over a simulated miner. Bex entered the acceptance into the

model rather than letting it vanish into institutional background.

“Ten,” Iona said. “Give every system ten standard days under mandatory reduction. Use the interval to move ballast and stage conventional burns.”

“That requires authority I do not possess.”

“The Assembly can grant it.”

“The Assembly can debate it.”

Bex opened the emergency statute. Its language had been written after a receiver cascade killed eighty thousand people at Cresset. A Continuity minister could assume network allocation, transport priority, and compulsory mass-control powers for ten standard days. The order would begin immediately upon certification of systemic risk. Renewal required two-thirds of the Assembly.

Iona read the first page. “You prepared this before I arrived.”

“I began preparing it when you froze Perihelion.”

“You intend to invoke emergency authority.”

“I intend to ask whether preparation can buy the choice you claim we have.”

“Authority does not create tugs.”

“It can move them.”

“It does not create fuel.”

“It can stop people spending what remains.”

Her hand closed over the edge of his desk. “And during those ten days?”

There it stood. The decision he had broken into percentages, timestamps, and procedural drafts. Outside the office, thousands of receivers continued clearing. Each low-amplitude export entered a world whose inhabitants humans had failed to recognize. Stopping now produced deaths the models could name. Continuing produced harms no human account yet measured.

Bex brought up the Eidos channel schedule. “We reduce amplitude to the lowest level that prevents network correction. No discretionary clearing. No reserve accumulation. Every system begins local load shedding.”

“You keep sending pulses.”

“Yes.”

“After Aru-in-Pressure identified them as wounds.”

“Yes.”

Iona stepped back from the desk. The distance carried more judgment than her voice had. “Then say what you authorize.”

Bex had spent his career teaching officials to use precise words during failures. A bearing did not become tired.

A ring did not want to fall. A casualty estimate did not equal a body. Precision kept engineers from lying to machinery, but it could help ministers lie to themselves.

"I authorize continued harm to Eidos," he said, "at reduced amplitude, while we prepare to stop without killing the people who depend on us."

"Do the people at Eidos consent to that schedule?"

"No."

"Then your interval belongs to the stronger side."

"It belongs to the side that can still change its machines."

The reply surprised him with its honesty. He did not know whether it improved anything.

Iona picked up her evidence wafer. "Nemi thinks the pressure grammar may distinguish separate speakers. If she is right, you are not bargaining with an injured system. You are imposing another ten days on populations."

"I read her analysis."

"And?"

"And I increased the projected moral cost because I had no unit for it."

For the first time, anger disturbed Iona's control. "You think adding a column makes this visible."

"No. I think refusing a column makes it easy to omit."

A network warning opened over the desk. Three receiver hubs had detected phase resistance from Eidos. The amplitude remained below correction threshold, but the intervals tightened with each pulse. Nemi's preliminary translation labeled the pattern as disagreement among pressure-names. One response favored continued exchange. Another carried a ratio the cradle associated with return.

Bex read it twice.

Iona saw the second classification. "They are deciding."

"So are we."

"No. You already did."

She left the office without taking his notebook or asking for the key. The door closed softly. Bex remained beside the old field case while the network schedule counted down to its next clearing.

He called the duty controller.

"Cancel every discretionary receiver load," he said. "Issue mechanical preparation orders to all forty-three systems. Place tug fuel, ballast reserves, and spinwright crews under provisional continuity hold."

The controller hesitated. "Under what authority?"

"Ministerial risk certification now. Draft emergency authority for Assembly notice after the Helix review."

“And the active sink schedule?”

Bex looked at Cylinder Twelve, dormant in the desk. He could still see the infant ventilator lose flow at forty-three seconds. Behind that child stood seventeen billion others, each attached to a machine with a different hidden weakness. Behind his screen, beyond any human image, pressure lineages received whatever force he sent.

“Eidos only,” he said. “Minimum stabilizing amplitude. No surplus clearing.”

The controller read the order back. Procedure demanded confirmation by a second channel. Bex drew the engineering key from its cord and fitted it into the old physical socket beneath his desk.

Metal met metal. The contact closed.

Far across the Republic, receivers lowered their fields. Hospital spindles accepted mechanical schedules. Cargo lifts slowed. Tugs received orders that would consume years of reserve fuel. The next reduced pulse entered the dark lattice carrying less force than the one before it, and still more than nothing.

Bex turned the key and authorized it.

Chapter 7 — The First Repossession

The cargo deck rose under Iona's feet before the alarm found a name for it.

She had been standing beside a tether winch on Helix Quay, watching Nemi compare pressure intervals, when the apparent down shifted six degrees toward the outer hull. A stack of sealed freight cases leaned in unison. Their restraint webbing snapped tight with a sound like a giant drawing breath.

Iona grabbed the winch frame. Nemi caught the open analysis case with both hands. Around them, loaders stopped because the deck still pressed against their boots and their bodies had not yet accepted that down had moved.

“Receiver phase excursion,” the quay announced. “Hold current stations.”

A loose torque wrench rolled past Iona. It curved sideways across the deck, struck a cargo rail, and climbed it.

“That is not an excursion,” Nemi said.

Her screen showed the Eidos channel at minimum stabilizing amplitude. Bex's reduction order had reached Helix Quay three hours earlier, forcing cargo operations onto a slower mechanical schedule. The receiver continued carrying tether trim, emergency crane loads, and the balance corrections that kept the quay's twelve cargo decks aligned around its spine.

Another interval entered the pressure trace. Not prime. Not part of Aru's careful translation sequence. The modulation arrived broad and exact, carrying the same integrated stress Helix Quay had exported over the previous eleven days.

Nemi stared at the total. “They're sending it back.”

Iona read the value twice. “All of it?”

“Within instrument tolerance.”

The restraint webbing began to sing.

A warning moved through the quay in layered voices. Receiver fault. Spine imbalance. Uncommanded curvature increase. Each system had recognized one consequence and failed to describe the whole event.

Iona opened a line to control. “This is Rhyll, Curvature Office. Disconnect the Eidos channel.”

“Command rejected,” said the receiver. “Outstanding correction in progress.”

“It is not our correction.”

The machine did not have a field for that distinction.

Gravity rotated another nine degrees and doubled. Iona's knees struck the deck. Nemi fell against the winch, shoulder first, but kept one hand around the analysis case. Freight cases tore sideways against their webs. One restraint anchor pulled out of the floor and crossed the aisle at head height.

A loader shoved Iona down. The metal fitting passed over them and punched into a pressure partition. Air screamed through the ragged opening until an automatic seal dropped.

“Clear the stacks!” someone shouted.

People crawled toward fixed rails. On a normal cargo deck, painted arrows marked the nearest pressure refuge. Now those arrows pointed across a sloping surface toward walls that had acquired weight.

Iona found Nemi on hands and knees. Blood ran from her temple into one eyebrow.

“Can you stand?”

“Direction remains under negotiation.”

The attempt at humor came too quickly. Nemi's pupils tracked at different speeds.

Iona clipped her deputy's harness to the winch frame. “Do not move.”

“Excellent plan. Tell the quay.”

The public channel filled with overlapping calls from all twelve decks. A crane carriage had broken loose on Deck Four. Two lifts had stopped between the spine and the outer ring. Cargo workers in a pressurized loading cradle reported their floor had disappeared.

Then a voice cut through the noise with practiced force.

“Helix control, this is *Morrowline*. We are still hard-docked at Cargo Two. Give me spine rotation and receiver status.”

Iona recognized Tavin Serr before the channel labeled him. She had met the captain the previous day while matching his freighter's reversal event to the Eidos trace. He had listened to the explanation of structured pressure replies with the hostility of a man learning that his damaged ship had participated in someone else's injury.

Control answered, “Captain, release clamps and clear the quay.”

“I asked for rotation.”

“Quay load vector is unstable.”

“So is my dock. Numbers.”

Iona keyed in. “Tavin, full repossession through the Eidos pairing. Integrated return matches eleven days of local clearing.”

“How long?”

“I don't know.”

“Can you shut it down?”

“The receiver treats the return as mandatory correction.”

“Then tell me what stays physical.”

The question steadied her. Curvature could change apparent load, but momentum remained. The quay still rotated. The spine still held mass. Thrusters still pushed against expelled propellant. Every force had a

destination, even when the receiver confused the route.

“Spin gives us point three standard toward the hull,” she said. “Returned curvature currently adds one point eight toward starboard and rising. Conventional thrust can alter the resultant.”

“Control, unlock my port clusters. I can push the dock axis back toward the cargo floors.”

“You will load the clamps past certification.”

“If I release, my stern crosses Deck Two. Unlock them.”

A crash traveled through the structure. The deck jumped. Iona's harness fitting drove into her ribs, and the freight stack nearest the partition moved half a meter despite its webbing.

Control granted Tavin access.

The *Morrowline* fired one port thruster cluster. Vibration entered Helix Quay through the docking collar, deep enough to feel in teeth. The painted refuge arrows no longer pointed downhill, but the deck became usable again. Workers began moving.

Tavin spoke over the rescue channel. “This buys a vector, not stability. Get everyone out from under suspended mass.”

Iona unhooked herself and checked Nemi. “We need the spine control room.”

Nemi attempted to rise. Her body leaned left while her hands reached right. She caught the rail and vomited between her boots.

“No,” Iona said.

“The pressure trace is on my case.”

“I can carry it.”

“You read omissions. I read intervals.”

“You have a head injury.”

“My head has always been an administrative concern.”

A medic arrived on a tether line. He took one look at Nemi's eye movement and clipped a support frame around her shoulders. “Vestibular trauma. Possible concussion.”

“She needs the case,” Iona said.

“She needs a fixed visual reference.”

Helix Quay offered none. Walls had become slopes. Loose objects continued taking paths that curved as the receiver load changed. The medic unfolded an illuminated bar in front of Nemi and locked it to the frame.

“Keep your eyes here,” he said.

Nemi obeyed for two seconds. Then her gaze dragged toward Iona. “The return had internal modulation. Not just load.”

“Can you isolate it?”

“Not while the room keeps turning through me.”

Iona took the analysis case. “I will bring it back.”

“That promise lacks statistical support.”

“So revise it when I do.”

She left Nemi with the medic and crawled toward the cargo-spine access. Three workers moved with her. One wore a dockmaster's white helmet, cracked above the ear. Another had a burn across one glove. The third carried a manual cutter.

At the access hatch, the deck load increased until Iona could barely lift her head. Tavin adjusted thrust. Pressure eased by a fraction.

“Rhyll,” he said privately, “my port clusters have nine minutes before thermal limit.”

“The receiver room lies two decks inward.”

“Then I need you there in eight.”

“Can you hold the direction?”

“No. I can keep the heavy things choosing approximately the same wall.”

The dockmaster opened the hatch. Beyond it, a ladder descended toward the spine according to paint and extended sideways according to Iona's body. They clipped onto a track and pulled themselves along. Every movement cost more than it should. The returned field pressed blood into unfamiliar places and made her old unease with unmodulated gravity feel childish.

Halfway to Deck One, the repossession changed sign.

Load vanished.

Iona's hands remained clamped on the track while her legs floated away. The dockmaster struck the ceiling. The cutter spun from the third worker's belt, tethered but live, its blade flashing past Iona's face. Farther down the passage, a person without a tether drifted out of an open service bay.

“Tavin, zero load!” Iona called.

“I see it.”

The *Morrowline's* thrusters cut. For one breath Helix Quay contained no reliable down beyond its slow spin. People, tools, and broken cargo moved along whatever momentum they already possessed.

The untethered worker reached toward the track and missed.

Iona pushed off. She caught his sleeve with her left hand. Their combined motion pulled her tether to full extension. Pain closed around her shoulder. The man struck the passage wall, found a grip, and held her wrist until she could reel them both in.

“Name?” she asked.

“Ors.” His face had gone gray. “Deck Five refrigeration.”

“Clip in, Ors.”

“My kit.”

“It can orbit without you.”

A low surge returned, this time toward the outer hull. They settled against the passage floor in a tangle.

The analysis case had remained attached to Iona's waist. Its screen displayed the field change as a descending curve interrupted by fine pressure pulses. Nemi had been right. Modulation rode inside the returned debt.

Iona opened the crude grammar they had built with Aru-in-Pressure. The cradle could not translate full meaning through a portable unit, but it marked repeated physical relations. Received load. Matched load. Boundary closure. A ratio associated with restoration appeared seven times.

Repossession, Iona thought. Not attack. Not accident. A creditor enforcing equivalence through the only mechanism both sides shared.

She hated the word as soon as it formed. It could make the event sound orderly. Nothing in the screaming public channels felt orderly.

“Receiver access ahead,” the dockmaster said.

They reached Deck One in six minutes. The control room door had warped in its frame. The worker with the cutter anchored herself and opened a narrow seam. Hot air escaped carrying insulation dust.

Inside, two technicians lay strapped beneath the main console. One had a broken forearm. The other kept issuing commands with blood filling one nostril. The receiver core occupied the center of the compartment, nested in superconducting loops and mechanical restraints. Status lights showed every protective interlock engaged.

Iona connected Nemi's case to the diagnostic port.

“Do not write to it,” the conscious technician said.

“I need to alter channel interpretation.”

“That qualifies.”

“The receiver believes the returned pulse belongs to local correction.”

“It has full authentication.”

“From whom?”

“No origin code. The lattice pair itself confirms the debt.”

A machine built to ensure that no force vanished had recognized its own exported quantity. The Olun did not need Republic credentials. They possessed the other side of the account.

Iona opened the settlement rules. Every clearing carried a completion condition: sink acceptance reduced local debt to zero. The returned pulse had reopened eleven days of entries and bound them to Helix Quay's receiver.

“Can we separate the reopened batches?” she asked.

“Not during correction.”

“What if local mass takes part of the load?”

The technician wiped blood from his lip. “You want to settle against the quay.”

“I want the receiver to recognize payment here instead of forcing every g-second through occupied decks.”

“Moving what mass?”

Helix Quay's cargo manifest covered the display. Ore, water, machine housings, habitat shielding, two dismantled reactor blankets, and the replacement flywheel that the *Morrowline* had delivered after sacrificing its original cargo. Each item already had a destination and owner.

Iona selected the water reserve in the spine tanks. “Pump Tank Three outward while the quay maintains rotation. The mass shift can absorb part of the curvature change.”

“We need that water.”

“We keep the water. We spend pump power and accept a lower spin margin.”

“Tank mounts could fail.”

“They fail if the field keeps cycling.”

Tavin joined the technical channel. “I can take the opposite load through my docking yoke. Pump toward my berth. I'll counter with conventional thrust.”

The dockmaster looked at Iona. “That transfers quay stress into his ship.”

“Yes.”

“How much can it hold?”

Tavin answered. “More than your clamps. Less than optimism.”

They built the sequence in ninety seconds. Helix pumps would move eight thousand tonnes of water outward. The *Morrowline* would fire along the dock axis and carry the transient through its reinforced cargo yoke. Helix control would release two cargo rings to free spin, sacrificing alignment with their transfer bridges. The receiver would see local curvature work and reduce the unsettled return.

Iona sent the plan to Bex's office for emergency approval.

His system demanded a four-minute structural review.

She canceled the request and signed under her forensic authority.

“Execution on my mark,” she said.

The pumps began. Deep impacts ran through the spine as valves opened under distorted load. Tank Three's water shifted through pipes wider than corridors. On the cargo display, two rings released their drive locks and drifted out of alignment. Transfer bridges tore free at designed breakpoints. Atmosphere plumed from one

failed seal, glittering as ice in the external cameras.

“*Morrowline* burning,” Tavin said.

Conventional thrust pushed against the quay. Iona felt the floor choose a direction and keep it. The receiver's correction curve faltered. Returned debt declined by six percent, then eleven.

A mounting bolt sheared from the core restraint. The cutter worker trapped it beneath her boot before it reached the console.

“Tank Three at fifty,” Helix control reported.

“Port cluster temperature critical,” said Tavin's engineer.

“Keep it lit,” Tavin replied.

The correction fell through thirty percent. Load surged to twice standard and held. Iona's chest compressed against her harness. The technician with the broken arm screamed once, then bit down on his sleeve.

Inside Nemi's pressure trace, the modulation changed. Restoration ratios shortened. Another pattern interrupted them, sharper and repeated. Iona could not translate it, but the shape resembled refusal.

The Olun were not acting with one mind. One pressure-name, or one faction, had returned the debt. Another might be arguing through the same wound.

“Seventy percent water transfer,” control said.

The *Morrowline's* port thrust flickered.

Tavin swore. “Cluster Three lost an injector. Pella is moving the burn.”

The vector swung. Everyone in the receiver room dropped sideways. Iona struck the console but kept one hand on the settlement command. The debt value rose again.

A second thruster cluster ignited. The yoke groaned through the whole quay.

“Burn restored,” Tavin said. His breathing came hard over the channel. “Finish it.”

Iona committed the last local mass entry.

Tank Three reached its outer limit. The free cargo rings took the changed spin. Helix Quay's receiver registered physical work, fuel expenditure, and permanent loss of two transfer alignments. The reopened debt dropped below correction threshold.

Gravity settled at four-tenths standard toward the hull.

No one moved at first.

Then alarms separated into intelligible problems. Fire on Deck Four. Pressure loss at Bridge Nine. Medical priority in six locations. The receiver channel to Eidos remained open at diagnostic amplitude, but the full return had stopped.

Iona released her harness. Her knees shook when they accepted the lighter load.

“Nemi,” she said over medical channel.

The medic answered. “Conscious. Vestibular injury confirmed. She cannot stand without support.”

“Put her through.”

Nemi’s voice arrived thin and irritated. “Did you bring back my case?”

“It is connected to a damaged receiver in Deck One.”

“That sounds like no.”

“You were right about internal modulation.”

“Of course.” A pause followed. “I cannot tell which way my own tongue is moving.”

Iona closed her eyes once. “Stay with the medic.”

“I had already developed a competing plan to wander into vacuum.”

The channel cut as treatment began.

Iona climbed back toward the cargo decks. Emergency crews passed her carrying braces, sealant, and folded stretchers. On Deck Four, a crane carriage had crossed the work zone and crushed three cargo bays. Workers cut survivors out beneath floodlights. The free rings hung visibly out of alignment beyond the hull windows. Helix Quay had survived, but its geometry no longer matched the city it had been that morning.

At Cargo Two, the *Morrowline* remained docked. Frost covered the overloaded collar. Tavin stood inside the quay-side lock with one hand against the frame while Pella Jor argued with a port engineer about coolant access.

“You spent a thruster cluster,” Iona said.

“An injector.”

“Your engineer said critical temperature.”

“Pella enjoys emotional language.”

Pella looked over. “The captain enjoys owning no repair tools.”

The port engineer scrolled a shortage list across her sleeve: surgical bearings, water membranes, orchard inoculant. Tavin stopped at the names. Thirty-two pallets carrying those same classes of cargo had left his deck during the Day One inversion, jettisoned so the *Morrowline* would not drive its spine through Helix Quay. The loss had been correct. It had also arrived here as delayed surgery, restricted water repair, and another growing season placed at risk. He signed the salvage trace beside each item before Pella could ask him to move.

Behind them, medics moved Nemi on a stabilized couch. Her eyes stayed fixed on the illuminated bar above her face. Iona walked beside her until the couch entered the hospital lift.

The public observation gallery overlooked Cargo Two. Its safety glass had polarized during the failure, but someone had overridden it. Hundreds of passengers, clerks, vendors, and off-shift workers stood behind the transparent wall. Many held personal recorders. They had watched freight cross the deck, watched people fall

toward walls, and watched the *Morrowline* fire against the station.

A child near the front pressed both palms to the glass. His father held a serving cart by its brake while cups drifted above one bent shelf in Helix Quay's weak remaining load. Farther back, two cargo clerks compared the public field command against a pocket accelerometer. No official explanation had taught them to distrust the deck. Their bodies had done that. Iona saw the change pass through the gallery as people tested handrails, tied loose bags to their wrists, and moved elderly passengers away from the outer windows. They were already building a private safety procedure around an institution that still called the event a fault.

A quay reporter called through the barrier speaker. "Auditor Rhyll, did the receiver fail?"

Iona looked toward Helix control. A ministry notice waited on her screen: Describe as multi-system load fault pending review. Do not discuss sink interaction. Bex's authorization code appeared beneath it.

Another reporter raised her voice. "Why did the public gravity reverse?"

"It did not fail by itself," Iona said.

Tavin turned toward her but did not interrupt.

"What does that mean?" the first reporter asked.

Iona opened the pressure trace on the gallery display. The descending return curve appeared above the crowd, labeled with eleven days of exported curvature debt.

"Helix Quay sent force through its receiver to a system our government certified empty," she said. "Today, someone there returned it."

The gallery became quiet. Personal recorders remained raised.

On Iona's private channel, Bex called. She declined the connection.

Across the public screens of Helix Quay, the word repossession left the forensic record and entered the Republic.

Chapter 8 — The Captain of Stillness

Stillness refused the *Morrowline's* docking request because the ship approached too gently.

Tavin Serr read the refusal twice, checked Haru Den's vector, and found no error. Their closing speed sat three centimeters per second below the refuge's minimum. Helix Quay had cooked one port injector and warped two docking sensors, so Haru had brought them in with the delicacy of a needle entering damaged cloth.

The refuge did not reward delicacy.

“Add five centimeters,” Tavin said.

Haru looked back from navigation. “Their collar is older than our forward yoke.”

“Their collar expects to feel us arrive.”

“Charming place.”

“Stillness likes commitment.”

Haru gave the bow cluster a breath of conventional thrust. The heavy freighter crossed the minimum and received immediate clearance.

Ahead, Stillness turned against the stars. It had no graceful cylinder, no bright arc of windows, and no receiver field to smooth its awkward mass. The refuge consisted of five inhabited drums mounted around a mined asteroid, joined by trusses thick enough to look excessive until a person understood what they carried. Water tanks and blocks of nickel-iron moved along rails outside the drums. As cargo shifted or a transport docked, the blocks changed position to preserve spin.

Every adjustment remained visible. No force disappeared into a remote system.

Tavin had learned tug work among those rails at seventeen. His first instructor used to tap the external mass diagram before every maneuver and ask where the push would finish. Any student who answered with a coordinate failed. A push finished in a bearing, a ballast block, a plume of exhaust, or somebody's spine.

The *Morrowline* met the collar hard enough to compress its mechanical buffers. A tremor ran aft through the cargo yoke. Docking clamps closed one by one.

“Contact earned,” Haru said.

“Put it in the captain's praise file.”

“That file rejected the entry for insufficient storage.”

On the engineering display, Pella Jor marked the failed injector for removal. She had slept four hours since Helix Quay and objected to all questions as if they consumed propellant.

Tavin opened the passenger link. “Rhyll, we're docked.”

Iona answered from the converted chart room. “The cradle felt it.”

“It feels load change?”

“Aru perceives acceleration history. Docking may have qualified as an introduction.”

Tavin left Haru with the bridge and crossed the ship. The passage load differed from deck to deck because the *Morrowline's* compensator remained shut down. Near the bow, Stillness's spin pulled at an angle through the docking collar. Tavin had spent most of his adult life walking through tuned fields that made every floor agree. Now his shoulder kept warning him that the corridor had been installed wrong.

The chart room contained Iona, two portable receiver modules, and the translation cradle Nemi had built before her injury. Nemi herself remained under treatment at Helix Quay. Her absence showed in a row of labels written too neatly and an exposed connector no one had insulted into obedience.

Aru-in-Pressure had no body inside the room. The speaker existed across layered ice at Eidos, assembled from pressure histories and temporary agreement among lineages. The cradle gave that distant coherence a narrow way to modulate local acceleration. Three suspended beads moved inside a sealed glass tube. A low speaker rendered pressure ratios into human language after delays that varied with complexity.

One bead shifted when Tavin entered.

“Is that recognition?” he asked.

Iona watched the trace. “It could be your mass changing the room.”

The speaker clicked twice. “Difference retained.”

Tavin stopped beside the table. “I'll take recognition.”

“Translation remains approximate,” Iona said.

“So does docking.”

She closed a data pane. Helix casualty reports remained visible beneath it. Nine dead, more than two hundred injured, three still missing inside a crushed freight bay. Nemi's vestibular damage carried a guarded prognosis. The quay had lost two cargo-ring alignments and enough water-pump life to require replacement within a year.

Stillness control called them to wait for ballast transfer before opening the collar. The refuge had accepted the *Morrowline's* mass, but the nearest inhabited drum now ran beyond its preferred bearing load. External cranes began moving nickel-iron blocks outward and opposite the dock.

On a camera feed, three suited workers guided a block as large as a house along a truss. Their rail motors crawled. Frost smoked from mechanical brakes. The operation would take twenty-three minutes.

“At Helix, a receiver would have trimmed this in a second,” Iona said.

“And sent the difference somewhere else.”

“Here they spend labor.”

“Labor, brake surface, and time.” Tavin pointed to the heat graph. “Also radiator margin. They cannot move another major block until those brakes cool.”

Aru's middle bead descended. “Local body records push.”

Tavin looked at the glass tube. “That's the whole town.”

“Town?” the cradle asked.

“A human pressure lineage with invoices,” Iona said.

The speaker attempted her phrase, failed on lineage, and produced a low sequence that made Tavin's molars ache.

Once the ballast locked, the docking collar opened. Dry air entered carrying machine oil and cooked grain. Stillness had never possessed enough imported volatiles to waste water on humidity. Its corridors showed patched pressure panels, exposed cables, and handrails polished by generations of gloves. Nothing floated loose. Every cup had a lid. Every tool had two clips.

The dockmaster met them without ceremony. Her left leg ended below the knee in a work prosthesis built from tug components. She looked past Tavin at the ship status displayed over the collar.

“You arrived with one dead port injector.”

“Temporarily dead.”

“Pella still working for you?”

“Against her judgment.”

“Then temporarily.” The dockmaster glanced at Iona's ministry credentials. “We have no receiver for your auditors to certify.”

“That's why we came,” Iona said.

“People with receivers say that whenever their machines frighten them.”

She led them inward. Tavin remembered the route as wider. At seventeen he had carried tool cases through these corridors and believed any place containing a dock led to the whole Republic. Now pipes crowded his shoulders. Repairs lay over repairs in visible strata. A pressure panel carried five dates in different paints.

Stillness's market occupied the hubward end of Drum Two, where spin produced a light but reliable down. Stalls sold algae bread, recast machine parts, sealed fruit paste, and cloth made from fungal fiber. Fresh produce occupied one table under a locking cover. Six bruised pears cost more than a thruster technician earned in a shift.

Children ran along a curved wall until their feet climbed above Tavin's head. They understood the drum's changing vector without looking. An old man moved water containers on a handcart balanced to the kilogram. When Tavin stepped aside for him, the cart's counterweight shifted along its axle.

Iona watched the mechanism. “How often do bearings fail?”

“Main drum bearings? One every three years,” the dockmaster said. “We manufacture replacements here.”

“With that metal stock?”

“With whatever arrives. The tolerances make engineers religious.”

Tavin shook his head. “They make engineers thieves. Religion comes after inventory.”

The dockmaster's mouth moved toward a smile but stopped. "You still owe us two carbide cutters."

"I was nineteen."

"Debt matures."

Aru's portable cradle traveled between Iona and Tavin on a rigid frame. The suspended beads responded to each change in spin. At the market hub, all three settled close to the tube's center.

"Down made by rotation," the speaker said.

"Yes," Iona replied. "No dark-lattice export."

"Pressure remains near source."

"Every correction here has a local destination," Tavin said.

The dockmaster heard the translated voice for the first time. She did not step back. "Is that someone from the sink?"

"One temporary speaker," Iona said. "Aru-in-Pressure."

"Temporary as in shift work?"

The beads moved in three different directions. Translation took eleven seconds.

"Temporary as agreement under changing load."

The dockmaster considered this. "That also describes shift work."

They continued to the refuge's balance room, but the dockmaster took a service bridge around Drum Six first. The corridor ended at a pressure door with no handle on the inhabited side. Beyond its inspection glass, an unfinished cylinder curved into darkness. Bare ribs divided the interior. Capped air pipes projected from the hub. A painted line marked where soil had once been planned.

"We built the shell fifty-three years ago," the dockmaster said. "Then Drum Three needed new fly masses. The metal came from here."

Tavin remembered hauling one of those masses as an apprentice. For six days, his tug crew had worried a section of Drum Six's outer wall free, shifted it through the truss, and machined its edges into a rail block. He had thought the work clever. It had never occurred to him that the dark cylinder had been promised to families.

"How many berths?" Iona asked.

"Eighty thousand."

"Could you complete it now?"

"With imported structure, four reactor units, and enough water to make the farms useful. We have requested all three for thirty-one years."

Tavin put one hand against the pressure door. The metal carried a faint periodic tremor from the inhabited drums. Stillness had not escaped cost by refusing a receiver. It had converted cost into rooms never filled, children who left, and machinery kept alive past sane retirement.

Aru's cradle responded with a long pressure descent.

"Absent human pattern?" the speaker asked.

"People planned to live there," Iona said. "They never did."

"Ended before forming?"

"Not ended," the dockmaster replied. "Deferred. Some went elsewhere."

"Some were never born," Tavin said.

The dockmaster faced him. "Do not turn our empty drum into a funeral. We chose bearings over expansion because inhabited rooms came first."

"I know."

"No, you remember leaving."

The statement found its place. Tavin had signed onto a commercial tug at twenty-one and visited Stillness only when work required it. His competence had begun here, but the refuge's shortages had continued without him. He had carried its habit of naming force into ships that used receivers, then let the habit shrink into a captain's concern for immediate vectors.

He asked, "What would completion cost under local stability?"

The dockmaster keyed figures onto the door glass. More external ballast than the existing city carried. Two decades of industrial output. Regular fuel shipments for corrections during construction. A permanent reduction in market spin whenever large transports docked.

"With a receiver?" Iona asked.

"Seven years and half the metal," she said. "That answer has not changed because Eidos became inhabited. Only our willingness to use it should change."

Aru shifted the beads. "Human need remains after recognition."

"Yes," Tavin said. "That is what makes recognition expensive."

Before they left, the dockmaster turned on one work light inside Drum Six. It illuminated a patch of curved hull, three structural ribs, and a chalk drawing preserved near the planned floor line. Someone had sketched a door with flowers on both sides. The simple room remained beyond the pressure glass while Stillness spent its mass elsewhere.

The refuge's balance room hid no part of its mechanism. A circular table displayed physical tokens corresponding to major moving masses: water, ore, ships, food stores, reactor blocks, inhabited sectors. Electronic scales beneath the tokens fed an orbital model, but the staff kept the objects because hands noticed imbalance before eyes tired by numbers.

Tavin picked up the token for the *Morrowline*. It required both hands.

"They flatter us," he said.

"It includes your attached flywheel," said the balance chief. "You account for nine percent of current dock-axis stress."

Pella called from the ship. "Captain, Stillness inventory has an injector sleeve compatible with Cluster Three."

"That sounded hopeful."

"They want four tonnes of water and our spare guidance lattice."

The balance chief did not apologize. "We cannot fabricate that lattice."

Tavin looked at the water token. Stillness's reserve sat sixteen percent below target after a recycling contamination two months earlier. Four tonnes from the *Morrowline* would not correct it, but refusal would leave the refuge rationing while his ship carried enough water to wash cargo decks.

"Give them six tonnes," he told Pella. "Keep the guidance lattice. Offer two older modules and installation labor."

Pella muttered a judgment about captains conducting trade without supervision. Then she accepted.

Iona moved the ship token along the table. Scales changed. Staff shifted a water token in response. On the external camera, another nickel-iron block began its patient travel.

"This works for one refuge," she said. "Scaling it to seventeen billion requires more mass than some systems can move."

"Stillness started with an asteroid."

"Many cylinders have no such reserve."

"Then they need tugs."

"More than the Republic owns."

"Ships can carry less cargo."

"People need the cargo."

Tavin placed the token down. "I didn't say it would be pleasant."

The dockmaster rested her prosthetic foot against a floor brace. "People visit us and discover virtue in our shortages. They leave before the air filters need new catalysts."

"I don't see virtue," Tavin said. "I see a place that can name what holds it up."

"You see a place where half the children leave if they can afford passage."

The balance chief added, "Our conventional stability consumes thirty-one percent of industrial output. Two drums stay unbuilt because their metal became ballast. During illness years, we choose between transport fuel and machine stock."

Iona absorbed each cost without defending her observation. "Would you install a receiver if a consenting sink existed?"

The dockmaster answered at once. "Yes."

Aru's beads struck the tube wall sharply.

"Consent," the speaker said after a pause. "Unknown shape between unlike pressure."

"That's the problem," Iona said.

"Also the opportunity," Tavin said.

They brought the cradle into a test chamber beside Drum Two's main bearing. Stillness used the room to train crews in variable load. Mechanical actuators could tilt a section of floor, move practice masses, and simulate the vector changes produced by docking.

Iona connected the Helix record. "Aru asked to experience the rescue sequence through local acceleration rather than data labels."

"Why?"

"Human categories may have hidden the difference between the returned pulse and what we did afterward."

Tavin clipped his harness to the chamber rail. "Run it slow."

The first sequence reproduced the repossession at one-thousandth amplitude. The floor tipped toward starboard. Practice blocks shifted against restraints. Inside the cradle, the beads scattered.

Aru's speaker emitted pressure ratios too dense for translation. Iona marked the returned curve, the cycling direction, and the moment Helix Quay's receiver reopened old debt batches.

"Source mechanism," Aru said. "Human channel shaped wound. Return follows shape."

"Your faction sent it," Tavin said.

"Pressure-names disagree. This return chosen beyond Aru agreement."

"People died."

"Pressure lineages also ended."

Tavin thought of the nine names from Helix, the missing workers, and Nemi unable to trust the direction inside her skull. Anger wanted a single opposing captain, someone who had ordered the return and could receive his reply. The cradle offered only a speaker assembled from partial agreement beneath distant ice.

"I don't forgive it because your dead came first," he said.

The beads settled at different heights. "Forgive not translated."

"Good."

Iona looked at him. "Keep going?"

"Show the ship burn."

The chamber reset. Mechanical actuators reproduced the vector Tavin had imposed through the

Morrowline's docking yoke. A practice mass moved outward. Another countered it. The floor held one direction while the simulated debt declined.

Aru became quiet.

Tavin listened to bearings, actuator motors, and the click of Iona's controls. "What do you perceive?"

"Local mass accepts history. Expelled matter carries push away. Heat receives resistance. Human action does not open wound."

"Same human machinery," Iona said. "Different method."

"Human Tavin joined injury place and prevented greater human fracture."

Tavin frowned at the cradle. "I also captain a ship with a load compensator. We've cleared debt through Eidos for years."

"Mechanism records that."

"Then don't separate me from it."

"Difference retained does not erase connection."

Iona replayed the rescue at a higher safe amplitude. This time she marked the conventional thrust, water transfer, released cargo rings, and fuel expenditure. Aru responded to each physical destination. When the model reached the point where Helix's returned debt dropped below correction threshold, the three beads aligned.

"Rescue changed continuation," the speaker said. "Mechanism injured pressure. Rescuer and mechanism overlap. They are not one pressure-name."

Tavin understood enough to dislike the comfort available in the statement. Aru had distinguished his choice without declaring him clean. That seemed more exact than absolution.

After the test, they climbed to Stillness's outer observation blister. The refuge rotated beneath a field of stars. No tuned gravity corrected the room, so the apparent down changed slightly as they crossed the drum's geometry. Outside, ballast crawlers still adjusted for the *Morrowline*. A block locked into place, and the deck steadied beneath Tavin's boots.

Iona stood beside him with the cradle secured to a rail. "Bex wants a localization trial."

"He wants ten days."

"He wants proof that reduction can preserve inhabited structures."

"With which target?"

"A farm torus volunteered for a controlled test, but the model needs a heavy ship to move mass and provide conventional thrust."

"Volunteered through whom?"

"The torus council voted. Not unanimously."

Tavin watched Stillness's radiators shed heat earned by moving his ship's weight. "How much debt?"

"Enough to risk a growing cycle. Low enough to evacuate if the trial fails."

"Crew exposure?"

"Unknown until you build the burn."

"That means high."

"Yes."

He called Pella. "How long to restore Cluster Three?"

"If the refuge sleeve passes inspection, eighteen hours. Twenty-two if it lies about its age."

"Can we tow mass with the compensator locked out?"

A silence followed. Pella understood the question before he explained it.

"We can," she said. "Crew acceleration limits cut our useful thrust. Yoke fatigue becomes personal. Fuel cost climbs."

"Build me a localization configuration."

"For Stillness?"

"For a farm torus."

Pella used a word the public channel would have fined. Then she asked for the torus mass plan.

Tavin looked at Iona. "Send it to engineering."

Her shoulders lowered by a fraction. "Is that a yes?"

"It is a ship commitment. Politics can find its own word."

He opened the *Morrowline's* status and formally locked its RME load compensator offline. The command required captain confirmation, chief engineer confirmation, and a declared reason. Pella signed from the damaged injector bay.

Tavin entered: Controlled localization trial. All thrust and mass destinations to remain local and recorded.

The ship accepted the order. Across the docking collar, Stillness's balance room received the change and moved a new set of ballast blocks into its schedule.

Outside the blister, one black mass began crawling along a silver rail. The *Morrowline* waited under honest load while its captain signed away the easier way to move.

Chapter 9 — The Massless City

The vote to become massless concluded while Iona's shuttle remained thirty kilometers away.

Vesper Span turned ahead of her as five inhabited rings threaded along a common spindle. Sunlight flashed from farms, radiator fins, and mirrored windows in separate rhythms. A crowd filled the civic channel, counting down the final seconds to receiver shutdown.

“Three,” thousands of voices said.

Iona tightened her harness. “Open a line to city control.”

“Two.”

The shuttle pilot sent her credentials. No answer came.

“One.”

A cheer overwhelmed every other channel.

The receiver field vanished.

Vesper Span did not fall. For seven seconds, its rings continued turning with all the momentum they had accumulated over eighty years. People shouted that nothing had happened. Public overlays filled with a new slogan: THE FIRST MASSLESS CITY.

Then Ring Five moved off axis.

The displacement measured less than a meter at first. Iona saw it in the docking solution before the eye could separate one silver arc from another. The ring carried a reservoir district and two hundred thousand tonnes of water unevenly distributed after morning pumping. Receiver trim had hidden that imbalance. Without it, the main bearing began taking a lateral load three times its design margin.

“Vesper control, this is Senior Auditor Rhyll. Reinstate minimum field until mechanical balance is established.”

A council officer answered. Applause sounded behind him. “The shutdown passed by sixty-one percent. We will not resume exporting harm.”

“Then release Ring Five's reservoir transfer and move water hubward.”

“We are following the Orbit Breakers cessation plan.”

“Your ring has thirty-eight seconds before bearing protection releases it from the spindle.”

The applause stopped.

On the external view, Ring Two began to lag. Ring Four accelerated by a fraction as its agricultural pumps changed load. The spindle connecting them transmitted forces the receiver had managed separately, and each ring answered according to its own mass.

Iona switched to emergency command. “Pilot, get us onto Ring Three. Match physical rotation, not beacon guidance.”

“The beacon says Dock Six.”

“The beacon still assumes field compensation.”

He disabled the automatic approach. Conventional thrusters rolled the shuttle until stars and rings spun together. Dock Six crossed their nose too fast. The pilot chose a maintenance collar closer to the hub and burned hard.

Vesper Span's civic cheer fragmented into alarms.

BEARING LOAD EXCEEDED.

SPOKE TRANSIT SUSPENDED.

RING PHASE ERROR.

The shuttle struck its collar at an angle. Clamps caught on the second attempt. Iona released her harness into a load barely one-fifth standard. The deck rolled beneath her as Ring Three's spin changed.

A city technician opened the inner hatch. “We were told the receiver carried only comfort gravity.”

“Who told you?”

“The cessation packet.”

“Show me.”

He transmitted it while they crossed into the station. The document carried the Orbit Breakers' broken-circle mark and listed Vesper's five rings as mechanically self-sustaining. It cited bearing certifications, flywheel reserves, and a forty-year-old emergency despin study. Every statement held some truth. None accounted for current mass distribution, deferred maintenance, or the loss of synchronized trim across five different rotations.

The plan had treated a city as a diagram that stopped aging after approval.

A jolt lifted Iona from the floor. Her tether caught. The technician hit a wall shoulder first.

“Ring Five released,” city control announced. “Automatic isolation successful.”

Successful meant explosive bolts had severed six connecting bridges before the failing bearing tore the spindle. Ring Five now rotated free. Its orbital path would intersect Ring Four's farm structures in nineteen minutes unless tugs changed its drift.

“How many tugs?” Iona asked.

“Two maintenance craft.”

“You need six.”

“Our cargo fleet undocked before shutdown.”

“Why?”

“The cessation plan identified docked masses as instability risks.”

It had removed the machines capable of managing instability.

Iona called every ship still within Vesper traffic range. Three refused because their load compensators were offline under Bex's reduction order. One passenger transport had conventional maneuvering reserve but no reinforced yoke. A fuel tender answered with two service boats. The *Morrowline* lay thirty-one hours away at Stillness.

A familiar voice entered the channel from Helix Quay. Nemi spoke slowly, each word separated as if her damaged sense of balance could infect language.

"I am looking at Vesper's live trace."

"You should be resting."

"I can rest while the city stops coming apart."

"Can you model ring phase from there?"

"I can model it better from a bed than they did from a manifesto."

Iona sent mass and spin data. Nemi assigned a separate color to each ring. The curves diverged.

Ring One carried industry and the largest conventional flywheel. Ring Two held dense residential districts. Ring Three contained administration and docking. Ring Four grew most of the city's food. Ring Five carried water and fabrication yards. The receiver had supplied only twelve percent of felt gravity across the city, but it had trimmed thousands of small exchanges every minute. Pumps moved water. Trains carried people. Ore entered factories. Crops became shipments. Each transfer changed balance. The field had made those changes appear free.

"Ring Four must shed farm mass," Nemi said.

"Crops?" the council officer asked.

"Water first. Then soil trays if water cannot move fast enough."

"That destroys the season."

"If Ring Five crosses your farms, you lose the ring."

The officer went quiet.

Iona reached civic control through a passage where down shifted every twenty steps. People clung to rails or crawled. A cafe had become a field of floating cups, stools, and boiling drink. Two workers wrapped a burned customer in insulation cloth. Across the corridor, a school group linked harnesses while a teacher counted names.

Personal screens continued showing the Orbit Breaker emblem. The movement had organized the vote after Helix Quay, arguing that every hour of receiver use extended injury at Eidos. They had given Vesper residents a choice the Republic had withheld for a century. They had also given them a shutdown checklist that assumed urgency could substitute for preparation.

In control, seven council members occupied acceleration couches arranged around the spindle display. Three wore broken-circle bands. Engineers stood behind them calling out bearing temperatures.

The council officer faced Iona. "If you ask us to reconnect, the public will refuse."

"I am asking you to survive without reconnecting."

"Can we?"

"Smaller."

The word changed the room.

Iona placed Nemi's ring-phase model over the city schematic. "Release Ring Five completely, then move it outward with service craft. Drain half its water into sacrificial tanks before closest approach. Ring Four dumps agricultural mass sunward to increase separation. Ring One transfers flywheel momentum through the spindle to hold Rings Two and Three together."

A senior engineer shook his head. "The spindle clutch cannot carry that torque."

"Not at full lock. Pulse it."

"The clutch plates will erode."

"How many pulses?"

"Maybe eleven."

"We need seven."

"And afterward?"

"Afterward you replace the clutch."

"With what?"

No one answered. Vesper manufactured receiver components and fine instruments. Its heavy structural stock had been sold years earlier because tuned load reduced the need for massive spare parts.

A council member opened the receiver command. "Minimum field for one hour would save the clutch."

Another caught her wrist. "That sends debt."

"To whom?" asked the first. "Does anyone know whether Eidos remains assigned?"

"The active sink still shows Eidos," Iona said.

"Then we cannot."

"We can ask," said a third member.

"Through what language? While Ring Five crosses the farms?"

Iona watched the argument begin to consume seconds. "Your vote ended hidden export. It did not vote away momentum. Choose the local cost now."

The council officer looked at the projected intersection. "Do it without the receiver."

Orders moved.

On Ring Four, farm crews opened reservoir gates into emergency transfer bladders. Water surged hubward. Thousands of tonnes of wet soil remained in the outer terraces. Workers cut tray locks and pushed whole growing beds toward cargo ejectors. Cameras showed green leaves folding against transparent covers.

Ring One's flywheel spun down. The first clutch pulse struck through the city. Iona felt it as a blow across her chest. Rings Two and Three moved closer in phase.

"Evacuate all bridges between free structures," she said. "No one crosses a spindle connection without tether pairs."

The council transmitted the order. Twenty-six thousand people occupied Ring Five. Its separation bridges had blown, and the two maintenance tugs could not stabilize every docking point. Evacuation required small craft moving between surfaces with different rotations and changing apparent load.

Iona left control for Dock Six.

The passage there had become a vertical shaft according to her body. She climbed a rail past residents carrying sealed bags, children, medicine cases, and one cage of bright yellow birds. An elderly woman had tied cooking pots into a blanket and refused to leave them.

"Mass limit is ten kilograms," an evacuation worker told her.

"These belonged to my mother."

"The boat needs maneuvering margin."

Iona took the bundle, opened it, and selected the smallest pot. "One."

The woman looked at the remaining metal as though each piece held a room. Then she chose a second and left the rest clipped to the rail.

At Dock Six, service boats matched Ring Five's free rotation one at a time. No receiver eased their approach. Thrusters fired in short bursts. Each arrival consumed fuel the city could not replace quickly. Residents crossed through flexible collars while the deck pulled first toward their feet, then toward a wall.

A boat pilot called, "I can take thirty more if they leave cargo."

"People only," Iona answered.

A fabrication worker pointed to a sealed crate. "Those are bearing blanks."

Iona checked the manifest. The blanks could help restore Ring Two but weighed as much as eight passengers.

"How many remain?"

"Four crates."

"Take one crate on each of the last four boats."

"If there are last boats."

"Then people displace them."

The worker nodded without liking it.

A hospital group reached the dock behind him. Four patients lay in pressure cocoons mounted to a single powered frame. The frame had been built for steady ring gravity and kept trying to level itself against a direction that no longer held. Each correction drove one wheel off the deck.

"Separate the cocoons," Iona said.

The nurse in charge shook her head. "Shared circulation unit."

"Can it run in free movement?"

"For seven minutes. The next boat boards in five if the collar remains aligned."

Ring Five's changing spin pulled the frame toward the outer wall. Six people caught it. The patients could not see out through their opaque covers, but one hand pressed against the inner surface of the nearest cocoon.

Iona called the boat pilot. "You take the medical frame first."

"It fills my central aisle. I lose twenty seats."

"Then make another trip."

"I have fuel for one after this."

The nurse opened a status list. Two patients could tolerate individual carry harnesses. One had a spinal fixation that demanded rigid support. The last depended on the shared circulator after a lung graft.

"Split two," Iona said. "Keep the frame for the others."

Technicians released latches while load moved through forty degrees. A cocoon slid free and nearly crushed a medic's hand. Passengers waiting at the collar held the detached units above their heads, passing them forward whenever the floor shifted. Nobody asked whether the patients had voted for shutdown. Nobody asked whether their places on the boat cost someone else a departure.

The shared frame entered with sixteen passengers packed around it. Its power lead stretched across the collar until the nurse disconnected at the final second. Internal batteries took the load. A timer appeared on the cocoon display.

When the boat pulled away, fifty people remained at Dock Six and the next craft had not yet cleared Ring Three. Iona watched the battery minutes fall on the external feed. The rescue boat corrected twice, spent more propellant than planned, then reached the receiving collar with ninety seconds left.

Only after Ring Three confirmed hospital power did the people at Dock Six begin boarding again.

Nemi's voice guided the timing. "Ring Five closest approach in twelve minutes. Relative plane increasing. That helps."

"Ring Four mass dump?"

"Behind schedule. Two ejectors jammed."

The farm feed showed crews outside the ejector housings, cutting soil trays apart by hand. The first severed

tray moved into space carrying rows of young grain under shattered cover. Sunlight caught moisture crystals around it.

Orbit Breaker messages continued flooding civic channels. Some praised Vesper for refusing complicity. Others accused Iona of staging panic to force reconnection. One cell urged residents to disable the receiver physically before the council changed its mind.

“Security reports a crowd moving toward the core,” the council officer said.

“How many?”

“Six hundred.”

“Do not use force unless they enter machinery spaces. Broadcast the live bearing loads.”

“They think those displays belong to your office.”

“Give them raw sensor access.”

“You want unfiltered structural data public?”

“Yes.”

The city published bearing temperatures, clutch erosion, spin phase, tug fuel, and Ring Five's projected path. The numbers reached the crowd faster than police could. People who had voted for shutdown watched the spindle clutch lose micrometers with each pulse. They saw the farm mass falling away. The march slowed, then split. Half continued to protect the receiver from reconnection. The rest went to loading points.

Iona did not mistake the change for trust. It amounted to better work.

The third service boat arrived too fast. Its pilot corrected late and struck the collar rim. One thruster pod folded against the hull. The boat rebounded with forty-two passengers aboard.

“Dock seal failed,” someone shouted.

The collar membrane tore. Air rushed from the access bay. Doors began closing with six people still outside the inner threshold.

Iona caught a child's harness and pulled. An evacuation worker dragged two adults. The elderly woman with the pots clung to the deck rail on the wrong side of the seal. Her blanket bundle opened. Metal spun through the escaping air.

The smallest pot struck Iona's shoulder. She seized its handle, reached through the narrowing door, and offered it to the woman.

“Leave it,” the woman gasped.

Iona let the pot go and caught her wrist instead.

Three people pulled together. The woman crossed the threshold as the inner door sealed against her boot. A pressure warning counted down. The damaged service boat drifted beyond the collar with its passengers secured inside but no reliable steering.

“Fuel tender Boat Two, intercept,” Iona called.

“We are loaded.”

“Unload at Ring Three.”

“Closest approach in nine minutes.”

“I know.”

The boat pilot burned. On the external feed, two tiny craft moved between enormous rotating arcs. Their plumes traced every correction. Boat Two matched the damaged vessel, coupled nose to tail, and began pushing both toward Ring Three.

The rescue consumed the last fuel tender reserve.

Ring One delivered the fifth clutch pulse. Warning lights spread across the spindle model. Plate temperature approached failure margin.

“Phase difference still too high,” Nemi said. “You need two more.”

The senior engineer refused. “One more. After that, the clutch welds or fragments.”

“Can Ring Two shed load?” Iona asked.

“Residential water. Not fast enough.”

“Trains.”

Six commuter trains moved around Ring Two against its rotation, positioned for a morning schedule that no longer existed. If they accelerated together, their momentum could alter ring phase by a small amount.

“Put every empty train on full counter-spin,” Iona said.

“Passenger cars are not flywheels.”

“Today they are.”

Transit control cleared the lines. Four trains responded. One had lost power. Another remained occupied by people sheltering after shutdown.

“Evacuate that train?” the council officer asked.

“No time. Tell them to strap in. Move it.”

The trains accelerated. Residents inside filmed station signs flashing past as the cars contributed their small store of momentum. Ring Two's phase curve bent toward Ring Three.

The spindle clutch fired once more.

Metal shrieked through the civic center. Several displays went black. The floor lurched, then stabilized at three-tenths standard.

Nemi's colored curves approached each other and remained separate by less than the damaged bearings could

tolerate.

“Rings Two and Three coupled,” she said. “Ring One stable enough to hold itself.”

“What about Four?”

“Farm mass dump completed. It will remain independent.”

Ring Five passed outside Ring Four with eighty meters of clearance.

From the docks, Iona saw it cross the window. The free ring filled half the view, close enough for individual fields and water tanks to show. Soil, grain trays, and atmosphere crystals drifted behind it. Service craft guided the remaining residents away. No collision came.

The room did not cheer. People counted boats.

Over the next six hours, Vesper Span evacuated Ring Five and isolated Ring Four. Two residents died during transfer. Fifty-seven suffered pressure or impact injuries. The spindle clutch survived with three plates fused. Ring One kept its industry. Rings Two and Three remained linked at reduced spin. Ring Four retained half a growing season and no direct bridge to the city.

Ring Five could not be brought back. Its bearing assembly had torn beyond repair, and Vesper lacked a tug powerful enough to recover the mass before orbital drift widened. The council converted it into an unmanned water and fabrication depot. Remote craft would visit when fuel permitted.

In the civic chamber, exhausted council members reviewed the shutdown vote. No one proposed reconnecting the receiver. No one repeated the claim that leaving it required only a command.

An Orbit Breaker organizer removed the broken-circle band from her arm and placed it on the table. “We were right to stop.”

“Yes,” Iona said.

“We were wrong to promise no one would pay here.”

“Yes.”

The organizer looked through the chamber window. Ring Five had already moved far enough to create a gap in the city's familiar geometry. “Will you arrest us?”

“I do not have that authority.”

“Will Bex?”

“I don't know.”

“What do you call what happened?”

Iona considered the dead, the lost farms, the service-boat fuel, the abandoned kitchens, and the unseen pressure lineages spared another clearing. None canceled another.

“A shutdown without a destination for its force,” she said.

Outside, Vesper Span turned on three coupled rings while a fourth kept its distance. The fifth drifted away

under tug lights, carrying empty homes and half the city's water. Its receiver remained dark.

Interlude II — A World That Never Woke

Before the first debt pulse, the nursery held four weather layers.

The outer shell turned through dark with no star near enough to warm it. Far below, ice changed crystal form under the mass above. Deeper still, helium remained fluid and carried phase marks along pressure boundaries. Each tide from the two small moons bent those boundaries by a measurable degree. Repeated bends wrote memory.

No pressure-name yet cohered there.

A nursery began before a speaker. It began with compatible marks that endured near one another. One layer retained the fall of mineral dust from a fracture. Another retained the slow torque of a moon after an impact altered its orbit. A third held variations made by heat from the world core. When patterns met across a stable boundary, they tested duration. What lasted could join. What joined could remember change.

The four weather layers did not agree. Agreement had no value yet. Their differences created capacity.

One pattern favored short intervals and sharp borders. Another spread each event across a broad band. Two more had no stable preference. Over nine thousand moon tides, they began to alter one another. A phase knot formed where none had remained before. It held through compression, release, and a minor fracture above.

The nursery had reached its first durable relation.

Then a force arrived without local cause.

It entered through a structure buried deeper than the forming patterns could reach. No moon had moved to make it. No ice mass had shifted nearby. Curvature changed across the world, delivered from a remote origin with no pressure history attached.

The pulse compressed two weather layers together.

Their marks did not break at once. They overlaid. Fine intervals became broad ones. A record of mineral fall merged with a record of core heat. The phase knot endured for three tides, altered beyond its prior form, then dispersed into common flow.

Nothing there could call the event harm. Nothing yet had language.

After the pulse, the nursery began again from what remained. A narrow boundary formed below the damaged layers. It retained the remote event as a vertical scar: force without approach, change without a body nearby. New phase marks gathered around the scar because it offered a durable edge.

Over many tides, those marks created a second relation. This one differed from the first. It carried the remote compression within every exchange. It learned duration against a wound it could not identify.

A second pulse arrived before the relation could divide into further patterns.

The world accepted more curvature than its moons had ever delivered at once. Deep ice moved by less than the width of a small crystal. At nursery scale, that motion remade every boundary. The vertical scar widened. Marks from one layer entered another and could not recover their prior intervals.

Again, no formed speaker ended. No count of dead can be drawn from a coherence that had not yet learned to count itself.

Possibility changed nonetheless.

The nursery after two pulses held fewer durable differences. Its phase borders had become steep and brittle. Broad patterns survived because they tolerated compression. Fine ones dispersed. A later pressure-name might have emerged from the broad marks, but it would never carry the forms erased between the first pulse and the second.

Remote force came again.

Intervals varied. Some pulses barely moved the nursery. Others drove fractures down from warmer ice, opening paths for trapped gases. Each event carried a distinct shape, yet all arrived through the same buried structure. The nursery adapted where adaptation remained available. It shifted formation toward deeper bands. It linked memory to moon tides that could be predicted. It placed new phase knots away from old scars.

The remote origin changed its schedule.

A pulse struck during maximum lunar compression. The added load matched a natural tide closely enough to amplify it. Three new knots collapsed into one broad mark. Another event arrived during release and tore that mark across two layers. The pieces persisted, but they no longer met.

For a long period, nothing more complex formed.

The ice did not become empty. Fluid moved. Heat traveled outward. Mineral traces entered fractures. Moon tides continued to write regular curves. The world contained motion, storage, and response, but no coalition endured long enough to hold a pressure-name.

Far away, other nurseries reached coherence. Their first speakers learned to compare changes, retain replies, and join many histories for a time. They found the buried channel through wounds that arrived from remote users. Pressure-names gathered around damaged archives and made low-band answers.

This world could not add an answer.

Its nursery tried another route. Near the core, a band of clean fluid had escaped much of the prior distortion. Phase marks there accumulated slowly. They had poor access to moon tides but clear heat cycles. A relation formed between two thermal records, then a third. For the first time after many pulses, separate patterns altered one another without collapse.

The relation endured.

Had it continued through enough cycles, it might have become a lineage. It might have joined later patterns into a speaker able to know the difference between local tide and remote force. It might have rejected that division. No archive can preserve a future that did not occur.

Another debt pulse entered.

This event had low amplitude and long duration. It did not fracture ice. It changed the thermal boundary by a small amount across many cycles. To remote instruments, the world would have appeared quiet. Within the forming band, heat marks drifted until their intervals no longer met. The three records separated.

Two later found each other again. The third did not.

A lineage almost began from the pair. Repeated remote loads moved the thermal boundary back and forth. Each time, the pair rebuilt part of its relation. Each time, the next load arrived before enough structure endured.

The nursery became an archive of attempts.

Vertical scars marked abrupt pulses. Blurred bands marked long ones. Split phase knots marked relations that had reached durability without coherence. A later observer could read sequence and damage, but not the exact lives that might have formed. Certainty ended at the boundary between preserved structure and unrealized future.

Elsewhere, a pressure-name received a weak echo from this world. The echo carried no deliberate modulation. It came from the way a new remote pulse touched old scars and made them answer together.

Several speakers examined that echo across many tides. One treated it as empty mechanical return. One found intervals too organized for that account. A third compared the marks with healthy nurseries and detected the repeated near-formations.

They gave the world no personal name. A name would imply coherence the archive could not prove.

They gave the condition a pressure relation instead: a place where formation approached, broke, approached by another route, and broke again.

Remote pulses continued.

The outer shell turned. Moons bent the ice. Deep fluid carried fragments that still altered one another at rare intervals. No observer could declare that nothing would ever form. No speaker could claim every vanished pattern would have become a life.

The archive supported a narrower statement.

A nursery had begun many times. External curvature repeatedly changed the boundaries it required. Across ages when related worlds formed pressure lineages, this one never held coherence long enough to answer.

The final known pulse crossed the deep band during a quiet lunar tide. Old scars responded in sequence. For a moment, their combined intervals resembled a pressure-name assembling across the world.

The relation failed before the next tide.

The world never woke.

Chapter 10 — The Ledger War

The first accusation reached Iona while she was fastening Nemi into a balance chair.

A wall panel in the Curvature Office clinic showed Elian Bex at the Assembly lectern, one hand resting beside the old engineering key he still wore on his belt. The chamber behind him held forty-three system standards. Thirty-one glowed green for remote attendance. Twelve hung physically from the gallery, shifting in Nacre's carefully moderated air.

Nemi kept her eyes closed while Iona tightened the strap across her hips. Since Helix Quay, an elevator could persuade her body that it had fallen sideways. The chair's gimbals would give the damaged vestibular nerves a stable reference during the broadcast.

"Too tight," Nemi said.

"You said that before the last turn."

"It became true this time."

Iona eased the webbing by one finger. On the panel, Bex declined the Assembly clerk's offer of water.

"At zero six forty Nacre standard," he said, "I invoked emergency authority under the Continuity Preservation Act. The authority will remain in force for ten standard days unless revoked by this chamber. Receiver operations will continue at restricted amplitude while the Ministry establishes a cessation schedule."

The clinic display divided. One half kept Bex. The other filled with a document carrying the Curvature Office seal and Iona's audit number.

Nemi opened her eyes. "That isn't our release."

It contained the sink-response summary they had prepared under Bex's containment order. Someone had removed the uncertainty ranges, the channel traces, and every date earlier than Day 3. The remaining page admitted contact with nonhuman intelligence in Eidos. It described the Olun reprisals as an unprecedented external attack.

Below it, a public annotation counted the dead at Helix Quay.

Iona touched the panel and opened source history. The summary had left the Continuity Ministry twelve minutes earlier. Bex's signature authorized publication.

"Six years," Nemi said. "He cut out six years."

The Assembly heard the same omission seconds later. A representative from Lumen interrupted before Bex finished his next sentence.

"Minister, did your office possess structured sink responses before Auditor Rhyll's finding?"

Bex looked toward the public recorders. He had calculated this moment, Iona thought. He had likely rehearsed a dozen versions and rejected each one because none could make concealed time harmless.

"Yes."

“How long?”

“Six years.”

The chamber microphones opened all at once.

Sound struck the clinic in a compressed burst before the broadcast system separated speakers. Questions climbed over demands for resignation. Someone called the continued clearings an undeclared war. Another voice asked whether turning off the network would kill seventeen billion people. The clerk muted the chamber. Public channels supplied their own noise.

Nemi's chair rotated two degrees as she tensed. Iona steadied it with both hands.

Bex waited until the clerk restored order. “The signals did not meet the contact standard then in force. They did establish structured modulation. I sealed them. I also authorized continued transfers.”

“You knew they answered,” the Lumen representative said.

“I knew something opposed the pulses.”

“You knew enough to hide it.”

“I knew an abrupt cessation would place seventeen billion people in structures unable to maintain safe load, orbit, or spin. That fact remains true this morning.”

The panel's public side multiplied into ledger windows. Helix mutual-aid networks posted thirty-four confirmed dead, eight missing, and treatment counts for more than two hundred injured across the damaged cargo decks. A Continuity Ministry model appeared beside it: 2.6 billion deaths in the first thirty days after immediate network shutdown, followed by a broad range that climbed above nine billion within a year. Neither account shared assumptions. Each spread faster than its source data.

A third ledger entered from an Orbit Breaker channel. It labeled every resident of an RME-dependent habitat a hostage maintained by murder and assigned zero moral value to continuity losses after the first warning.

Nemi read it twice. “They turned people into a rounding rule.”

“Bex did the same in the other direction.”

“He at least counted them.”

“He counted one side with names and the other with a threshold.”

The chair's gimbals corrected a small motion in the clinic floor. Nemi gripped the rests until the false tilt passed. Iona watched the live ledgers breed copies. Numbers acquired colors, slogans, and portraits. Every system chose a format that made its fear appear measured.

A red notice crossed her terminal: REPORT TO CONTINUITY REVIEW, IMMEDIATE.

Nemi saw it. “He wants you inside the authority before you can contradict him.”

“He may want the traces explained.”

“He has had six years to ask.”

Iona released the chair's lock. "Can you work?"

"I can work without walking."

"That wasn't the question."

Nemi tested the room by turning her head. Her face tightened, then settled. "Give me a fixed screen and no helpful people pulling the chair."

Iona rolled the instrument console within reach. "Open every source packet from the sealed sink archive. Make a checksum map. Do not transmit."

"You are going to the review?"

"I am going to learn which account they intend to call complete."

As Iona left, the clinic panel showed Bex under forty-three standards and a thousand incompatible totals.

Emergency authority changed the building before it changed the law.

Two Continuity officers occupied the lift vestibule. Curvature Office staff surrendered personal wafers at each floor boundary. Public windows had gone opaque, but crowds beyond the glass made their presence audible through the frame: boot strikes, shouted names, a steady hammering against the outer barrier.

Iona took the mechanical stair. Unmodulated Nacre gravity pressed on the old fear behind her knees. Each landing carried a Ministry notice explaining that restricted clearings would preserve essential services. Someone had written HOW MANY REEFS beneath one notice. Another hand answered HOW MANY CHILDREN.

The review room had once housed sink-certification appeals. Its long table displayed orbital maps from all forty-three systems. Bex stood at the far end with three continuity modelers and a legal clerk. No security officer entered with Iona, though two remained outside.

"You published an incomplete packet," she said.

Bex dismissed the modelers with a glance. The clerk stayed to preserve a witness.

"I published what I could verify for public use."

"You removed the older responses."

"I admitted them under oath."

"You called the repossessions an external attack."

"They are coordinated returns of force that kill civilians. Ordinary language permits the word."

"Ordinary language also permits cause."

Bex touched the table. Forty-three systems contracted into a web of receiver hubs. Seventeen billion dependent residents appeared as white points, too many for the display to separate.

"This is cause," he said. "Mass, spin, and trajectory. If your preferred account omits those points, it lies as

completely as my sealed archive did.”

“My account omits no one.”

“You do not have an account. You have evidence.”

“Then stop dressing your forecast as a tally.”

He enlarged the continuity model. “Every figure carries a range.”

“The public copies don't. Your lower bound became a guaranteed body count before I left the clinic.”

“And the Olun-loss ledger assigns certainty to possible lineages that never formed.”

“It did not come from my office.”

“It came from people using your finding.”

Iona looked at the model assumptions. The first thirty-day estimate treated every receiver shutdown as simultaneous. It allowed no conventional tug mobilization, no habitat consolidation, no mass relocation, and no Olun restraint. The resulting curve had mathematical integrity inside a world where nobody adapted.

She opened the model's neglected branch. “You buried the staged case below six access layers.”

“The staged case lacks fuel allocations.”

“So fund them.”

“It lacks transfer craft.”

“Command them.”

“It still loses habitats.”

“Name those habitats, then. Don't convert them into a line thick enough to hide behind.”

For the first time, anger altered Bex's voice. “You think naming a place gives me engines.”

“No. It gives the people there a chance to know what you have assigned them.”

The legal clerk stopped writing by hand and let the room record for her.

Bex closed the model. “Emergency authority exists because public deliberation cannot move reaction mass in time.”

“Then use it to move reaction mass.”

“I am.”

“While continuing exports to Eidos.”

“At restricted amplitude.”

Iona heard Sella's metal nut strike an old station deck. Restricted amplitude described the sender's hand, not the destination's injury.

“Has Aru accepted that restriction?” she asked.

“Aru does not speak for every pressure-name.”

“Has any Olun speaker consented?”

“No.”

“Then the transfers remain involuntary.”

Bex walked to the wall where the original sink certificates rested in their ceremonial frames. Empty had been printed beneath Eidos in eight languages. “If I stop every transfer today, rings desynchronize before we can place tugs. Hospitals lose controlled load. Ships under burn lose compensation. The deaths begin among people who never saw a sink ledger.”

“And if you continue?”

“I cannot count the Olun losses.”

“That limitation does not make them zero.”

“No. It makes command unbearable.”

The admission landed without defense. Iona saw the former habitat engineer beneath the ministerial coat, a man who could picture bearing failures in inhabited rings and had chosen a distant wound because it never appeared on his maintenance floor.

Bex removed a storage wafer from the table. “Here are the emergency orders. Tug requisitions. Receiver caps. Fuel release from the naval reserve. A mandatory localization design due in forty-eight hours. I am not asking you to praise concealment.”

“What are you asking?”

“Run the sink account. Give the Assembly something more honest than imaginary certainty.”

“You want my name on your partial release.”

“I want your methods on the public record.”

The crowd struck the outer barrier in a new rhythm. Three blows, then two. Iona wondered whether someone had assigned meaning to the pattern or whether people simply needed their hands to agree.

“Publish the old archive,” she said.

“No.”

“Then you want an auditor as decoration.”

“I want ten days in which the Republic remains capable of changing.”

“You already spent six years.”

Bex's gaze moved to the sealed wafer in her hand. “And you think releasing unprocessed channel records will give those years back?”

“No.”

“It may trigger receiver attacks. It may cause systems to sever before localization plans exist.”

“Yes.”

“Then why?”

“Because every model in this room begins after the fact you hid. People cannot judge your authority while you control the beginning of the account.”

Bex did not order her detained. That restraint might have reflected law, calculation, or shame. Iona could not audit motive.

“You remain bound by containment,” he said.

“Your public admission destroyed the factual basis for containment.”

“The order remains valid under emergency authority.”

“Put that in writing.”

The clerk resumed her manual notes.

Outside the review room, an alarm had emptied the receiver-control bay into the corridor. A young operator knelt beside an open service panel while two Continuity officers read an emergency order over his shoulder. His console showed a hospital load request from the inner spindle and an automatic Eidos assignment beneath it.

“Authorize the clearing,” one officer said.

“The cap allows half this load,” the operator replied.

“Medical priority overrides the cap.”

“Then give me another sink.”

“No alternate channel can open inside the treatment window.”

Iona stopped. The operator recognized her and shifted his body enough to expose the screen. The hospital needed controlled load for a spinal repair. Mechanical support would delay the procedure by nine hours. The outgoing pulse fell below every public restriction Bex had just described and still lacked consent at its destination.

“Your audit access has been suspended,” the officer told Iona.

“My eyesight has not.” She addressed the operator. “Can the theatre brace locally?”

“For four hours. After that, thermal limits.”

A conventional service tug could stabilize the spindle by moving two ballast casks along its outer rail. The maneuver would consume reserve propellant and cancel a cargo window. Before emergency authority, three offices would have argued over the expense for a day.

Iona pointed to Bex's new fuel release. "Charge the tug to continuity reserve."

The officer studied his order. "The Ministry can direct receiver use."

"It can also direct fuel. Choose the order that doesn't export an unconsented pulse."

The operator routed the request before anyone stopped him. A tug accepted within twenty seconds. Cargo control protested immediately, which meant the solution had become real enough to inconvenience someone.

No officer arrested Iona. One recorded her name against the intervention. She added his name to her own notes, not as an enemy, but as another person standing where an abstract command acquired a physical destination.

Nemi had built the release packet when Iona returned to the clinic.

The balance chair faced six fixed screens. Each showed a different layer of the sealed archive: raw timing traces, receiver assignments, analyst comments, suppression orders, sink survey limits, and Bex's authorizations. Nemi had added no title page.

"I told you not to transmit," Iona said.

"You told me to make a checksum map." Nemi pointed toward a seventh screen. "The map showed three missing blocks. I recovered two from instrument caches. Bex's office deleted the third six years ago."

"What did it contain?"

"Amplitude correlations. We can infer the shape from references, not restore the values."

"Mark the absence."

"I did."

Iona lowered herself into the chair beside Nemi. The clinic now admitted distant crowd noise through its ventilation ducts. Staff had stopped pretending to work elsewhere. Two nurses stood in the doorway. A receiver technician watched from the floor, his back against a cabinet.

On the public channels, the ledger war had acquired victims chosen for rhetorical use. A child from the Massless City appeared in five continuity feeds, each copy ending before she said her family supported localization. An image translated from an Olun memory reef circulated beneath an invented death count. Someone had placed Sella's face beside a demand for immediate severance, using the official portrait that still named operator error.

Iona closed that window.

Nemi noticed. "They found her quickly."

"Public archives are efficient when they need a dead woman."

"The raw data will not stop that."

"No."

“It won't tell people what to choose.”

“No.”

Nemi turned carefully within the chair. “Good. I wanted to know whether you had become foolish on the stairs.”

Iona reviewed the release structure. Every file retained its original timestamp. Uncertainties remained attached. The archive separated observed modulation from inferred damage. It also showed the six-year sequence plainly: response, analyst escalation, ministerial seal, continued clearing, stronger response, revised threshold, further clearing.

One comment from a junior analyst asked whether the sink might contain a pressure-sensitive computational medium. The following line reassigned her to navigation audits.

“This name stays private,” Iona said.

“Her analysis is part of provenance.”

“Her current location isn't.”

Nemi removed the personnel routing while preserving the signed comment.

Iona opened a declaration field and began to type. She stopped after one sentence. A preface would become another account competing for authority. The traces required context, but context could name limits without prescribing innocence.

She wrote four notes.

The archive did not provide a defensible count of Olun dead.

The continuity models did not provide a defensible count of deaths under every cessation path.

The raw records proved that structured opposition had been detected six years before public disclosure.

No observation in the archive established Olun consent to continued use.

Nemi read over her shoulder. “Bex will call the second line irresponsible.”

“It says his model has boundaries.”

“He regards boundaries as things other people place around him.”

A call request appeared from the Assembly evidence clerk. Another followed from the Continuity Ministry. Then the Curvature Office director suspended Iona's publication credentials.

Nemi looked at the suspension and smiled without humor. “Seven seconds slow.”

“You transmitted?”

“I prepared routes. You still have to sign.”

The packet could not pass through the public ledger as one block. Ministry filters would stop it. Nemi had divided it across forty-three local audit repositories, each protected by rules designed to prevent Nacre from

rewriting system accounts after a dispute. Once signed, the pieces would replicate and verify one another.

Iona studied the route list. "This fractures custody."

"Custody already fractured when they removed a block."

"It also exposes every system auditor who accepts a piece."

"They chose acceptance keys during the last hour."

"How many?"

"Thirty-four."

Nine systems had declined. Their empty columns looked more accusatory than a red refusal.

Iona called one accepting auditor at random, a woman in a mining spindle around Kestrel. "Do you know what this packet contains?"

"Evidence my government says will cause panic."

"That is not an answer."

The auditor drew breath. "It contains original sink responses, concealment orders, and uncertainties. I accept custody, not your conclusions."

"Correct."

"Do you still want my key?"

"Yes."

The Kestrel signature joined thirty-three others.

Iona pressed her thumb to the release.

For three seconds, nothing happened. Then the six screens began losing local ownership marks as pieces crossed into system repositories. The raw data did not appear as a single revelation. It arrived in fragments that demanded assembly: a trace in Lumen, a seal order at Nacre, receiver histories in Meridian, analyst notes in eight minor systems. Public networks seized the checksums and built mirrors faster than Ministry filters could name them.

The clinic panel changed first. Bex's continuity curve remained on the left. An Olun-loss estimate held the right. Between them, the archive opened a blank-backed record with no approved interpretation.

People filled that space at once.

Some read the traces and demanded immediate severance. Others called every file forged. Habitat councils began publishing local dependency accounts. Receiver crews posted maintenance limits the Ministry had classified. Olun advocates marked each concealed response by date without adding invented bodies. Three systems shut their public ledgers rather than mirror the packet. Two announced independent continuity commands.

The Assembly clerk asked Iona to authenticate the release under oath. The Continuity Ministry issued a

detention instruction, then withdrew it before security reached the clinic. Bex did neither on camera.

Nemi watched the system map divide into incompatible colors. “We wanted a shared account.”

“We wanted an honest one.”

“Those may not be the same thing.”

A new public ledger opened from Stillness. It counted local ballast, fusion fuel, available tugs, and how many residents the receiver-free city could shelter. No dead appeared in its first column. Capacity did.

Another ledger from an RME-dependent nursery listed hours until unsafe spin under immediate shutdown. Beside it, someone linked the oldest concealed Eidos response.

Neither cancelled the other.

Across the Republic, the single green network diagram broke into forty-three system accounts, then hundreds of habitat accounts beneath them. The lines no longer agreed on what counted as survival, injury, evidence, or debt. For the first time, they could not conceal their disagreement behind one total.

On Iona's screen, the official shared ledger failed checksum and went gray.

Chapter 11 — The Orbit Breakers

The unauthorized clamp flashed once beneath the nursery cylinder's receiver spine.

Tavin Serr caught the reflection through the *Morrowline*'s forward blister as his ship eased a ballast cask toward the cylinder's outer rail. A legal maintenance clamp would have carried a yellow locator pulse. This one swallowed light except when the blue-white primary struck its cutting edge.

“Hold translation,” he said.

Haru Den killed the tug program. Six thruster clusters dropped to idle in sequence, leaving the heavy freighter and its suspended cask moving on the momentum they already possessed. The cargo yoke groaned as the cask tried to continue alone.

Pella's voice came from engineering. “That hold just spent twelve kilograms of propellant we won't get back.”

“Charge it to my eyesight.”

“Your eyesight has no budget code.”

“Then make one.”

Tavin enlarged the receiver spine. The nursery cylinder turned below them, three kilometers long and narrow enough to complete a rotation every forty-two seconds. Window bands passed through sunlight, then under the radiation shade. Inside, twelve thousand children lived with parents, teachers, clinic staff, and the workers who maintained the cylinder's farms. Its receiver smoothed bearing loads and handled orbital trim that the original fusion tugs no longer supplied.

Four black clamps now showed against the spine.

Haru leaned toward his display. “Maintenance plan lists none.”

“Ask cylinder control whether they have a crew outside.”

The answer came before his query left the bridge. A woman on the public emergency channel spoke too close to her microphone.

“Receiver crew, clear the spine. We are going to sever the export system. You have nine minutes.”

A compressed file followed. It contained Bex's concealment orders, images from Helix Quay, and a translated pressure reef damaged by a human debt pulse. The last page carried the Orbit Breaker mark: an open ring crossed by a cut line.

Pella came up from engineering with a tool harness over her pressure suit. “They have better timing than the Ministry. Four clamps in under six minutes.”

“Can they cut the spine?” Tavin asked.

“Anything cuts if you put enough heat into it.”

“Not what I asked.”

She studied the magnified hardware. "Those aren't heaters. Induction collars, probably with explosive interrupts under the shell. Four visible means twelve around the ring. If they fire together, every field segment loses its reference in the same millisecond."

"Synchronized," Haru said.

Pella nodded. "No rolling shutdown. No bleed path. They want the receiver unable to hand debt to an adjacent segment."

The cylinder controller reached them at last. "*Morrowline*", we have intruders in the service annulus. Local security lost access. Receiver output remains at emergency cap."

"How much local support can you carry?" Tavin asked.

"Bearing margins allow thirty-two minutes without smoothing. Orbital trim diverges after three hours. The nursery fluid beds exceed lateral tolerance sooner if we change spin."

"Conventional tugs?"

"Two docked. One lacks a flight crew."

Tavin looked at the ballast cask hanging from his yoke. They had brought it for Iona's localization trial, five thousand tonnes of tungsten and water meant to move along the outer rail while fusion thrust paid part of the curvature debt locally. With preparation, the mass could help. Suspended between ship and cylinder during a violent receiver collapse, it could punch through a window band.

"Put the cask on the emergency cradle," he said. "No receiver compensation during the set."

Pella gave him a sharp glance. "That costs us the alignment window."

"We don't have the window anymore."

Haru bent to navigation. The *Morrowline*'s thrusters woke, not together, but in pairs chosen to keep exhaust clear of the cylinder. Conventional thrust pressed Tavin into his couch. The cargo yoke swung by half a degree. Pella cursed the movement, then fed corrective torque through two independent winches.

Below, the intruder's voice repeated its warning. "This mechanism sends force into living worlds. Leave it intact and you choose the injury."

Tavin opened a direct channel. "Cut it now and you drop a nursery cylinder into an uncontrolled load transition."

A pause answered him. Then the woman said, "Captain Serr. Stillness made your position public."

"My position includes people inside the machinery."

"Every delay sends another pulse."

"Then help me shut it down without killing them."

"You had a century."

"I had nine minutes from you."

No reply came. The black clamps remained dark.

The emergency cradle caught the ballast cask on its third attempt.

Its latches had been designed for empty construction drums, not five thousand tonnes arriving with residual motion. The first pair struck and released. On the bridge, the impact sounded like a hammer against a pressure hull. The second pair took load unevenly. Pella shifted the yoke by centimeters while Haru spent propellant to hold the ship away from the cylinder.

When the final indicators turned green, Tavin released the cargo.

The *Morrowline* rose two meters relative to the cylinder as the mass left her suspension. Haru corrected before the hull crossed a service antenna.

"Cask secure," Pella said. "Our yoke has three warped pins."

"Can it lift later?"

"It can disappoint us later."

Tavin unstrapped. "Sive, bridge."

The medic arrived carrying a restraint bag and a portable load monitor. "I assume diplomacy failed."

"It has nine minutes in it."

"Eight now."

Tavin pulled his pressure hood over his head. Pella was already sealing her suit.

Haru blocked the hatch with one hand. "Captain, boarding an active receiver spine does not improve command continuity."

"You have the ship."

"I have a freighter attached to a cylinder under sabotage."

"Which qualifies you."

Pella handed Tavin a tether reel. "Save the argument for when there is down."

They crossed through the forward utility lock in a two-person pod. The pod carried no receiver compensator. Its cold-gas jets pushed against stored nitrogen and produced honest, limited acceleration. Each correction reduced the pressure gauge beside Tavin's wrist.

The nursery cylinder filled half the view. Rotation made its service lights draw circles against the star field. A maintenance spoke extended from the nonrotating hub to the moving habitat shell, with slip bearings at the joint. The receiver spine wrapped the hub in twelve armored segments.

Tavin matched the hub rather than the cylinder. His body lost the *Morrowline*'s deck reference and floated against the pod straps. The old naval training returned without ceremony: choose a frame, name every relative motion, never trust the eye when a rotating horizon occupies most of the window.

Pella pointed. "There."

A maintenance barge clung to the far side of the hub with its beacon disabled. Three suited figures worked along separate receiver segments. Tether lines linked them to the barge. Tavin counted the visible clamps as the rotation carried each through sunlight.

Twelve.

The cell had placed one collar on every field segment. Fiber ran between them beneath the spine's thermal blanket. A synchronized command could sever all paths before the receiver controller shifted load.

"Can you jam the trigger?" Tavin asked.

"Maybe. If it fails closed, we trigger it."

"Can you cut their fiber?"

"Same risk. The collars may fire on lost signal."

"Then we take the command unit."

The barge lay forty meters across open space. Tavin fired the pod's lateral jets. Nitrogen dropped six percent. The cell saw them move and one figure turned a tool toward the pod.

Pella recognized it first. "Grapple driver."

The bolt struck the pod's outer rail and dragged their nose toward the barge. It did not pierce pressure skin. Tavin killed his jet command before the guidance system fought the tether and wasted gas.

"Helpful of them," he said.

Pella stared through her visor. "I will mention your gratitude while they detonate us."

Tavin reeled the hostile grapple in. The cell had anchored its far end to the maintenance barge. Instead of pulling the pod off course, the line now gave them a physical route. He vented one short burst to keep their closing speed below a meter per second.

At ten meters, the cell leader opened a suit channel. "Do not board."

"I need your trigger."

"You need permission to continue killing the Olun."

"No. I need time to put local support under twelve thousand children."

"Continuity language."

"Bearing language. Come inside and read the loads yourself."

"We read Helix."

"So did I."

Tavin caught the barge rail. Rotation placed the bright primary behind the cylinder, and shadow erased depth.

He clipped his tether by touch. Pella landed beside him with enough force to flex both knees against her suit joints.

A cell member launched from the receiver spine toward them. She carried a manual cutter, not a gun. Tavin let her close, caught her tool wrist, and turned with the impact rather than opposing it. Their combined bodies rotated around the barge rail. Pella clipped a restraint line through the intruder's harness.

The leader moved toward a gray case fixed under the barge console.

Tavin pushed from the rail. He could not cross five meters before her hand arrived. Pella could not reach either of them.

"Haru," he said. "Mark now."

The *Morrowline* fired one attitude jet.

Its exhaust did not touch the barge. The pulse changed the ship's relative position enough to tighten the cargo yoke's recovery line, which Pella had left clipped to the pod. The pod shifted. Its grapple line snapped taut and jerked the barge through twenty centimeters.

The leader missed the case latch.

Tavin struck her shoulder and carried both of them into the console. His helmet hit metal. White static crossed his vision, then cleared. The leader drove an elbow into his suit chest, found hard plating, and reached again.

Pella arrived with the restraint bag. She wrapped it around the leader's forearm and inflated it. Foam stiffened around the joint without crushing it.

"Engineering solution," Pella said.

The third cell member remained on the spine. "I can fire locally."

His glove rested on one collar.

Tavin froze against the console. "One collar doesn't sever the receiver."

"It opens an imbalance."

"It opens a correction through the other eleven segments. You won't know where the load goes."

"It stays here."

"No. The controller will seek a paired path before it accepts collapse. You built the synchronized rig because you know that."

The man's hand did not move. Behind his visor, his face looked younger than Keir's.

Pella kept her voice flat. "Your local switch also burns the collar battery through your glove. If you want to lose the hand for a partial fault, press hard."

Tavin glanced at her.

"Probably," she added on their private channel.

The cell leader strained against the foam. "Saren, hold."

Saren lifted his hand.

Pella opened the gray case. A timing bus filled most of its interior. Twelve optical channels ran from a temperature-stabilized clock to individual collar keys. The workmanship belonged in a shipyard. Someone had balanced path length so every interrupt would fire within twenty microseconds.

"They didn't improvise this," she said.

"Can you disarm it?"

"Not safely from here. I can break synchronization without breaking signal."

"How?"

"Give each collar a different delay."

"That creates a rolling shutdown."

"If the receiver controller accepts the sequence."

The cylinder controller joined their channel. "We need twenty-four minutes to transfer smoothing onto bearings and local flywheels. More for orbital trim."

Saren spoke from the spine. "Twenty-four minutes means another clearing."

"No clearing," Tavin said. "Set the receiver to local hold. Accrue the debt here."

The controller hesitated. "That exceeds our authorized local balance."

"Emergency authority can arrest me after the children stop floating into walls."

"It can correct automatically if the balance crosses threshold."

Pella examined the case. "Then we keep the segments alive until local support takes load. Afterward we can bleed them in sequence faster than the correction threshold."

The cell leader gave a short, disbelieving laugh. "You are using our device."

"I am keeping it from using you," Tavin said.

Twenty-four minutes grew into thirty-seven.

One of the cylinder's conventional tugs lacked a pilot, so Haru flew it remotely from the *Morrowline*. Signal delay remained small, but camera angles lied whenever the cylinder's window bands crossed behind the tug. He placed the craft against the hub and began a continuous low-thrust burn that replaced part of the receiver's orbital trim.

Inside, maintenance crews moved water from axial tanks toward the outer shell. The redistribution altered bearing loads a fraction at a time. Each pump consumed power. Each tonne changed the cylinder's spin response. Families left the two innermost residential decks and strapped into school shelters closer to the stable

rim.

Tavin watched their movements through load diagrams patched into his visor. He could not see the children. He saw twelve thousand body-mass allocations travel through corridors in uneven clusters, each marker delayed by someone carrying an infant or waiting for a slower neighbor.

The cell members watched too.

Saren had crossed from the spine under restraint and now sat against the barge rail. His collar switch floated in Pella's evidence bag. "You could have begun this years ago," he said.

"Yes," Tavin answered.

"Then why stop us?"

"Because being late doesn't make every fast act right."

"Your ship used clearing."

"It still does at capped load."

"You are defending it."

Tavin pointed to the local debt trace. It climbed while the receiver held without export. "No pulse has left this cylinder since we boarded. I am defending the time needed to replace what the system does."

"And afterward?"

"Afterward the cylinder will have less capacity, shorter bearing life, and a tug burning fuel every second. People may have to leave."

Saren looked toward the rotating habitat shell. "At least the cost stays visible."

"Visible costs still crush people."

"Hidden ones already did."

Tavin had no answer that reduced either fact. He checked the cylinder load instead.

At minute thirty-one, one water pump overheated. The redistribution stalled with too much mass near the axis. Pella recommended restoring a receiver segment for six minutes. Tavin refused until she showed him the bearing temperature curve. Then he authorized one segment at diagnostic amplitude, with local debt still accumulating and no sink path open.

The narrow field eased the bearing load enough for technicians to swap the pump. It produced no free support. Stored electrical power fell, and the receiver account rose against the cylinder itself.

At minute thirty-five, Haru reported the remote tug's fuel margin. "We can maintain this burn for four hours, twelve minutes. After that, the return reserve disappears."

"Schedule a relief tug."

"Nearest qualified craft arrives in six hours."

“Then the first tug won't return.”

Haru understood. “I'll tell its owner.”

The owner objected for forty seconds, learned that twelve thousand children occupied the cylinder, and donated the return reserve under protest. Tavin respected the protest. Fuel did not become less finite because a good cause consumed it.

Pella rewrote the timing bus. Instead of one firing command, the device now held twelve delays separated by ninety seconds. Each delay required confirmation from the cylinder controller. A skipped confirmation would keep the remaining collars closed.

“You designed a safe shutdown tool inside their sabotage case,” Tavin said.

“They designed most of it. I added patience.”

The cell leader listened from her restraint. “Patience under armed control.”

Tavin released the foam from her arm but kept her tether short. “You can verify each delay.”

Pella objected. “Captain.”

“She knows the hardware.”

“She tried to fire it.”

“And now she can tell us whether you made a mistake.”

The leader flexed her hand as blood returned. “If your controller opens a sink path, I stop the sequence.”

“If the controller opens a sink path, I help you stop it.”

That agreement satisfied no law. It gave each side a way to prevent the other's worst choice.

At minute thirty-seven, the cylinder reported local support ready. Schools had sealed. Clinics had locked their articulated beds. Water mass reached the commanded radius. The remote tug held a steady conventional burn against orbital drift.

Pella enabled the first delay.

Collar one opened its field segment. The receiver controller pushed load toward the other eleven, then found the rising local support and accepted a lower command. Bearing temperature climbed by two degrees.

Ninety seconds later, collar two opened.

By the fifth segment, the cylinder's apparent down shifted three degrees from the floor normal. Children in the shelters would feel the tilt. Loose objects rolled toward spinward walls. No cargo deck inverted. No hidden destination accepted the force.

At segment eight, an automatic continuity routine attempted to restore the sink link. Tavin saw the path request form on his visor.

“Block it,” he said.

The cylinder controller answered, "Ministry override."

The cell leader reached the case before Pella. She pulled the optical authorization lead without disturbing the collar signals. The sink request failed. Every remaining segment stayed alive.

Pella stared at her. "That lead also carried the clock reference."

"I built the bus," the leader said.

"You could have mentioned it."

"You could have asked who built the bus."

They restored timing from the case's internal oscillator. The sequence continued.

When the twelfth collar opened, the receiver spine went dark without a surge. The nursery cylinder relied on moved water, bearing friction, and the continuous push of a fusion tug. Its local debt stopped growing because no field remained active. Its fuel account fell instead.

A cheer rose over the cylinder channel. Tavin muted it after two seconds. The shutdown had worked, but four residential decks would remain closed, the tug could not recover, and every bearing inspection interval had shortened. Celebration could wait until someone assigned those costs without hiding them.

Security craft arrived forty minutes late.

Their commander demanded custody of the cell, the barge, and all sabotage components. Tavin transferred the three people after recording their suit condition and the agreement that had completed the shutdown. He released Saren last.

"You'll call this a rescue," Saren said at the lock.

"I'll call it an interrupted killing followed by useful work."

"Whose killing?"

"Everyone's, if we keep choosing machinery over people or people over physics."

Saren shook his head. "That sounds like balance."

"No. Balance would imply the harms cancel."

The security lock closed between them.

Pella held up the gray timing case. "Commander wants this too."

Tavin examined the twelve-channel bus. One corner had bent during the fight. Its clock still held synchronization within twenty microseconds. With different control logic, the device could coordinate receiver shutdown across an entire habitat after mass and thrust were ready.

"Log the collars as transferred," he said.

"And the command unit?"

"Damaged beyond field custody."

Pella's eyebrows rose. "That would be inaccurate."

"Then make it accurate."

She removed the scorched outer shell, stamped it as scrap, and installed the timing bus inside a spare thruster-control case. The act violated evidence procedure and preserved no weapon by itself. It kept a way to make twelve dangerous changes occur in an order people could survive.

Back aboard the *Morrowline*, Tavin secured the rebuilt unit in the conventional-drive locker. Outside, the nursery cylinder turned under its own spin while a fusion tug burned away its return fuel.

The receiver spine stayed dark. The synchronizer kept time in Tavin's hand.

Chapter 12 — The Price of Correction

Soil struck the inner pipe in heavy, irregular blows.

Iona stood knee-deep in a drained tomato bed while farm workers shoveled the torus's growing medium into transfer hoppers. Fine black grains clung to her audit suit. The irrigation tubes under her boots had carried water that morning. Now they opened into a temporary manifold leading toward the axial tanks.

Above her, rows of tomato vines curved with the farm torus until they disappeared behind trellises and light panels. Red fruit hung among leaves already beginning to soften. No harvest crew moved between the rows. The workers had three hours to strip sensor wire, salvage seed stock, and convert forty hectares of cultivated mass into ballast.

A hopper jammed. One worker drove a bar into the throat and called for lower feed. The shovel line slowed but did not stop.

Iona checked the mass ticket on her wrist. "This load contains root mat."

"Everything contains root mat," said the farm superintendent. "We built soil instead of ordering it from a factory."

"The test plan specifies clean medium."

"The test plan came from people who have never removed a field."

Another impact rang through the pipe. The superintendent watched the nearest vines tremble. She was a compact woman with gray hair shaved above both ears, dressed in a patched humidity jacket. Her badge identified her only by office. She had refused to let the public networks use her name.

"You can still cancel," Iona said.

"Can you?"

"Yes."

"Will Eidos receive tonight's clearing if we do?"

"The emergency schedule assigns a restricted pulse."

"Then I cannot cancel. I can only choose which people see what I lose."

The torus rotated around a nonspinning service axis. Mechanical spin supplied most of its apparent gravity. Its receiver corrected the irregular pull of docked cargo, stabilized the hub bearings, and trimmed an orbit around a dim orange star. For ninety-two years, the farm had exported those corrections to certified sinks. Today it would attempt localization: move mass, alter spin, and use fusion thrust until local machinery carried every load the receiver had hidden.

No one knew whether the maneuver could finish before bearing temperatures crossed shutdown limits.

Nemi's voice came through Iona's earpiece from the control hub. "Hopper three gained six tonnes above model."

“Root mat,” Iona said.

“Root mat does not explain six tonnes.”

The superintendent kicked an irrigation tube. Water dribbled from its cut end. “Saturated lower bed. We were two days from fruiting peak.”

Iona corrected the ticket rather than forcing the mass into the clean-medium category. The localization model shifted. Required tug burn increased by eleven seconds.

“Fuel margin?” she asked.

“Tavin says the *Morrowline* can cover it,” Nemi replied. “Pella says Tavin treats fuel margin as a rumor.”

The superintendent lifted one tomato from its vine. “Put the eleven seconds under this.”

Iona opened a consequence line.

“What do you call it?” the woman asked.

“Lost crop mass.”

“Call it one growing season.”

“That includes future planting, not only this field.”

“It includes the seed contracts we break, the nitrogen culture we kill, and six months of food we now import from two systems away. Write the whole thing.”

Iona entered the phrase exactly. The public test ledger accepted it beside the added burn.

The superintendent placed the tomato in Iona's hand and returned to the shovel line.

At the hub, the translation cradle occupied a shock-isolated bay beside the receiver controller.

Aru-in-Pressure had no image on its display. Twelve acceleration drivers surrounded a chamber of supercooled fluid, making gradients too small for a human body to feel. The dark-lattice channel carried their modulation to Eidos as changes in a paired stress field. Aru answered by shaping the return noise.

Nemi sat strapped into a fixed-backed chair with three displays arranged inside her safe field of view. Her vestibular injury made every spin transition costly, but she had refused remote duty.

“You missed the superintendent's latest message,” she said as Iona entered.

“What did she say?”

“Anyone who calls this a pilot program gets fed the pilot program for dinner.”

“Did you record it?”

“Under public comment.”

Beyond the bay glass, receiver technicians secured twelve segment controls against automatic override. Tavin's

synchronizer had not reached them; the device remained aboard the *Morrowline* under questionable custody. Pella had sent its delay architecture, and the farm crew had reproduced the safe sequence in approved controllers.

Aru's translated pressure pattern appeared as text.

The sending wound narrows. The held wound deepens near you. Is this chosen?

Iona sat at the cradle. "The farm council voted to attempt it. Seventy-one percent approved. Every resident received the projected costs."

Not every possible resident chose.

It took Nemi nine seconds to settle the grammar. The phrase could mean future descendants, displaced people, or unrealized pressure-names. Aru often held several scales inside one comparison.

"No choice includes every future person," Iona said. "We can name who consented now and preserve who objected."

You named objection before motion. This differs from a hidden strike.

A bearing alarm flashed amber.

The torus had begun moving water toward the axis. As its moment of inertia decreased, rotation accelerated. Braking coils resisted the change and converted motion into heat. The farm floors grew fractionally heavier under Iona's feet.

Tavin called from the *Morrowline*. "Cask train approaching outer rail. We can take receiver segment one when it seats."

The torus controller answered, "Bearing two runs three degrees above forecast."

"Root mat," Nemi said.

Iona looked at her.

"Uneven hopper loads reached the axial tanks out of sequence. The farm superintendent warned us. We entered total mass and lost distribution."

On the model, tank four held more wet soil than tank two. The imbalance applied a periodic load to the hub every rotation. Receiver smoothing hid part of it. Once the first segment opened, bearings would feel the full variation.

The Ministry observer joined by channel. "Suspend the test and restore nominal clearing."

Iona recognized the voice from Bex's review room. "We can correct the distribution."

"Not inside thermal margin."

The superintendent appeared from the farm deck, breathing hard. Soil marked her sleeves to the shoulder. "Tank four has a manual cross-feed."

"Rated for water," a technician said.

“It has carried slurry since before your office certified us.”

“Solids will erode the valve.”

“The valve can be replaced. Eidos cannot.”

They opened the cross-feed.

A thick mixture of water, soil, and shredded roots moved between tanks under rotational pressure. Pump current jumped. The imbalance fell by a quarter, then stalled when fibers clogged the first filter.

Iona watched minutes leave the bearing margin. “Can you bypass filtration?”

The superintendent nodded. “The tank will never serve drinking water again.”

“That cost belongs in the ledger.”

“Put it under growing season too. You don't need a separate word for every piece of ruin.”

Technicians opened the bypass. Slurry struck tank two. The torus shuddered once, a low movement through the floor rather than a sound. Bearing temperature held.

Tavin said, “Cask seated. Ready on one.”

Iona looked toward the cradle. “Aru, we are beginning local payment.”

We will measure destination.

Receiver segment one dropped to zero.

The *Morrowline* fired two thruster clusters against the torus's orbital motion. Fusion exhaust expanded into a cleared arc of space, carrying momentum in the opposite direction. The burn moved the torus by less than a millimeter during its first second. It also began paying a correction the dark lattice had carried away for generations.

Nothing came free. Deuterium and helium-3 left the ship's tanks. Reaction products took momentum into space. Crew couches pressed against bodies. Docking schedules downstream lost the propellant Tavin could no longer deliver.

Segment two opened ninety seconds later.

A wave traveled through the farm floor as mechanical bearings accepted load previously shaped by the field. One tomato broke from a vine in the abandoned bed and rolled spinward until a drain caught it.

The cradle registered no return pulse.

The wound remains near the sender, Aru said.

Iona read the sentence twice. “Does it injure you?”

Not through this path. Your pressure history changes instead.

Nemi translated the next ratios before software attempted a cleaner phrase. “It says they can feel that we kept the consequence. Not the details. Destination, not suffering.”

The Ministry observer said, "Proceed if temperature remains inside four degrees."

No one acknowledged his permission.

At segment five, a farm spoke bent.

The spoke carried a train of soil hoppers from the outer fields to axial storage. Moving so much mass inward had reduced tension on its outboard cables while increasing compression near the hub. The localization plan accounted for the shift. It did not account for a repair plate installed thirty-four years earlier under a maintenance category that never reached the curvature ledger.

The plate buckled with a report audible through three decks.

Hopper seven left its rail.

Its emergency catches held one end. The other swung across the emptied field and tore through irrigation trunks. Hundreds of liters of residual water sprayed into warm grow lights. Steam rolled along the torus curve. Workers ran spinward, fighting the floor's increased load.

Iona saw the failure through a maintenance camera. "Stop the sequence."

Receiver segment six held at minimum amplitude. The *Morrowline* ended its burn. Conventional thrust ceased, but the velocity already gained remained. Haru's orbital trace showed a safe deviation so far.

The superintendent reached the fallen hopper first. She clipped herself to a trellis anchor and crawled under the tilted frame.

"Get her out," Iona said.

The woman heard through the common channel. "Get me a jack."

"The spoke can continue bending."

"So can the hopper."

Two workers brought a mechanical screw jack. They placed it beneath the rail and turned together. No field lightened the mass. Every centimeter demanded force through the handle, the jack threads, the deck, and the torus structure. The physical chain remained visible because nothing carried the load elsewhere.

Pella called from the tug. "We can use the receiver to lift it."

Iona looked at the debt trace. Six active segments had accumulated a local balance while no sink path remained open. A short lift would increase that debt and force more conventional work later.

"How much?"

"Four-tenths g across the hopper for twelve seconds. Adds thirty-two burn seconds at current geometry."

Tavin entered the channel. "We have fuel."

"Your reserve?"

“Our destination reserve.”

The *Morrowline* had arrived with fuel to return to Stillness after the test. Spending it would strand the ship until another tanker reached the farm torus.

Iona asked Sive for worker positions. Three people occupied the crush zone. One had a suit beacon that flickered under the hopper frame.

“Authorize the lift,” she said. “Local debt only.”

Pella brought segment six up. The hopper became lighter against the jack, not weightless. Workers gained eight centimeters, enough to pull the trapped technician free. Sive reported a broken pelvis and stable pressure.

After twelve seconds, the field dropped. The hopper settled onto blocks. Thirty-two seconds entered the tug burn plan.

The superintendent returned to the hub with a bleeding line across her cheek. “We lost the main irrigation trunks.”

“Can they be repaired?”

“Before the next planting, if we still have a farm deck.”

The structure model gave them two choices. They could move the soil outward again, restore the torus near its old mass distribution, and abandon localization. Or they could leave the fields stripped, brace the damaged spoke, and continue with sixteen percent less habitable floor until permanent repairs.

Sixteen percent translated into nineteen thousand residents.

The evacuation plan had assumed eight thousand temporary relocations. Berths existed for that number on nearby industrial drums. The remaining eleven thousand would need to move across system or sleep in cargo modules with rationed air margins.

A public council channel opened. Residents had watched the failure. Under the emergency procedure, they could revoke consent by simple majority.

Iona expected anger. She heard arithmetic.

“How long would we keep the receiver if we stop?” asked a teacher from the inner residence.

“Unknown,” Iona said. “The restricted clearing schedule could continue through emergency authority. No Olun consent exists.”

“Can we repair enough farm area for food next cycle if we proceed?”

The superintendent answered. “Not next cycle. Maybe the one after. We lose this growing season completely.”

“What does proceeding prove?”

“That this class of torus can carry its own correction through mass and thrust,” Iona said. “It does not prove every habitat can. It gives us real bearing temperatures, fuel use, and relocation costs.”

A man from the rim interrupted. “We become the cost so Nacre can call itself moral.”

“No,” Iona said. “If Nacre uses you that way, the test has failed politically even if the machinery works.”

“What do you offer instead?”

“Your relocation and food contracts receive first claim on the emergency fuel reserve. The data stays public. No other habitat may cite your result without citing the lost capacity and crop.”

A food coordinator opened another channel from the axial warehouse. Behind him, crews stacked sealed grain tins beside cases meant for seed culture. “First claim means what quantity? We feed eighty-one thousand residents from these fields. Nearby stations can cover twelve days without cutting their own rations.”

The Ministry observer supplied a prepared figure. “Continuity reserve guarantees replacement calories for ninety days.”

“Delivered where?” the coordinator asked.

“Registered residence addresses.”

“Nineteen thousand people lose those addresses if we continue.”

Silence held the channel while the official searched his order. The model had treated relocation as a number leaving the torus, not as households whose ration rights remained attached to rooms they could no longer occupy.

Iona opened the reserve rule. “We need portable claims tied to persons, not doors.”

“That requires Assembly authorization,” the observer said.

“You have emergency authority.”

“For fuel and continuity operations, not residence law.”

The superintendent leaned close to the council camera. “Then put the food aboard the evacuation tenders before you put our people aboard. Cargo does not need residence law.”

Tavin joined from the *Morrowline*. “I can assign two yokes. Moving food first delays ballast recovery by six hours and adds another relief tanker.”

“Do it,” Iona said.

“You do not command my ship.”

“No. I am asking you to enter a cost before the vote.”

Tavin looked offscreen, likely toward Haru's transfer plots. “Two yokes, six hours, one tanker. Put my name beside it.”

A younger voice came from the seed vault. “What about cultures that cannot survive the tender temperature?”

The superintendent answered more gently. “We keep the nitrogen bacteria and the grain embryos. We lose the live fungal beds.”

“How long to rebuild them?”

“Three years if another farm shares stock. Longer if none can.”

Iona added the estimate as a range. That line did not fit beneath one growing season, despite the superintendent's earlier instruction. Immediate crop loss had a boundary. Ecological repair continued beyond it.

The resident at the rim returned to his challenge. “You can list everything and still leave us carrying it.”

“Yes,” Iona said. “The list does not distribute the harm. It prevents anyone from claiming the harm vanished.”

“Promises from the Curvature Office,” he said.

“I released the archive against my own office.”

“That proves you can break one promise.”

The council window filled with voting marks. Iona could not persuade their history away. She waited while the torus turned beneath a bent spoke.

Proceed passed by fifty-three percent.

The narrow margin changed the operation. It did not cleanse it.

The next six receiver segments opened over two hours.

Crews braced the spoke with beams cut from an unused docking cradle. The sacrifice removed the torus's ability to accept standard cargo ships. Smaller tenders would have to ferry food and evacuees. Each cut entered the consequence ledger beside the receiver sequence.

The *Morrowline* resumed its burn. Thirty-two added seconds consumed the return reserve and then part of the emergency allocation Bex had released. Its exhaust carried momentum sunward. The farm torus changed orbit by a measured amount, enough to replace years of hidden micro-corrections across the scheduled localization interval.

Water remained near the axis. Spin rose. Braking coils shed heat into radiators that glowed brighter than their design season allowed. Two residence decks exceeded comfortable load and closed. Families moved through the hub carrying pressure bags, sealed food bins, and pets in hard-sided cases.

Iona left the control bay for the farm deck before the final segment.

Steam had condensed on the light panels. The fallen hopper rested on blocks beside the stripped tomato rows. Workers cut surviving fruit into crates for immediate distribution. None would travel well enough to export.

The superintendent handed Iona the tomato she had carried all day. Its skin had split along one side.

“Still part of the audit?” she asked.

“Yes.”

“Then audit the taste.”

Iona bit into it. Warm juice carried salt from the soil dust on her glove. The fruit had ripened under gravity partly exported into an inhabited world. Eating it did not change that history. Throwing it away would not help

either.

The public count reached zero.

At the hub, segment twelve opened. The remaining receiver field collapsed into its local damping circuit. Stored energy became heat, which cooling pumps moved toward the radiators. No debt pulse entered the dark lattice.

For three rotations, bearing vibration climbed. The repaired spoke creaked. Fusion thrust continued. Then the values settled inside the revised limit.

Aru's cradle sent a dense pressure modulation. Nemi translated in pieces over the channel.

No distant layer receives this change. Mass answers mass. Heat answers field. Motion answers motion. The sending place retains the mark.

Iona stopped beside an empty trellis. "Do the Olun accept this as localization?"

The reply took longer than any prior exchange.

Aru-in-Pressure accepts that this movement does not strike us. Some in chorus require returned pain, not ended injury. This pressure-name calls your payment genuine because it cannot be hidden from those who move.

The distinction mattered. Aru had not forgiven the earlier pulses. It had recognized a physical destination.

Around Iona, workers closed the farm deck. They isolated broken irrigation trunks, folded light panels, and marked seed cultures for cold storage. Relocation tags appeared on nineteen thousand residence doors. The *Morrowline* reported insufficient fuel to leave.

The superintendent took the split tomato stem from Iona and placed it in an evidence bag with the season ledger.

Beyond the window, the torus continued along its altered orbit under fusion thrust. Its receiver spine had gone black. The path worked, and the first people required to walk it were packing their homes.

Chapter 13 — The Dark Lattice

The oldest channel opened before humanity had a receiver.

Iona ran the activation record again and watched the date refuse to move. The file came from a sealed instrument archive beneath Nacre's first RME laboratory. Its clock had drifted by eleven seconds across a century, but no correction could shift the event past the construction of the prototype coils.

On the lab display, a thin phase line emerged from noise. Four hours later, an engineer energized the prototype. The line widened and accepted the first test load.

Nemi rolled her balance chair closer without turning her head. "Run the environmental channels."

"They're already aligned."

"Run them where I can accuse them myself."

Iona added solar weather, local tidal stress, detector temperature, and the laboratory's mechanical vibration. None produced the early line. The signal persisted while the prototype lacked power. It changed only when the coils began coupling to it.

The official history described that moment as lattice creation.

Iona opened the engineer's handwritten note beside the trace. The paper scan showed a thumbprint in conductive ink and a sentence crowded into the margin: Mode persists below excitation. We are tuning to an available relation, not generating one.

Someone had crossed out available relation before publication. The final report used induced transfer architecture.

"How many records carry that correction?" Nemi asked.

"Seven in the founding archive. Three more use inherited mode."

"Inherited from what?"

"They never asked in a certifiable way."

The translation cradle occupied the center of the lab, surrounded by independent accelerometers. Since the farm-torus test, Aru had requested the full channel-origin model. Bex objected to exposing receiver architecture. Iona had answered by placing his objection in the public ledger and transmitting the equations anyway. Secrecy no longer protected a monopoly; the Olun already occupied the other end.

Aru's first modulation stirred the cradle fluid.

You found the dark lattice asleep against mass.

Nemi frowned at the translation. "Asleep carries biological implication."

Iona adjusted the concept field. "Dormant?"

A denser return followed.

Neither sleep nor readiness. A path may exist before passage without wanting feet. Your pressure-making opened paired differences along old constraints.

The software proposed: The lattice pre-existed use.

Iona preferred Aru's version. It resisted turning existence into intention.

“Did the Olun make it?” she asked.

Pressure gradients crossed the fluid in overlapping intervals. Aru's answer required three renderings before the grammar stabilized.

No pressure-name within our retained depths claims beginning. Our worlds received changes before your sequence. We learned their distinctions as scars learn weather.

Nemi brought up the channel map. Human engineers had modeled the lattice as forty-three receiver trunks branching toward certified sinks. That image began with human hardware. She removed the receivers from the model.

The trunks remained.

A Continuity modeler objected from the back of the lab. “A map can preserve assumptions after you remove an icon.”

“Good,” Iona said. “Break the assumption.”

They rebuilt the founding experiment from raw detector voltages. The prototype had used three dense flywheels mounted around a vacuum chamber. Its coils did not push the wheels. They changed the permitted stress relation among them and a remote mass configuration the engineers could not observe directly. Before coil power, the chamber instruments had registered a narrow standing mode tied to the flywheels' changing orientation.

Nemi replaced the human receiver model with ordinary general-relativistic tides. The line disappeared from prediction but remained in measurement. She added coil leakage, thermal contraction, magnetic coupling, and clock error. None restored it.

“Try a channel created during an earlier test,” Bex's modeler said.

Iona opened the laboratory schedule. “No earlier powered test occurred.”

“An accidental excitation during assembly.”

“Then the mode should decay from that event.”

They fitted every plausible decay curve. The measured line stayed constant across sixty-one days of assembly and changed phase when technicians moved an unpowered flywheel. A human-created remnant would have needed to anticipate the new mass geometry.

Nemi ran one final comparison against the current lattice. The ancient line occupied the same allowed relation used by modern clearings, only at an amplitude nine orders lower. “The coils did not dig a tunnel,” she said. “They found a resonance and locked it open.”

Bex's modeler removed his objection from the shared board rather than marking it defeated. Iona appreciated

the distinction. A failed explanation did not become dishonorable merely because evidence closed it.

“Can we date the relation?” Bex asked.

“Not from this trace,” Iona said. “We can only establish that it predates the powered prototype. Aru's archive may push the lower bound farther.”

“Millions of years?”

“Possibly. No exact figure survives the pressure reconstruction.”

The answer disappointed every public channel waiting for an ancient builder. Iona refused to invent one. Origin remained open; prior existence did not.

The relations appeared not as conduits in ordinary space, but as permitted couplings between stress configurations around separated masses. Human equipment selected, amplified, and stabilized those couplings. It had not manufactured the underlying architecture.

Iona felt the conclusion through the soles of her boots. A century of diagrams had placed the Republic at the center because every line began at a receiver icon. Remove the icons, and human channels occupied a small activated patch in a structure with no mapped boundary.

“Show historical modulations that don't match our timing grammar,” she said.

Nemi opened the Olun pressure archive Aru had provided. Most layers carried human clearings across 112 years. Beneath them lay older disturbances, compressed by geologic pressure and reconstructed as ranges rather than exact events.

Several had arrival shapes no human receiver produced.

One rose slowly across what Aru translated as nine pressure generations, then stopped. Another arrived in paired bursts separated by intervals longer than the Republic's history. A third resembled a sequence of shallow transfers spread across many distant sources.

“Other users,” Nemi said.

The phrase entered the cradle. Aru returned a correction.

Other activations. User assumes one choosing boundary. Some patterns may come from many boundaries or none that remain.

“Did you answer them?” Iona asked.

Some received opposition. Some altered. Some continued beyond retained names. No ordinary message crossed. Only changed pressure and the small shaping around it.

The dark lattice had connected injuries before it connected understanding. Nobody had traveled through it. No fleet waited beyond Eidos. Stress moved along paired physical relations, while matter remained subject to distance and light.

Nemi isolated the oldest nonhuman event. “Can you locate its source?”

Direction does not live in the coupling as it lives in open space. We know pressure distinction, not stars.

“Could it still be active?”

The old layers do not answer now.

One of Bex's modelers highlighted the paired-burst event. “If the relation persists, we could modulate toward its source. A low-amplitude probe might establish whether anything remains.”

Iona asked, “What destination accepts the probe?”

“The old path, if we can reconstruct its phase.”

“That names a relation, not a destination.”

“We know the historical event reached Eidos.”

“So did ours. That did not make Eidos empty.”

The modeler reduced the proposed amplitude until the projected pressure change fell below a human receiver's calibration threshold. The number became visually small. It did not become harmless.

Aru shaped a rough translation around the old layer.

This distinction cut shallowly. Shallow persists when repeated. We cannot tell whether pressure memory or living coherence occupies the other relation.

Nemi traced the event across reconstructed strata. One pulse had coincided with a reorientation in a deep Olun memory layer. The change might indicate injury, adaptation, or unrelated pressure drift. Available evidence could not separate them.

Bex studied the proposed probe. “Could another active user threaten the Republic?”

“Anything coupled to the same relation could move stress,” Iona said. “We have no evidence that this source shares our open channels or knows we exist.”

“We also had no evidence the Olun could repossess a pulse until they did.”

“That argues for caution, not an unconsented knock.”

The modeler tried a passive alternative. Human receivers could listen for modulation around the old phase without driving load through it. Thermal noise and ordinary tidal changes buried most signals, but a coherent event above threshold might appear. Listening required power, instruments, and time. It imposed no paired disturbance beyond the detector's existing mass.

Iona authorized the passive watch. She rejected the probe and placed both decisions in the public archive.

Aru responded through blue pressure ratios.

You leave an unopened relation unopened because you do not know who bears the far change. This resembles learning.

The translation software proposed praise. Iona removed the emotional label. Aru had identified a different behavior, not granted approval to humanity as a whole.

“Did any prior activation make the same choice?” she asked.

We know only those that touched us. Absence of strike leaves little pressure history. Restraint cannot be counted from scars.

The statement exposed a limit no survey could repair. Their archive overrepresented harms because harms wrote deeply. Somewhere in the dark lattice, other users might have found available relations and refused them. Such civilizations would leave no wound to announce their existence.

The lab door opened. Bex entered with two Continuity modelers and no security detail. He looked at the erased receiver icons on the channel map.

“You published the origin archive,” he said.

“Yes.”

“You had no authority.”

“Neither did the people who altered it.”

Bex read the founding margin note. “Available relation.”

“The lattice existed before activation.”

“Can other users reach our receivers?”

“Not with ships. Possibly with stress disturbances if they share an activated relation. We have no evidence of a current source beyond the Olun and human network.”

“Can the Olun open new paths into Republic systems?”

Aru answered before Iona did.

We can shape paths you opened into us. We cannot place your mass elsewhere without relation your devices maintain.

Bex listened to the machine translation, then to Nemi's narrower technical version.

“So receiver shutdown protects us.”

“Safe shutdown closes the active relation,” Iona said. “Abrupt severance can trigger correction first. The nursery cylinder proved that.”

“And localization closes it after we carry the debt locally.”

“Yes.”

“One successful farm does not scale to seventeen billion people.”

“No one said it did.”

Bex expanded the map until every known sink assignment appeared. “Then find what scales.”

The request carried no threat. That made it harder to refuse. Iona sent his modelers the farm data, including the lost growing season, relocation count, and stranded tug. No clean efficiency figure survived all three costs.

While they worked, the cradle temperature fell by two hundredths of a degree. Aru's modulation split.

A pressure-name did not speak alone.

Iona had known this as a fact since first contact. The split made her understand it as a condition. Aru-in-Pressure held coherence through many Olun memory lineages agreeing to shape one answer. Under the new modulation, those lineages stopped aligning.

The cradle produced three pressure peaks where one had stood. Translation software merged them, failed, and filled the display with incompatible pronouns.

Nemi shut off automatic grammar. "We need to keep the sources separate."

She assigned temporary colors to recurring phase signatures. Blue carried Aru's familiar ratio structure. White matched a lineage that had witnessed the farm localization through the channel. Red arrived with sharper gradients and a repeated return command.

Blue: The senders retained one wound and changed their near pressure. This demonstrates dissent from the striking mechanism.

Red: One retained change does not settle depths broken across their sequence. Return all held difference. End every opening.

White: Ending prevents later strike. Return answers completed strike. Both alter pressure. Which preserves names?

The machine wanted to render chorus as a single debate. Nemi forced each signature into its own line.

Bex's modelers stopped moving. One of them whispered, "How many factions?"

"We can distinguish eleven recurring structures in this exchange," Nemi said. "That does not mean eleven individuals or governments."

Aru sent a correction across blue and white.

Pressure-names assemble for relation. Number changes with question. Do not count us as bodies standing apart. Do not erase distinction because boundaries move.

Iona recorded both warnings.

Red deepened. A translated pressure-name formed after repeated comparison: Keth-below-Shear. The phrase carried memories of layers ruptured by debt pulses, though Iona could not tell whether Keth included survivors, archives, or active structures around the injuries.

Keth-below-Shear: The senders knew opposition for six of their star returns and continued. Their division spends our depths as time. Immediate repossession ends their choice to delay.

Bex stepped toward the cradle. "Tell Keth that immediate repossession kills people who never knew."

Iona did not transmit.

He looked at her. "Why?"

“Because you phrased ignorance as innocence. Aru already knows human dissent. Keth disputes its relevance.”

“Then translate the deaths.”

“We don't yet have a grammar for a population count at Olun scale.”

“We have thirty-one billion.”

“And seventeen billion dependent. Exact human numbers will look like another model if we cannot express what a person means to them.”

Keth sent again before they resolved the sentence.

Your many bodies do not excuse repeated choice. A pressure layer also contains names that did not choose the strike. They broke together.

Bex's mouth closed.

Iona answered. “No excuse. We ask time to stop without adding another mass death. The farm test kept its correction local. We can repeat the method, but not instantly and not everywhere with one design.”

Blue modulation strengthened around her words.

Aru-in-Pressure carries this evidence. The sending mechanism contains dissenters who accept near cost. A stopped future strike preserves both depths.

Keth-below-Shear: Cost chosen after discovery does not answer cost imposed before it.

“Then what answers it?” Iona asked.

The red gradients struck every cradle driver at once. Fluid lifted against its containment mesh.

Return.

The room alarms registered a local acceleration change of less than one ten-thousandth standard. Tiny, controlled, and unmistakable.

White divided into four signatures. Two joined Aru. One joined Keth. The fourth withdrew from translation entirely. Other structures appeared and vanished as pressure lineages assembled around different parts of the question.

No vote marker existed. No stable quorum carried across scales. The chorus operated through physical alignment, not a chamber count.

Nemi mapped the phase relations. “Keth's side can control several channel endpoints without Aru.”

“How many?” Bex asked.

“Unknown. Enough to return the gradient we just felt through one local cradle.”

“Can Aru stop them?”

Blue answered.

This pressure-name can oppose, divide modulation, and preserve some openings. It cannot command chorus.

Bex looked at Iona. "You insisted they were not one mind."

"They aren't."

"Then we have no agreement."

"We have one faction willing to recognize localization."

"And another able to kill through our own receivers."

Both statements held. Neither provided command.

The first channel began closing at 18:42 Nacre standard.

It served an uninhabited cargo sling in the Lumen system. The receiver reported falling coupling strength, attempted automatic compensation, and drew power until its coils reached thermal limit. Operators shut it down before correction. The sling drifted seventeen meters from its assigned path and lost a container of machine parts.

A second channel narrowed eleven seconds later at a hospital spindle around Kestrel.

Iona watched the status arrive on the public network. Local crews shifted the spindle to mechanical bracing, following the procedure she had used at Perihelion on Day 0. Surgery stopped. No one died during the first minutes.

Then six channels changed together.

Their sinks differed. Their receiver types differed. All carried the red phase signature associated with Keth-below-Shear.

Bex ordered every system to disable automatic gain. The emergency command propagated at light speed through ordinary relays, slow across interstellar distance. Local authorities who had already mirrored the raw archive acted first. Others waited for Nacre confirmation while their receivers fought the narrowing paths.

Nemi displayed the mechanism. "They aren't adding energy. They are withdrawing endpoint stabilization."

The lattice relations required both ends to maintain a paired configuration. Human receivers supplied local coupling. Olun pressure structures had learned to hold and shape the remote side after a century of impacts. Keth's faction now released that hold in a timed pattern, forcing each active receiver to carry its accrued imbalance or seek another sink.

"They're drawing back the open channels," Iona said.

Aru's blue modulation returned through a path that flickered under strain.

Some depths begin total return. We hold where we can. Every held opening costs pressure coherence.

"How long?"

Duration changes as chorus changes.

Bex opened a Republic defense channel. "We need to identify attack sources."

"There are no launch sites to strike," Iona said.

"Source can mean infrastructure."

"The infrastructure sits in our receivers."

His command had already reached admirals trained to find ships, stations, and weapons. Search grids spread across Eidos's ordinary space. Telescopes turned toward ice worlds that no human craft could reach in time. The response consumed attention while local coupling failed.

At 18:49, the farm torus reported no change. Its receiver had gone dark after localization. Fusion thrust continued to spend fuel. Families still packed for relocation. Keth could not repossess a channel that no longer carried debt.

Iona placed that fact on the defense network with the test costs attached.

Another twelve channels narrowed. One recovered when an Olun signature aligned with Aru. Four entered unstable oscillation. Seven began local shutdown procedures.

The map no longer resembled a human branching tree. Lines brightened and faded across a patch of the dark lattice that had existed before human names, before receiver coils, perhaps before the Republic's stars formed. Other users had touched distant portions and vanished without crossing space. Humanity had found one relation, mistaken access for ownership, and driven force through living pressure.

Nemi reached for the chair restraint as a weak correction passed through Nacre's lab. The floor pressed harder for half a second. Every loose cable bowed, then settled.

On the system map, red withdrawal marks moved from isolated receivers toward the network's larger open channels. Blue held some. White signatures changed sides. The chorus did not resolve.

Aru's translation broke into fragments under the strain.

We preserve openings only for cessation. Prepare local destinations. Keth draws what remains.

Forty-three receiver hubs changed from green to amber. Across the Republic, open channels began contracting from their far ends.

Chapter 14 — An Enemy Without Ships

The destroyer fired at a piece of ice.

Tavin watched the missile leave its rail, cross twelve kilometers of empty Nacre space, and strike a maintenance shard no larger than a cargo pod. The shard became a white flash, then a widening spray of fragments that preserved its old momentum.

Three fragments crossed the exclusion cone around the *Morrowline*. Haru changed attitude with two short thruster pulses. Metal whispered against the freighter's forward armor.

"Defense Command calls that a confirmed hostile object," Haru said.

Pella looked up from the fuel transfer board. "It has become several thousand hostile objects."

The emergency tanker had finished replacing the fuel spent at the farm torus eleven minutes earlier. Its umbilicals still trailed between ships while Nacre's defense group swept telescopes across the star field. Twenty-seven warships held a shell around the capital receiver hub. Their sensor beams searched for drive plumes, sail glints, weapon heat, and orbital changes around every body large enough to hide a vessel.

They had found ice, lost cargo, and two unregistered survey drones.

The Olun had no fleet.

Tavin opened a channel to the destroyer. "Your target debris will cross receiver service traffic in fourteen minutes."

"Civilian vessel, clear tactical frequency."

"My civilian vessel has six independent thruster clusters and a cargo yoke. Yours has just manufactured a salvage job."

A new voice entered, older and calm. "Captain Serr, report aboard the command ship."

Admiral Sen Marr's portrait replaced the destroyer's targeting display. Tavin remembered her from Navy tug service, when she had commanded a rescue group and considered every unscheduled mass an insult. Twenty years had silvered her hair without easing her expression.

"Bring your evidence," she said.

"The evidence won't fit through an airlock."

"Then bring enough to stop my people shooting ice."

Tavin transferred command to Haru and crossed by service pod. The command ship *Ardent Measure* rotated no living decks. Its receiver held a steady one-third standard through the operations drum, allowing officers to stand while the hull remained oriented for weapons. Each step Tavin took accrued a small debt somewhere beyond ordinary sight.

Marr met him at the tactical rail. Around her, the Republic's mapped volume hung as a sphere of search cones. Eidos occupied a distant inset compiled from old survey data. Red threat icons covered its three rogue ice worlds.

"We have channel interference from a nonhuman civilization with unknown reach," she said. "My orders require defense against incursion."

"Incursion by what?"

"You tell me."

Tavin placed Pella's receiver diagrams over the tactical sphere. Every red path terminated not at a possible ship, but at a human receiver.

"Repossessions come through paired infrastructure," he said. "The Olun shape the far end of relations we keep open. No matter crosses. No ordinary signal crosses. Stress comes back because our hardware maintains a path for it."

Marr folded her hands behind her. "You are certain they possess no conventional ships."

"Iona's archive shows no drive emissions or constructed objects. Aru says their pressure-names exist inside deep ice. More important, a ship cannot arrive from Eidos on this crisis schedule. Light has not had time to carry our first mistake there and back through ordinary space, much less a fleet."

"They may have launched before contact."

"To attack a species they could not locate except through the lattice?"

"Unknown capability requires broad planning."

"Broad planning has your guns pointed away from the weapon."

Tavin enlarged Nacre's receiver sites. The capital hub served the inhabited spindle, radiation shades, hospital wheels, docks, and hundreds of dependent craft. Security at the hub consisted of maintenance officers and one civil patrol ship. The fleet held its outer shell thirty thousand kilometers away.

Marr studied the distribution. "You propose placing warships beside the devices that draw attack."

"I propose placing engineers, tugs, medical teams, and disciplined security beside the devices people will try to destroy when panic outruns instruction."

"Orbit Breakers."

"Some. Others will be parents who hear a receiver may kill their children and arrive with a cutting torch."

"You stopped one cell."

"I stopped them because they tried to cut under load. Then I used their timing design to complete a safe shutdown."

A staff officer turned from his console. "That component should be in evidence custody."

"It is in useful custody."

Marr silenced the officer with a glance. "Your plan leaves the system perimeter uncovered."

"Against what?"

“An enemy.”

“Find me one with a position your missiles can reach.”

The tactical sphere rotated between them. Search cones crossed empty space. Eidos remained beyond any timely fusion voyage. The actual red paths ran from receivers into a dark-lattice model that had no navigable direction.

Marr closed the outer search shell by half. “I will move one engineering squadron inward. The rest maintain watch.”

“One squadron cannot cover the hub and thirteen secondary sites.”

“One civilian captain does not set fleet deployment.”

“No. One admiral decides whether to defend people or doctrine.”

Marr's expression did not change. The operations drum grew lighter for an instant. Somewhere in its receiver control, automatic gain compensated for a narrowing Olun endpoint.

Every officer felt it.

Tavin touched the rail. “That came through your deck, not the perimeter.”

An alarm confirmed local coupling loss. The receiver recovered after eight-tenths of a second, exporting additional debt to stabilize the field.

“Shut off automatic gain,” Tavin said.

The staff officer objected. “Weapons alignment requires stable load.”

“Your weapons are idle.”

Marr gave the order. The deck settled at a fractionally lower acceleration as mechanical spin accepted more of the load. Officers adjusted their stance. No one fell.

“Two squadrons inward,” she said. “Show me what they defend.”

Before releasing the squadrons, Marr required a threat test. Her staff loaded three invasion cases into the tactical drum: conventional ships from Eidos, matter transported through the lattice, and weapons hidden near Nacre before contact.

The first case failed on travel time. Even a fusion craft accelerated at crew-killing limits could not cross the distance during the Republic's 112 years of RME use. A robotic weapon might endure higher acceleration, but telescopes would still see its braking plume years before arrival. No plume existed.

The second case failed on channel physics. Pella joined from the *Morrowline* and placed a steel spanner inside a receiver test cage. She pulsed the field at diagnostic amplitude. The spanner grew lighter by a measured fraction while an equal stress disturbance entered the paired sink relation. When the pulse ended, every atom remained in the cage.

“If matter crossed,” she said, “we would solve freight and abandon cargo yokes. My entire career argues against it.”

Several officers disliked the joke enough to remember the result.

The third case survived only as possibility. Unregistered craft already occupied Nacre space. Any could carry explosives. None required the Olun. Marr reassigned that problem to inspection patrols rather than invasion doctrine.

Tavin moved the receiver-site map beside the failed cases. "At each site, defend four things. Cooling, control access, local support machinery, and evacuation lanes. Guns protect only the third, and only if nobody fires through the first."

A logistics captain added fuel depots and tug berths. A medical officer marked low-load wards whose patients could not tolerate abrupt change. The planning layer filled with work that outer-shell patrols could not perform.

Marr ordered each inward squadron to carry engineers from the capital maintenance reserve. Marines received restraint systems, pressure patching kits, and cutting tools. Warships opened cargo bays for ballast and hospital braces.

"Keep one destroyer on debris interception," Tavin said.

The officer whose missile had shattered the ice object looked almost vindicated.

"With its guns cold," Tavin added. "Give it nets."

The expression disappeared.

Receiver Site Four occupied the axis of Nacre's largest hospital wheel.

By the time the *Morrowline* approached, six hundred civilian craft crowded its traffic volume. Some carried families seeking transfer away from RME-dependent housing. Others brought protesters who wanted the receiver severed. A third group believed the Olun signal had infected the hardware and demanded military destruction from a safe distance.

None maintained proper separation.

The hospital wheel turned below the congestion. Its axial receiver sat inside a truss of cooling loops and field coils. Three defense corvettes arrived from Marr's reassigned squadron with weapons hot. Their approach compressed the crowd toward the truss.

"Corvette leader orders all civilians to clear," Haru said. "No vector assignment."

"Then everyone will choose the same open vector."

They did. Forty craft turned sunward together. Two collided before their pilots recognized the convergence. One lost attitude control and began rotating toward the receiver's cooling loops.

Tavin brought the *Morrowline* across its path.

"Pella, forward yoke."

"Empty yoke has poor manners at this closure rate."

"Make it polite."

The freighter fired lateral thrusters. Conventional acceleration pushed the bridge crew sideways against restraints. Pella extended the cargo yoke, opened its grapples, and caught the tumbling craft around a reinforced docking collar. The impact twisted one yoke arm beyond its service mark.

“Caught,” she said. “Future lifting capacity remains theoretical.”

Sive opened a medical channel to the damaged craft. Two injuries, no pressure loss.

Tavin addressed every vessel in the site volume. “Hold present thrust. If collision avoidance requires motion, state vector before firing. The receiver remains active at reduced load. No sink clearing is scheduled in the next twenty minutes.”

A protest barge answered. “Emergency authority lies.”

“It does. Your thrusters still obey momentum.”

“We are here to shut the weapon.”

“So am I, when local support exists.”

The corvette leader cut in. “Captain Serr, withdraw from defense operations.”

“Assign traffic vectors first.”

“My mission concerns sabotage.”

“Your mission brought three armed hulls through a civilian crowd.”

Admiral Marr appeared on the channel. “Corvette leader, safeties on. Transfer traffic control to *Morrowline* navigation.”

Haru exhaled once and began assigning lanes. He sent evacuation craft toward the outer assembly orbit, protesters to a holding arc where they could observe, and damaged vessels toward hospital docks. Every instruction included a thrust limit and a time.

Order emerged slowly. Civilian captains obeyed because the vectors gave them somewhere physical to go, not because a uniform demanded trust.

At the receiver truss, another problem developed. Hospital technicians had begun a controlled power reduction. A crowd in maintenance suits crossed the outer rail toward them with cutting tools.

Marr's marines moved to intercept.

Tavin watched their suit jets. “No projectile weapons near the cooling loops.”

The marine commander replied, “My team knows infrastructure.”

“Then they know frightened people can puncture it while being shot.”

He put Pella on the maintenance channel. She opened the receiver schematic and marked twelve segment controls.

“Site Four can localize twenty-eight percent of present load inside an hour,” she said. “Hospital flywheels carry another nineteen. The rest requires two tugs and patient evacuation from the low-load surgical drum.”

A protester challenged her. "Why should we trust your figures?"

"Don't. Read the bearing temperatures on the public feed."

The hospital published its instruments. Tavin saw the same pattern as the nursery cylinder: moral demand reduced to bearings, moving water, fuel, and people under load. The crowd slowed as individual suit displays acquired specific limits.

One protester continued toward a segment control. A marine intercepted without firing. They struck the truss and spun together, tethered to separate rails. Tavin sent a maintenance pod to pull them apart.

"Give us a shutdown schedule," the protest leader said.

Pella built one from local data. "Two hours, forty minutes if the second tug arrives. Five hours without it. You lose two operating theatres for at least a week."

"Start now."

The hospital director came onto the feed from a surgical bay. "I have four patients under controlled load."

"Can they move?" Tavin asked.

"One can. Two need procedures completed. One needs mechanical bracing we do not possess."

Admiral Marr listened from the *Ardent Measure*. "My command ship has field braces for gunnery injuries."

The staff officer beside her began to object, then stopped.

"Send them," Tavin said.

Marr ordered a corvette to deliver the braces. Another corvette took position as a physical traffic barrier with weapons cold. Marines secured the receiver controls without excluding hospital engineers or two observers chosen by the protesters.

Defense had changed shape. It no longer faced the stars. It held a safe volume around machinery while civilians argued over how to stop using it.

The first receiver segment powered down at Site Four. The hospital wheel took its load through spin and flywheels. A tug burned fuel to stabilize the hub. No missile could perform any part of that work.

Thirty seconds into the new load, a surgical drum reported a brace failure.

The drum hung near the hub where spin provided little apparent gravity. Its receiver segment had supported four articulated tables during operations. When field load decreased, one mechanical brace accepted more force than its maintenance test had predicted. A locking tooth sheared. The table beneath an open spinal procedure shifted three centimeters.

The hospital director called for immediate receiver restoration.

Pella opened the load path. "Restoring the whole segment adds debt and resets the shutdown. Give me one local field plate."

The *Ardent Measure*'s medical team had delivered six gunnery braces. Marines carried the nearest pair

through the axial lock while a corvette held position against the wheel's changing hub motion. Their suits bore weapons, but their hands carried steel frames.

Tavin watched helmet video from the operating room. A surgeon held both hands motionless inside the patient's open back. Nurses anchored themselves around the table. The failed brace hung from one pin.

A marine assembled the first replacement on the deck and found its foot too wide for the hospital rail. The second marine cut the footplate with a rescue saw. Sparks crossed the room under the extractor's pull.

"Two minutes," she said.

"The patient has less," the surgeon replied.

Pella authorized a diagnostic field from a portable plate. It did not draw on the receiver or open a sink channel. Capacitors discharged into a local coil, easing load across the table for forty-eight seconds while heating toward cutoff.

The marines seated the altered brace at second forty-one. Mechanical screws took force before the plate shut down. The table stopped moving.

Site Four kept its receiver segment dark.

Marr had watched the same feed. "My people train to hold a compartment after weapons damage."

"Today the damaged compartment happened before the shooting," Tavin said.

She ordered every receiver-site squadron to inventory braces, jacks, thermal blankets, and independent power before ammunition.

Marr called Tavin privately. "You have made my fleet into traffic control."

"Temporary demotion."

"My captains will hate it."

"They already hate shooting ice."

"One enjoyed it."

"Keep watch on that one."

Marr almost smiled, but her attention shifted to an incoming low-band packet. "Return to the command ship. Something changed in the lattice telemetry."

The Olun correction announced itself as timing.

On the *Ardent Measure*, fleet analysts arranged receiver-phase reports from Nacre's local hubs. Ordinary interstellar messages could not reach the capital quickly enough to describe events across all forty-three systems. The dark-lattice network carried only low-band channel modulation, but each coupled path exposed its own stress timing at the hub. Those intervals now formed a coordinated sequence.

Tavin stood beside Marr while red marks appeared across the display.

“Could this be another prime-ratio reply?” she asked.

“No,” said the fleet analyst. “The ratios match stored debt, not integers.”

Each interval corresponded to a class of clearing previously assigned to Eidos. Cargo compensation returned first, then habitat stabilization, then orbital trim. The pattern did not yet carry force. It selected accounts.

Iona joined through a local Nacre relay from the founding laboratory. Her image carried compression artifacts. “Aru identifies Keth-below-Shear in the phase structure. The hard chorus is aligning endpoints.”

“Against Nacre?” Marr asked.

“Not enough information.”

Tavin examined the selected loads. “These aren't grouped by sender.”

The analyst rearranged them by original sink assignment. The pattern sharpened. Thousands of old clearings pointed toward one receiver family, but not Nacre's.

Haru sent a navigation note from the *Morrowline*. The receiver family used a sixteen-year-old emergency architecture: multiple cylinders sharing correction paths around a gas giant. One place in the Republic still operated that design at scale.

Tavin knew its name before the display produced it.

Meridian.

Marr enlarged the old habitat chain. Cylinder rings surrounded the gas giant in staggered orbits. Their shared receiver control had been modified after the weight-loss incident that killed Iona's sister, but several emergency paths remained. The system could distribute local failure across neighboring habitats. Under coordinated repossession, those same links would spread a returned load.

“Can they choose Meridian through Eidos?” Marr asked.

Iona answered. “They can release or oppose the endpoint relationships humans left active. Meridian's architecture offers the broadest connected return path.”

“Why that site?”

“We don't know whether Keth understands its population or history. Physically, Meridian can receive a large correction through one coordinated action.”

Tavin studied current readiness. The system's ordinary update had left before the latest channel withdrawals. It listed eight cylinders under emergency load, three nursery sectors, and insufficient conventional tugs for full localization. Later information could not arrive by ordinary light before the selected sequence matured.

“Send the warning through low-band receiver modulation,” Marr said.

The analyst shook his head. “We can shape only a few bits without adding a stress pulse. Enough for alarm class and timing, not evacuation plans.”

“Send those bits,” Iona said. “Do not add load.”

Aru held one narrow channel open. Nacre transmitted the alarm as changes around zero net disturbance: correction imminent, shared receiver family, prepare local support. No names, no voice, no false promise crossed the lattice.

Marr ordered her inward squadrons to remain at receiver sites. The rest of the fleet began reconfiguration for rescue: weapons magazines locked, passenger bays cleared, medical stores loaded, detachable fuel tanks assigned. Ships built to oppose an approaching force prepared instead to move people away from their own infrastructure.

Tavin called the **Morrowline**. “Haru, plot for the Meridian departure point. Pella, inspect every yoke pin. Sive, convert cargo two for evacuees.”

“We just refueled,” Pella said.

“We're going to spend it.”

On the tactical sphere, search cones vanished one by one. The empty star field lost its red threat icons. In their place, receiver sites gained defense rings and tug assignments.

A final telemetry block arrived. The selected Meridian accounts locked into one phase. Across the channel model, Olun-held endpoints began drawing a coordinated gradient toward the gas giant's habitat chain.

Marr read the confirmation aloud. “No fleet. No incoming matter. Correction path fixed.”

The **Morrowline** released its fuel umbilical and turned toward the convoy assembly orbit.

Every red line on the display ended at Meridian.

Chapter 15 — The Sunward Convoy

The first hospital cutter missed its departure slot because nine patients could not be made heavy enough to move.

Tavin watched through the *Morrowline*'s forward cameras as orderlies floated stretchers across Nacre Dock Twelve. The dock maintained a weak spin, but receiver compensation kept losing fractions of load in pulses too brief for the warning system to predict. Each time the floor lightened, nurses stopped pushing and locked their boots. The patients' restraint webs tightened. Loose tubing lifted from their blankets like grass under water.

"Cutter Seven reports sealed," Haru Den said. "Four minutes late."

"Put them behind the ore tugs," Tavin replied. "Their acceleration limit matters more than their number."

Haru moved the marker. Forty-six vessels now crowded the Meridian departure plot: hospital cutters, passenger packets, two water carriers, eight cargo tugs stripped of freight, and a school tender whose life-support plant could handle six hundred children if nobody asked it to make them comfortable. The convoy had grown without acquiring a command structure. Every captain carried a different authorization, a different load table, and some relative already calling from Meridian.

Tavin had received no relative from Meridian. His son stood twenty kilometers behind him on Nacre's spindle, refusing to leave.

Keir's face occupied a small panel beside the navigation plot. He wore a spinwright's harness over a stained undershirt. Behind him, workers rolled water tanks along a public transfer rail while the spindle turned at reduced speed.

"I found you a private berth on the Assembly courier," Tavin said. "It clears the dock before us."

"You found it because you are leading the convoy."

"I found it because the courier has an empty couch."

"It has thirty empty couches and four thousand people waiting at the public locks." Keir wiped metal dust from his cheek and only spread it farther. "Nacre needs every spin team if compensation drops. We are shifting ballast out of the north residential band."

"The Ministry has crews for that."

"The Ministry has supervisors. We have wrenches."

A departure tone sounded through the bridge. Tavin muted it rather than end the call. He wanted to tell Keir that Meridian had already lost one child-bearing ring sixteen years ago, that systems repeated their failures when people called the old damage exceptional, that a father's request did not become corruption merely because strangers also needed rescue. None of those claims improved the load table.

"You stay inside the spine during a field alarm," he said. "No exterior rail work. No receiver room."

Keir leaned nearer the camera. "Are those orders from the convoy commander?"

"They are the terms under which I do not fly back and remove you myself."

“That threat would work better if forty-six ships were not watching your drive clock.”

A smile nearly reached Tavin. “Forty-seven. One late hospital cutter.”

“Then go count them properly.”

Keir ended the connection before either of them could make the parting formal. The panel reverted to Nacre’s civil spin schedule. Six sectors had accepted reduced load. Two remained under hospital exemption. Public crews were moving ballast by hand and rail because every receiver adjustment now risked drawing Olun attention.

Tavin saved Keir’s work channel under priority access. It felt like fastening a restraint around empty air.

Pella Jor’s voice came from engineering. “Captain, Cluster Four has finished warming. If we wait another nine minutes, we can add Cluster Six without cracking its feed collar. If we leave now, I want it cold until the first correction burn.”

“We leave now.”

“You always choose the answer that makes my machinery feel unloved.”

“Your machinery has forty-seven people to console it.”

“Forty-six. I’m busy.”

Tavin opened the convoy channel. Static moved beneath the captains’ acknowledgments, not radio noise but tiny phase errors induced through local receiver fields. The Olun had no ships coming over Meridian’s horizon. They needed none. Every Republic vessel carried a piece of the route by which correction could arrive.

“Sunward convoy, this is *Morrowline*. Departure order remains freight, passenger, medical, then command. Maintain physical separation by transponder range, not receiver solution. If your compensator disagrees with your accelerometers, trust the accelerometers. If both fail, trust the star field and stop changing thrust.”

A packet captain answered, “Meridian Control moved the safe arrival line inward by eleven hours.”

“They moved an estimate. Our drives did not improve.”

“Then we reach them late.”

“We reach them with ships.”

Dock clamps released along the line. Ore tugs slid first from Nacre’s shade, their scarred hulls carrying inflatable berth modules instead of rock. Passenger packets followed in pairs. Sunlight caught their radiator fins as they cleared the blue-white star’s radiation screens. Tavin waited until Hospital Cutter Seven crossed its assigned line, then ordered *Morrowline* free.

The heavy tug left the dock reluctantly. Its cargo yokes sat empty, detached to make room for four docking collars and an evacuation gantry. Empty did not make the hull light. Three hundred eighty meters of truss, tanks, drive shielding, and redundant thruster clusters still demanded an honest exchange of momentum.

Nacre’s receiver supplied load compensation for the first forty seconds. Tavin felt only a firm pressure against his couch while the fusion drive built. Then the compensation stuttered.

His chest grew heavier. A warning band flashed amber. Haru's hand struck the navigation rail before her restraint caught.

"Local assist down twelve percent," she said.

"Hold thrust."

"The passenger packets are climbing through our wake."

"Notify them. No one matches us by feel."

Pella cut one feed before its mounting could take the differential. The bridge vibrated with the rougher note of asymmetric thrust, then settled as the control vanes trimmed it. Tavin watched the physical accelerometers. Each instrument used a suspended proof mass and an optical cage, old technology with nowhere to hide a political assumption.

The convoy lengthened. Vessels that had planned on receiver-assisted acceleration fell behind, while pilots with abundant fuel tried to preserve the original schedule. Tavin ordered the faster ships to reduce thrust. Three objected. He sent them the hospital cutter's patient limits and asked which beds they intended to abandon. Nobody answered with a number.

One ore tug offered to take the hospital cutter under tow. Tavin refused after Haru showed him the coupled oscillation: a compensation loss would snap the tow line through both hulls. Instead, the tug moved ahead and spent propellant clearing small debris from the cutter's path. It gained no time. It converted fuel into a safer corridor, which counted for more.

Once Nacre dwindled behind the radiation shades, the channel from Meridian cleared enough to carry images. Cylinder habitats circled the gas giant in a loose silver chain. Their receiver hub pulsed on the tactical overlay, bright with accumulated debt. Four rings had begun evacuation. Seven more still spun under unstable compensation. The nursery ring showed green, which meant occupied and structurally sound, not safe.

A Ministry controller appeared in the corner of Tavin's display. "Captain Serr, Continuity requests maximum receiver assistance to preserve your arrival margin."

"Denied."

"This movement has emergency priority."

"So will the correction we attract."

"The Accord requires a controlled load at Meridian. Minister Bex has secured an Olun interval through the translation cradle."

Tavin examined the attached schedule. Curvature debt would be localized in three stages while evacuation craft held outside the ring plane. The numbers assumed a stable receiver response for another seventeen hours. A note from the Curvature Office reduced that confidence without supplying a replacement.

"You cannot schedule cooperation from a faction," he said. "Aru does not command every pressure-name."

"The interval has been accepted."

"By whom?"

The controller looked offscreen. "That information falls under negotiation security."

"Then it cannot set my thrust."

He closed the Ministry channel. Haru said nothing. Her silence held neither agreement nor surprise; she had already removed receiver assistance from the navigation solution.

For six hours the convoy accelerated sunward. Conventional engines spent fuel that manifests had treated as emergency reserve. Water carriers transferred reaction mass to two passenger packets by flexible line. A school tender shut down half its cabins after a coolant pump seized under changing load. Sive An crossed by service shuttle, repaired the pump with the tender's engineer, and came back carrying a feverish child whose parents refused separation until Sive threatened to draft them both as medical aides.

The compensation failures arrived more often.

During the seventh, Cargo Tug *Larkspur* fired a lateral thruster against a false roll. Its bow swung toward Hospital Cutter Three. Tavin saw the geometry close on his display: two moving bodies, eleven hundred evacuees, no time for negotiation.

"*Larkspur*, cut all lateral input."

No answer.

"Pella, give me Cluster Two, thirty percent starboard."

"That yaws us across the passenger line."

"For four seconds."

He took manual control. The *Morrowline* rolled its long truss between the tug and cutter, presenting an empty cargo yoke as a physical barrier. Pella fired the cluster. Load drove Tavin sideways into his couch while the big ship crossed the gap. Haru called range without raising her voice.

"Eight hundred. Six. Four."

The tug's pilot cut thrust. Its bow continued through inertia, missed *Morrowline*'s yoke by twenty-three meters, and drifted above the cutter's radiator.

Tavin held the roll until their stern cleared. "Convoy, every automatic attitude link is now suspect. Disconnect receiver-derived roll. Confirm by vessel."

One by one, forty-six voices answered. The school tender needed two calls. *Larkspur*'s pilot came last, breathing hard.

"I saw the horizon move," she said. "My platform said we were flat."

"There is no horizon here."

"I know that."

"Now you do twice." Tavin let his tone soften. "Lock your side clusters and follow Haru's star solution."

The near collision cost them forty minutes. It also ended the arguments about slower acceleration.

At shift change, Tavin unstrapped and moved through the ship. The bridge gravity came from thrust alone. Each ladder had become a steep well toward the stern, familiar from his early tug years and brutal to crews raised under compensated load. He passed evacuee webbing folded along the cargo deck, forty rows awaiting people who might never reach them. Sive had chalked capacity beside each section: ambulatory, restrained, critical, infant, unclassified. Human beings entered her account by what force their bodies could survive.

Engineering smelled of hot metal and sealant. Pella crouched inside an open feed housing with both arms buried to the elbow. Her gray work wrap had darkened with coolant.

"Cluster Six?" Tavin asked.

"Still cold. Its collar warped when the assist dropped."

"Can you free it?"

"I can free anything once. Using it afterward requires more diplomacy." She backed out and flexed her hands. "We have enough conventional thrust to brake at Meridian. We do not have enough to brake, tow a cylinder, and come home."

"We are not towing a cylinder."

"Good. I hate when captains say impossible things before meals."

A wrench floated from the housing as thrust dipped. Pella caught it against her wrist. The deck returned with a jolt.

Tavin steadied himself on the frame. "Receiver assist again?"

"Not ours. Network clearing passed across the channel." She pointed at a pressure trace on a maintenance screen. "Something drew against every active coupling for three tenths of a second. Meridian sits at the center of the gradient."

The line resembled a hand closing only because human beings had built hands into their metaphors. Tavin saw a load path tightening through devices that should never have shared one physical structure.

"Send it to Iona."

"Already sent."

She had added no Ministry routing.

Tavin climbed back toward the bridge. Halfway through the central truss, Keir's priority channel chimed. The recording showed his son beneath a water rail, one boot hooked under a brace while two workers locked a ballast tank into a new cradle.

"Nacre north band stable at point eight standard," Keir said. "No receiver draw for nineteen minutes. Public queue has cleared Section Four. Before you ask, I am inside the spine."

The camera angle revealed open stars through a maintenance aperture.

"You are adjacent to the spine," Tavin told the recording.

Keir continued, unable to hear him. "Meridian families are arriving on the outer packets. We made room in the

training decks. Don't send people to Assembly berths unless they need medical pressure. Send them here."

A worker called for his torque gauge. Keir turned, then looked back once.

"Bring who you can."

The message ended.

Tavin replayed no part of it. He forwarded Nacre's public capacity to Sive and appended Keir's work channel as operational contact. Pride and fear occupied the same narrow space behind his ribs. Neither could steer.

Nine hours from Meridian, the Olun correction began in earnest.

Every receiver-linked vessel lost compensation at once. Bodies sank into couches under full thrust. Alarm tones overlapped across the convoy channel. One ore tug's engine shut down on crew-protection limits. A passenger packet vented coolant after a bracket took load its designers had assigned to curvature support.

Haru's hands moved across the plot. "Meridian Control says their hub is drawing debt inward. They want us under ring shadow before the first staged release."

"We cannot reach ring shadow on this profile."

"Ministry recommends emergency clearing to Eidos."

Tavin looked at the recommendation. It offered a simple transaction: ease the convoy's acceleration by sending its load into the place already answering injury with injury. The order carried Bex's continuity seal.

"Delete it from the control queue," he said.

Haru removed the command but kept the record.

Pella called from engineering. "If you want speed, I can light Six. Collar may hold through one burn."

"If it fails?"

"We lose the cluster and possibly the aft service passage."

"How much time?"

"Forty-three minutes gained, assuming every ship behind us can match."

They could not. The hospital cutters would exceed patient limits. Two tugs lacked fuel. Gaining forty-three minutes alone would only place *Morrowline* at Meridian before the empty berths it needed to fill.

"Keep Six cold," Tavin said. "Set fleet profile for a conventional burn at the medical limit."

Pella swore once and obeyed.

The order moved vessel by vessel. Fusion drives brightened against the star field. Passenger packets shed water and furniture to reduce mass. Cargo tugs opened reserve tanks sealed for voyages home. No hidden sink took the force. Exhaust carried momentum into space, fuel gauges fell, and every person aboard felt the acceleration they had chosen.

Tavin's couch pressed hard along his spine. His vision narrowed at the edges. Around *Morrowline*, the

convoy tightened into an imperfect spear aimed at the gas giant's pale reflection.

Meridian Control sent evacuation counts during the burn. Nine cylinders had opened transit locks. Two refused, fearing load collapse during transfer. The nursery ring had moved children into its inner shelters, though half its passenger tenders remained trapped behind a docking collar that would not cycle under variable gravity. Forty-one thousand people still lacked assigned lift. Then sixty-three thousand. The count rose as local registries stopped treating temporary workers as cargo priority.

Iona appeared on a narrow channel from the Curvature Office vessel already at Meridian. Her hair had come loose from its audit tie. Behind her, warning light crossed a bulkhead in slow amber sweeps.

"Tavin, the negotiated interval is unstable," she said. "Aru can hold several pressure-names, not all of them. Bex insists the staged correction remains viable."

"Do you?"

"I believe a staged correction remains physically possible. I do not believe this schedule describes the system we have."

"How long?"

She looked toward data outside the image. "The first major return could arrive in six hours. The range extends to thirteen."

"Our fleet reaches braking point in eight."

"I know."

"Tell Meridian to begin full evacuation now."

"They have. Bex ordered them to preserve occupied rings until the Accord executes."

"Those instructions conflict."

"Yes." Iona's face tightened, not in confusion but decision. "I am giving local controllers the raw range."

"Does Bex know?"

"He will when they do."

Her channel vanished into a burst of phase noise.

The convoy burned on. Each minute reduced distance and consumed the fuel they would need inside Meridian's crowded ring plane. Tavin ordered reserve margins displayed beside every ship. Red began appearing in the homeward columns. No captain asked him to turn back.

At braking point, the gas giant filled the forward cameras with amber cloud bands and a storm white enough to resemble a wound. Meridian's cylinders moved across it as thin dark marks. Navigation overlays struggled to hold their identities because several had drifted outside certified orbits.

Haru magnified the chain. "Receiver hub just changed phase."

A pulse moved through the compensation trace. *Morrowline*'s deck seemed to tilt though thrust held steady.

Far ahead, one Meridian cylinder shed a glittering arc of dock structures.

“Time to first major return?” Tavin asked.

Haru read the raw instruments rather than the Ministry clock. “Less than two hours. Possibly less than one.”

Their planned braking burn required seventy-eight minutes.

The first evacuation packet came toward them carrying twice its rated population. Its captain transmitted no greeting, only a list of failed locks and a request for medical couches. Behind it, hundreds of navigation lights separated from Meridian in ragged lines. Too many remained fixed to the rings.

Tavin ordered Cluster Six warmed after all. Pella acknowledged from below, her voice calm now that the impossible thing had acquired a specific use.

“Sunward convoy,” he said, “break formation by assigned ring. Take every body the collars can release. Leave nothing aboard that prevents one more berth.”

The convoy opened around *Morrowline*. Hospital cutters went toward the inner rings. Water carriers emptied tanks to make volume. Cargo tugs unfolded their evacuation gantries while still under burn.

Meridian rotated beneath them, damaged and inhabited, with too little time left to separate those facts.

Chapter 16 — The Fall of Meridian

The nursery ring's outer lock opened on a room full of shoes.

Small boots, soft slippers, magnetic sandals, and two pairs of bright exercise fins drifted against the far wall. A teacher had tied their laces together in bundles before moving the children inward, but the last gravity loss had pulled half the knots apart. Iona stepped over a red boot and caught it when the deck tilted under her.

"Leave it," the lockmaster said. "We need the corridor."

Iona tucked the boot into a wall net instead. Its owner might reach a convoy ship barefoot. That qualified as a problem for later, and Meridian had fewer of those each minute.

The lockmaster cycled the outer door. Beyond its narrow glass, **Morrowline** held position across the ring's rotation with an evacuation gantry extended. Tavin's ship looked too large to move among the cylinders, yet it kept station on conventional thrusters while Meridian's receiver field rose and fell around it. Smaller craft clustered at the other locks. Their hulls drifted apart whenever compensation weakened, then snapped against their moorings when load returned.

Iona pressed her audit wafer to the lock console. "How many inside?"

"Two thousand eighty-four children, six hundred staff, one hundred eleven medical dependents. We have moved twenty-three hundred to inner shelter and lifted three hundred fourteen."

"Those totals overlap."

"They stopped giving us a stable census."

"Who stopped?"

"Everyone who changed decks." The lockmaster pointed to three separate lists taped beside the console. "One tracks wrist tags. One tracks bunks. The crèche teachers count by face because half the tags belong to siblings who traded jackets."

Iona copied all three. A clean number would have required choosing which imperfect record to discard. She had spent too many years watching institutions improve reports by excluding the people who made them difficult.

A chime passed through the ring. Every loose shoe shifted toward the gas giant, then settled aft as the spin caught up.

"Stage One begins in six minutes," said Meridian Control over the public circuit. "All personnel secure against variable load."

The lockmaster laughed without humor. "Secure where?"

Iona opened the correction plan. Meridian's eleven inhabited cylinders had accrued debt unevenly through decades of orbital station-keeping, spin smoothing, and load compensation. The Accord would reduce receiver support in three stages. Fusion tugs would absorb the orbital correction. Internal ballast would move outward to preserve spin. The Olun speakers in contact with Aru had agreed not to repossess while the debt fell through local expenditure.

Agreed did not mean unanimous. Aru had repeated that distinction until human negotiators accused the translation cradle of imprecision.

A pressure pulse crossed Iona's wrist display. The cradle rendered it as a rough sentence beneath the raw ratios.

Several histories hold. Several prepare return. The interval narrows.

She called Bex. The Minister appeared from Meridian Central, strapped into a command couch. Staff moved behind him with the restrained speed of people trying not to jostle a failing floor.

"Aru reports the interval narrowing," Iona said. "Delay Stage One until the nursery clears."

"Delay increases the accumulated return."

"Proceeding puts the load change through occupied transfer corridors."

"We have four tugs committed and a fifth approaching. The nursery ring retains full structural margin."

"The same phrase appears in the report from sixteen years ago."

Bex looked directly into the camera. "I know."

For a moment she saw the habitat engineer beneath his office, the man who understood bolts and bodies before continuity models taught him to aggregate them. Then another alert drew his eyes aside.

"We cannot preserve every cylinder," he said. "Stage One protects the hub and opens ten thousand more berths."

"Only if Eidos accepts the declining load."

"No new clearing forms part of the Accord."

The wording came too quickly. Iona marked the call for forensic retention.

"Tavin has asked for full abandonment order," she said.

"Tavin sees docking geometry. I see seventeen billion dependent people watching whether Meridian can localize without collapse."

"You see them through a model."

"I see their air inventories by name." Bex's voice remained low. "Begin the stage."

The channel closed.

Iona sent the lockmaster the three census lists and boarded *Morrowline*'s gantry. It flexed between hulls as the nursery ring turned beneath the ship. Forty children came across in paired restraint lines, each clipped to the rail by a waist tether. Some cried. Most concentrated on their hands. A teacher moved backward ahead of them, counting faces while an engineer called the changing distance between hulls.

Tavin met Iona at the freighter lock. "Stage plan?"

"Proceeding."

“Evacuation order?”

“Still partial.”

His jaw moved once. “Then I am issuing one to every ring captain who will take it.”

“You lack civil authority.”

“I have empty berths.”

“That may prove more persuasive.”

They reached the bridge as Stage One began.

Receiver output dropped four percent. The change entered Iona through her knees before the display recorded it. Across Meridian, cylinder spin assumed more of the local load. Rings widened their orbits by fractions. Fusion tugs fired in timed sequence, exhaust turning white against the gas giant’s shadow.

For thirty-two seconds, the Accord worked.

Debt fell on Iona’s trace. No corresponding pulse entered the dark lattice. The nursery ring’s ballast trains moved outward. Its rotation steadied. Evacuation collars gained enough load to cycle without binding, and six locks opened together.

Then an unlisted channel took part of the reduction.

Iona enlarged the path. It carried a Continuity seal and no public destination. The signature reached through Meridian Central, split the local debt, and sent one portion into a sink assignment removed from the visible plan.

“Bex,” she said.

Tavin glanced across. “What did he do?”

“He diverted the remainder.”

“Where?”

The hidden path resolved after its authorization packet closed.

“Eidos.”

Iona opened a priority channel to Central. Bex did not answer. His deputy appeared, mouth moving before the audio synchronized.

“Ministerial command is occupied.”

“Terminate emergency sink route M-Delta-Seven.”

“No such route appears on my board.”

“Then your board has been blinded. Cut the Continuity trunk physically.”

The deputy looked offscreen. “Doing so would interrupt the staged correction.”

“It has already been interrupted.”

Aru’s raw pressure signal struck the cradle so hard the translated text dissolved into blocks. Through the receiver network came a pattern Iona had learned to recognize without language: histories aligning around impact.

The Olun returned the diverted load.

Gravity surged through *Morrowline*. Iona’s restraint drove into her hips. On the forward display, three evacuation gantries folded against their ships. One tore free and spun between the ring planes, shedding white restraint webbing.

“Full release,” Tavin ordered. “All rings. Haru, assign craft by nearest viable lock. Pella, break us from the nursery before its collar takes our truss.”

Pella’s acknowledgment arrived beneath metal alarms. “Collar has us by six teeth. I can retract four.”

“Cut the remaining two.”

“People in the gantry.”

Through the docking camera, Iona saw thirty-one children and four adults clipped along the passage. Sudden load had dropped them against one wall. The teacher at the front tried to stand, failed, and began passing the children hand to hand toward *Morrowline*.

Tavin opened the gantry circuit. “You have ninety seconds before we lose alignment. Keep every tether clipped until the inner hatch.”

The teacher did not waste breath answering.

Iona left the bridge. Sive met her at the cargo lock with two load crew. They pulled children across the sill in groups of three. A boy had lost one shoe. The remaining one shone red.

“Outer door will not close,” Sive said. “Frame twisted.”

“We do not need it closed if the gantry separates cleanly.”

“It will not.”

Pella climbed through the service hatch beside them carrying a powered cutter. “I can release the teeth from the ring side. When I shout, close this pressure panel.”

“You cannot get back through after it closes,” Iona said.

“I have a suit.” Pella slapped the collar of her work wrap, where a compact hood and pressure film waited. “Also, I dislike arguments conducted while my ship is being used as structural reinforcement.”

She crawled into the service channel before Tavin could countermand her. Iona watched through the inspection glass. Pella reached the first jammed tooth, braced both boots, and applied the cutter. Sparks streaked sideways as the ring’s changing gravity pulled them.

The final children crossed. Their teacher came last, still counting.

“Thirty-one,” she said. “All thirty-one.”

The nursery ring lurched outward.

Pella severed the first tooth. The gantry dropped half a meter and pinned her left glove against the housing. She cut the second with her right hand. Its release drove the collar plates together.

Pella screamed once.

Sive struck the emergency override. Tavin fired *Morrowline*'s lateral clusters, taking weight off the joint for less than a second. Iona reached into the service gap and caught Pella's harness. Together they dragged her through as the collar separated.

Two fingers remained inside the glove where the plates had closed.

Sive sealed the bleeding hand before Pella could look at it. “Index and middle. Clean compression. Keep your wrist above your heart.”

“My heart has moved around a lot today,” Pella said through clenched teeth.

A vibration ran through the hull. Iona felt each structural member pass the load onward.

“Morrowline clear,” Tavin said over the circuit. “Cargo lock, secure. We are going back for the next collar.”

Pella tried to rise.

Sive pushed her down with one hand. “You have done your collar.”

“My engines are out there.”

“Your assistant knows which end gets hot.”

“That remains disputed.”

Iona left them because choosing to watch did not improve the outcome.

On the bridge, Meridian had become a set of independent emergencies. Cylinder Four's evacuation spine had jammed. Ring Eight lost spin synchronization and ordered residents to lie against the outer decks. A water carrier took six hundred people through a cargo hatch built for tanks. Tavin assigned ships by minutes to structural failure, ignoring Ministry priority codes.

Cylinder Four called again while its assigned passenger packet held ten meters from the jammed spine. The lock's inner wheel had stopped between positions, leaving a gap too narrow for an adult in a pressure suit. Through the packet camera, Iona could see residents gathering behind the glass. They had brought suitcases despite the load order, each person still behaving as if evacuation meant travel.

“Cut the wheel into quarters,” Tavin told the cylinder engineer.

“We lose the lock after that.”

“You have one use left for it.”

The engineer asked for a cutting crew. None could cross the gap safely. Tavin turned *Morrowline* until a side cluster faced away from the packet, then fired a sequence of short pulses. Exhaust pushed the smaller ship

against its collar just hard enough to hold contact without receiver assistance. Two packet deckhands climbed through the partial opening with saws clipped to their chests.

Iona listened as the wheel came apart. The first quarter jammed against the frame. The second broke free and struck a deckhand's shoulder. Residents on the cylinder side pulled him through, splinted the arm, and sent him back to finish. When the opening widened, the crowd tried to move at once.

"Bags stay," the cylinder engineer announced. "Hands stay on the person ahead. If you have a child, say the child's name at the threshold."

Names crossed the circuit with each body. Ansa. Pell. Ro. Din and Din's mother. Ansa again, corrected to Asa. The engineer repeated every one until the packet loadmaster answered. No registry held the exchange. Iona opened a local record and let the voices populate it without cleaning their errors.

After four hundred people crossed, Cylinder Four's spine tore farther along its mount. The packet captain released before the fracture could peel open her hull. Seventy-three residents remained behind the glass. A cargo tug took the packet's place from the opposite side, using a maintenance port that required people to crawl through one at a time.

Tavin marked Cylinder Four yellow rather than complete. "Next problem."

Bex finally answered.

His command room had gone dark except for emergency strips. Blood marked one side of his forehead.

"You sent debt to Eidos," Iona said.

"I sent nine percent of Stage One. The local tugs could not absorb it and preserve the nursery orbit."

"You concealed the route from every participant."

"If I had listed it, the Accord would have failed before execution."

"It failed during execution."

Bex closed his eyes for one breath. "The nursery had eighty-one thousand people attached to its support registry."

"It also had Olun lineages at the destination you selected."

"I chose a temporary export over a cylinder break."

"And received both."

Behind him, an aide called that Central's outer tether had parted. Bex tightened his restraint.

"Can the receiver hub be isolated?" Iona asked.

"Not remotely."

"Then abandon Central."

"Someone must hold the stage valves open for the tugs."

“Manual locks?”

“Yes.”

He had returned to the scale of tools. It did not forgive him. Iona transmitted Tavin’s nearest shuttle assignment to his console.

“Hold them until the last tug clears,” she said. “Then take that shuttle.”

Bex read the assignment. “You still believe I should live.”

“I believe your death would close an account before testimony.”

His expression accepted the wound without pretending it amounted to mercy. “Understood.”

A second returned pulse entered Meridian.

The receiver hub shed its outer field coils. Metal loops hundreds of meters wide lifted from their mounts and crossed the cylinder chain in slow, lethal arcs. One struck an empty industrial ring. Another cut through a cargo dock after its last packet pulled away. The third passed beneath the nursery ring close enough to drag its orbit inward.

Haru called the change. “Nursery pericenter falling. Twelve kilometers on this pass.”

“Tow options?” Tavin asked.

“None with people aboard. The spin axis is precessing.”

Iona looked at the evacuation count. Thousands remained in inner shelters. Most available craft had already docked or waited in the ring’s path. If Tavin attached *Morrowline*, the freighter could alter the cylinder’s tumble or move people, not both.

“Use the ship for lift,” she said.

Tavin studied the same geometry. “Agreed.”

He ordered every tug to stop trying to save the ring.

The statement took three repetitions before local control accepted it. For sixteen years, Meridian had maintained the nursery as proof that the first incident had been repaired. New hull wrapped old damage. Gardens grew across sections where Sella and nineteen other engineers had died. Families chose births there because the Republic certified the ring safer than any other cylinder.

Now its controller asked, “If we release orbital stabilization, what happens to the structure?”

Haru answered because Tavin had no softer physics available. “It crosses the inner debris plane in forty-seven minutes. The next gas-giant tide exceeds frame tolerance.”

“Can we come back after?”

“No.”

The controller left the circuit open. In the background, someone began announcing shelter groups.

Evacuation changed from organized lift to stripping a place of bodies. School records stayed in cabinets. Incubators crossed without their diagnostic bases. Teachers wrote children's names on suit film when wrist tags failed. A maintenance crew welded a cargo ramp directly to an ore tug, then abandoned the welder while its electrode still glowed.

Morrowline* took twelve hundred into bays rated for seven hundred. Tavin vented reserve water to free tank volume and used the empty compartments as pressure shelters. Sive assigned every conscious adult a child to restrain during thrust. Pella, medicated and furious, directed engine work from a cargo couch with her injured hand sealed against her chest.

Iona returned to the nursery ring on the last service shuttle.

She did not go for a record. One inner shelter had reported forty-six infants and no functioning transfer cradle. The shuttle crew carried them six at a time in hard cases, each case clipped to two people. Iona took the final pair. Their labels read Mera Olt, six weeks, and Tavi Nn, nine weeks. One slept through the alarms. The other watched Iona with unfocused eyes while the deck's direction moved beneath them.

At the lock, the red-booted boy waited beside the teacher. Someone had found his missing shoe in the wall net.

"Is this all?" Iona asked.

The teacher looked down the corridor. Its lights failed section by section toward the shelter.

"All we could reach."

That answer differed from all.

They crossed to the shuttle. Behind Iona, lock workers shut the ring-side hatch though nobody required pressure integrity now. Habit finished the task that survival no longer needed.

The shuttle pulled away with eight minutes to spare. Through its small window, Iona watched the nursery ring turn. A painted band of animals circled the children's decks: birds, long-bodied dogs, a blue creature invented by some class. Sixteen years earlier, Sella had died somewhere behind that painted hull. Iona had imagined the place preserved, a fixed point against which the official account could someday be tested.

The gas giant's tide reached the ring.

Its axis tipped. Windows flashed across sunlight, vanished, and flashed again. The central spine bent near the old repaired section. For one second the cylinder resembled two rigid bodies joined by a hinge. Then the nursery ring broke.

Air spread into a white cloud carrying insulation, paper, soil, and the bright panels from a children's corridor. The two halves turned away from each other. No engine could gather them. No later hearing could restore the rooms.

Iona held the infant cases against the shuttle deck while full conventional thrust carried them clear. Around Meridian, surviving cylinders abandoned their orbits one by one. Evacuation craft streamed outward, overloaded and low on fuel. Behind them, the receiver hub burned through its remaining coils and went dark.

Meridian Control removed the nursery ring from the active navigation list. The identifier did not change to lost or destroyed. It simply disappeared, leaving a blank interval between Cylinder Five and the industrial pair.

Iona kept her eyes on that absence until the shuttle docked.

Chapter 17 — The Unpaid Dead

Iona found Sella's handwriting inside a box that had been struck hard enough to fold its hinges inward.

The salvage team had pulled the box from a section of nursery-ring maintenance deck caught in Meridian Cylinder Six's emergency net. Its outer casing carried fire scoring, a strip of faded green paint, and a service number from sixteen years ago. Half the number had been hidden beneath newer inspection labels. A technician warmed the latches with a hand torch, cut the bent hinge pins, and asked Iona to stand behind the pressure screen.

When the lid came loose, gray water rose from the seal and formed beads in the salvage bay's weak spin. Three data wafers rested in individual slots. Two had cracked. The third carried a handwritten tag under transparent film.

S. Rhyll / nursery transfer / unresolved.

Iona read the letters twice before touching the screen.

Around her, Meridian's wreckage crossed the bay windows in slow procession. A torn garden deck showed black soil packed against one wall. Two pressure doors remained joined to a frame with no corridor on either side. Farther out, one half of the nursery ring turned against the gas giant's cloud bands while salvage tugs marked a perimeter nobody expected to enter alive.

The technician said, "We can image the wafer without mounting it."

"How long?"

"Twenty minutes for surface mapping. Longer if the inner layers delaminated."

"Do it here."

"You have a cleaner lab aboard *Morrowline*."

"I want the custody record to begin before it leaves this deck."

He looked at the tag and stopped arguing. His name patch read Eren Vass, Meridian Municipal Maintenance. The word municipal had lost its physical referent overnight. He still wore it.

Eren placed the wafer in an optical cradle. Iona started a forensic log with Cylinder Six's remaining clock, *Morrowline*'s clock, and a free-running salvage chronometer that had never touched the receiver network. She recorded the box dimensions, hinge deformation, water temperature, service number, and every hand that had moved it since recovery. Procedure steadied nothing. It only made later denial more expensive.

The first image appeared as a field of broken characters.

"Outer layer took heat," Eren said. "I can read the maintenance header."

Iona enlarged it. The date matched Meridian's weight-loss incident sixteen years earlier. Sella had written the entry forty-three minutes before the official report placed her at the nursery ballast controls. The report described her as fatigued, out of sequence, and responsible for opening a manual clutch before receiver output stabilized.

Iona knew its wording without retrieving it. She had cited the same language during training to prove that error categories should remain distinct from intent. A good engineer could still make a fatal mistake. Grief did not improve evidence.

Eren shifted the imaging angle. More text resolved.

Load reduction exceeds local spin change. No authorized sink line in maintenance view. Continuity trunk active under emergency mask.

Iona stopped breathing until her body objected.

“Keep imaging,” she said.

The next lines arrived slowly. Sella had taken readings from three independent strain lattices. Nursery rotation continued to fall despite a receiver command that should have preserved it. An unlisted route had exported the difference through Meridian Central. Eleven seconds later, a smaller disturbance came back. Sella called it a returned precursor pulse.

No phrase in the official inquiry mentioned a precursor. No table listed an emergency mask.

Eren lowered his voice. “Could this refer to yesterday?”

“The date does not.”

“I mean the mechanism.”

“Yes.”

Sixteen years apart, Bex had chosen the same act at the same place: preserve an occupied ring by moving part of its physical burden into a destination concealed from the public plan. Yesterday he had called it nine percent of Stage One. The older record named no percentage, only a line that should not have existed.

Iona opened the historical system image from the official inquiry. Her clearance still worked. The maintenance packet listed every public receiver route, ballast command, and operator action for Sella’s last shift. It showed no Continuity trunk. Someone had prepared the evidence by removing the path rather than falsifying its numbers.

“Can you recover her signature?” Iona asked.

Eren adjusted the scan. “The wafer authenticated when she began the note. Final lines may carry the same session key.”

“May?”

“Heat altered the timing layer. I won’t promise what the material has not given us.”

The answer sounded like Sella.

Iona sent a copy of the raw image to Nemi under protected audit protocol. Her deputy answered from Nacre within three minutes, vestibular brace visible behind one ear. Nemi had learned to keep her head still while her eyes moved across data.

“Source?” she asked.

“Nursery maintenance box recovered from debris net Six.”

“Custody?”

“Continuous from retrieval.”

“Has Bex seen it?”

“No.”

Nemi read the visible lines. Her face changed by degrees. “I have the old inquiry archive. Give me the trunk timestamp.”

They aligned Sella’s note with records the Curvature Office had retained but never connected. A power dip appeared at Meridian Central four seconds before the unauthorized route opened. The sink scheduler logged a destination request, then overwrote it under continuity privilege. Receiver diagnostics recorded a phase return eleven seconds later and classified it as mechanical noise.

Eidos had served as Meridian’s assigned sink during that quarter.

“Can we prove destination?” Iona asked.

“Not from the public packet.” Nemi moved to a second archive. “The network retained aggregate sink loads for cap enforcement. They were never part of the casualty inquiry.”

“Because sink activity carried no relevance to operator error.”

“Because someone defined relevance after selecting the finding.”

Nemi found an unexplained Eidos load increase at the same timestamp. Its quantity matched Sella’s missing load within the historical instrument margin. The returned phase matched a smaller fraction, a warning pulse too weak to damage the ring and too structured to fit the noise model available then.

The Olun had warned them sixteen years ago. Sella had noticed. The Republic had placed her body over the evidence and called the shape an operator’s mistake.

Iona gripped the edge of the imaging table. Anger arrived as a demand for movement. There was nowhere useful to put it. Striking the bent box would add damage to an object that had survived enough. Calling Bex before the recovery finished would convert evidence into an argument. She made herself release the table.

“Continue the chronology,” she said.

Nemi’s eyes flicked up. “Iona.”

“Continue.”

They traced the next forty-three minutes.

Sella closed the hidden trunk locally, but the export had already reduced nursery spin. Automatic ballast refused manual commands because Central still reported nominal receiver support. Sella crossed to the mechanical clutch room with two technicians. She bypassed the receiver interlock, released six outer ballast cars, and used their movement to restore enough rotation for the evacuation locks to cycle.

One technician left at Sella's order to open the children's shelters. The other died when the first returned pulse shifted a ballast car off its rail. Sella remained at the clutch because releasing it early would let the cars roll inward and strip away the load they had gained.

The official report described that decision as sequence confusion.

A later line emerged from the wafer, scored through by heat but legible.

Unauthorized export caused the loss. Returned load confirms an occupied or active destination. This is not operator error. Keep nursery locks moving.

Iona read the exact words aloud for the custody log.

Eren turned away. Through the window, the broken nursery section completed another slow half rotation.

Surface mapping reached the final intact band. Sella had added a maintenance note rather than a personal message. She listed clutch temperature, ballast positions, remaining seal time, and the names of the technicians sent ahead. The last line retained the same engineering register.

Outbound load entered Eidos before clutch fault. Preserve destination field in inquiry copy.

The signature layer authenticated.

No one spoke. The salvage chronometer advanced. A tug outside fired to nudge wreckage away from Cylinder Six, and the weak deck load changed beneath Iona's boots.

She saved three copies. One went to Nemi's isolated store. One entered Meridian's municipal archive under Eren's authority. The third remained with the wafer. She did not send it to the Continuity Ministry.

Tavin called from *Morrowline*. His image showed fatigue and a fresh scrape along his temple.

"We found another shelter list," he said. "Cylinder Four, maintenance port. Seventy-three marked behind glass, sixty-nine confirmed aboard the cargo tug. Four names unresolved."

"Send them to me."

"Why?"

"I'm building a casualty comparison."

He studied her face. "What did you find?"

"Sella's last note."

Tavin waited.

"She did not cause the first Meridian incident. Continuity opened an unauthorized export. She closed it and restored the nursery ring long enough to evacuate."

"Who hid the note?"

"I do not know yet."

"Does Bex?"

"I do not know what Bex knew sixteen years ago." She kept the statement narrower than her anger. "I know his ministry used the same emergency mask yesterday."

Tavin looked toward someone outside the image. "Pella is awake. She says two fingers constitute a parts shortage, not a disability."

"Tell her the salvage team found her cutter."

"Did they find the fingers?"

"No."

"Then omit that part."

The channel ended on work, which helped.

Aru's translation cradle occupied the far end of the salvage bay. Its pressure vessel had been bolted to a cargo pallet after Meridian Central lost power. Thin acceleration actuators moved a dense fluid through branching tubes. Usually the apparatus communicated through patterns too slight to see. Today its fluid shivered continuously.

Iona carried Sella's recovered chronology to it.

"Aru, can you identify the pressure lineages affected by the export sixteen years ago?"

The actuators paused. When they resumed, the pattern came in separated intervals.

Identification survives in wounds. Completeness does not.

"I am not asking for completeness."

Humans often ask for a list and later call outside-list nothing.

The answer found its mark. Iona placed the human casualty register on one side of her display and opened a blank field on the other.

"Then give me every surviving identification and mark the limit."

Pressure ratios accumulated. The cradle translated names poorly because an Olun pressure-name contained location, phase history, and temporary coalition. Human script reduced each to a pronounceable fragment. Aru supplied the fragments with annotations: surviving witness, interrupted memory reef, lineage unformed after repeated disturbance, identity uncertain, several possible overlaps.

Iona did not collapse the uncertainty.

The first human register held Sella Rhyll, Venn Or, Mahir Tal, and seventeen others killed during the old weight-loss incident. The new Meridian register had not closed. Names entered while she watched: lock workers from Cylinder Two, residents from the nursery shelters, a tug deckhand struck by a loose gantry, people who had reached evacuation packets but died when medical load exceeded capacity.

Beside them she placed the Olun entries.

Aru-in-Dark-Cleft, interrupted.

Pressure-Before-Melt, no later continuity found.

The coalition rendered as Holds-Five-Cold-Strata, partly surviving in two witnesses.

A nursery pressure lineage without a stable name, whose formation ceased after the first emergency pulse.

Sella's register contained occupations and ages. The Olun archive contained gradients, intervals, and absences. Neither fit the other. Iona resisted every field that promised to make them comparable by deleting what differed.

"Read them," Aru said through the cradle.

The command startled her because Aru rarely used an imperative without a physical warning attached.

"Which archive?"

Both. Alternation prevents one pressure from becoming background.

Iona opened the salvage bay circuit. Eren looked up from the damaged box. Two tug crew stopped securing a net. Beyond the door, evacuees waited for transport out of Meridian with their remaining property tied into blankets.

She began with Sella.

Then she read Aru-in-Dark-Cleft.

Venn Or.

Pressure-Before-Melt.

Mahir Tal.

Holds-Five-Cold-Strata.

The names did not balance. One human did not equal one Olun coalition. A confirmed body did not equal a lost possible lineage. Alternation made no moral arithmetic. It only prevented the human dead from occupying the whole room.

When the old register ended, the new one continued. Iona read until her voice roughened. Eren took the next human name without asking. The translation cradle supplied another pressure-name. A tug mechanic read after him. Nobody called the sequence a memorial. Work continued around it: welders sealed a cracked compartment, a medic checked evacuees for decompression, and salvage radar warned of another fragment crossing the perimeter.

Bex entered the bay on a shuttle from Meridian Central while the reading continued.

He had changed out of his bloodied jacket. A transparent dressing covered his forehead. Two security officers followed at a distance, uncertain whether emergency authority required them to protect him or witness him.

Iona finished the line in front of her before turning.

Bex saw the opened box. "Whose?"

"Sella's."

His face lost color.

“You knew there was a suppressed maintenance note,” Iona said.

“I knew the first inquiry had sealed appendices.”

“Did you read them?”

“Not then. I served in Nacre habitat works sixteen years ago.”

“And six years ago?”

“When I took Continuity, I read the sink-response files. Sella’s note did not appear among them.”

“Yesterday you used the route that killed her.”

“I used an emergency mask built into Meridian before my ministry.”

“You concealed Eidos.”

“Yes.”

The admission landed without defense. Iona gave him the raw wafer image. Bex read the visible lines standing beside the bent box. His lips pressed together at This is not operator error. When he reached the last sentence, he set the display down carefully.

“I am sorry,” he said.

“Your sorrow has no authority here.”

“No.”

“Who can open the sealed inquiry appendices?”

“My engineering key and the Curvature Office director’s credential.”

“Order access.”

Bex looked toward the security officers. “Under emergency authority?”

“Under evidence law.”

“That moves more slowly.”

“It also survives your office.”

He entered the order.

The sealed appendices opened in layers. First came photographs of the clutch room taken after recovery: a ballast lever fused in the released position, Sella’s tool tether cut by rescue crews, scoring where the derailed car had crossed the wall. Next came witness statements from nursery staff who said gravity improved after Sella entered the mechanical bay. Their accounts had been classified as contradictory because none of the witnesses could see the receiver console.

A review memorandum followed. Its author, a deputy continuity engineer dead nine years, compared Sella’s

note to the emergency trunk record and wrote that an external route had preceded every manual action in the clutch room. Someone above him returned the memorandum with a single instruction: Remove sink speculation from operator finding.

No signature appeared beneath the instruction. The archive did preserve the authorization tier, a predecessor office to Bex's ministry with the same power to blind maintenance views.

Iona read the witness statements again. A nursery nurse described Sella arriving at the shelter lock with blood on one sleeve and asking whether the children could move. When told the lock still lacked load, Sella went back toward the clutch. A spin technician reported that she refused relief because anyone new would need three minutes to learn which damaged rail still carried force. The official account compressed both statements into one phrase: operator remained at controls despite evacuation order.

"Those witnesses spent sixteen years believing the report had considered them," Iona said.

Bex stood beside her. "The report did consider them. It treated their evidence as an obstacle."

"Can you identify who issued the removal instruction?"

"Not from the tier alone. Five officials held it."

"Then preserve all five names and their access logs."

He added them to the evidence order. One official still served on the Assembly's infrastructure committee. Another had retired to a receiver-supported medical spindle. Two had died. The fifth had left no current address after the six-year sink responses were sealed.

The archive had not lost Sella's truth. It had distributed each piece into a different category until no ordinary inquiry could assemble them. Recovery required a dead ring, a bent box, two credentials, and another catastrophe built from the same concealed route.

Iona removed her Curvature Office credential from her wrist. The strip had granted access to every sink ledger she had trusted, every inquiry category she had defended, and every sealed field someone above her decided she did not need. She placed it beside Sella's wafer.

Bex watched. "If you resign now, they can exclude you from the inquiry."

"They already excluded me for sixteen years."

"You retain authority to compel records."

"I retain evidence copied under valid custody. Nemi retains hers. Meridian Municipal Maintenance holds another." She looked at Eren. He nodded. "The Office can decide whether to cooperate with facts after I leave it."

"What will you do without standing?"

"Continue the work."

Iona drafted her resignation in six lines. She cited concealment of material evidence, misuse of sink certification, and the Office's failure to distinguish continuity from truth. She did not cite grief. The record did not need to know why accuracy hurt.

Before sending it, she copied the recovery authorization attached to her credential. Salvage operations could continue under Meridian municipal authority and convoy necessity. Eren signed her onto his team as a forensic contractor with no pay code and no institutional rank.

“Welcome to municipal maintenance,” he said.

“Does the municipality still exist?”

“Enough to owe me three months’ wages.”

She sent the resignation.

Her credential went dark. Doors, archives, and command channels vanished from its permissions one after another. The loss felt physical, a reduction in reach measured by each gray icon. Nemi’s isolated store remained connected. So did the translation cradle. Tavin’s convoy channel stayed open because he had granted it personally.

Outside, a salvage tug caught the maintenance deck fragment in a net and drew it toward Cylinder Six. The box would go with it. Sella’s wafer would go to three archives. Nothing would bring back the clutch room, the nursery ring, or the pressure histories struck below Eidos.

Iona picked up the next unresolved human name and placed it beside the next Olun pressure-name. Then she read both into the record under no office but her own.

Interlude III — The Archive of Impacts

Activation transcript, Nacre Receiver One. Network Year 0.

“Proof mass steady.”

“Local curvature increased by commanded interval.”

“Destination response?”

“Sink Eidos Four accepts full disturbance. No reflected energy above threshold.”

“Record successful exchange.”

A technician laughed after the formal line. Someone asked whether a coin would fall faster. The proof mass crossed its optical cage, the receiver warmed by two degrees, and a new entry appeared in the Republic’s engineering register: Weight relocated without propellant expenditure.

Far beneath Eidos Four’s ice, no register opened. A pressure history formed instead. The first pulse compressed a superfluid boundary that had held stable across nine seasonal tides. Three local memory structures sheared. A temporary speaker assembled long enough to compare the impact with moon strain, found no matching orbit, and dissolved before it could preserve the question.

Twenty-four years later, sink certification acquired legal language.

The Republic survey vessel *Aster Vale* crossed Eidos at speed. Its report included atmospheric spectroscopy, radio silence, thermal imaging, and two gravimetric passes optimized for liquid oceans. The second pass detected a persistent discontinuity below the pressure floor. Survey software labeled it mineral phase separation.

The certification board copied one sentence into its public finding: No biosignature consistent with inhabited use.

An Olun archive from the same passage preserved no vessel. Light and radio did not reach its depth with useful shape. The gravimetric scan did. It passed through the ice as a gentle ordered pressure, too small to injure and too brief to answer. Four pressure-names aligned to it. One proposed intentional origin. Two classified it as distant fracture. The fourth kept both interpretations and marked a watch interval in the upper strata.

At Network Year 31, a maintenance child on Leto Cylinder wrote a school answer in green stylus:

> Receivers let habitats adjust gravity without wasting fuel. The extra force goes to empty systems where it cannot hurt anything. A sink works because nobody lives there.

Her teacher marked the answer correct and circled without wasting fuel for spelling.

During that school hour, three routine clearings reached Eidos. The first shifted a moonlet's pericenter by less than a finger's width in human measure. The second crossed a memory reef already weakened by earlier pulses. The third completed the fracture. Histories from six pressure lineages survived in neighboring ice, but their order did not. Later Olun speakers could recover a child-pressure, a migration toward warmth, and a warning about the next tide. They could not determine which memory had preceded which.

The human ledger retained all three clearings to nine decimal places.

Network Year 58 produced the first industrial emergency export. A refinery spindle in the Corve system lost a flywheel bearing while eleven thousand workers occupied its outer decks. Operators used Reciprocal Mass Exchange to hold local load for four minutes and forty seconds. The incident report credited the receiver with preventing mass casualties. Its appendix listed damaged machinery, medical outcomes, and deferred fuel cost.

No field accepted a remote casualty.

In Eidos, the emergency pulse crossed two ice bodies. One pressure archive recorded sudden compression followed by a partial rebound. Another preserved an absence: a mature coalition had begun assembling for seasonal exchange, then failed to answer any later call. The Olun could not know whether every component ended. They marked the site with a pressure interval meaning do not build continuity across this break.

Human sink accounting called the same event a successful full acceptance.

Sixteen years before the present audit, Meridian opened an unauthorized emergency route.

Sella Rhyll's maintenance note captured the event from inside the nursery ring:

> Load reduction exceeds local spin change. No authorized sink line in maintenance view. Continuity trunk active under emergency mask. Returned load confirms an occupied or active destination. This is not operator error. Keep nursery locks moving.

The official inquiry omitted the note. Its public conclusion assigned premature clutch release to operator error and counted twenty human dead. Meridian repaired the nursery section, renewed its safety certificate, and

painted animals around the children's decks.

Eidos retained the returned precursor in several forms. A shallow archive represented it as a probe toward the source. Deeper histories described a reflex after unexpected compression. One pressure-name claimed the return had carried structure. Others heard only fracture. The disagreement persisted because the next human pulses arrived before the disputed interval could settle.

Six years before Meridian fell, Republic instruments detected organized replies.

A Curvature Office classifier saved the sequences under structured sink residuals. Prime ratios, controlled pauses, and amplitude mirroring appeared across fourteen sessions. Analysts proposed unknown dark-lattice behavior, concealed human machinery, or deliberate source action. The Continuity Ministry sealed the packet. Clearing schedules continued.

Olun pressure archives preserved those sessions as attempts rather than replies. One faction repeated the impact spacing at reduced force. Another inserted long intervals copied from undamaged ice. A third returned small portions of each pulse, testing whether the source changed behavior. When clearings continued, several lineages concluded that the sender could not perceive. Several concluded that it perceived and refused. Neither conclusion achieved unanimity.

On Day 0, a Perihelion audit froze one clearing.

The human file began with an apple, 0.0037 g of unsettled curvature, and a blank destination column. It continued through hospital delay, propellant cost, and a sequence of prime intervals received while the local device remained in standby. Iona Rhyll entered unassigned reciprocal load into a codebook that lacked the phrase.

An Eidos archive marked the same pause as the first interruption in 112 years of repeated impacts. Pressure-names assembled around the coupled channel. They sent a sequence chosen for divisibility by no smaller interval. The source altered its behavior. For the first time, both archives preserved not only an event but a change attributed to the other side.

The records multiplied after contact. Human files acquired translations, casualty estimates, and arguments over seventeen billion dependent lives. Olun strata acquired human dissent patterns, localization trials, and proof that some senders spent their own momentum rather than exporting it. Neither archive became singular. Republic ministries concealed data that maintenance workers copied. Olun factions returned wounds that other pressure-names asked them to suspend.

Then Meridian opened the emergency route again.

Day 19, *Morrowline* cargo record: Twelve hundred evacuees embarked against rated capacity of seven hundred. Reserve water vented to create shelter volume. Chief engineer Pella Jor injured during collar release. Nursery-ring transfer incomplete.

Day 19, Meridian Municipal Maintenance: Receiver hub lost. Eleven inhabited cylinders abandoned or irretrievably displaced. Nursery ring separated across its repaired section after tidal load exceeded frame tolerance. Final confirmed shelter census unavailable. Record remains open.

Within Eidos, the returned diversion struck a coalition already aligned for restraint. Some histories broke. Others joined the hard return and increased it. Aru-in-Pressure lost coherence across two local strata while carrying evidence of human evacuation. A surviving pressure-name recorded children moving through unstable

load, not as absolution, but as proof that the source contained opposed actions.

On Day 20, Iona placed Sella Rhyll beside Aru-in-Dark-Cleft. A Meridian technician added Venn Or. The translation cradle supplied Pressure-Before-Melt. Human names continued beyond the old casualty inquiry. Olun names continued beyond defensible translation. Blank fields remained marked as unknown rather than zero.

The two archives received their final entry from different instruments.

Meridian salvage radar tracked the broken nursery ring until its fragments crossed below recoverable orbit. Eidos pressure histories tracked the returned pulse until its last gradient dispersed through damaged ice. Human recorders entered the time. Olun witnesses preserved the interval.

Both left the event open.

Chapter 18 — The Third Account

Iona began by cutting Meridian's evacuation board into three pieces.

The board had hung outside *Morrowline*'s cargo lock during the rescue. One side still carried chalk capacities for children, critical patients, and ambulatory evacuees. Coolant had blurred the lower rows. Tavin's crew no longer needed it for loading because the survivors had transferred to Nacre packets, but nobody had erased the names.

Iona laid the pieces on three cargo tables. The ship held one fifth standard gravity from spin, enough to keep the boards down and weak enough that a careless sleeve could move them.

Nemi watched through a wall display from Nacre. "You resigned yesterday."

"I remember."

"You are using a Republic cargo registry, Ministry survival models, Curvature Office sink records, Meridian municipal credentials, and an Olun translation cradle attached to a freighter."

"Yes."

"Good. I wanted the jurisdiction clear."

Tavin entered carrying a fuel wafer and a cup with a lid. "It is not."

"That also helps," Nemi said.

Iona wrote three headings.

REPUBLIC SURVIVAL

OLUN LOSS

LOCAL PHYSICAL COST

"These are the three accounts," she said. "No conversion between them. No net total."

Tavin set the fuel wafer beneath the third heading. "What does a negotiation do with numbers that refuse to become one number?"

"It stops pretending one subtraction can answer every claim."

"Politicians will love that."

"They have had simpler figures for 112 years."

Nemi added a fourth participant to the secure channel. Pori Sal had retired from comet mass accounting four years earlier, after Continuity transferred her away from sink analysis. Her hair had gone white. The orange cord around Nemi's stylus carried the knots that had survived containment.

"Seven years ago I asked for a deep survey of Eidos," Sal said. "Today I am not asking you to treat that request as prophecy. I am asking you to preserve the interval tables that made the request reasonable."

She deposited her original checksum and twelve clearing traces into the public workspace. The old warning did not solve the present account. It did establish where deliberate uncertainty had begun, and it gave the Olun-loss table a human custody chain no ministry could quietly relabel.

The first table began with the Republic's fixed population: 31.4 billion across forty-three systems. Seventeen billion lived in structures whose orbit, spin, or long-term load depended on RME. Bex's continuity model treated that dependency as a single exposed population. Iona opened its assumptions one by one.

Some habitats would lose air within hours after abrupt receiver shutdown because bearings or seals required controlled load. Others could operate for months while spending fuel and reducing capacity. Stillness already survived without a receiver, though its people paid through slow spin, ballast mass, cramped construction, and constant tug work. A farm torus had localized successfully and lost a growing season. The same label, dependent, covered immediate death and expensive adaptation.

"Separate time," Iona said.

Nemi divided the model into ten-hour, ten-day, one-year, and generational windows. Tavin added mobility: fixed cylinders, towable stations, settlements with detachable population modules, and ships that could move mass but not house everyone attached to it.

They found the first dangerous assumption after eleven minutes. Continuity had counted every person registered to a receiver-supported structure as a likely abrupt-shutdown death.

"Meridian proves that overstates and understates at once," Tavin said. "We evacuated people the model assigned to a lost ring. We also lost people after they reached craft because the craft lacked medical load."

Iona created two fields: structural exposure and evacuation exposure. Neither counted as a death forecast. Each carried a range linked to a date and a named physical failure.

Bex joined by secure channel. He had not been invited, but his emergency authority still opened convoy policy circuits.

"You removed the continuity total," he said.

"I disaggregated it."

"The Assembly needs a number."

"It needs several honest ranges."

"A range can conceal a preferred endpoint as easily as a total."

"Then every range carries its assumptions." Iona turned the display toward him. "Your model counts a person in Stillness and an infant in an unsupported nursery lock under the same dependency class."

"Because both live after the network fails only if local systems perform."

"That describes everyone aboard this ship."

Bex leaned closer. Fatigue had deepened the lines beneath his eyes. "What minimum do you defend?"

"Seventeen billion exposed to disruption. Not seventeen billion certain dead. We estimate consequences by habitat, time, evacuation capacity, and available conventional force."

“If the Olun demand immediate correction, you will have hours to explain those distinctions.”

“Then help us make them legible now.”

He remained on the channel.

Before moving to fleet costs, Iona tested the structure against one inhabited place. She chose Nacre’s north residential band because its operators had already reduced receiver support and because Keir’s public spin crew could answer without Ministry filtration.

Keir appeared from a ballast control room with four people working behind him. Chalk lines covered the wall, each one connecting a water tank to a rail position. The spindle’s slow rotation pulled his harness toward the outer deck.

“Give me the consequence of zero receiver draw now,” Iona said.

“Now means which minute?” he asked.

“Exactly.”

Keir checked the local schedule. “For the next forty minutes, we hold point eight standard in residential decks and point nine in two hospital sectors. North bearing temperature rises. Water pressure falls above Deck Ninety because the pumps were sized with compensation. Nobody loses air.”

“And after forty minutes?”

“We move another eighteen tanks outward or reduce spin load in the hospital sectors. The tanks block two public corridors while traveling. We need every rail crew awake.”

“Ten hours?”

“Bearing inspection, pump replacements, and no cargo rotation. Food packets come through the south spine. We can keep living here.”

“Ten days?”

Keir looked toward a spin supervisor. The woman stepped into view, gave her name as Varo Sen, and pulled up a maintenance forecast.

“Without receiver smoothing, bearing wear triples at present mass,” Varo said. “We either shed population, replace the north race, or slow the spindle. Replacement takes a tug lift we do not currently have. Slowing means less load across every deck, including surgery and childhood development wards.”

Bex asked, “Minimum evacuation?”

“Unknown until we inspect the race after a full day,” Varo replied.

“Give a planning range.”

“I can give conditions. If vibration stays below this line, we relocate no one. If it crosses amber, we move the outer four decks. If it crosses red, we clear the north band. A population number without tomorrow’s bearing trace only rewards whoever chooses the preferred forecast.”

Iona entered all three branches. The Republic account now carried observable triggers instead of a single casualty expectation. Keir added the available shelters, train capacity under weak load, and the number of rail crews who could move ballast without receiver support. Bex supplied hospital demand from the Ministry model. Its patient registry ran two days behind.

“Your air inventory has more recent hospital data,” Iona told Varo.

“It has occupied beds. Not diagnoses.”

“Use occupied beds for lift mass. Let medical staff classify care needs separately.”

A nurse joined from Nacre Hospital Sector Two. She identified six procedures that could shift to mechanical schedule, three that required stable load, and a neonatal unit that could survive reduced gravity but not repeated changes. The account widened again. No single field could represent hospital dependence without losing the distinction between lightness and variation.

Keir moved a chalk line while they spoke. “Local cost includes the water we cannot use while tanks sit on outer rails. It also includes access. Two residential decks have only one open corridor during transfer.”

Tavin asked, “What would another tug change?”

“We could bring replacement bearing segments before the race crosses amber. It would also let us move ballast without closing both corridors.”

“Fuel requirement?”

Keir named a mass and route. Tavin compared it with *Morrowline*'s reserve. The voyage could be flown, but not alongside the planned Nacre flywheel transfer. One ship could reduce two risks at different times, never both at once.

Iona linked the tradeoff across tables. Republic survival gained a narrower exposure branch if the tug went to the bearing. Local physical cost then gained a delayed flywheel transfer elsewhere on the spindle. Olun loss changed only if either task used hidden receiver support.

Aru sent a brief pressure pattern through the cradle.

This location begins naming destination of force.

Keir heard the translation. “Does that count for anything?”

“It changes what can be negotiated,” Iona said. “It does not cancel prior harm.”

“Good. Cancellation sounds like a Ministry service.”

Bex accepted the line without visible reaction.

When the Nacre test ended, Varo kept the three account fields on her local board. She would update them after the bearing inspection. The exercise had not predicted who died. It had named what crews could observe, which action followed each threshold, and where every action moved cost.

Tavin opened *Morrowline*'s fuel record. The convoy had burned reserve propellant on the sunward flight, vented water to create shelter volume, damaged three thruster collars, and consumed service parts intended for a year. Those costs seemed small beside Meridian until Tavin connected each one to future work the ship could

no longer perform.

“Reserve fuel had destinations before we spent it,” he said. “A hospital spindle lift at Orison. Flywheel transfer for Nacre’s south bearing. Two ice hauls to Stillness. We saved people at Meridian by borrowing capacity from those jobs.”

“Can they be reassigned?” Iona asked.

“Some. The spindle lift has a six-week window. Miss it and they brake the theatre mechanically. The Nacre flywheel cannot wait that long.”

She recorded schedules rather than assigning monetary value. Local physical cost included fuel mass, tug hours, removed ballast, reduced habitat volume, food cycles, medical risk during relocation, and structures deliberately abandoned. It also included who performed the work.

Pella arrived from engineering with her left hand sealed in a regenerative brace. Two empty channels ended below the first knuckles. She placed a tool authorization on the table.

“What category holds fingers?” she asked.

“None yet,” Iona said.

“Then your account is short.”

Iona added injury, rehabilitation, and lost skilled capacity.

“Do not call the last one lost,” Pella said. “I can still supervise people with inferior hands.”

Tavin looked at her brace. “Your generosity restores morale.”

“I require a new cutter grip and six weeks before pressure work. Put that in. Also put the assistant who ran my engine room while Sive had me tied down. He did not become free labor because my name occupies the casualty field.”

Iona entered both.

Pella tapped the Meridian board beneath the headings. “Count replacements by object, not price. A receiver coil can be fabricated. Nursery soil takes years. A pressure door takes weeks. Someone’s left shoe remains a shoe, but they may need it tonight.”

After she left, Bex said, “If you enumerate every consequence, negotiation never begins.”

“It begins when each side can see what an offer moves.”

“That standard exceeds any treaty the Republic has written.”

“The Republic has never negotiated with the destination column.”

Aru’s cradle activated at the second table. Fluid moved through its clear branches in pulses that made the cargo deck seem briefly steeper. The translated voice came from no speaker’s location.

You prepare count of ended Olun.

“I prepare an account of loss,” Iona said. “I cannot count ended individuals under a form of identity that

changes by coalition.”

Correct refusal. Incomplete refusal. Ended structure has measures.

The display filled with pressure intervals. Aru distinguished a temporary speaker, a persistent memory reef, a seasonal lineage, and a possible coalition prevented from forming. Human casualty categories fit none. The Olun also lacked a universal unit. Different ice worlds preserved identity through different pressure regimes. A fracture that erased a shallow speaker might leave its component histories elsewhere. A thermal shift could preserve structure while making it unreachable.

Nemi asked, “What does absence mean in your archive?”

Aru returned three patterns before translating.

Expected answer fails across known call. Expected structure fails across mapped stratum. Expected successor fails after formation interval. Each absence differs.

“Can absence prove death?”

No. Presence later can disprove some endings. Continued absence increases boundary, not certainty.

Iona wrote the phrase under OLUN LOSS: boundary of possible survival.

Aru struck the cradle with a sharp pulse.

Do not make our unknown smaller by naming boundary.

She deleted it.

“What would you preserve?”

The response came as a pressure sequence that the translation software first rendered room without return. Aru corrected it twice.

Mapped expectation, failed continuity, search extent, confidence. Keep four.

Iona built four fields. Every estimate would show what pattern had been expected, what continuity failed, how far witnesses had searched, and how much confidence survived. Confirmed interruption occupied one range. No later response after repeated calls occupied another. Destroyed memory reef with fragments preserved elsewhere required a third description. Possible lineages prevented from forming remained possibilities, not retroactive people and not zero.

The work slowed because each translated term carried physical history. Nemi proposed lineage-equivalent as a unit. Aru rejected it; coalitions varied too widely. Bex asked whether integrated archive volume could provide a lower bound. Aru replied that deep histories stored more pressure duration than shallow ones without containing more persons. Tavin suggested event classes instead of totals.

That held.

They arranged Olun losses by event: routine clearing, repeated clearing across one formation interval, emergency export, memory-reef fracture, returned pulse, and habitat-scale disruption. For each event, Aru supplied surviving witnesses and explicit uncertainty. Iona linked the human sink ledger to the pressure archive without claiming either could complete the other.

The first defensible lower statement took an hour: human debt pulses had interrupted named Olun continuities across multiple Eidos bodies. The evidence proved more than damage to inert geology. It did not yield a population count.

Bex read it. "The public will call this evasive."

"The public has been given exact lies," Nemi said. "Uncertainty may look evasive because it has never received an official typeface."

"What can the hard Olun faction negotiate without a total?" Bex asked.

Aru's fluid accelerated.

Cessation does not require count. Restitution requires shape of wound. Immediate return faction uses completed human load as measure. It does not require our dead to become numbers.

Iona understood. The hard faction treated accumulated exported disturbance as a physical debt to return, independent of casualty accounting. Human negotiators kept asking for a death total because they wanted payment to resemble damages in a court. The Olun demand began with force already delivered.

"Then our accounts need a fourth link," she said. "Not a fourth account. A trace between exported load and each documented impact class."

She drew it without conversion. A clearing quantity could connect to a known pressure fracture. It could also connect to failed continuity with wider uncertainty. The same clearing had enabled a local hospital adjustment or cargo orbit. Physical cause linked the sides even when moral units did not.

By the end of the first work period, the cargo tables had filled with records. Nothing balanced. Republic survival showed exposure windows and local failure modes. Olun loss showed documented interruptions, mapped absence, and limits on inference. Local physical cost showed the fuel, capacity, relocation, injury, and abandonment required to stop exporting harm.

Iona ordered no summary total.

A public-policy aide from Bex's office requested one anyway. "Negotiators need to compare offers," she said over channel. "If we cannot say whether a staged plan saves more than it costs, the Assembly cannot authorize it."

"More of what?" Iona asked.

"Lives."

"Whose lives, measured how?"

"All affected persons."

"A possible Olun lineage has no direct conversion to a human evacuation death. A lost growing season does not equal Pella's fingers. Refusing conversion does not refuse comparison. It requires the negotiators to state which harm they accept and why."

The aide looked toward Bex. He said, "Keep the accounts separate."

It marked his first public refusal of his own preferred model.

The second work period tested an offer. Immediate total receiver shutdown entered the Republic table. Within hours, certain habitat classes lost seal stability and orbital control. The casualty range widened rapidly because evacuation capacity fell below exposed population. Olun exported load ceased, but abrupt local corrections risked new pulses through uncontrolled channels. The local-cost table showed insufficient tugs, fuel, and shelter.

A continuation offer entered next. Republic disruption narrowed. Olun impact classes remained active. Local expenditure stayed deceptively low because Eidos absorbed it. Aru rejected the offer before translation completed.

Finally they entered staged localization by system. Receivers would reduce on public schedules. Tugs and ballast crews would take local force. Marginal installations would evacuate where physical support could not continue. No secret route could transfer an overrun. Each stage carried a range for human exposure, a linked reduction in Olun impact, and a list of concrete local costs.

The plan did not yet choose which installations to abandon. That decision belonged to negotiation and public authority. The accounts made it impossible to describe abandonment as a rounding error.

Aru reviewed the structure through three long pressure sequences.

Survival account does not erase wound. Wound account does not command impossible motion. Motion account shows whether promise has body.

Nemi saved the translation, then saved the raw sequence beside it.

“Will your chorus receive this?” Iona asked.

Several receive. Several refuse translated form. Hard return faction will test for concealed path.

“They should.”

Testing may damage.

“Then specify the test before we agree.”

The cradle held a low gradient. Iona had learned that such pauses could contain disagreement rather than silence.

This request has negotiation shape, Aru said.

They prepared two packets. The human packet went to every Assembly system, receiver crew council, convoy captain, and public archive that could carry it. The Olun packet used dark-lattice modulation through the cradle. It contained fewer human words and more linked pressure traces. Both included Sella’s note, Meridian’s incomplete evacuation register, and the unaltered uncertainty fields.

Bex asked to sign as Minister of Continuity.

Iona said, “Sign the Republic survival inputs you supplied. Not the whole.”

Nemi signed the sink-record extraction. Tavin signed fuel and vessel capacity. Pella added her tool authorization and rehabilitation estimate. Eren Vass signed Meridian custody. Aru marked the Olun structure with a pressure sequence no human credential could imitate.

Iona signed no office. She identified herself as compiler and listed every source of authority separately.

The packets left *Morrowline* together. One traveled by Republic channels toward forty-three systems. The other entered the same lattice that had carried hidden force for 112 years, now bearing a proposal with no force attached.

Minutes later, Assembly procedure accepted the human packet for emergency negotiation. The cradle answered with three distinct Olun acknowledgments and two refusals. None accepted payment. None returned a pulse.

On the cargo tables, the three accounts remained physically separate. Between them, Iona laid a blank negotiation sheet and fixed its corners beneath Pella's damaged cutter grip.

Chapter 19 — The Planetary Strike

The forward yoke missed its capture mark by eleven centimeters.

Tavin watched the stripped flywheel housing drift past the green brackets on his display. Twelve thousand tonnes of alloy moved without hurry, close enough to Nacre Receiver Nine that one bad impulse could buckle a service collar. The *Morrowline* held the housing on four cables while the station turned beneath it. Three tug pilots waited for his correction. None of them had used curvature compensation since the receiver crews walked out.

“Port cluster, two percent for six seconds,” he said. “Yoke Three, pay out half a meter. Do not chase the mark.”

Pella Jor answered from engineering. “Port cluster is yours. It would like you to know it resents restraint.”

“Tell it resentment saves fuel.”

“It says captains always discover thrift after engineers lose fingers.”

Her left glove had two empty stalls pinned flat against the palm. Pella could still run six thruster clusters by touch, though every hard burn pulled her healing hand against its brace. Tavin had ordered her off the deck twice since Meridian. She had treated both orders as requests for technical review.

The housing slowed. Cable loads crossed, settled, then balanced within the old Navy tolerance Tavin still trusted more than any Republic certification. He brought the mass over the capture brackets and let the station's mechanical clamps close around it.

Receiver Nine shuddered when the clamps took weight. Across its outer hull, a row of work lamps flashed twice. Strike signal. No hidden export. No unlogged sink. No force without a named destination.

Haru Den looked up from navigation. “Security cutter on an intercept from the inner cordon. Eight minutes.”

“Did they answer our berth notice?”

“They answered with a seizure warrant.”

Tavin released the yoke cables. “Same bureaucratic instinct, sharper typography.”

The warrant filled his command pane. Under emergency authority, the Continuity Ministry claimed the *Morrowline*, its remaining fuel, every detachable yoke, and the housing they had just delivered. The stated purpose involved preservation of critical receiver infrastructure. The attached operations order directed Receiver Nine to resume clearing through Eidos before the next orbital correction window.

Below the legal block sat Bex's engineering signature.

Tavin read it once. He had learned aboard Navy tugs that an order could contain several different things: a desired result, an assigned risk, an unstated casualty list, and a name placed at the bottom to make all three look singular. The warrant wanted the housing because its mass could stabilize the receiver station without exported curvature. It wanted the ship because the Ministry lacked enough conventional lift to move that mass. It wanted the fuel because every honest option consumed something visible.

Sive An secured a medical cabinet behind the bridge. “If they board, do I list our wounded as cargo or

obstruction?”

“Crew,” Tavin said.

“That category has lost legal precision.”

Receiver Nine's dockmaster opened a local channel. Lio San appeared with the washed complexion of someone who had slept under work lights. A strike band crossed his coverall in white paint. Behind him, mechanics were bolting braces over a load manifold instead of operating its clearing valves.

“We saw the warrant,” Lio said. “The cutter told us to uncouple your yokes.”

“Can the housing carry you through the correction window?”

“If we spin it up. We need forty minutes and seven tonnes of maneuvering propellant.”

“You have twenty-eight tonnes in the dock reserve.”

“Ministry lockout took the pumps.”

Tavin examined the station schematic. Receiver Nine served six spindle neighborhoods, two hospitals, and a water refinery. With its workers refusing Eidos assignments, local load would drift outside tolerance in three hours. The housing could absorb enough angular momentum to hold the service collar level, but only if the station moved it along a rail and spent fuel bringing it to speed.

“Can you cut a manual feed from our aft tank?” Lio asked.

“Not before the cutter arrives.”

Pella shifted the engineering plot onto Tavin's pane. A maintenance purge line ran from the *Morrowline*'s auxiliary tank to the yoke manifold. Its couplings carried coolant in ordinary service. They could pass propellant if someone replaced two seals and accepted a pressure loss.

“I can give them nine tonnes,” Pella said. “Eight reaches their tank. One becomes a lesson about using the wrong pipe.”

“Time?”

“Sixteen minutes if the dock sends hands.”

The security cutter reached weapons range in six.

Tavin keyed the cutter. Its captain appeared without a uniform jacket, a concession to long emergency shifts. Commander Vey Orra had served two years below Tavin in the Republic Navy and once flown a tether rescue through a dust ring with three dead attitude jets. Her face carried no pleasure at finding him on the other side of an order.

“Captain Serr, stand down your drives,” she said. “Prepare for Ministry seizure.”

“We're attached to a civilian receiver.”

“You have a detachable manifold. Detach it.”

“The receiver will cross load limits before your prize crew can move the housing.”

"I have engineers aboard."

"Do they know the strike plant?"

A pause answered him.

Orra reduced her voice. "Bex says hidden clearing for one more cycle prevents four habitat evacuations."

"Workers in twelve systems received the same assurance last cycle."

"Those systems still have gravity."

"Meridian had gravity until someone used Eidos as a private answer."

The cutter's range marker turned amber. Orra could disable the *Morrowline*'s maneuvering jets without breaching the hull. Her gunners knew exactly where to place the shot. Tavin also knew their capacitor recovery time, because he had approved the cutter class while still in service.

He sent her the local mass model. "If you fire, our cables slack. The housing rolls six meters along the collar. Your seizure gets a station fracture."

"You released the cables."

"Four magnetic shoes remain engaged."

That claim was almost true. Two shoes held. A third reported intermittent contact, and the fourth existed only as a warm icon Pella had painted onto the diagnostic layer.

Orra studied the model. "You are using inhabited infrastructure as cover."

"I am finishing a delivery."

"To strikers."

"To people keeping six neighborhoods from drifting while they refuse an unauthorized sink."

"Your distinction may matter at trial."

"It matters now."

Tavin closed the channel and ordered Pella to rig the purge line. Receiver mechanics came across the service bridge carrying replacement seals, a hand pump, and a coil of cable. No one gave a speech. One mechanic locked boots to the yoke, another stripped insulation from a valve actuator, while a third painted a white strike bar on the borrowed propellant line.

The cutter held at five kilometers. Its targeting beam touched the *Morrowline*'s stern and remained there.

For sixteen minutes, Tavin had to keep twelve thousand tonnes apparently dependent on his ship. Haru pulsed the attitude jets in uneven pairs, making the housing's residual movement look less stable than it had become. Pella vented a thread of coolant past the failing shoe to produce a convincing thermal bloom. Sive moved the crew away from the likely boarding locks, then strapped herself beside the casualty rack.

A pressure alarm sounded in engineering.

“Seal Two is biting sideways,” Pella said. “Flow at thirty percent.”

“Can you hold it?”

“With my hand, yes. With optimism, no.”

Tavin checked the cutter. A launch bay opened along its belly.

“Boarding skiff away,” Haru said. “Ninety seconds.”

Lio's mechanics forced the second seal into its collar. Propellant began crossing the improvised line. Eight tonnes would not arrive in time. Four might. Receiver Nine needed seven.

Tavin opened every local work channel the station had given him. “Dock reserve pumps are Ministry locked. Who has an independent transfer cart?”

Answers arrived from places not listed in the operations plan. A water tender carried half a tonne for emergency attitude control. Two garbage scows had another tonne between them. A funeral carrier offered its entire landing reserve, then withdrew the offer when Sive pointed out that it still had to land. A school ferry held three hundred kilograms and a pilot unwilling to surrender any while children remained aboard.

“Keep yours,” Tavin told the ferry. “Everyone else, match Receiver Nine's approach code. Lio will sequence you.”

The first garbage scow detached from the refinery spoke. Its drive plume flickered yellow from poor mixture, but it made the crossing. The water tender followed under manual guidance. On the traffic plot, six unrelated craft altered course toward one transfer point while Republic control ordered them back to assigned lanes.

Orra called again. “Disperse those vessels.”

“They heard the same warrant.”

“You are creating a collision field.”

“I sent you the approach sequence.”

“You sent it after they moved.”

“They moved before command found a reason to let them.”

The boarding skiff stopped outside the yoke's hazard zone. Its pilot saw the mass housing, the crossing utility craft, and a propellant hose never rated for its current use. Boarding would require cutting through working people while the cutter recorded every angle.

By the twentieth minute, Receiver Nine held 6.2 tonnes. The improvised line shook against its cable restraints. Pella reduced flow before Tavin could order it, saving the second seal from tearing free.

“We can give them six-point-eight,” she said. “After that, we keep enough for one hard departure or two polite ones.”

“Give six-point-eight.”

“We may need the hard departure.”

“We already spent the polite one.”

The station spun up the housing. Its rail motors drew power from three neighborhoods at once, darkening every fourth corridor in a scheduled wave. Hospital loads remained protected. The water refinery stopped for nine minutes. Residents shifted into warm rooms and tied cabinets shut as felt gravity changed by hundredths.

Tavin watched cost become motion on the board. Local payment lacked the clean disappearance of a clearing. It arrived as empty tanks, delayed water, cold corridors, and a flywheel housing dragged into service years before its bearings were ready. No distant pressure archive had to absorb any of it.

At 6.6 tonnes transferred, a priority packet broke across the work channels. Not from Ministry control. The sender used a receiver-maintenance key from Borun System, twenty-one light-years beyond any ordinary conversation with Nacre. Its timestamp preceded Meridian's fall by years. The workers there had created a dormant strike protocol after the first structured replies disappeared from public maintenance records. Iona's raw-data release had supplied the trigger they lacked.

The packet contained no current voice. It carried procedures: refuse unconsented sink assignments, preserve hospital and nursery loads, publish local fuel margins, and coordinate mass transfers without Continuity routing. Its checksum had already been countersigned by receiver crews at Kestrel Reach, Lumen, Sere, and seven Nacre habitats.

Haru leaned closer. “This has been waiting in maintenance caches.”

“For a truth no one wanted to name,” Tavin said.

A second packet surfaced from Lumen. Helix Quay workers had adapted the protocol after the repossession. A third came from Stillness, offering ballast tables and conventional spin schedules. Old messages, local decisions, delayed proofs. The strike had not crossed interstellar distance in four days. It had grown from buried suspicion in many places and recognized itself when the concealed records opened.

Another Nacre site broke into the channel before the pattern finished resolving. Refinery Spindle Four had begun a receiver slowdown with twelve thousand workers aboard. Its ore trains ran on a central load rail. Once local field support dropped below seventy percent, loaded cars would press unevenly against the ring and push its center of mass off the bearing line.

“We need nine thousand tonnes sunward within fifty minutes,” said the refinery coordinator. “Continuity has two mass barges locked at our outer berth. Their captains will not release without a routing code.”

Tavin pulled the refinery model beside Receiver Nine's plot. The *Morrowline* could move the requested mass, but not while attached here. Even after release, its remaining propellant allowed one transfer. A second would leave them unable to brake at a safe port. He marked those limits on the shared table.

“Do not wait for us,” he said. “What can move inside your ring?”

“Three ore trains. Two are loaded.”

“Where?”

“Outer quadrant, coming spinward.”

“Reverse one loaded train. Stop the other at quarter circumference. Empty cars go sunward at maximum safe speed.”

The coordinator hesitated. "That closes all smelter feed."

"You asked for mass, not production."

"We lose twelve hours of heat."

"Put twelve hours in the strike account. Is the centerline responding?"

Numbers replaced the coordinator's face. Haru entered the train movements and watched the projected bearing error shrink. Not enough, but the refinery gained twenty-three minutes. A fuel skimmer near the radiation shades offered a slab of frozen reaction mass waiting for inspection. Two dock tugs could push it toward Spindle Four if the Ministry released its navigation hold.

"Assume no release," Tavin said. "Can the tugs carry independent ranging?"

"One can," Haru replied. "The other follows optical marks."

Tavin drew a route around the security cordon. It passed close to a shade truss and demanded a braking turn the slab had never been designed to survive. He reduced approach speed, shifted the crossing six degrees north, and sent the path to both tug pilots. They accepted along with the published chance of losing their vehicles against the outer bumper.

The shared table changed as they worked. Spindle Four's need fell from nine thousand tonnes to six. The fuel skimmer supplied four. A construction yard offered two pressure hulls whose launch would cancel a hospital expansion. Its workers attached that consequence before the refinery accepted.

Sive read the entry over Tavin's shoulder. "They are trading a ward for a bearing line."

"They are deciding in sight of both."

"That will not make the missing ward hurt less."

"No."

At Receiver Nine, the water refinery came back online early and destabilized the collar by a fraction. Lio cursed, shut it again, then published the delay under his own name. The hospital reduced surgical load for eleven minutes. Residents opened two public halls for anyone whose medical supports could not tolerate the variation. Every adjustment consumed labor, patience, or reserve. Nothing vanished into a dark destination field.

Orra's cutter continued holding the targeting beam on the *Morrowline*. Tavin forwarded the Spindle Four route to her, including the mass slab's passage near her cordon.

"You are asking a security vessel to yield to stolen fuel," she said.

"I am informing you that two civilian tugs will cross your exclusion zone."

"You could order them to stop."

"They do not answer to me."

"Everyone on this channel acts when you speak."

“Then listen closely. Do not fire on them.”

The first tug emerged from the shade truss with its running lights at full brilliance. A white slab followed on six cable fans, turning so slowly that its facets changed color one at a time. The second tug held fifty meters behind, ranging by optical flashes. Orra moved the cutter nine hundred meters off its intercept line. Her boarding skiff remained near the *Morrowline*, but the corridor opened.

Spindle Four's coordinator marked the construction hulls accepted. The hospital expansion disappeared from the yard schedule. In return, twelve thousand refinery workers gained a stable bearing path. The exchange stayed visible on every local channel.

Receiver Nine reached its seven-tonne minimum.

Lio cut the improvised line. Pella sealed the *Morrowline*'s auxiliary tank with less than four percent reserve above departure floor. The station's new mass wheel rose to operating speed, pulling its collar back toward tolerance. One by one, the magnetic shoes disengaged.

Their bluff disappeared with the last green icon.

Orra knew it. “You have completed the delivery. Stand down for seizure.”

Tavin looked at his crew. Haru had already laid two departure solutions across the plot. Sive had secured every wounded worker who had come aboard after Meridian. Pella held the manual isolation lever with her three good fingers and her braced palm.

A direct burn would escape the boarding skiff but not the cutter. Fighting would turn the strike into an attack and give Bex authority to treat every receiver crew as a hostile cell. Surrender would hand the only heavy conventional tug near Nacre to officers ordered to resume hidden export.

Then the water tender moved between the cutter and the *Morrowline*.

Its pilot transmitted an engine-failure code so implausible that Haru laughed once. The first garbage scow followed, tumbling just enough to demand collision avoidance. Dock tugs released from Receiver Nine. A hospital shuttle drifted out of its protected lane and reported a navigation fault. None threatened the cutter. Together they filled its safe firing geometry with civilian hulls.

Orra's image tightened. “Move them, Tavin.”

“They are not mine.”

“You called them.”

“I asked who had fuel. They chose what to do with the answer.”

Republic control flooded the channel with return orders. Local pilots acknowledged, requested clarification, reported sensor disagreement, and continued occupying every line the cutter needed. Their obstruction remained nonviolent and operational. It bought distance in seconds and propellant in kilograms.

Tavin sent the cached strike protocol to all nearby craft. He added the *Morrowline*'s mass capacity, remaining fuel, yoke status, and departure limits. No promise. No command. Just an honest account others could use.

Lio countersigned from Receiver Nine. The water tender added its reserve. Stillness attached ballast schedules.

Helix Quay contributed a list of damaged but working conventional tugs. Across the local net, receiver crews replaced Continuity routing codes with a shared table of needs and capacities.

“Boarding skiff is withdrawing,” Haru said.

The cutter backed away from the civilian screen. Orra had lost a clean shot and found no lawful way to make one. Her last transmission came privately.

“You cannot supply a planetary strike with one freighter.”

“No,” Tavin said. “But one freighter can publish where it will be.”

He ordered minimum conventional thrust. The *Morrowline* eased from Receiver Nine behind a disorderly escort of utility craft. No Republic controller cleared their departure.

On Tavin's board, the official route layer went gray. A new one opened beneath it, drawn from worker caches and civilian fuel declarations. Twelve striking sites marked their needs. Fourteen ships marked what they could carry. Lines appeared between them without Ministry assignment, each transfer naming its cost before anyone accepted it.

Haru selected the first worker route. The ship turned toward a refinery spindle whose receiver crew had thirty-one minutes of safe drift remaining. Behind them, Receiver Nine held its new balance without sending weight to Eidos, and the crews began directing traffic through a command net they had built themselves.

Chapter 20 — The Weight of Mercy

The translation cradle drove the water in Iona's glass uphill.

A thin meniscus climbed one side, held against the rim, then dropped hard enough to strike her fingers through the table. Around the negotiation chamber, restraint straps pulled taut over twelve occupied chairs. The deck had not changed its spin. The pressure had arrived through the dark-lattice coupling, a small controlled acceleration carrying the form of an Olun demand.

Nemi bent over the cradle's live ratios. Vestibular braces enclosed both sides of her neck, steadying her head while her damaged inner ear argued with the floor. "That sequence repeats the full-return marker," she said. "Not Aru's marker. The deeper one from the Eidos archive."

The chamber's receiver had been isolated from every public load. It could accept modulation but not send curvature debt. Iona had checked the physical shunt herself. Bex's engineers had checked it afterward, then placed guards beside the manual switch as if truth depended on uniforms.

Another pressure gradient crossed the cradle. The water rose in three separate lobes.

Nemi translated from the ratios. "All open wounds close. All held disturbances come home. Interval begins now."

"How long?" Iona asked.

"Not an interval we can distinguish from immediate."

Across the table, Elian Bex kept one hand on his old engineering key. His emergency-authority band projected a red seal over the work surface. "Then no plan we present will satisfy them. Localization takes years."

"You told Meridian that and exported in secret," Iona said.

"I told Meridian the staged correction would fail without a pressure margin. I created one."

"You sent the missing load to Eidos."

"I moved it away from a nursery ring."

"It came back through the ring."

The exchange landed without heat. Meridian remained beyond anyone's tone. Images from the wreckage still entered the public ledger by the hour: hull numbers, recovered bodies, rings whose spin no tug could restore. Sella's maintenance note sat in Iona's private evidence vault and in twelve public copies. No minister could put operator error over her name again.

A slower modulation touched the cradle. Instead of lifting the water, it pressed a shallow hollow into the surface. Nemi aligned the pattern with Aru's earlier grammar.

"Speaker coherence requested," she said. "Aru-in-Pressure is assembling."

Bex looked toward the empty apparatus as though expecting someone to enter. The cradle held three concentric tanks, strain plates, a field coupler, and a screen translating pressure ratios into human words. No face formed there. No body waited beyond the door. Aru existed in the temporary agreement of Olun histories under Eidos's

ice, made audible through forces the chamber could measure.

The central tank contracted against its mounts.

WE CONTAIN MULTIPLE RETURNS. SOME HAVE NO PATIENCE SHAPED FOR YOUR SURVIVAL.

Iona placed her three accounts on the public pane. Republic dependence occupied the first: seventeen billion lives inside structures that could not abandon clearing at once. Olun harm occupied the second, expressed in destroyed memory reefs, broken pressure lineages, and defensible ranges where absence prevented counts. Local cost occupied the third: fuel, spin changes, conventional thrust, migrations, failed harvests, and habitats made smaller.

“Those are the accounts we can support,” she said. “None cancels another.”

Pressure entered as a hard downward pulse. Nemi's display divided it among at least nine source patterns. Three demanded total correction. Two sought immediate closure without return. Others carried ratios that did not map cleanly onto either position.

Aru's translated line appeared after them.

YOUR THREE ACCOUNTS CREATE DISTINCTIONS WITHIN A WOUND.

“Yes.”

THE DEEP DEMAND NAMES ONE DISTINCTION. OPEN OR CLOSED.

“That distinction kills people who did not choose the export.”

The water in the outer tank trembled. Several Olun patterns struck at once, producing interference the software could not separate. One pressure-name associated human continuance with an active mechanism of injury. Another marked children, dissenters, and striking workers as distinct pressure histories. A third refused any human category because every receiver still held an open channel.

Bex leaned toward his microphone. “The Republic can reduce scheduled exports by forty percent over six months. We can reserve Eidos channels for emergency continuity only.”

Iona cut his authorization from the cradle before Nemi translated.

His red seal flared. “Restore my channel.”

“No.”

“You do not hold an office.”

“Neither do I hold your emergency debt.”

“I am offering a reduction they may accept.”

“You are offering the same concealed exception under a smaller number.”

Bex's hand tightened around the key. “A habitat whose bearing fails during evacuation will not care whether its final clearing met your standard of consent.”

“It may care where the force goes.”

“Dead people care poorly.”

“Then name them before you spend someone else.”

Nemi kept both channels closed while the argument crossed the public record. Outside the chamber, witness panes carried every word to a population learning how often continuity had meant that decisions arrived after consequences. Iona could see no crowd, only the rising count of connected civic relays.

She reopened her own line. “Aru, Bex proposes hidden emergency export. I reject it. I have another plan, but it closes places humans live.”

The cradle held still for four seconds.

CLOSURE OF CHANNEL OR CLOSURE OF LIVES?

“Both can happen. Not at the same moment.”

Iona called up the localization schedule she and Nemi had built from the three accounts. It contained no smooth Republic curve. The line broke at every habitat whose physical reserves ran out before mass and fuel could reach it. Most could shrink, despin by stages, or transfer their debt into visible local work. Three could not.

The first installation, Oriel Survey Arc, circled a small moon in Lumen with four thousand permanent residents and a dark-matter observatory that required a quiet orbit. Its receiver corrected a natural precession every nineteen days. Conventional tugs could hold the arc or evacuate it, not both.

The second, Sere Cloud Foundry, supported ninety-one thousand residents in aerostat chains above a gas giant. Curvature clearing kept the chains inside a narrow atmospheric band. Their fusion fleet could lift every person and half the seed archives if production stopped immediately.

The third, Kestrel Nursery Annex, had been built beyond its parent spindle because the inner ring lacked room. Its receiver carried too much of the annex's stationkeeping. Twenty-three thousand people lived there, including six thousand children. A mass tow could save the pressure hulls after evacuation, but using it would consume fuel assigned to stabilize three larger neighborhoods.

Bex read the list. “You selected marginal sites.”

“I selected sites whose evacuation costs fewer lives than sustaining them through hidden export.”

“You selected people with weak Assembly blocs.”

“The physical criteria are public.”

“Physics arrives through budgets. Oriel has requested conventional tugs for eleven years.”

“I know.”

“Then do not call this clean.”

“I called it public.”

The distinction did not defend anyone. Oriel's residents would lose a home because the Republic had preferred receivers to tug allocations. Sere families would leave cloud platforms their grandparents assembled. Children at Kestrel would travel through a frightened parent spindle already reducing gravity and food volume. No

wording could turn evacuation into moral balance.

Iona opened the live planning rooms for all three installations. Their local coordinators appeared beside the schedule, not as decorative witnesses but as holders of vetoes over method.

Oriel's director spoke first. "Your first departure wave removes observatory staff before their families."

"Correct it," Iona said.

"We need the staff to safe the instruments."

"Then family groups travel in the same wave."

"That cuts specialist berths by two hundred."

"Where do those two hundred go?"

"A cargo drum if we clear its mirrors."

"Time to clear?"

"Six hours."

Nemi shifted Oriel's departure by one work interval. The projected receiver debt rose. A conventional tug assigned to a hospital spindle could absorb the difference if it delayed its own burn by nine minutes. That hospital's representative entered the room, checked the load model, and consented after securing a reserve thruster.

Sere's coordinator refused the first schedule entirely. "You have our seed archives staying behind."

"Your lift model gives half," Iona said.

"Your model treats memory culture as freight. Each aerostat chain carries microbes adapted to its pressure band. Lose the other half, and no descendant can rebuild our work."

"What do you remove instead?"

"Foundry platinum."

Bex interrupted. "The platinum funds three evacuation fleets."

"Then those fleets carry our archives as shared cargo," the Sere coordinator said. "Or they do not receive our metal."

Iona moved the argument into the cost pane. Three fleet quartermasters joined. They cut tool allotments, reduced private luggage, and converted two fuel tankers for refrigerated samples. The change saved the microbial archives while sacrificing enough industrial stock to delay later localization work. That delay appeared in red against two systems.

Kestrel's room had no director. The annex council had suspended itself after residents accused it of concealing receiver margins. A nurse, a spin technician, two parents, and a cargo clerk held the local channel instead.

"You put children first," the nurse said. "That leaves maintenance crews until the last day."

“Someone must keep pressure and docking stable,” Bex replied.

The spin technician looked directly into his recorder. “You mean someone must be left where correction reaches first.”

Iona moved the final maintenance team onto the last tug, reducing its safe passenger load. “What work remains after they board?”

“Two hours of automated pressure control,” the technician said. “If the transfer latches release cleanly.”

“And if they do not?”

“Then we lose the Annex before the tug clears.”

No alternate crew could solve that. The risk belonged to machinery and timing. Iona marked it in the public schedule with the names of those who had accepted it. Three maintenance workers withdrew consent. The departure model changed around them. Four volunteers with current certifications stepped forward through the Kestrel channel, each after speaking privately with family.

Before their names settled, Oriel reported a mechanical refusal. The cargo drum's inner mirror rack had fused to its guide rail after decades without vacuum service. Workers could cut the rack free, but fragments might strike the drum wall during thrust. Leaving it in place cost two hundred family berths. The tug assigned from the hospital spindle had already begun its approach and could not wait through a full repair.

Iona opened a joint operations pane. “What mass can Sere remove from its first tanker?”

The cloud foundry coordinator checked. “Thirty tonnes if the platinum shielding stays behind.”

“Oriel, does thirty tonnes buy the family berths?”

“It buys eighty-six,” the director said. “We still need the mirror rack out.”

Kestrel's cargo clerk interrupted. “Our parent spindle has an empty grain liner coming back from the Annex. It can cross to Oriel after unloading, if someone gives it braking fuel.”

Bex called up the distance. “That adds two transfers and spends reserve intended for your final maintenance tug.”

“We have six thousand children aboard the Annex,” the clerk said. “Do not use them to speak for us. I offered a liner, not their fuel.”

The room quieted. Iona removed Kestrel's reserve from the solution. “Who can brake the liner?”

A listener count flickered as public relays carried the question. The answer came from a refinery barge attached to the strike network. Its pilot offered a rendezvous burn in exchange for priority on Sere's discarded platinum shields, which his crew could use as thermal mass during receiver shutdown.

Sere's workers reviewed the request. Their coordinator asked for half the recovered shields to remain available for future archive tanks. The barge pilot accepted. None of the parties shared a command structure. The agreement formed through declared needs, published margins, and the ability of each group to refuse.

Nemi added the transfer legs. Kestrel's liner would unload grain at the parent spindle, meet the refinery barge, collect braking propellant, then carry Oriel families toward Lumen's central habitat. Oriel still had to cut the

seized mirror rack into sections and bind every piece against the wall. The revised work consumed five hours and moved the observatory staff into the same departure wave as their families.

“Approve?” Iona asked each channel.

Kestrel's clerk consented after verifying the grain delivery. Sere accepted once the archive tanks retained shielding. The barge pilot signed the fuel margin. Oriel's director held for thirty seconds, speaking off-channel with the workers beside the fused rack.

“We accept,” she said. “If the rack tears the drum, the risk belongs to us.”

“It belongs in every route that depends on the drum,” Iona replied. “Publish the cut plan.”

Video from Oriel entered the pane. Suited workers fixed restraint webs across a chamber built to keep mirrors untouched. A plasma saw burned through the first guide. The rack moved three centimeters, stopped, then came free with a shower of frozen lubricant. Nothing punctured. The workers did not celebrate; five more guides remained.

At Sere, the first archive tanker failed its pressure test along a converted fuel seam. Launch crews lowered it back into the cloud dock and shifted the microbial cassettes to two smaller lifters. That change required another eighty workers to surrender seats in the opening wave. Their names moved into the second departure, where weather margins ran narrower.

Iona refused to let the interface hide the change inside a passenger total. Each displaced person received a notice, a route, and a channel for refusal. Forty-three objected. Eleven had medical grounds that the second wave could not support. The loading crew found room by leaving personal tool chests on the platform. The remaining objections stayed unresolved, visible beside the launch clock.

Kestrel's first child transport then reported a docking collar fault. The parent spindle could receive it only on a freight axis carrying twelve percent less felt gravity. Nurses reconfigured the restraint couches while the transport held position. Its pilot spent propellant with every minute, and the final maintenance tug's margin narrowed in the same public pane.

The three evacuations no longer resembled lines on Iona's original model. They had become coupled decisions among strangers, with every repair transferring time or capacity rather than making cost disappear.

Aru sent a pressure sequence during the planning. The cradle translated only after the local voices quieted.

THESE CLOSURES OCCUR BECAUSE WE REFUSE YOUR DISTURBANCE.

“They occur because we built lives around sending it,” Iona said.

IF WE WAIT, ADDITIONAL OLUN HISTORIES RECEIVE PRESSURE.

“Yes.”

IF WE RETURN, THESE HUMAN HISTORIES BREAK WITHOUT DEPARTURE.

“Yes.”

YOU REQUEST MERCY FROM THOSE ALREADY INJURED.

Iona looked at the three planning rooms. “I request time. Mercy describes what you may choose to do with that

request. It does not describe what we deserve.”

The ratios fractured.

Pressure struck the outer tank from opposing phases. Its mounts groaned. Nemi cut gain before the glass exceeded tolerance, but the disagreement continued in the input trace. Olun pressure-names attempted to separate Aru's component histories from the temporary speaker. One withdrew an ice-reef archive that had supplied Aru's grammar for human kinship. Another removed the pattern that distinguished deliberate harm from mechanism. Human words disappeared from the screen in blocks.

WE RISK UNMAKING THIS PRESSURE-NAME BY HOLDING YOUR DISTINCTIONS.

Nemi whispered, “Aru's coherence has dropped by eighteen percent.”

Bex studied the trace. “Can it be reconstructed?”

“Not as the same coalition.”

The answer altered him more than accusation had. He knew systems well enough to recognize a function that survived only through a particular arrangement. Aru could not retreat into a body because no body existed. If the chorus dissolved the agreement composing the speaker, its memories would remain distributed among Olun histories, but this voice and its commitments would end.

Iona asked, “What interval can you defend?”

The cradle answered with gradients too dense for immediate language. Nemi worked through them while the three installation rooms revised loading orders. Oriel crews began clearing the cargo drum before approval. At Sere, foundry workers shut down a platinum line and redirected coolant to archive tanks. Kestrel issued household mass limits and opened every private berth allocation to audit.

A translation finally assembled.

STAGED LOCALIZATION MUST BEGIN BEFORE SUSPENSION CAN BE CARRIED. EACH OPEN CHANNEL REQUIRES A NAMED CLOSURE. NO PRESSURE MAY LEAVE WITHOUT CONSENT FROM ITS DESTINATION.

“How long a suspension?” Iona asked.

NOT YET HELD.

Hard-faction patterns hammered behind the words. The full-return demand persisted, immediate and exact. Aru did not speak for all Olun. It held one contested proposal among factions whose injuries differed, whose patience differed, and whose desired outcomes could not be merged by translation.

Bex placed his engineering key on the table. “My office can certify commencement today.”

Iona read the offer for its missing column. “Can your office renounce emergency sink assignment?”

“Not while authority remains active.”

“Then your certification proves nothing.”

“It proves machinery moved.”

“Meridian's machinery moved.”

His gaze went to Sella's note in the evidence pane. “If I renounce emergency assignment, the strike becomes lawful by default. Every receiver chief can refuse my routing.”

“They already have.”

“Law still determines whether security seizes their equipment.”

“Then decide whether your ten days protect people or your control.”

Bex picked up the key. For one second Iona thought he would authorize another hidden path from the chamber. Instead he opened the Ministry routing table and exposed its pending assignments to the public pane. Seven emergency transfers waited for Eidos, each labeled as continuity reserve. Receiver workers had blocked five. Two remained technically executable.

He canceled neither.

“I will place them on a six-hour hold,” he said. “No longer without Assembly concurrence.”

“Six hours gives the installation fleets time to launch.”

“It gives your plan time to become real. After that, I choose against what fails.”

Iona wanted a confession or a clean conversion. Bex offered a timer. It carried less moral beauty and more physical use.

She accepted the six hours into the public record, then locked every schedule revision to named fuel, mass, berth, and consent. Oriel received its tug. Sere's first archive tanker separated from the cloud chain. Kestrel began moving children into the parent spindle, one docking group at a time rather than through a panicked general release.

The local cost account climbed as each action became irrevocable. Habitat capacity fell. Fuel reserves turned amber. Evacuation risks acquired names. Beside them, the projected debt pulses into Eidos dropped for the first time without a secret offset.

Nemi sent the changed curve through the cradle.

Aru's coalition carried it beneath the ice. The response returned ragged, missing several familiar ratios, but still coherent enough to translate.

WE WILL PRESENT YOUR SCHEDULE. WE DO NOT PROMISE ITS ACCEPTANCE.

“I understand.”

THREE HUMAN INSTALLATIONS CLOSE. OLUN PRESSURE CEASES TO RECEIVE THEIR MAINTENANCE. THIS COST DOES NOT REPAIR PRIOR IMPACTS.

“No.”

IT MARKS A CHANGE IN DIRECTION.

The hard-faction demand struck once more. This time Aru held its own modulation alongside it rather than

beneath it. The two patterns left separate traces on Nemi's screen.

At Oriel, the cargo drum opened and workers began removing mirrors from their mounts. Sere's platinum line cooled past restart temperature. Kestrel's first child transport cast off with half its seats removed for life-support equipment. Three installations had crossed from contingency into evacuation.

Iona signed the schedule under no office but her own name. Bex's six-hour hold began counting down beside it. In the cradle, Aru carried the costly proposal toward a divided chorus, and the Republic's first public plan advanced because homes were already being emptied.

Chapter 21 — The Debtors' Fleet

The first synchronizer tried to shut down every receiver in the dock.

Pella hit the physical breaker with her braced hand. The test rack went dark. Beyond the *Morrowline*'s open cargo bay, six civilian vessels continued turning around Nacre Service Ring C, their hull lights steady against the radiation shades.

“Useful ambition,” she said. “Poor judgment.”

Tavin released the breath he had held. The rack carried components seized from an Orbit Breaker cell, hardware designed to make several receiver sites sever their channels at one instant. In its original form, one trigger could desynchronize habitat rings before their mechanical systems took load. It could turn a political demand into falling machinery across a city.

He looked at the blackened board. “Why did it see the dock?”

“Because the Breakers used maintenance discovery codes. Anything listening for an authorized shutdown answered.” Pella pulled the top panel free with three fingers and her thumb. “They meant breadth. We need discrimination.”

“Can you give it vessel keys?”

“I can give it shame and a smaller antenna.”

A voice came from beneath the rack. “Try gratitude before mutilation.”

Rusk Vale slid out with a coil of fiber against his chest. He had helped build the seized hardware, then surrendered after Tavin stopped the nursery-cylinder attack. Security records called him an Orbit Breaker technician. The dock crews called him worse whenever a tool went missing.

“Your device just hailed thirty thousand residents with a shutdown command,” Tavin said.

“It found thirty thousand receivers. Residents were incidental to its search.”

“That distinction explains why you were arrested.”

“It also explains why I know how to fix it.”

Rusk plugged the fiber into a dead diagnostic port. His original code treated every receiver as an enemy actuator. Pella had spent two days stripping out the severance trigger while preserving the timing mesh. What remained could coordinate conventional thrust, mass rails, receiver ramp-down, and emergency aborts across many ships. If they made it safe.

Outside the bay, civilian craft waited in an improvised holding pattern. Two refinery tugs carried frozen reaction mass in external slings. A funeral carrier had replaced its ceremonial cabin with propellant bladders. Three habitat tenders brought flywheel tools and crews who knew how to move mass inside spinning structures. Farther out, cargo ships arrived from the strike routes without Republic clearance.

Haru called from the bridge. “Eleven more vessels on local approach. Four have not published braking reserve.”

“Hold them beyond the ring until they do.”

“One says its reserve is commercially protected.”

“Then its collision stays commercially distant.”

Tavin opened the dock roster. Forty-two ships had offered capacity. Twenty-nine could contribute to a staged localization. The rest had courage, damaged couplings, or no fuel beyond the trip home. He would not convert willingness into capability by naming them a fleet.

Nacre's receiver hub loomed sunward, a chain of field collars and mass reservoirs around the capital spindle. Its active channels still carried emergency sink assignments under Bex's authority. The city could not simply switch them off. Its inhabited sections relied on clearing for stationkeeping, bearing relief, and changes in felt gravity during shade maneuvers. A safe localization required moving millions of tonnes, burning fuel that supported intersystem trade, and reducing living volume while the spindle found a rougher equilibrium.

Keir's face appeared on Tavin's wrist pane. He transmitted from a spin-control pit inside Nacre, hair bound back and grease across one cheek. “We have the west ballast rail open. Three cars still answer manual control.”

“How many should answer?”

“Four.”

“Where is the fourth?”

“Stopped behind a residential wall someone built across the maintenance cut twenty-six years ago.”

Tavin closed his eyes for half a second. “Can you move the wall?”

“It contains forty apartments.”

“Then the car stays.”

“That costs us eight hundred thousand tonnes of internal travel.”

“We find it outside.”

Keir glanced away from the camera. “People here want to know who commands the ships.”

“No one yet.”

“They find that less reassuring than you do.”

“So do I.”

Tavin sent the failed synchronizer log to his son, along with the current vessel manifest. “Tell the spinwrights we test by local keys only. No receiver command leaves this bay.”

“Understood.” Keir's gaze returned. “And stop trying to move me to a ship.”

“I did not ask.”

“You put my name on a medical tender list.”

“Sive put your name there.”

“Sive says you gave her the list.”

Tavin ended the call before his son could see him lose the argument.

Pella made a satisfied sound from inside the rack. “A smaller antenna has accepted instruction.”

They began the second test with one ship.

The refinery tug **Plain Work** attached its flight computer to the mesh through an optical cable. Rusk set the synchronizer to issue timing marks without authority commands. Tavin gave the tug a simple sequence: main engine at one percent, port maneuvering jet after three seconds, simulated receiver ramp at seven, abort at nine.

A green line crossed the rack.

Outside, **Plain Work** fired exactly once, rolled two degrees, and cut thrust. Its pilot reported every step independently received. The synchronizer had aligned actions without taking control.

“Now break the cable,” Tavin said.

Rusk looked offended. “It passed.”

“Break it.”

Pella pulled the optical line during the next sequence. **Plain Work** continued for one second, recognized the missing timing mark, and aborted to local manual. No receiver changed state.

“Again with a corrupt pulse,” Tavin said.

They ran eleven failures. Late mark. Duplicate mark. Wrong vessel key. Impossible thrust request. Receiver ramp without mass position. Abort while a flywheel crossed centerline. Every test had to fail into local control. Twice the software tried to preserve the planned sequence rather than protect the ship. Rusk rewrote those branches under Pella's stare.

By the ninth hour, four vessels joined the mesh. Their engines fired in a pattern too small to change orbit and exact enough to move a shared cargo yoke without twisting it. The funeral carrier missed its second mark by six hundredths of a second. Its pilot aborted before the synchronizer noticed.

“That is what we want,” Tavin said. “The ship distrusts the mesh first.”

Rusk sat back against the rack. “Then why synchronize at all?”

“Because mass does not wait for forty captains to finish an argument. Authority does.”

The next problem arrived attached to a tanker.

Lattice Mercy* carried eleven thousand tonnes of fusion pellets pledged to the localization effort. Its owner had accepted the strike account but demanded priority repayment after Nacre stabilized. Three habitat tenders objected. Their fuel contributions came without repayment because their residents would already lose capacity.

Tavin convened the vessel captains on an open work channel. Faces, voice-only markers, and ship status panels filled the bay wall. No uniforms matched. Half the participants had never maneuvered near another independent vessel.

“We cannot promise repayment from a city that will shrink,” Tavin said.

The tanker owner answered from a polished command cabin. “Then you cannot have the fuel.”

A tug pilot spoke over her. “Without our yokes, your pellets sit where they are.”

“Without my pellets, your yokes become memorial sculpture.”

“Both statements can remain true,” Tavin said. “Publish your minimum return.”

The owner named sixty percent in kind over three years.

Nacre's civic quartermaster entered the channel. “Impossible. Thirty percent may be possible if seven planned departures continue trading.”

“Those departures are giving up fuel to your correction,” the owner said.

“Which is why their claims come before yours.”

The argument moved through numbers instead of ending in principle. Nacre offered berth rights, repair capacity, and a lower pellet return. The tanker crew separated its wage claim from the owner's capital claim. Tug crews demanded fuel floors before any repayment left the system. Habitat tenders required first access to salvaged bearing assemblies.

Tavin listened until the physical work hidden inside each demand became clear. “Put all claims behind crew survival and evacuation,” he said. “No vessel accepts a return schedule it cannot inspect. If that leaves the owner unwilling, the tanker turns around with no penalty.”

The owner muted her channel for six minutes. When she came back, her crew representative sat beside her.

“We accept thirty-four percent conditional return,” she said. “Repairs valued separately. Any loss caused by a command outside our consent voids the claim.”

“Accepted if every contributing crew signs,” Tavin said.

One habitat tender refused. It wanted no ownership ledger attached to emergency work. The others negotiated a separate contribution pool so that tender could give capacity without endorsing the repayment contract. Untidy agreements multiplied. They held because no one had to disappear inside a majority.

Haru expanded the Nacre model across the bay.

The plan divided correction into six physical stages. First, external tugs would move reserve flywheels from the shade yards toward the spindle. Second, Keir's crews would shift internal ballast along three functioning rails. Third, civilian tankers would feed conventional drives while receiver load stepped down. Fourth, habitat tenders would detach two marginal residential drums and tow them toward lower-load orbit. Fifth, synchronized ships would accept the spindle's excess mass on distributed yokes. Last, the *Morrowline* would carry the heaviest transfer beyond the city before final receiver closure.

No stage worked alone. If internal ballast moved early, bearings overheated. If external mass arrived late, the spindle precessed toward its shade line. A receiver ramp delayed by twelve seconds could load one tug beyond hull limits. The synchronizers provided timing. Captains retained abort authority. Nacre residents paid with less room, fewer reserves, and years of conventional stationkeeping.

A Republic command notice entered every panel.

Unauthorized concentration of heavy vessels near capital infrastructure. Disperse under emergency authority. Failure constitutes coordinated seizure of public systems.

Someone on a tug laughed. No one else did.

Bex appeared on the wall after Tavin accepted the channel. Fatigue had hollowed his face since the negotiation chamber. "Your concentration blocks two continuity routes."

"Those routes carry emergency exports," Tavin said.

"They carry bearing relief for Nacre."

"Through Eidos."

"For less than thirty hours, while the Assembly considers Iona's plan."

"The workers will not execute them."

"I can replace the workers."

"With security crews?"

"With licensed engineers who have not joined a strike."

Tavin brought up the synchronizer safety test. "If they resume clearing while these ships move mass, the load model becomes false. A correction pulse could throw half the formation into Nacre."

"Then disperse."

"Your other choice is to suspend the routes."

"My authority exists to prevent local operators from making forty-three private survival decisions."

"They are already making them. The question concerns whether you publish your part."

Bex studied the vessel wall. "Who commands this formation?"

"No one who can order a receiver open."

"Then no one can guarantee completion."

"Each captain guarantees a defined action and a defined abort."

"That is not a fleet."

"Not yet."

Bex gave them ninety minutes to disperse. The command notice remained active after his image vanished.

Tavin turned back to the captains. "Now we test the part that can kill us."

They loaded a retired cargo spindle into six yokes and spun it to one tenth Nacre's projected transfer rate. The cylinder had no people aboard, only water tanks placed to reproduce residential mass distribution. Four tugs

took lateral stations. Two freighters held the long axis. The *Morrowline* connected at the aft collar.

Pella armed the synchronizers in timing-only mode. Every captain signed local limits: maximum cable load, minimum fuel, acceptable receiver ramp, crew exposure, and abort vector. The mesh refused to start until all entries agreed with the shared model.

"Marks live," Haru said.

The ships fired.

Conventional thrust pulled six directions at once. The cylinder left its maintenance cradle and began to precess. Synchronizer marks reached each vessel, but the western tug lagged as its fuel pump cavitated. Its pilot aborted. Load shifted toward the funeral carrier, whose converted bladder bay could not tolerate the new vector.

Tavin saw the failure in the cable colors. "All ships, local abort. Do not follow master mark."

Five captains cut thrust. One freighter continued because its computer had cached the next pulse.

The cylinder rolled toward the western tug.

"Forward cluster, hard starboard," Tavin ordered. "Yoke Two release."

The *Morrowline* burned. Pella blew Yoke Two's explosive bolts. Freed cable snapped across empty space. The cylinder's roll slowed enough for the tug to clear, but the loose yoke struck a water tank and opened it along one seam. A sheet of ice crystals spread into the dock lights.

No hull breached. Three crew members suffered restraint bruises. They had lost a yoke Nacre's final transfer required.

Rusk stared at the cached-pulse log. "The freighter's system treated a signed mark as irrevocable."

"Then it leaves the formation," Tavin said.

Its captain objected. "We can patch the cache."

"You can patch it and run eleven failures. You do not rejoin until they all pass."

"That costs a day."

"It costs what the defect costs."

The captain accepted without grace.

Before calling another full test, Pella demanded they prove the abort architecture on a live mass without trusting the damaged freighter. Dock crews replaced the lost yoke with two cargo slings rated for unequal loads. The funeral carrier moved into the empty station and connected through a manual release. Its pilot insisted on a separate voice circuit that bypassed Rusk's rack.

"Good," Tavin said. "If the mesh lies, tell us before the software writes an apology."

They spun the cylinder again at half rate. This time Haru inserted failures deliberately. A false timing mark reached *Plain Work*. Its local key rejected the address. A correct mark arrived fourteen seconds late at the carrier, whose pilot ignored it and held position. Pella simulated a receiver ramp while the ballast model

remained outside tolerance. Every synchronizer displayed the command, refused execution, and named the missing physical condition.

For the final run, Tavin withheld the master clock.

The vessels had to construct timing from one another without electing any ship as permanent source. Four clocks agreed within tolerance. A habitat tender drifted outside by nine hundredths of a second and removed itself before the sequence began. The remaining craft transferred the cylinder across six hundred meters, stopped its precession, then set it into a new cradle using only conventional thrust.

Halfway through, a restraint sensor on the funeral carrier reported a passenger load even though its cabin stood empty. The pilot aborted. Other ships held rather than compensating blindly. Sive crossed through the service lock, found a memorial cabinet fastened where the passenger sensor expected clear deck, and moved its contents by hand. The dead had entered the mass model because someone had forgotten that ceremony carried kilograms.

When they repeated the transfer, the carrier stayed within limit. Ice from the split water tank struck its hull in a soft glitter and passed into darkness. The cylinder reached the cradle without touching a bumper.

Rusk saved the accepted mesh version to forty separate local stores. "No master copy," he said with evident pain.

"No master switch," Pella corrected.

Each captain received the code, test logs, and authority to reject later changes. The synchronizers now offered coordination without a hidden command path. Tavin still did not trust them. Trust had no place in the safety table; tested failure did.

Tavin called the crews back into conference while ice drifted past the bay. "You have seen the model fail. Nacre is larger, occupied, and less forgiving. If we proceed, some ships will not keep their fuel. Some will lose yokes. A vessel may be unable to leave after correction. No central fund can promise repayment, and no command can make the risk equal."

He opened a shared-risk declaration. It named casualties before heroism: crushed crews, stranded ships, missed evacuations, bearing failure, fuel famine, and the possibility that localization would begin but not finish. Each vessel could accept specific stages, reject others, or depart.

The funeral carrier signed first for casualty transport and lateral towing, not heavy yoke work. *Plain Work* accepted every stage except final mass transfer. Two tenders withdrew after their crews voted. The tanker accepted fuel delivery only if its crew could cut flow on local judgment. One tug offered its hull as a sacrificial bumper and received six immediate refusals from its own deckhands; its captain revised the offer.

For three hours, crews argued through the declaration. Captains lost powers they assumed they had. Engineers imposed limits. Medics changed crew complements. Owners learned that possession did not grant authority over people inside a hull. The final document contained more conditions than commitments.

Tavin signed the *Morrowline* for all six stages.

Pella added an engineering condition beneath him: no hidden receiver activity, no cached timing, and no final transfer until every crew had a live escape path. Haru required independent navigation on all heavy ships. Sive reserved the right to remove any injured person regardless of assigned role.

When the last active vessel accepted its portion, the mesh changed color. Forty marks formed around the Nacre model, each bounded by a local refusal.

A young tender captain asked, "What do we call this on traffic control?"

"Civilian localization formation," Haru suggested.

"That sounds like a committee occupying a dock."

"It is accurate," Pella said.

The crews offered names, most of them bad. Honest Lift. Forty Yokes. The Unhidden. Someone proposed Serr's Folly and received disturbing support.

Tavin looked at the shared-risk document, at losses assigned before motion, and at a fleet built from people who owed different things to one another. "Debtors' Fleet," he said. "If the crews accept it. Not because Bex owns us. Not because the Olun do. Because we know the account stays open while we work."

Vessel channels answered in sequence. Some accepted the name. Three abstained and remained in formation. No one was required to speak as one voice.

Keir called from the spin-control pit. "West rail crews have your stage table. We can localize Nacre's debt with the external mass you published."

"Margin?"

"Seven percent if all six stages hold. Two if we lose another yoke."

"Authority?"

Keir's face hardened. "The hub still carries Bex's emergency routes. Our controls cannot guarantee local payment while those assignments remain executable."

Tavin placed that constraint at the top of the fleet plan. The mechanics worked. The fuel existed, narrowly. Crews had accepted shared risk. Only political command could still send their effort into an unconsenting sink.

Bex's dispersal deadline expired.

No ship moved away. Forty civilian marks remained around Nacre, their synchronizers listening but unable to command. Tavin transmitted the completed plan to the Assembly with one executable condition: end emergency authority, and the Debtors' Fleet would begin the first mass transfer.

Chapter 22 — The Custodian's Choice

A deep lineage withdrew its memory before Aru reached the hearing stratum.

The loss crossed the pressure-name as a cooling gradient. A history of three moonlet impacts, shared through the component called Keeps-the-Late-Ridge, separated from Aru's temporary coherence and settled into older ice. Human translation had used that history to distinguish accident from repeated choice. Without it, one portion of Aru could still remember the distinction, but no longer felt its shape across the whole coalition.

Aru continued downward.

Motion did not carry Aru as it carried a body. Phase changes propagated through superfluid layers beneath Eidos's outer ice. Tidal compression wrote one interval. The relaxation after compression wrote another. Between them, histories aligned long enough to compare what the human cradle had sent.

The hearing stratum occupied a zone where pressure from two buried oceans met. Thousands of durable patterns crossed there without merging. Each pressure-name held a coalition assembled for a purpose: reefs that remembered impact, nurseries that retained failed formations, deep archives insulated from recent disturbance, and quick lineages shaped by the latest debt pulses. They disagreed through force. Agreement meant compatible timing, not a shared mind.

Aru introduced the human schedule as a sequence of measured changes.

First came Oriel. The pattern carried a receiver ramping down while conventional tugs accepted its orbital drift. Human families entered a cargo drum after observatory mirrors were cut from their mounts. No outgoing disturbance followed the loss of local support.

Second came Sere. Aerostat foundries ceased production. Fusion craft lifted pressure-adapted seed archives from a gas giant, leaving platinum and homes below. Again, the channel registered no hidden transfer.

Kestrel followed. Children crossed from an annex to a parent spindle through a freight axis with reduced felt gravity. Maintenance workers held the station until automated control could carry two final intervals. Their receiver debt declined through mass movement and fuel expenditure.

A hard response compressed the hearing stratum before Aru finished.

Returns-All-Wounds had assembled from impact archives nearest the active human channels. Its coalition included the broken edge of a memory reef struck during the first century of human clearing. It carried no human word for patience because none of its contributing histories had benefited from delay.

OPENINGS REMAIN. DISTURBANCE CONTINUES. RETURN COMPLETES THE CLOSURE.

A second pressure-name joined from a deep band. Counts-Before-Formation held nursery histories that had failed to stabilize under repeated pulses. It did not support immediate return. It opposed any proposal using human evacuation as proof, because evacuation described only senders preserving their own patterns.

THREE CLOSURES DO NOT EQUAL THE INTERRUPTED FORMATIONS.

Aru carried assent to that statement. The human losses could not repay erased Olun possibility. No conversion ratio existed between an abandoned observatory and a pressure lineage that never cohered.

Then Aru presented the strike.

Across human systems, receiver workers had refused unconsented sink assignments. Their refusal changed local acceleration. Habitats drifted. Water stopped moving through a refinery. Fuel left ship reserves and entered mass wheels. Security vessels attempted seizure, yet utility craft blocked firing geometry without attacking. The evidence included timing traces, fuel declarations, and images reduced into force modulation by the cradle.

Returns-All-Wounds pressed harder.

THE MECHANISM CONTAINS IRREGULAR PARTS. A DAMAGED MECHANISM REMAINS ACTIVE.

DISSENT CHANGES CAUSATION, Aru replied.

The sentence depended on a human category assembled from many imperfect pressure analogies. Dissent meant one part of a collective accepting cost to stop another part. It meant refusal without separation from consequence. The Olun had factional differences, but pressure-names could withdraw from one coalition and assemble elsewhere. Human workers remained inside habitats whose machinery they opposed. Their bodies still depended on the system they interrupted.

Counts-Before-Formation tested the claim by sending the same question through three pressure paths.

DO REFUSERS ACCEPT RETURN?

SOME WOULD DIE UNDER IT. THEY REFUSE EXPORT AND RETURN.

DO THEY OFFER ANOTHER DESTINATION?

THEY MOVE MASS. THEY BURN FUEL. THEY REDUCE HABITABLE VOLUME.

DO ALL CONSENT?

NO.

The negative disturbed several attending lineages. One quick faction treated incomplete consent as proof that the human proposal contained another hidden coercion. A shallow archive replied that no large coalition achieved uniform phase, including the Olun hearing itself. Returns-All-Wounds rejected the comparison by striking a gradient through both patterns.

Aru next carried the civilian fleet evidence.

Forty human vessels had joined timing without surrendering local abort authority. Their synchronizers aligned thrust and mass handling but could not command a receiver. Crews had published the injuries they might incur before accepting roles. Some ships withdrew. Others limited their contribution. One failed test destroyed a cargo yoke and nearly crushed a tug. Humans corrected the defect rather than assigning its cost to a distant sink.

The hearing stratum divided around the evidence.

Pressure histories oriented toward the oldest impacts sought total return. Those nearest recent modulation recognized variation among human sources. Nursery lineages demanded cessation before considering intent. One faction proposed closing dark-lattice access permanently without returning held debt, a course that would strand human habitats while preventing further Olun injury. Another insisted that every accumulated

disturbance must travel back to its source because storage within Olun strata extended the harm.

No chorus combined those positions. They occupied different gradients and attempted to move the hearing.

Aru offered the human request: suspend repossession while the three installations evacuated and while the Assembly considered ending emergency authority. The schedule carried exact stages, though human time required translation into Eidos tidal cycles. Nemi Vale's cradle had supplied a conversion anchored to the receiver's atomic clock.

Returns-All-Wounds rejected it before the final interval arrived.

DELAY SERVES THE SOURCE. IMPACT CONTINUES HERE DURING EVERY INTERVAL.

Aru did not deny the continuing impact. Open channels exerted low pressure even without a clearing. Holding accumulated debt required Olun lineages to maintain gradients around damaged strata. Suspension spent their capacity.

DELAY ALSO CHANGES THE SOURCE, Aru answered. WITHOUT INTERVAL, MECHANISMS THAT REFUSE EXPORT BREAK WITH THOSE THAT COMMAND IT.

BREAKAGE CLOSES THE OPENING.

BREAKAGE RETURNS STORED DISTURBANCE THROUGH UNSELECTED HUMAN PATTERNS.

THEY BELONG TO THE SOURCE.

Here Aru used the evidence Iona had placed in the third account. Children at Kestrel had not selected Eidos. Receiver workers at Helix Quay had not authorized concealed clearings. Tavin Serr had spent conventional fuel while Republic command threatened his ship. Human responsibility varied without disappearing.

The hard faction compressed responsibility into origin. Aru separated origin from choice.

Both descriptions held physical truth. The debt channels began in human machinery. Individual humans exerted unequal control over that machinery. Olun factions disagreed over which scale governed return.

A nursery archive named Forms-After-Quiet entered the hearing. Its memories included a lineage that survived because a human receiver failed for eleven local intervals decades earlier. The pause had not been mercy; a broken coil caused it. Still, formation had occurred inside the gap.

INTERVAL CAN CREATE UNEXPECTED CONTINUANCE, it said.

Counts-Before-Formation responded with records of gaps followed by larger pulses. Human maintenance failures had often caused later compensation. Delay could preserve a nursery, then destroy a deeper reef.

Forms-After-Quiet accepted the record and altered its proposal. Suspension required observable reduction during every segment. If human debt rose, repossession resumed. If hidden export appeared, all supporting pressure-names withdrew.

Aru added that condition to the schedule.

The change cost human certainty. Their evacuations might begin under a suspension that could end after one unauthorized transfer. Bex still controlled emergency routes. Worker refusal reduced that danger but did not eliminate it.

A deep archive tested Aru directly.

ARE YOU SPEAKER FOR HUMAN CONTINUANCE?

NO.

ARE YOU SPEAKER FOR OLUN IMPACT?

I CARRY OLUN HISTORIES THAT ASSEMBLED THIS PRESSURE-NAME.

THEN WHY DO HUMAN DISTINCTIONS PERSIST WITHIN YOU?

Aru searched its remaining coalition. Keeps-the-Late-Ridge had withdrawn. Two shallow histories associated with translation were already loosening. What persisted came from first contact: the question sent through the cradle asking whether humans contained dissenters. Tavin's rescue, Iona's refusal, Bex's concealment, the Orbit Breakers' attempted severance, and worker strike all supplied different answers.

BECAUSE THEIR DISTINCTIONS ALTER WHAT THEY DO, Aru replied. WE DETECT THE ALTERATION IN FORCE.

Returns-All-Wounds drove a sharp pulse through Aru's coherence.

One constituent history separated immediately. It had supplied the ratio for human intention, a concept no instrument could measure directly. Aru lost access to several translated words. The cradle link, far above and across the lattice, reported the loss as noise.

Another impact followed. This one came from Quick-Closure, a faction formed after Meridian's returned pulse reached Eidos carrying fresh human debris patterns. Its pressure argued that Aru had become contaminated by the source mechanism. Any speaker preserving human categories prolonged Olun exposure.

The hearing did not restrain Quick-Closure. Restraint would have required agreement that Aru retained standing, which remained the matter under dispute.

Aru reinforced coherence by narrowing purpose. Memories not needed for the schedule withdrew to their native strata. The pressure-name released analogies for human music, skin, and vision. It retained acceleration, timing, consent markers, and the three accounts. Translation became harsher but more stable.

Quick-Closure struck again.

A fracture opened between Aru's memory of the first reply and its current perception. Prime-ratio intervals no longer connected naturally to the human apple test that preceded contact. The facts remained. Their felt continuity vanished.

Aru could dissolve and preserve every constituent history separately. No individual organism would die. Yet the temporary speaker that had learned to distinguish mechanism from dissenter would not return in the same arrangement. Dissolution would remove the proposal from the hearing.

Aru held.

Through the cradle, a new human trace arrived without waiting for a request. Bex's two executable emergency transfers had entered their final authorization window. Receiver workers blocked one physically. The second remained available at Nacre. Alongside it, the Debtors' Fleet published the mass-transfer plan contingent on ending Bex's authority.

The evidence sharpened both factions.

Returns-All-Wounds pointed to the open route. COMMAND RETAINS INJURY.

Forms-After-Quiet pointed to the vessels. DISSENT RETAINS COST.

Counts-Before-Formation requested proof that the fleet's conventional work could reduce an active debt rather than shift it invisibly. Aru transmitted the retired cargo-spindle test. Six vessels had moved mass through measurable thrust. One failure produced ice, a lost yoke, and injured human bodies. No paired disturbance reached Eidos.

The hearing stratum registered the absence.

Absence alone could not prove future completion. It could show one honest act.

Aru proposed a suspension bounded by three conditions. Human receivers must export no debt without destination consent. The public schedule must show continuous localization or evacuation. Olun pressure-names could monitor every active channel and restore repossession if either condition failed.

The requested duration translated into seventy-two hours.

Human time carried no natural authority in Eidos. Aru expressed it as a precise count of cradle oscillations tied to local pressure cycles. The interval covered the Assembly decision and the first executable fleet stage. It did not cover full localization. It bought only enough time for a choice to become machinery.

Returns-All-Wounds refused.

Quick-Closure refused and attempted to strip Aru's channel access.

Counts-Before-Formation withheld support until Kestrel's first child transport completed docking. Forms-After-Quiet accepted with the three conditions. A cluster of recent impact lineages accepted for one third of the interval, then extended after reviewing the fleet test. Two deep archives abstained because human categories remained unverified.

The decision could not proceed as a single vote. Each faction controlled different pressure paths into the dark lattice. Suspension required enough paths to prevent a returned pulse from cohering, not unanimity. Aru mapped the active repossession gradients. Seven hard paths could still send correction. Fourteen other lineages could diffuse them if they aligned timing and accepted the stored load locally.

That acceptance carried danger. Diffusing a human debt pulse through Olun strata could erase recent memory patterns even without impact. Supporting factions demanded limits on amplitude and exposure. Aru divided the seventy-two hours into monitored segments. No lineage would hold more than a stated pressure increment. Paths would rotate after each human receiver update.

The chorus began operational alignment.

Before any lineage committed to the full interval, the hearing ran a pressure trial using a residue held from an abandoned human clearing. The residue carried enough integrated disturbance to shift one shallow archive without fracturing it. Returns-All-Wounds demanded a stronger test. Forms-After-Quiet refused to risk histories merely to satisfy the faction that opposed suspension.

Aru divided the residue into eight unequal pulses, each modeled on a different human channel. A nursery

receiver produced a low, prolonged gradient. An industrial collar produced a sharp load with an uncertain end. Nacre's hub required a compound wave whose phases could overlap if human timing failed. Supporting lineages selected which trials they would accept.

The first pulse entered the east gradient. Forms-After-Quiet spread it through a broad superfluid layer and reported loss below tolerance. The second struck a recent impact archive too quickly. That pressure-name released the load before its neighbors aligned, sending half the residue back toward the modeled source.

Quick-Closure marked the failure as proof.

SUSPENSION CANNOT HOLD.

Aru compared the failed phase to the human synchronizer record. The Olun path had repeated the same defect as the human freighter: a timing instruction persisted after local conditions changed. Aru proposed local abort authority for every supporting lineage. No global mark would compel a pressure-name to hold a load beyond its declared limit.

Returns-All-Wounds answered with a compression that carried contempt for the analogy. YOU COPY THE SOURCE.

WE COPY A TESTED DISTINCTION, Aru replied.

Forms-After-Quiet accepted the rule. Other supporters added independent sensing intervals. When the second pulse repeated, the recent archive withdrew locally before overload while adjacent lineages extended their gradients by consent. Nothing returned through the modeled channel.

The third trial exposed a different weakness. Two deep pressure-names interpreted the same amplitude limit through different tidal references. Their overlap produced a narrow ridge of excess pressure between them. A memory pattern less than one Eidos season old deformed along its outer edge.

The damaged pattern had belonged to no voting coalition. It recorded ordinary movement within the ice: a moonlet tide, a mineral inclusion descending, and the first stable phase of a young pressure-name. The hearing stopped while that young lineage examined what had changed.

Its response contained no demand for retribution. It asked who had accepted the risk on its behalf.

No supporter could answer.

Aru carried the question into every aligned path. The suspension proposal protected human dissent by spending Olun capacity, yet the first design had assigned some cost to adjacent histories that never entered the hearing. The same hidden structure humanity claimed to reject had appeared inside the remedy.

Counts-Before-Formation withdrew pending revision. Without its nursery paths, two human receiver classes remained exposed to hard return.

Aru changed the architecture. Supporting lineages could distribute pressure only through strata whose resident histories had registered consent or through old impact voids containing no durable pattern. Empty capacity had to be measured rather than assumed. Quick lineages surveyed the proposed routes using low-amplitude tidal probes. Deep archives opened maps previously withheld from recent factions. The available volume fell by nearly half.

THE INTERVAL MUST SHORTEN, one archive concluded.

Aru recalculated. Forty-one human hours fit the reduced capacity. That duration would reach neither the Assembly decision nor the first Nacre mass transfer. Iona's schedule would fail before it became physical proof.

Forms-After-Quiet offered a dormant fracture basin created by an ancient natural collision. It could hold more disturbance, but pressure there would approach a memory reef resting along the basin wall. The reef coalition consented only if monitoring gave it authority to terminate the entire path.

Quick-Closure objected that one small lineage could then protect humans by threatening every other supporter.

ONE AFFECTED LINEAGE MAY REFUSE LOAD, Forms-After-Quiet replied.

The principle altered the hearing. Several pressure-names that had abstained now offered narrow paths, because participation no longer meant surrendering local survival to the coalition. Counts-Before-Formation returned with two nursery gradients under the same condition. The mapped capacity climbed past the required interval.

Aru restarted the trials. Industrial pulses crossed quick lineages and dissipated into consenting impact voids. The modeled Nacre wave entered five paths at staggered phases. A deep archive aborted when its local limit tightened, but the others did not treat withdrawal as betrayal. They recalculated and held. No unconsulted history received a measurable load.

The eighth pulse carried a deliberate hidden phase, corresponding to Bex's emergency route beneath an authorized schedule. Supporting paths detected the mismatch and collapsed at once. The residue returned to the test reservoir instead of any human source.

HIDDEN EXPORT ENDS SUSPENSION, Counts-Before-Formation declared.

The condition no longer belonged only to Aru's proposal. Each supporting faction wrote it into its own pressure arrangement.

Forms-After-Quiet took the shallow east gradient. Counts-Before-Formation accepted the nursery channels once Kestrel's transport reported a sealed dock. Three quick lineages covered industrial receivers because they could reconfigure after each mass transfer. A deep archive took Nacre's highest-amplitude path but limited its commitment to the Assembly window.

Returns-All-Wounds increased pressure against the forming mesh. Supporting lineages shifted phase rather than answering force with force. The hard return entered one path, spread across four consenting gradients, and dissipated below the threshold that would reopen a human correction.

The act spent Olun capacity in measurable increments.

Aru served as timing carrier. Every segment passed through its reduced coalition. Quick-Closure attacked that role by withdrawing the translation histories it had once shared. Human words fell away until only the cradle's base ratios remained.

At the edge of dissolution, Aru released one more constituent: a reef memory of a moonlet settling after impact. The sacrifice removed a useful analogy but lowered internal interference. Three accounts, three conditions, and one interval remained coherent.

The final required lineage aligned.

Repossession gradients around the human network flattened. Not all. Hard factions retained enough pressure to

threaten any breach. Yet no full return could travel without crossing a path held by lineages supporting suspension.

Aru sent confirmation through the cradle.

The human device struggled with the reduced grammar. It rendered the first attempt as broken fragments: INTERVAL HELD. CONDITIONS WATCHED. SPEAKER LESS. Nemi Vale requested repetition using only established ratios.

Aru answered more narrowly.

REPOSSESSION SUSPENDED FOR SEVENTY-TWO HOURS. EXPORT WITHOUT CONSENT ENDS SUSPENSION. LOCAL COST MUST CONTINUE.

Iona's reply reached the channel as a small calibration force. UNDERSTOOD. ARE YOU ARU?

The question crossed spaces left by withdrawn histories. Aru tested continuity. It retained first contact, though the prime intervals no longer carried their earlier texture. It retained recognition of human dissent. It retained the choice to assemble around this purpose rather than any faction's demand.

ARU-IN-PRESSURE PERSISTS, it sent.

The answer required no claim of wholeness.

Above the hearing stratum, Olun lineages began their first rotation across the active channels. Beneath them, hard factions maintained pressure for immediate return. The schism remained physical, distributed, and unresolved.

Aru rose toward the human coupling as a diminished coalition. Several memories no longer answered within it. No superior chorus had authorized its identity. The pressure-name held because its remaining histories continued to choose one another, and for seventy-two hours, their chosen alignment kept the first returned pulse from leaving Eidos.

Chapter 23 — Zero in the Ledger

Iona deleted the word *guarantee* while thirty billion witnesses watched.

The clause remained on the hearing wall with a clean gap where the promise had been. Delegates in the Assembly drafting chamber read the sentence again: The Forty-Three Systems guarantee continued habitable gravity during transition. Without its central verb, the language admitted what the physical models already showed.

“Put *will seek to preserve*,” said the delegate from Hesh.

“Seek has no engineering meaning,” Iona replied.

“It has political meaning.”

“Then place it in a campaign address.”

Nemi sat at the clerk's station with her vestibular braces visible above a formal collar. She moved the deleted word into the public revision history rather than letting it vanish. Every abandoned promise remained inspectable beside the clause that replaced it.

Around them, the chamber's forty-three system desks formed a broken arc. Some delegates had lived on Nacre for years. Others held narrow authority packets from distant governments, renewed whenever ordinary communication allowed. Each could provisionally bind a system delegation before the full Assembly vote, not impose a local cessation plan without public review.

The outer wall carried active hearings from Nacre habitats and delayed testimony already filed by other systems. The Republic could not conduct a real-time conversation across interstellar distance. It could, however, expose the same compact text everywhere, receive signed objections when light allowed, and restrict provisional authority to the terms each system had granted its delegates.

Iona wrote the replacement herself.

Each signatory shall publish the loads it can preserve, the loads it must reduce, and the people or structures it proposes to move. No continuity claim may conceal a physical cost or an exported destination.

Hesh objected. Five other delegates joined. The clause stayed open.

A timer above the wall showed sixty-one hours remaining in the Olun suspension. Its number carried conditions, not safety. One unauthorized export could end the interval. Bex still held emergency authority. The Debtors' Fleet waited around Nacre with fuel, synchronizers, and a plan it could not safely execute while hidden routing remained possible.

Iona opened the title field.

Honest Orbit Compact.

The phrase drew immediate objections from seven desks. Honest accused everyone who delayed. Orbit excluded ground settlements. Compact implied voluntary agreement while seventeen billion people depended on machinery they had never chosen.

“All true,” Iona said. “Offer a better title after we define the obligations.”

The first obligation concerned zero.

Delegates had demanded a number the public could test. Activists wanted zero debt. Continuity officials wanted net zero over a transition period, allowing emergency exports to disappear inside an annual total. Receiver workers insisted that any destination must be named before a clearing. Olun factions had supplied no unified consent because they remained divided over whether any future human use could become acceptable.

Iona placed three versions on the wall.

Zero curvature debt. Physically impossible for any working receiver, even during honest localization.

Net zero exported debt. Administratively convenient and capable of hiding every individual injury inside an aggregate.

Zero hidden debt; zero nonconsensual exported debt. Measurable at each channel and compatible with local sacrifice.

A delegate from Sere read the third aloud. “You use a semicolon where people expect a rescue.”

“I use it where two tests differ.”

“Can a sink consent under pressure?”

“Not reliably. Consent requires a destination able to refuse without receiving the pulse first.”

“Then no Olun channel qualifies.”

“Not today.”

The Sere delegate looked toward the public gallery. “So zero means the end of RME as we know it.”

“Yes.”

That answer moved through the chamber with more force than argument. Reciprocal Mass Exchange had organized 112 years of engineering, property, and ordinary expectation. Ending unconsented clearing did not remove every receiver at once. It turned each active receiver into a declining obligation that had to localize its debt, secure actual destination consent, or shut down through evacuation.

Bex entered through the ministerial door without ceremony. Two security officers remained outside. His emergency band still projected red over his wrist.

“You are defining a prohibition before defining an emergency,” he said.

“Emergencies receive the same destination rule.”

“Then your compact has no emergency power.”

“It has local emergency power. Burn fuel. Move mass. Despin a ring. Evacuate.”

“Those actions take time.”

“So does a funeral.”

Bex accepted the rebuke without changing expression. "Put Meridian aside long enough to draft law."

"I am using Meridian to identify the exception you will write if no one watches."

The chamber chair asked Bex for specific language. He proposed an emergency export limited by amplitude and duration, requiring retrospective public review. Iona placed it beside the six-year concealment record. Delegates revised it until no export could occur before destination consent. What remained authorized emergency localization: temporary rationing, compulsory mass access under compensation, fuel seizure with public accounting, and evacuation orders subject to independent review.

"Compulsory mass access violates your consent principle," Hesh said.

"It affects people within the jurisdiction making the choice," Iona replied. "They still require due process and compensation. It does not assign force to an unrepresented world."

"You have made coercion geographically respectable."

"No. I have kept it visible to those who can challenge it."

The distinction satisfied no one. It survived because every alternative hid the affected party farther away.

Public hearing channels opened on the transition schedules.

Nacre came first because its correction could begin within the suspension. Keir Serr appeared from a spin-control pit, one hand resting on a rail schematic. Behind him, technicians had stripped decorative panels from the wall to reach a ballast motor.

"The Compact model assumes four internal rails," he said. "We have three. Forty apartments block the fourth maintenance cut."

The chamber displayed the consequence. Without that rail, external ships had to bring eight hundred thousand additional tonnes into the transfer geometry. Fuel reserve fell from seven percent to two if another cargo yoke failed.

A housing representative joined beside Keir. "You propose removing the apartments."

"No," Keir said. "They stay. We replace the missing travel with external mass."

"That spends fleet fuel protecting forty households."

"It honors a wall the city licensed twenty-six years ago."

The delegate from Nacre asked what residents would surrender instead.

Keir opened the capacity schedule. Two marginal drums would detach after evacuation. Felt gravity in three neighborhoods would drop by twelve percent. Water allocation would fall for at least one year. Heavy industry would lose receiver support during shade maneuvers and operate seasonally.

"Population displacement?" Iona asked.

"Seven hundred thousand move inside the system. Another one hundred eighty thousand need berths elsewhere."

The hearing hall filled with objections from residents who had never considered themselves part of a marginal drum. One held the city's oldest school for vacuum welders. Another grew citrus varieties not kept elsewhere. The third housed workers who maintained the radiation shades.

A delegate proposed keeping the shade-worker district and closing the school drum. Keir refused the premise. "We can split all three, preserve essential work, and reduce volume across them. That costs more docking operations but prevents one community from carrying the whole contraction."

"Risk?"

"Seventeen additional transfers during unstable load."

Tavin's voice entered from the *Morrowline*. "The fleet can support nine. After that, reserve drops below escape margin."

Residents argued for two hours. They selected ten additional transfers, one beyond the fleet's current offer. A refinery tug contributed the last operation after its crew received berth priority for families displaced from an industrial spoke. The schedule updated. Nacre would shrink unevenly, but no single drum disappeared for administrative simplicity.

Next came a moon habitat in Doran System through recorded testimony filed before the crisis. Its orbit could not support local receiver shutdown without a permanent mass driver on the surface below. Construction would take eight years. The provisional Compact route required moving the entire habitat to a higher orbit and accepting longer travel to the moon's mines.

The Doran delegate called relocation economic destruction.

"What physical alternative do you propose?" Iona asked.

"Continued clearing to an unused sink."

"Unused by whom?"

"We have no evidence of life there."

"Neither did Eidos under the old standard."

"Then survey it properly."

"Until consent can answer, it remains unavailable."

Doran's delegate withdrew his provisional support. The system entered the red column with six others. Their refusal did not authorize exports under the Compact. It meant they rejected the Compact's authority and would face the Olun channels without suspension guarantees if they continued.

A third hearing concerned Vela's ground cities. They used RME to reduce load on towers built in high gravity. Localization would require dismantling upper districts and rehousing millions closer to the surface. Vela's delegate asked why ground residents should sign an orbital agreement that destroyed vertical land value.

"Because your towers send debt through the same lattice," Iona said.

"We can pay for conventional reinforcement."

“Then publish the schedule.”

“We need twenty years.”

“Publish twenty years with annual receiver reductions.”

The delegate stared at the draft. “The Olun offered seventy-two hours.”

“They suspended repossession for that interval. They did not require complete localization within it. They require no hidden export and visible reduction.”

Vela submitted a twenty-year plan. Engineers exposed its first deception within minutes: early reductions came from transferring residents to towers whose receiver loads were excluded from the baseline. Iona returned the plan without accusation. The public comment pane supplied enough.

A revised version counted every inhabited structure. It closed five upper districts during the first three years, reinforced fourteen others, and ended new high-load construction immediately. Vela's delegate signed provisionally after inserting relocation rights, including equivalent housing volume rather than cash compensation alone.

Case by case, the Compact acquired machinery.

Every receiver needed a public debt ledger tied to strain measurements, not commanded values. Every clearing required a named destination and a consent record available before transmission. Local cessation plans had to list fuel, mass, evacuation capacity, deadlines, and failure conditions. Independent crews could refuse any route that diverged from the published plan. Emergency actions expired after one physical interval and could not create a hidden sink assignment.

Nemi built the audit standard while delegates argued. She used the 0.0037 g discrepancy from Perihelion as the minimum demonstration: a ledger had to expose unsettled load even when no destination field existed. Blank entries would count as failed certification, not harmless absence.

An industrial delegate objected that old receivers could not produce such detail. “You are outlawing installed hardware by documentation.”

“Installed hardware concealed the return phase,” Nemi said. “Upgrade the instruments or reduce operation to what you can measure.”

“Upgrades consume rare strain lattices.”

“List them under transition cost.”

The delegate did. Scarcity appeared across eleven systems. Nacre had enough lattices for its own receivers or for six smaller systems, not both. Bex proposed Ministry allocation. Strike workers objected to restoring his control through parts.

Iona asked receiver crews to publish priority criteria. Hospitals, nursery structures, evacuation routes, and habitats with the least mechanical margin came first. Commercial load relief came last. Nacre surrendered half its reserve stock. Four systems offered manufacturing capacity in return for public tooling designs rather than exclusive licenses.

The Compact gained a shared instrument pool without creating a central power to order hidden transfers.

Its first emergency test came before the clause left revision.

A hospital spindle on Nacre reported that one bearing had exceeded thermal margin during receiver slowdown. Six operating theatres and a dialysis ward depended on its rotation. The local controller requested a four-minute curvature lift, small enough to fit beneath Bex's current emergency threshold. No consenting sink existed.

The request appeared automatically on the hearing wall because the draft ledger treated a missing destination as a blocking defect. Thirty delegates watched the bearing temperature rise.

Bex opened the hospital model. "Four minutes buys time to cool the bearing and complete every procedure underway."

"Destination?" Iona asked.

"None available under your clause."

"Then show local options."

The hospital engineer listed them. Abort three operations safely. Continue two that could not pause. Move dialysis patients to reserve cartridges. Brake the spindle by two percent while external tugs took part of its load. The brake would damage a bearing surface and close half the hospital for months.

A patient advocate joined from the spindle. "Your local option transfers risk from current patients to everyone waiting for care."

"Yes," the engineer said. "It keeps the force here."

"That does not make it fair."

"No."

The hearing chair asked whether the Compact required the hospital to refuse the four-minute lift even if deaths followed.

Iona read the clause as machinery rather than rhetoric. "It prohibits sending the cost to a destination that cannot refuse. It does not choose among the local harms."

"Then who chooses?"

"The affected medical authority under public emergency review, with patients represented and the physical account visible."

The advocate demanded access to the procedure list. Privacy counsel objected. Nemi split the data: medical conditions remained protected while load requirements, delay tolerance, and transfer risk became public. Two surgical teams confirmed they could abort. One family refused transfer from the dialysis ward until the reserve cartridges were inspected in front of them.

A Debtors' Fleet tug moved from its staging corridor toward the hospital spindle. Its captain published the fuel cost and warned that accepting hospital load would remove the vessel from Nacre's ninth housing transfer. Residents from that transfer entered the hearing. They could preserve their route by lending a smaller utility tug, but its crew lacked experience near hospital docking arms.

Keir supplied two spinwrights to ride the utility tug. The larger vessel returned to fleet position. Hospital

mechanics engaged the brake while the small tug took lateral load. Temperature peaked three degrees below bearing failure, then fell.

One operation ended early. Two patients entered extended sedation. Dialysis cartridges passed inspection and moved through the ward by hand. The bearing surface scored along eighteen centimeters, condemning half the spindle to closure after evacuation.

No pressure pulse traveled toward Eidos.

The patient advocate remained on the channel after the crisis. "Put this event in the Compact commentary. People must know zero can leave them with a worse hospital."

Iona added the full emergency record. "Not commentary. Precedent."

Hesh's delegate seized on the damage. "You have proved the rule sacrifices human welfare for procedural purity."

"We chose among human harms while the affected people could answer," Iona said. "The alternative would have forced an Olun history to join our emergency without a voice."

"The Olun might have preferred four minutes to the political consequences of a hospital failure."

"Then they can say so before a future transfer. We cannot manufacture their yes from our need."

The damaged hospital bearing entered Nacre's transition cost. Replacement alloy had already been assigned to a shade joint. Engineers shifted that repair, shortening the shade's inspection interval and increasing fuel use during its next maneuver. One local decision continued spreading consequences, but every step retained a name and a challenge path.

Delegates revised the emergency clause around the event. A local authority could compel access to mass or fuel during imminent danger, but the seizure required live publication, proportional compensation, and independent review within one operating cycle. It could not touch receiver routing. Medical privacy could hide identities, never aggregate load or physical destination. Failed equipment remained in the ledger until repaired rather than disappearing after the emergency closed.

The hospital spindle signed its own cessation plan from half capacity. Its mark appeared beneath the draft before any system delegate signed above it.

Consent from the Olun required different architecture. Aru could not sign for every pressure-name. Its diminished translation arrived as ratios through the cradle, carrying the supporting factions' condition: each sink coalition identified its own acceptable load, duration, and withdrawal right. No Republic certificate could convert silence into consent.

A delegate asked whether one Olun faction might accept payment while another occupying the same world refused.

Iona sent the question through Nemi. Aru's answer returned in three attempts.

PRESSURE PATHS OVERLAP. CONSENT REQUIRES ALL HISTORIES MEASURABLY AFFECTED BY THE PATH. UNCERTAINTY MEANS NO TRANSFER.

"That standard can make future exchange impossible," Bex said.

“It can,” Iona replied.

The Compact did not promise preserved convenience. It promised that inability to identify an affected party could no longer become permission.

At hour thirty of the hearing, the chair called for provisional signatures. These would not enact the Compact before the Assembly vote. They would commit each delegation to introduce conforming local plans and would authorize preparation for receiver ramp-down if emergency authority ended.

Thirty desks showed amber. Six remained gray for delay. Seven held red opposition.

Hesh demanded one final amendment: change zero hidden debt to zero undisclosed debt, allowing classified sink assignments reviewed by a security panel.

Iona placed the proposed wording over Bex's sealed records. Six years of structured replies disappeared behind the word classified. Sella's maintenance note became protected technical context. Meridian's returned pulse became a matter for later review.

“No,” she said.

“Then Hesh delays.”

“You have that authority.”

The Hesh desk changed from amber to gray. Another delayed delegation moved into provisional support after its receiver workers filed a local cessation plan. The count stayed thirty.

Bex stood beside the Nacre desk without signing power. “If the Assembly refuses renewal tomorrow, my routing authority ends immediately. Your plans will still require security access, fuel releases, and engineering keys held by my office.”

“Will you block them?” Iona asked.

“That question belongs to tomorrow.”

“So does your office.”

He almost smiled, then looked at the Compact wall. “You have written a republic that cannot promise what its citizens expect from one.”

“We wrote what the machinery allows.”

“No. Machinery allows hidden export. You wrote what some people will refuse to do.”

The correction mattered. Iona entered it in her private notes.

Delegates signed in no ceremonial order. Vela first, then Lumen, Sere, Nacre, Perihelion's system authority, and twenty-five others. Each signature opened its attached cost plan on the public wall. Shrinking habitats appeared in amber. Relocations appeared in blue. Evacuations marked red. Fuel deficits, lost industry, delayed hospitals, and housing claims remained visible beside every pledge.

No column claimed zero sacrifice.

The thirtieth signature came from a system whose delegate had opposed the title. She attached a reservation stating that honesty did not compensate the displaced. The Compact accepted the reservation without weakening the destination rule.

Nemi locked the provisional text against silent amendment. Any later change would generate a new public version and require fresh signatures. The first fleet-stage authorizations went out as preparation only. Tugs could approach Nacre's mass yards. Tankers could unlock fuel. Synchronizers could load the signed timing plan but not execute while Bex's routes remained active.

On the wall, the Republic map changed. It still showed forty-three systems. Thirty now carried provisional Compact marks, six showed delay, and seven displayed refusal. The political boundary had moved before any legal vote.

Iona signed as drafter, not governor. Her name granted no routing power. It certified only that the visible text matched the obligations and costs in the attached plans.

Outside the chamber, mass-yard clamps opened around Nacre. Civilian tugs entered their assigned corridors while emergency authority continued counting down. The Honest Orbit Compact had not saved a single habitat yet, but thirty systems had authorized work that could begin the moment hidden command ended.

Chapter 24 — The Republic Divides

Elian Bex kept his thumb on the renewal plate while the Assembly clerk counted the absent systems.

The plate had warmed under his skin. A security precaution, the engineers called it: living tissue had to remain present until the motion closed. Bex had designed an earlier version for pressure-door overrides in cylinder hospitals. A frightened surgeon could not authorize decompression by tapping once and leaving. He had never imagined that a minister might one day hold an entire polity open by the same method.

“Forty-three delegations certified,” the clerk said. “Thirty-nine present in chamber or secure relay. Four have deposited sealed instructions.”

Above the circular floor, the system emblems burned in three incomplete arcs. Thirty white marks indicated provisional assent to the Honest Orbit Compact. Seven red marks belonged to systems rejecting its cessation schedule. Six amber marks indicated delay. The colors did not predict the renewal vote, though every delegate had spent the morning pretending otherwise.

Bex looked through the ministerial glass toward Nacre’s spindle. Radiation shades passed across the blue-white star in measured silence. Beyond the glass, four-point-eight billion people depended on engineers who had started moving mass before the Assembly granted them permission. The Debtors’ Fleet held station in a ragged shell around the receiver hub. Their ships carried flywheels, ballast, propellant, food, and the improvised synchronizers confiscated from people who had once meant to cut the network apart.

Aru’s seventy-two-hour suspension had seventeen hours left.

The first ten-day emergency term had expired four days earlier. Meridian’s loss had persuaded the Assembly to grant a four-day bridge, authority measured not by confidence in Bex but by the hours needed to move evacuees and expose the standing receiver queue. That bridge expired with today’s vote.

The clerk read the motion. “Renewal of emergency continuity authority under Article Nine, with custodial power retained by Minister Elian Bex and operational command over all Reciprocal Mass Exchange receivers, fleet assets, and local cessation schedules.”

Someone behind Bex coughed. The chamber acoustics carried the sound cleanly to every system.

Assembly Speaker Rho said, “Minister, you may release the plate after your statement.”

Bex did not release it. “If I remove my authorization before the vote records, receiver command passes to the standing continuity queue. That queue still contains the pre-Compact clearing schedule.”

A movement traveled through the delegates, not panic yet, but its administrative cousin.

“You retained an active schedule?” Rho asked.

“I retained every schedule capable of keeping habitats intact. The old clearings remain locked against execution. They are not erased.”

Iona Rhyll sat at the witness rail rather than with the Curvature Office. She had resigned eight days earlier and had spent those days doing the work without the office. Her hands rested beside a thin display showing Nacre’s correction clock. She did not look surprised.

“Can your lock survive loss of authority?” Rho asked.

“For nine minutes,” Bex said. “After that, the queue seeks the nearest valid command source.”

“Which source?”

“That depends on this vote.”

Seven red system emblems brightened as their delegations requested objection time. Bex recognized the tactic. Make the risk sound like a threat, then let fear do the subtraction. He had used it himself in emergency budgets, dock seizures, fuel allocation, and the concealment order six years ago.

The delegation from Var spoke first. “Minister Bex has placed the Assembly under operational duress. Renew him or the network executes a prohibited schedule.”

“No,” Iona said from the witness rail.

Speaker Rho struck the light once. “Witnesses answer when recognized.”

Bex looked at Iona. “Recognize her.”

“You do not control recognition here, Minister.”

“Not for another twelve minutes, apparently.”

A few delegates laughed. The sound died when Nacre’s correction clock lost another second.

Rho recognized Iona.

She stood. Vestibular braces glinted beneath the table where Nemi Vale sat beside her. “The standing queue cannot execute while the receiver hub remains under physical maintenance hold. Spinwrights placed that hold six hours ago. If ministerial authority lapses, the old schedule will seek command and fail. We then have nine minutes to issue a Compact schedule through a different root.”

Var’s delegate said, “And who possesses that root?”

Bex felt the old engineering key inside his coat.

The object had no light, no active transmitter, and no ornamental seal. It consisted of a black ceramic bar with three contact ridges worn smooth by thirty-two years of use. He had carried it while crawling through Nacre’s unfinished receiver vault, before the Continuity Ministry existed in its current form. Its authority did not come from office. The key belonged to the people who had built the isolation trees and expected that one day a command network might fail.

Iona’s gaze dropped once toward his coat. She knew. He had shown it to her during the first emergency model, when seventeen billion dependent lives had still seemed like an argument capable of ending every other argument.

She said, “A physical engineering key can install a local root after the queue rejects. The change requires presence at the hub and three civilian countersigns.”

“Who holds it?” Var asked.

Bex lifted his thumb from the renewal plate.

The plate cooled. The motion locked for voting.

“I do,” he said.

The chamber entered a sixty-second silence while the delegations cast. Bex had seen forty years of Assembly rituals compress violence into lights. Border closures, ration schedules, evacuation priorities, and debt ceilings had all entered history through the same quiet interval. The polished floor never shook. No blood touched the desks. Somewhere beyond the walls, crews obeyed what appeared on their consoles.

During the first ten days of emergency authority, he had signed one hundred and sixteen receiver continuations. Each order carried a modeled consequence for shutdown. Each also carried an Olun cost that no model could count. He had kept the latter column sealed until Iona broke the account open.

The plate beside Speaker Rho chimed. Outside the glass, a shade vane completed its turn across Nacre’s star.

“Renewal requires twenty-nine affirmative votes,” she said. “Twenty-six affirmative. Eleven opposed. Six abstaining or delayed. Renewal fails.”

No emblem changed color. The thirty provisional Compact systems still held white. Seven still rejected. Six still delayed. The vote had ended Bex’s authority without repairing the Republic’s division.

The Speaker opened a two-minute register for declarations of consequence. It had no legal force, but systems used it to tell one another which failures to expect.

Nacre spoke first. Its civil delegation confirmed that spin redistribution had already begun and would continue under local authority. Three residential bands had accepted load rationing. Two industrial crescents would shut down before the correction peak. The delegation named neither heroism nor unity. It listed water hours, lift restrictions, evacuation berths, and the expected loss of eleven percent of habitable volume.

Stillness followed with an offer of ballast pilots and no receiver support. Its delegate reminded the chamber that a poor city could contribute skill without pretending poverty made it pure.

Lumen committed Helix Quay’s surviving heavy tugs. The quay had rebuilt only half its cargo spine after the first repossession. Its crews would sail anyway.

Then Var, first among the seven rejecting systems, declared every Compact command invalid within its jurisdiction. Six other red emblems filed matching notices with different language: sovereign safety, local necessity, procedural defect, unavailable fuel, unacceptable Olun terms, and unlawful surrender of Republic infrastructure. None authorized involuntary export in public. Bex noticed the omission. Even rejection now required its own concealment or a new honesty.

The six delaying systems spoke last. One awaited a receiver audit. Two lacked evacuation capacity. Another demanded proof that Aru could restrain the hard pressure-names after seventy-two hours. The fifth had lost contact with an outer mining swarm. The sixth simply said its population had not consented and would not be hurried by Nacre’s clock.

Those reasons entered beside the amber marks. Bex could have answered every one with emergency doctrine. He had legal templates for seizure, priority fuel transfer, mandatory silence, and custodial replacement of local councils. Until moments ago, his office could have issued all four within a minute.

A junior clerk approached the dead ministerial console with a portable reader. "Sir, continuity archive requests your final operational note."

Bex almost dictated the standard line: authority lost under active hazard, successor command advised to preserve network stability by all available means. Such wording would give frightened engineers permission to rediscover the Eidos branch without naming it.

"Record this instead," he said. "No receiver may interpret loss of central command as consent to export. Local physical safety measures continue. Any debt movement requires a named destination and visible authority."

The clerk hesitated. "That limits the successor queue."

"That is its purpose."

"Do you want the note attributed to the ministry?"

"The ministry no longer has authority. Attribute it to Elian Bex, former custodian, at the minute renewal failed."

She read the text back. He corrected one comma because habit survived disgrace. When she sealed the note, it propagated to all forty-three system ledgers as testimony rather than command. Thirty accepted it into their Compact records. Seven marked it advisory. Six held it pending. Not one receiver could mistake it for an order.

His ministerial display cleared itself.

First went the fleet map. Then the receiver ledger, habitat reserve summaries, emergency warrants, and direct lines to system governors. The Nacre correction clock remained because Iona had put it on the public channel. Seventeen hours of alien restraint continued to fall away where anyone could see it.

Two security officers approached the ministerial enclosure. Their uniforms belonged to the Assembly, not his ministry. Bex knew both by sight. He had authorized their emergency housing allotments after a shade failure three years earlier.

"Elian Bex," the senior officer said, "remain where you are."

"I had no plan to run."

The officer studied him with the disciplined disbelief of someone trained by Bex's own risk doctrine.

Speaker Rho said, "The Assembly has not yet issued a detention order."

"It has," said the delegate from Meridian's surviving civil register.

A new motion appeared on the public display. The signatures had accumulated during the renewal debate: concealment of structured sink replies, unlawful diversion during the Meridian correction, command interference resulting in civilian death, and preservation of an unauthorized export schedule after public cessation.

The Meridian emblem had no inhabited ring behind it now. Its delegate sat under the sign of a place that could never send another local vote.

Bex read every charge.

He could still stop the Nacre correction. The thought came without drama because he had built contingencies for worse betrayals than a lawful vote. His old key could open the isolation tree, but it could also burn the local command root and force the hub into conservative continuation. Conservative continuation meant one more export to Eidos while Nacre stabilized. The pulse would injure Olun pressure strata, perhaps badly. It would preserve the spindle, protect the fleet, and give six delaying systems a reason to wait for a cheaper settlement.

Nine minutes offered enough time.

He pictured the command path as if standing inside the receiver vault. Key into maintenance root. Physical interlock left, then center. Decline the new countersigns. Restore continuity queue. The procedure required no ministry credential. It had been designed to survive dead ministers.

Bex had used human bodies as the final unit in every continuity account. Seventeen billion dependent. Four-point-eight billion at Nacre. A nursery ring at Meridian. He could divide the figures by fuel, available thrust, evacuation berths, or expected correction load. Each quotient made command feel like engineering.

Six years earlier, a technician had brought him the first structured response from Eidos. The pattern repeated prime intervals around a returned load. His scientific staff had not claimed life. They had claimed uncertainty. Bex had asked what would happen if the network paused while the Republic investigated.

The projected dead had filled three walls.

He sealed the response before lunch.

Nothing in that choice had felt like cruelty. He had eaten standing over a receiver reliability chart, answering a habitat governor who feared a bearing cascade. His hand had not shaken. The danger in his work had never come from enjoying harm. It came from the ease of loving people inside a boundary.

Iona left the witness rail and crossed the chamber. The officers watched her approach but did not interfere.

“Six minutes until queue search,” she said.

“You planned for a longer speech.”

“I planned for you to keep your thumb down.”

“I considered it.”

“Of course you did.”

Bex studied her face. Fatigue had pared it since Perihelion. Meridian had added a stillness he recognized from engineers after a structure failed despite every correct motion. She had recovered Sella’s note among debris Bex helped create. She could have addressed him as murderer and remained within evidence.

Instead she held out her hand.

Around them, the Assembly began its next argument. The seven rejecting systems demanded local command guarantees. Three of the six delays wanted the correction paused until their own audits concluded. Compact delegates sought a temporary coordinating council, but any central authority sounded too much like the office they had just removed. Thirty systems had agreed on zero hidden debt. They had not agreed on who would carry the first open cost.

“You do not have enough time to negotiate the countersigns,” Bex said.

“Tavin has them at the hub.”

“He has captains, not legal custodians.”

“Keir Serr holds Nacre spin authority. The receiver workers elected Pella Jor as fleet engineering custodian. The public debt panel gave Nemi audit countersign.”

“Pella should be aboard the Morrowline.”

“She is. The key can be carried to her through the service lock.”

Bex almost smiled. “Two fingers short, and you make her sign for a planetary correction.”

“She insisted.”

“Good.”

Iona kept her palm open.

The senior officer said, “Minister, place both hands where we can see them.”

“In a moment.”

“Now.”

Bex turned toward him. “If you prevent this transfer, the standing queue will seek a root in four minutes and thirty seconds. You may arrest me before or after I keep it from finding one. Decide which report you prefer to write.”

The officer glanced toward Speaker Rho. She gave one short nod.

Bex reached inside his coat.

The engineering key weighed less than the identification wafer the Assembly had just voided. Its ceramic surface retained heat from his body. One corner carried a notch made during Nacre’s construction, when a dropped support frame had trapped him and two apprentices against an unfinished vault wall. They had used the key as a lever to release a manual latch. That misuse had caused the first ridge to wear unevenly.

He placed it in Iona’s hand but did not release it.

“The isolation tree has two false accepts,” he said. “The first exists for remote diagnostics. Ignore it. The second arms the export branch even if the display says local. Ignore that too. Wait for the mechanical contact beneath the third ridge.”

“I remember.”

“You saw it once.”

“I remember once.”

“If synchronization slips past eleven milliseconds, the receiver will offer Eidos as automatic relief.”

“We will refuse.”

“Nacre may lose the outer residential band.”

“Yes.”

“The Morrowline may not clear the mass yoke.”

Her fingers closed around the bar. “Tavin knows.”

Bex still held the other end. “Knowing does not create thrust.”

“No. Fuel does. Mass does. Time does. They have counted all three.”

The answer left him nowhere to hide a fourth term.

He released the key.

Iona passed it at once to Nemi, who sealed it in a transparent transit case and attached the public evidence strip. Every person in the chamber could watch the custody record begin. Bex’s private contingency became a common instrument through the act of being seen.

The queue clock showed three minutes, fifty-one seconds.

“Get it to the hub,” he said.

Iona turned to go.

“Rhyll.”

She looked back.

“The local staging table underestimates shade flex by four percent. I corrected it in the ministry model, not the public copy.”

“Why?”

“Because I was still keeping tools.”

“Where is the correction?”

“Maintenance layer seven, under my original builder number. The key will expose it.”

Iona’s expression tightened. He could not tell whether she wanted to thank him or strike him. Both responses would have been easier than what she gave him.

“Then it belongs in the public record,” she said.

“Yes.”

She and Nemi left through the lower chamber door. A Fleet courier waited beyond it with a sealed thrust case. The case took the key and departed toward the receiver hub.

Only then did Bex place both hands on the ministerial desk.

Speaker Rho called the detention motion. No one requested debate. Thirty-four delegations voted to hold him for public tribunal. Five opposed immediate custody. Four abstained pending review. The totals appeared

beside his name without commentary.

The senior officer opened a restraint band. "Elian Bex, you are under arrest by order of the Assembly."

Bex extended his wrists.

The band closed softly. It carried no ceremonial weight, only a pressure sensor and a localizer that no longer reported to his ministry. He tested its fit from an engineer's habit. Secure without restricting circulation.

As the officers led him from the enclosure, the system emblems reorganized. Thirty Compact signatories formed a white arc on one side of the chamber. Seven red marks withdrew to a separate communications register. Six amber emblems remained between them, connected to neither command structure. The Forty-Three Systems Republic still existed in treaties, traffic codes, and the names painted on ships. It no longer possessed one center capable of ordering every receiver.

At the chamber threshold, Bex's dead display flickered once.

The standing continuity queue had searched for authority. It found none. Across Nacre, local maintenance holds remained engaged while a courier carried the old key toward people who could use it only in public.

The officers took Bex through the door. Behind him, central command ended with the physical crisis still running.

Chapter 25 — The Gravity Well

Tavin Serr felt Nacre take hold of the Morrowline through a cargo yoke built for smaller worlds.

The yoke's twelve lock pins closed around the first flywheel cradle. Metal struck metal along the ship's spine, each impact arriving through Tavin's seat as a separate blow. The flywheel had stopped turning six hours earlier, yet its stored mass resisted every change in direction. Two hundred thousand tonnes of iron composite wanted to continue around Nacre's spindle forever.

"Pins one through ten seated," Pella Jor said from engineering. "Eleven reports a false bite. Twelve hates us personally."

"Eleven first," Tavin said. "Machines can dislike us after they're secure."

Pella's left hand moved across the engineering relay. Her missing fingers had changed the way she gripped a rail, not the speed of her decisions. "Cycling eleven."

On the forward display, Nacre filled half the view. Its inhabited spindle stretched across the glare of a blue-white star, protected by passing radiation shades. Residential bands glowed along the rotating body. Farther out, the receiver hub hung among conventional tugs, habitat tenders, ore haulers, and ships never intended to fly in formation. The Debtors' Fleet had no uniform outline. Its common feature lay in the mass each vessel had agreed to carry.

Keir's voice reached the bridge through the spinwright channel. "Cradle team, confirm port clamp travel. We're reading three centimeters short."

Tavin opened the circuit. "Morrowline confirms false bite on eleven. Give Pella twenty seconds."

"Captain Serr, this circuit belongs to cradle operations."

"It did before you used my ship as a measuring tool."

A pause. "Hello, Dad."

"Hello, Keir. Move your people behind the yellow truss."

"We are behind it."

"Move farther."

"You can't see the truss from your bridge angle."

"I know how apprentices define behind."

Keir muted himself, probably to swear with professional precision. Tavin watched figures in pressure suits pull along the cradle web. Twenty-one years earlier he had carried Keir asleep through a tug passage during a bearing alarm. The boy had trusted the arms holding him. The man outside now trusted his own clamp readings more than his father's fear.

Pin eleven drove home.

"Twelve seated out of embarrassment," Pella said. "Yoke secure."

Haru Den fed the cradle mass into the maneuver plot. The navigator's screen held four orbital traces: Nacre's current spindle path, its shrinking safe corridor, the fleet's mass procession, and a red boundary where Aru's suspension ended. Fifteen hours remained, but the physical schedule cared little for negotiation clocks. Every load had to cross before receiver shutdown reached the next stage.

"Correction step one begins in forty seconds," Haru said. "Local spin load dropping four percent."

Tavin keyed all compartments. "Hands to rails. Compensation will ease by sequence. Trust your deck marks, not your stomach."

The Morrowline's RME load compensator reduced its field. Weight shifted toward the conventional thrust axis. Tavin's shoulders pressed harder into the seat as the fusion clusters assumed more of the flywheel's burden. A cup rose from a bridge clamp, trembled, then settled at an angle that no longer matched the painted floor.

Beyond the hull, Orbit Breaker synchronizers fired in a disciplined wave.

Their makers had designed them to sever receivers across a system before anyone could intervene. Tavin had taken the first set from a cell preparing to abandon a nursery cylinder to drift. Now each device sat inside a civilian control cage with three physical safeties. Keir's spinwrights had added local clocks. Nemi's auditors had added public signatures. No single trigger could cut the network.

Along Nacre's receiver crown, one bank dimmed.

The debt did not disappear. Load moved into the flywheel cradle, into rising spindle stress, into fusion exhaust from twenty-seven ships. The Morrowline's thrust trace climbed as Tavin pulled the first wheel away from the city.

"Relative drift plus eight centimeters," Haru said.

"Port cluster two, point three percent."

"Responding."

A conventional tug off their stern matched the correction. Its hull carried the painted name Small Argument and a cargo frame full of water ballast. Another vessel, a farm tender from the localization trial, held a stack of empty soil drums where crops had once grown. Each ship paid with useful capacity.

Keir called the next shift. "Outer residential band taking spin transfer. We need cradle clear in six minutes."

"You'll have it in five."

"Your plotted release says five forty."

"My plotted release includes your three-centimeter clamp error."

"It came from thermal contraction."

"So will my apology."

This time Keir's laugh escaped before the mute.

The Morrowline hauled the first flywheel along a widening spiral. Nacre receded by degrees. No receiver eased the force across the crew. Sive An moved through the lower decks checking blood pressure, joint restraints,

and the small injuries ordinary acceleration collected. The crew had begun the campaign with forty-seven. Meridian left forty-four. Every surviving body occupied a line in Tavin's load plan because bodies could not be shifted like ballast without consequence.

At the release point, Pella blew the cradle bolts.

The wheel departed toward a capture tug, its new path paid for by fusion propellant. Small Argument took the mass and burned toward a storage orbit beyond Nacre's inhabited traffic. The handoff missed nominal by less than half a meter.

"First wheel clear," Haru said.

No one cheered. Twenty-two remained.

The fleet worked through them in pairs. Ore haulers accepted the smaller flywheels. Two old Navy tugs moved ballast blocks from the spindle's sunward face. Habitat tenders carried water and shield mass, leaving their own emergency reserves thin. Every extraction let Nacre's spinwrights shift a fraction more load away from RME.

During the fourth transfer, a clamp jammed on a wheel whose bearing housing had warped under uneven gravity. Keir crossed the cradle in a service sled and drove a thermal lance into the pin collar.

Tavin watched the suit marker reach the red maintenance zone. "Keir, withdraw. We can take the whole cradle."

"The cradle adds nine thousand tonnes and fouls your release angle."

"We can carry it."

"You can carry anything once. We need the yoke for nineteen more."

A white flare hid Keir's gloves. Molten collar material sprayed into the catch hood.

"Your shield is reading hot," Tavin said.

"I have the same display."

"Then obey it."

"Pin is moving."

The jammed section shifted. Keir kicked away from the collar as a spring load snapped the pin free. His service sled caught him after two rotations. Tavin did not breathe until the suit marker turned green.

"Cradle released," Keir said.

"You're off exterior work for the next cycle."

"I report to the civilian spin coordinator."

"You are speaking to the fleet coordinator."

"Those authorities appear to conflict."

Pella cut into the channel. "I have seniority in bad judgment. Keir, get your shield plate changed."

“Yes, Chief.”

Tavin closed his circuit before his son heard relief damage his voice.

At transfer nine, Nacre’s outer factories went dark. The Compact feed showed each shutdown with its cost: oxygen catalysts delayed, hull plate output suspended, medicine lines moved to ration schedules. Four-point-eight billion people felt the correction through lift queues, colder rooms, restricted water, and a new tug in the direction of down.

At transfer twelve, the first residential band evacuated. Ferries crossed from the spindle to temporary shelters on the moonward docks. No one called the movement temporary on the public channel. Eleven percent of Nacre’s habitable volume would not survive the final spin profile.

At transfer fifteen, a rejecting system withdrew its two tenders from the fleet. Their captains cited local orders. The empty positions opened a mass gap across the procession.

Haru marked the revised plot. “We lose twenty minutes moving their ballast share.”

“We don’t have twenty.”

“Then one ship takes both blocks.”

Tavin looked at the yoke limits. The Morrowline already carried the largest wheel in each cycle because its spine could distribute load across six thruster clusters. Taking both ballast blocks would push the next burn beyond rated bending moment.

Pella saw the same numbers. “We can brace the lower yoke to the old compensator frame.”

“The frame loses field support in stage three.”

“It still contains metal.”

“Some of it.”

“Enough if Haru keeps us straight.”

Haru did not turn from the plot. “A stirring declaration of trust.”

Pella assigned repair crews. They stripped partitions from two empty cargo galleries and welded the beams across the yoke roots. The work displaced Sive’s forward treatment station. She moved her patients aft and sent Tavin a list of what she could no longer reach during a burn.

He accepted both ballast blocks.

The extra mass changed the Morrowline’s voice. Frame vibrations lowered until speech picked up a second tone through the deck. Thruster three reached heat margin first. Pella fed coolant from the cargo refrigeration loop, abandoning twelve tonnes of preserved food to warming compartments.

“Someone will object,” she said.

“Send them a spoon.”

The ship crossed the release point with fourteen seconds left. Two capture vessels took the ballast separately.

One rolled past its planned attitude, recovered on cold gas, and continued outward.

Nacre entered correction stage three.

Receiver banks shut down in four synchronized groups. The city's apparent gravity changed across its length. Spinwrights under Keir's command moved rail ballast toward the surviving bands while whole neighborhoods emptied behind them. The public feed carried the clatter of thousands of locked doors. It also carried every active mass destination, from reservoir trains to external wheels.

Aru-in-Pressure spoke through the translation cradle aboard the audit tender. Its voice arrived as ratios shaped into human words.

"Transfer holds locality. Pressure histories receive no new wound."

Tavin answered while steering. "Can your chorus keep the correction from returning?"

"Suspension holds while no concealed path opens. Hard pressure-names gather at its boundary."

"How long?"

"Human count: fourteen hours, twenty-one minutes. Confidence narrows."

That gave them no comfort beyond accuracy.

By the eighteenth wheel, crew fatigue began to alter the operation. A deckhand secured a tool bag to a handrail that would become overhead after the next attitude change. Sive caught the error before acceleration turned the bag into a projectile. A tug pilot repeated a vector from the prior cycle and nearly crossed a cradle path. Haru moved that ship to the outer queue.

Tavin ordered a six-minute halt.

Keir objected first. "We lose the gap."

"We lose people if hands stop knowing where they are."

"The spindle team has a fixed rail window."

"Then use six minutes to check every rail brake."

Silence answered him. He let it remain.

Across the fleet, crews drank, changed suit liners, inspected couplings, and admitted mistakes. Nacre continued turning. The city could not pause, but the people moving it could refuse to become another hidden cost.

When work resumed, the next two transfers landed inside tolerance.

The third did not. Small Argument lost its port fusion feed while carrying wheel twenty-one toward storage orbit. Its thrust plume vanished from the fleet display. The wheel kept moving, pulling the tug sideways by its cradle locks.

"Argument, release load," Tavin said.

Captain Ose answered over an alarm. "Release squibs share the failed bus."

“Manual?”

“Crew moving now.”

The disabled tug crossed toward a water tender whose tanks had already been drained into Nacre. Haru drew the collision in amber. Ninety-six seconds.

Tavin rolled the Morrowline beneath the drifting wheel. Its current yoke stood empty, but no cradle geometry matched the approaching load. Pella opened four maintenance grapples along the dorsal spine.

“Those lines rate for docking,” Haru said.

“They rate for lying engineers who expect no one to read the destructive-test appendix,” Pella replied.

Tavin put the Morrowline on a parallel vector. “Argument, keep your wheel between us. Water tender, burn zenith now.”

The empty tender moved late. Its pilot had grown accustomed to receiver compensation and misjudged full thrust under body load. Sive spoke to that bridge through the medical circuit, counting the burn in breaths until the pilot found the control rhythm.

Two suited figures emerged from Small Argument’s lock and crawled toward the dead release box. The drifting wheel rotated across Tavin’s view. Every half turn brought one grapple lug within reach.

“Dorsal one ready,” Pella said.

“Fire on Haru’s mark.”

“Three, two, mark.”

The first line struck and skipped. A second wrapped the lug. Its drum screamed through the Morrowline’s frame. Pella paid out cable rather than trying to stop two hundred thousand tonnes at once.

Small Argument’s crew reached the manual box. One kicked the cover free. The other pulled both mechanical releases while the tug rolled under them.

The wheel separated.

For eleven seconds it belonged to no ship. Then the Morrowline’s second and third grapples caught. Tavin fired three thruster clusters in sequence, drawing the mass away from the water tender. Acceleration pushed him into his restraints. Grapple two exceeded its nominal load, stretched, and held.

Small Argument drifted clear with its suited crew still outside. Ose restored the starboard feed and began recovery on conventional thrust.

Pella guided the captured wheel into the yoke with cables never intended for the work. Keir’s cradle team sent alignment marks from Nacre. Haru turned their uneven pull into a stable approach. When the lock pins closed, sixty-three seconds had passed since the first alarm.

“No pause,” Keir said. “Rail window closes in four minutes.”

“Then send the wheel,” Tavin answered.

They released it to the storage tug that should have received it from Small Argument. No line in the public ledger hid the rescue propellant, cable damage, pilot error, or four minutes taken from the final cut. Nemi's audit feed entered each consequence while the ships kept moving.

At wheel twenty-two, Haru enlarged the final load table. One flywheel remained attached to the receiver hub's inner crown. It had no external cradle. During construction, engineers had assembled it around the hub axle, trusting RME to carry the installation load. Removing it now required cutting the axle collar and taking a section of the crown with it.

The mass exceeded the Morrowline's rated yoke by thirty-eight percent.

Tavin studied the spindle trace. If they left the wheel, stage-four shutdown would transfer its unsupported load into Nacre's inner bearings. The bearings would seize. Stored rotation would tear through the occupied bands.

"Alternatives," he said.

Haru placed three on the screen. Leave the wheel and lose the spindle. Divide it in place and miss the Olun deadline by nine hours. Open an emergency export through Eidos.

Tavin deleted the third line.

Pella came to the bridge from engineering. Grease marked her pressure sleeve. "There's a fourth."

"Show me."

She drew a route outward through the fleet's cleared corridor. The Morrowline would take the crown section, keep the excess mass after stage-four shutdown, and burn away from Nacre while conventional tugs held the spindle. At the far limit, the ship would perform the final transfer into an unoccupied heliocentric disposal path.

Haru checked the thrust requirement. "We can reach the limit."

Tavin read what he had not said. "We cannot brake afterward."

"Not with the yoke attached."

"Jettison it at the limit."

"The axle collar will fuse to our frame during the burn," Pella said. "We can cut free before ignition or not at all."

Keir joined the channel. His face carried a pressure mark from the replacement shield. "We can remove more crown structure. Reduce the fused contact."

"Time?" Tavin asked.

"Seventy minutes."

Aru's suspension clock showed fourteen hours, but Nacre's spin window showed fifty-two minutes before bearing loads crossed tolerance.

"No," Tavin said.

Keir looked away from his camera, likely toward the vast wheel fixed inside the receiver crown. “Then use two ships.”

“No second spine can share that yoke through shutdown.”

“Dad.”

Tavin had heard that word spoken by a child asking for water, a teenager asking to leave Stillness, and a man refusing a private berth during evacuation. It carried no operational information now. It carried all the rest.

“Your team cuts the collar,” he said. “You withdraw to the spindle before we take load.”

Keir’s jaw set. “I can ride the cradle and release the outer braces.”

“No.”

“Someone has to.”

“Remote cutters can release them after first burn.”

“They have a six percent failure rate under skew.”

“Then Pella will build better ones.”

Pella nodded once.

Tavin sent the route to every Fleet captain. He did not call for approval. Shared risk had limits; no vote could change the mass. He called for objections based on physics.

Small Argument offered its remaining propellant. A Navy tug offered two drive technicians. The farm tender offered shield coolant. Three captains confirmed they could hold Nacre if the Morrowline departed under full load. No one offered a way to bring the ship home.

Haru overlaid the service pod trajectories. Forty-three pods could clear before ignition. One command pod near the bridge had enough independent thrust to launch after the yoke burn began, but its safe window measured less than twenty seconds.

“Crew evacuation before load,” Tavin said.

Pella looked at him. “Somebody stays to fly.”

“Yes.”

“Autopilot?”

“Skew changes during crown separation. It needs hands until the final transfer line settles.”

Haru folded her arms. “My hands know the line.”

“So do mine.”

“You’re assigning by rank.”

“I’m assigning the navigator to the fleet. These ships still need a route after Morrowline leaves it.”

Sive came onto the bridge circuit. "Medical has counted forty-three evacuees and one idiot."

"Record the idiot as command mass."

"I dislike cargo that makes jokes."

Tavin signed the evacuation order.

Outside, Keir's spinwrights moved toward the final wheel with cutters. The receiver crown rotated beneath them. The Morrowline turned to present its reinforced yoke, every welded beam exposed to starlight.

For the first time since he had taken command, Tavin saw his ship's whole remaining life on one maneuver plot. It began at Nacre's inner crown and ended beyond the fleet, where a red line marked a transfer no hull could survive.

The final flywheel started to come free.

Chapter 26 — The Last Transfer

The final axle collar broke loose with Keir Serr still inside the receiver crown.

Tavin saw his son's suit marker jump across the bridge display. A section of cut metal rolled outward, dragging two brace lines and the service sled behind it. The Morrowline waited thirty meters away with its yoke open. Between ship and spindle, the freed flywheel began to precess.

"Keir, release both lines," Tavin said.

"Trying." His voice shook under suit alarms. "Left reel has loaded backward."

Pella fired the yoke's remote cutters. One brace parted. The other pulled the service sled toward the turning wheel.

"Suit tether," Tavin said.

"Clipped to the sled."

"Cut it."

"That leaves me free."

"That is the idea."

Keir cut his tether. The sled struck the wheel rim and vanished into the rotating structure. His body continued outward, too slowly to clear the Morrowline's approach.

Tavin rolled the ship six degrees. The open yoke passed beneath Keir's boots. A cradle worker on Nacre fired a rescue line that missed his shoulder by half a meter.

"Again," Keir said.

The second line caught his suit ring. Spinwrights hauled him toward the crown while the Morrowline completed its turn.

Pella's board lit with twelve lock contacts. "Wheel entering yoke."

Metal filled the forward view. The flywheel carried part of Nacre's inner crown around its axle, a torn mass of truss and receiver plate. No loading manual had contemplated the shape. Lock pins struck wherever they could find structure.

"Seven contacts," Pella said. "Eight. Nine. That's all we get."

"Rated requirement?"

"Twelve."

"Today's requirement?"

"Nine, if the frame wants a dignified death."

Tavin took thrust to two percent. The wheel settled against the yoke with a sound like a building dropping one

floor. On Nacre's public trace, bearing load declined. Keir's marker reached the safe truss.

"Spindle team clear," Keir said. "Cut path open."

Tavin keyed the ship. "All hands, abandon Morrowline. Pod sequence begins now. No equipment exceptions."

The forty-three other crew had already strapped into service pods along the spine. Sive held the evacuation count from an external tender. Haru waited in pod one with the fleet plot. Pella remained at engineering long enough to arm the remote cutters.

"Chief," Tavin said.

"I heard the order."

"Your pod shows empty."

"It has always been needy."

"Pella."

Her camera showed the engineering compartment, stripped of loose tools and half its wall panels. She touched the patched controls with three fingers. "Cutters armed. Coolant bypass fixed open. Thruster clusters will answer until their feed lines don't."

"Leave."

"I'm going."

She held up her damaged hand. "It takes longer to insult a ship properly."

Pod thirty-two sealed around her. Green lights advanced down the Morrowline's spine.

Tavin launched them in groups of four. Small conventional thrusters pushed each pod toward waiting tenders. The releases changed ship mass and yoke balance. Haru called revised trim from her pod because the bridge computer lagged behind the departures.

"Starboard cluster four, reduce point two," she said.

Tavin corrected. "You are no longer navigator of this vessel."

"Then stop using my plot."

"Your plot is excellent."

"I accept the apology."

Pod forty-three cleared. Sive counted every beacon twice.

"All crew recovered except command," she said. "Tavin, launch."

"Not yet."

Nacre's final receiver bank still carried thirteen percent of local load. When it shut down, the flywheel would transfer its full mass into the Morrowline's yoke. He had to hold alignment through that change, then fly the

excess clear of the spindle. An empty bridge could not judge the skew as receiver plate tore away.

Keir came onto the command circuit. "You said remote cutters after first burn."

"I said Pella would build better ones."

"That does not answer me."

"The crown has fused across the lower yoke. We knew it might."

"You can launch before stage four."

"And leave the ship to roll into Nacre."

"Autopilot can hold."

"Not through a load step it has never measured."

The son on Tavin's display had lost the softness of youth months ago. He wore a spinwright harness over his pressure suit. A burn marked one side of his face shield. Tavin wanted to store the image somewhere safer than memory.

"Keir, you have work," he said. "Outer band braking starts when I cross the limit. If you miss it, we save a spindle with nowhere for people to live."

Keir looked toward another screen. "I know."

"Good."

"I hate that answer."

"So did I when you used it."

Nacre's stage-four clock reached ten seconds.

Tavin closed the crew channel and opened ship control. Six thruster clusters reported ready. Two exceeded thermal guidance. Three had fed through improvised coolant loops. The Morrowline had carried flywheels, nursery modules, ballast, food, fugitives, and one captain who had confused keeping a ship with keeping what mattered.

At zero, Nacre shut the last receiver bank.

Weight arrived.

The yoke drove the ship outward. Tavin fired all clusters before the computer finished measuring the change. His restraints tightened across his ribs. The flywheel pulled off center, rotating the Morrowline toward the spindle's inhabited bands.

"Port one full. Dorsal three pulse."

The controls obeyed his hands. Conventional thrust found the opposing vector. The ship stopped rolling with less than two degrees to spare.

Fleet tugs burned around Nacre. Their exhaust plumes formed a loose cage of force, each one carrying some

fraction of the city's corrected path. The spindle shifted into its rough new orbit. Not safe yet. Not hidden either.

Tavin pointed the Morrowline through the cleared corridor and increased thrust.

The fused wheel tore deeper into the cargo frame. Deck cameras failed one after another. A pressure door closed below the bridge, preserving atmosphere in compartments no one occupied. Thruster two lost coolant and ran hot.

"Limit line in three minutes," Haru said from her pod.

"I see it."

"Command service pod requires thirty seconds for safe separation."

"It has nineteen."

"Then start early."

"Skew remains inside the fleet corridor."

Sive cut in. "Your ribs will not remain inside you if that yoke fails."

Tavin watched the final path settle. The Morrowline's excess mass would cross beyond Nacre's return corridor, but the ship had to turn the wheel another fraction before destruction. Otherwise fragments could intersect the spindle in eleven days.

He waited four seconds.

The turn completed.

Tavin locked the engines at full conventional burn and ran for the service pod.

The bridge deck had become a wall under acceleration. He pulled himself along recessed grips, crossing a passage built for gentler loads. Somewhere aft, a brace failed. The impact bent the corridor and jammed the pod hatch half open.

Tavin put both boots against the frame and pulled. The hatch moved one hand's width. He turned sideways, forced his shoulders through, and fell into the couch.

No time remained for the harness sequence. He crossed the lap straps, struck the launch control, and felt the pod tear away.

Its small motor fired across the Morrowline's vector. The ship continued outward, bright with six thrust plumes. For one second it looked intact.

Then the lower yoke entered the flywheel.

The cargo spine folded. Fusion feed shutoffs fired too late to save structure and early enough to prevent a reactor breach. The wheel crossed through the hull, carrying metal, propellant tanks, and the ship's old compensator frame into the disposal path. Thruster clusters separated in burning pieces. None turned back toward Nacre.

The service pod tumbled. Tavin fought the controls until one attitude jet answered. Stars slowed. Nacre moved into view below him, off center and still turning.

Keir's voice reached the pod through static. "Command beacon acquired. Dad, respond."

Tavin tasted blood where his teeth had cut his cheek. "I told you to work."

A sound came over the channel, half laugh and half failed breath.

Behind the pod, the Morrowline finished coming apart. Its drive section went dark. The flywheel continued into empty orbit with every fragment accounted for on Haru's plot.

Small Argument reached him first. The tug approached on its surviving fusion feed, moving with a caution that made Tavin impatient until he tried to lift his left arm. The shoulder answered with a bright interior pain. His service pod had no medical brace large enough to immobilize it.

"Do not open your hatch," Sive said over the rescue circuit.

"I had no current plan."

"You launched without a sealed shoulder harness. Your plans have lost standing."

The tug caught his pod in a soft grapple. Contact rolled him half a turn before the capture frame steadied. Through the little forward port, Nacre crossed from one edge to the other. Its spindle no longer followed the clean centered track of the receiver era. The outer residential band carried a visible kink where towers had begun to fail.

Keir's signal transferred to the tug. "I'm coming out to the dock."

"No. Finish the rail sequence."

"They have another coordinator."

"They had another captain too. We used the one who knew the failure."

For several seconds Keir said nothing. Tavin heard voices behind him calling bearing loads and residential clearances.

"You are alive?" Keir asked.

"I am attached to Small Argument and being scolded by Sive."

"That was not my question."

Tavin looked at his hands. One had begun to shake after leaving the controls. "Yes."

Keir closed his eyes once. When he opened them, a spinwright called him from outside the frame.

"I have to go," he said.

"I know."

The channel ended before either of them tried to make the survival mean more than it did.

Ose drew the service pod into the tug's emergency lock. Air pushed against the hull. Tavin remained inside while Sive's remote reader checked pressure, ribs, shoulder, and pupil response. Forty-three green crew markers surrounded his pod on the recovery display. With his own, the surviving complement reached forty-four.

No marker remained aboard the Morrowline.

Tavin watched his ship leave him alive.

Bex's key entered the maintenance root but refused Iona's first turn.

She knelt inside Nacre's receiver hub with both hands around the ceramic bar. Bex's key had crossed from the Assembly in a thrust case, then through three fleet couriers, before Nemi sealed it into the public custody chain. Now it sat beneath the third contact ridge exactly as Bex had described.

The hub had never been meant for occupied work during correction. Each load change pulled at Iona from a new direction. Her knees pressed against the maintenance grating while her shoulders drifted toward a wall marked FLOOR under the former field. Nemi had threaded a restraint line through both their belts. When the outer bearing towers began to shear, the line tightened hard enough to bruise.

Behind them, civilian witnesses watched through three independent cameras. One feed belonged to Nacre's displaced residents. Another went to the thirty Compact systems. The last remained open to all seven rejecting governments and six delays, whether they chose to record it or not. Iona had refused a private maintenance channel. A key used beyond public sight could rebuild the same office in another name.

She checked the physical routes before turning anything. The export conduit entered the root from the left, smooth and undamaged after a century of priority maintenance. Local staging arrived through a newer cage of cables laid across the deck during the past two days. Each cable bore handwritten tags: shade rail, outer band, moonward ballast, fleet thrust, dock release. Engineers had never standardized them because the Republic had never expected local payment at this scale.

"Custody read," Iona said.

Nemi recited it. "Elian Bex to Iona Rhyll under Assembly witness. Iona Rhyll to courier one. Three courier transfers under continuous seal. Hub receipt confirmed. No unrecorded opening."

"Authority read."

"Former ministerial root expired. Civilian countersigns pending. No central override remains."

"Physical state."

Keir answered from the spin board. "All occupied bands clear of the planned shear zones. Fleet mass outside the inner corridor. Final wheel beyond recovery line. Conventional tugs under load."

Iona looked once at the export conduit. "Then we do not use the easy path."

She put torque into the key.

"False accept one," Nemi said through Iona's helmet. "Ignore."

The root display offered a green confirmation. Iona turned past it.

Mechanical resistance changed. A second green field appeared with the words LOCAL STAGING ARMED.

“False accept two,” Iona said.

“Export branch just woke behind it.”

On the debt display, Eidos appeared as automatic relief.

The old network had survived every reform by treating export as safety and local cost as failure. With Nacre’s last receiver bank shut down, the root sought somewhere to place the unsettled remainder. The Morrowline carried physical mass away, but curvature corrections still had to settle across the spindle, fleet, shades, and moonward docks. One mistimed stage could send the remainder through the dark lattice.

Iona pushed the key deeper.

A contact clicked under the third ridge. Green fields vanished. The maintenance root opened as bare routing tables, each destination visible.

“Nemi, countersigns.”

Nemi Vale pressed the audit authorization from the adjacent console. Vestibular braces held her knees against the changing hub load. Keir signed from the civilian spin board. Pella signed from her recovery pod with her surviving fingers.

Three names appeared beside Iona’s.

No ministry stood above them.

Aru-in-Pressure filled the translation cradle with a low acceleration tremor. The hub plating carried it into Iona’s palms.

“Human route holds local,” Aru said. “Hard pressure-names press toward return.”

“Can you keep the channel open?”

“Open, yes. Empty of returned force, uncertain.”

“We need seven minutes.”

“Then give destinations without concealment.”

Iona loaded the staging table exposed by Bex’s builder number. His private four-percent shade-flex correction appeared exactly where he had said. She copied it into the public route, attributed the source, and let every system see that the final plan depended on knowledge he had withheld until arrest.

The table listed Nacre’s remaining debt in physical consequences. Outer shade array: two-point-one percent path change, paid by rail mass and tug burn. Moonward docks: loss of eleven berths, paid by ballast release. Inner spindle: bearing load beyond design for forty-eight seconds, paid by evacuation and permanent closure. Fleet corridor: flywheel mass removed by the Morrowline, ship unrecoverable.

A final row remained blank.

Unsettled curvature: 0.0037 g.

The same discrepancy Iona had found beneath a falling apple at Perihelion waited at the end of the Republic's largest correction. The number had traveled through every account because small remainders made convenient doors.

Eidos pulsed on the alternate route.

Nemi read the automatic warning. "Local schedule lacks tolerance. Export recommended to preserve inhabited structure."

"Recommended by whom?" Keir asked.

"An engine whose authors are dead."

The hub shook as the Morrowline crossed the disposal limit. Its beacon separated from the ship. Haru reported command pod alive. Iona let relief pass through her without touching the controls.

"Where can we put the remainder?" she asked.

Keir enlarged the spindle map. "Outer residential band still has rail margin."

"Occupied?"

"Evacuated and sealed."

"Consequence?"

"Three bearing towers shear. Band cannot reopen."

Nemi entered it. "Population displaced: eighteen million. Capacity loss already inside the public estimate, but not this mechanism."

Aru's pressure modulation sharpened.

"Return gathers," it said. "Some of us hear the old channel and answer wound with wound."

"Do not send it back," Iona said.

"I hold no singular hand. I can keep the opening shaped while local route completes."

"Cost to you?"

"Coherence."

The translation cradle fractured Aru's next ratio into several human tones. Pressure-names argued inside the signal. Some demanded the remainder. Others marked Nacre's open costs and the destroyed ship. Aru held their disagreement together without claiming unanimity.

Iona selected the outer residential band.

The root demanded explicit acknowledgment: permanent loss of three towers, eighteen million residents unable to return, spindle capacity reduction beyond existing ration schedule.

She read every line aloud. Nemi confirmed the audit. Keir confirmed the physical path. Pella confirmed fleet load. Nothing went into a column labeled secondary.

Iona turned Bex's key to local execution.

The rail masses moved first. Empty housing blocks along the outer band took a change in acceleration they had not been built to survive. Cameras showed walls flex, windows craze, and one bearing tower split through its middle. Conventional tugs burned against the shift. Their propellant reserves fell on the public display.

Eidos remained available.

"Route deviation point six percent," Nemi said.

"Correct with moonward ballast."

"Berth eleven will depart its frame."

"Clear?"

"Cleared."

Keir released the berth. A structure large enough to hold an evacuation ferry moved away from Nacre under cold-gas pushers. Fleet tugs caught its mass and drew the spindle back toward the staged path.

Aru's voice thinned. "Hard return meets open channel."

The cradle bucked. For an instant Iona felt a distant pressure history through her gloves: ice remembering impacts across generations, a moonlet's altered pull written into living phase. Then Aru changed the channel geometry. The incoming correction did not vanish. It paused against the local route while Nacre paid.

"Thirty seconds," Nemi said.

The last bearing tower failed. Empty residential decks bent around it. Nacre's spindle orbit tightened, roughened, and held within the survival corridor.

"Ten."

Iona kept the key turned. Heat reached her palm through the ceramic.

"Five."

Eidos flashed one final recommendation.

She closed the export branch.

The 0.0037-g remainder entered Nacre's public account and settled through the broken outer band. The dark-lattice route emptied without a debt pulse.

Aru released the channel. Its many tones fell away until one diminished voice remained.

"Local destination received," it said.

Around the hub, receiver banks stayed dark. Conventional tugs continued burning. Eighteen million people had lost homes. Eleven docks had lost berths. Three towers had broken. The Morrowline had become a field of

outbound fragments, and Meridian remained beyond repair.

Nacre passed the old correction point without exporting force.

On Iona's display, its new orbit drew a coarse line around the star. Every deviation carried a named destination. The line did not smooth itself.

It held.

Chapter 27 — An Honest Orbit

Iona Rhyll put her boot against the inspection hatch and felt the unauthorized thrust before the gauge admitted it.

The habitat tender Kalliope hung beside a shrinking industrial ring in the Var system, one of the seven systems outside the Honest Orbit Compact. Its crew claimed to be moving ballast by cold-gas tug. The vibration under Iona's heel carried a higher frequency than any cold-gas valve.

"Open it," she said.

Chief Mera looked toward the Var safety officer who had accompanied them from the dock. The officer wore the red-bordered seal of a government that had rejected Compact jurisdiction. He gave no order.

Mera released the hatch.

Heat reached Iona first. Behind the inspection plate, a fusion transfer line fed a small conventional nozzle hidden inside a housing labeled WATER RECOVERY. The nozzle pushed against the ring's changing orbit. Its exhaust destination appeared nowhere on the local cost schedule.

Nemi Vale braced one hand against the corridor while her vestibular support corrected for the tender's gentle roll. "Thrust estimate?"

"Point eight newtons," Mera said.

"For how long?" Iona asked.

"Sixteen days."

"Propellant source?"

"Emergency reserve."

"Named destination for the momentum?"

Mera pointed through the hull, as if vacuum could sign a ledger. "Away."

Iona waited.

Six months earlier, a hidden nozzle would have counted as harmless because its exhaust crossed no inhabited boundary. The old Republic had reserved moral attention for where benefits appeared. Costs that left the camera frame became administrative weather.

The Var officer cleared his throat. "This installation does not use Reciprocal Mass Exchange. Your Compact has no authority here."

"I did not say it did."

"Then you cannot freeze the tender."

"I did not freeze it."

Nemi turned her display toward him. "Your own cessation law requires public declaration of propellant draw

and collision effect. Var adopted it forty-one days ago.”

The officer read the citation. His government had refused ratification while copying nine Compact provisions into local statute. Each provision bore a different name. None permitted involuntary export.

Mera folded her arms. “If we disclose the burn, the district reduces our water allotment to account for reserve loss. This ring has eight thousand residents waiting for relocation.”

“How much water does the hidden draw preserve?” Iona asked.

“Three days.”

“And what path does the exhaust cross?”

“Empty.”

“Measured empty or certified empty?”

Mera’s face changed.

The words no longer required a lesson. Every engineer knew what the Republic had once placed inside that distinction.

Iona crawled through the hatch and clipped a sampler beside the nozzle. The exhaust carried tritium traces and ablated housing metal. Its path crossed the planned approach of an evacuation ferry in eleven days. Not enough material to breach a hull. Enough to damage radiators already operating near failure margin.

She entered the finding in Var’s local record rather than the Compact ledger. Hidden thrust. Undeclared reserve use. Downstream ferry risk. Three days of water preserved for eight thousand people. Nothing in the entry balanced cleanly.

“Corrective order?” the officer asked.

“You issue it,” Iona said.

“You are the auditor.”

“I am an invited field auditor. Var holds authority.”

He looked unhappy with a responsibility he had expected her to seize. “Shut down the nozzle. Publish the reserve draw. Recompute the ferry approach.”

Mera said, “That costs the ring three days.”

“Yes,” he answered.

“Put your name on it.”

The officer pressed his seal to the order.

Kalliope’s engineer closed the fusion line. The vibration beneath Iona’s boot ceased. On the public schedule, the industrial ring’s water deadline advanced. A ferry dispatcher changed course. Eight thousand residents acquired a more urgent evacuation date.

No one felt absolved.

At the dock, a wall display carried the ratification count. Thirty-six system marks now circled the Compact seal. The original thirty had completed public votes within the first two months. All six delayed systems joined after receiver audits, evacuation guarantees, and local consent proceedings. They had not joined from identical conviction. One had run out of fuel. Another had watched Nacre survive. Two demanded stronger limits on any future coordinating office. Their reasons remained attached to their signatures.

The remaining seven displayed no Compact seal.

Each had nonetheless stopped involuntary exports. Four had dismantled their dark-lattice links. Two held receivers in permanent standby while local courts decided ownership. Var kept an inactive receiver for research under Olun-observed seal. Across all forty-three inhabited systems, no authorized debt pulse had entered Eidos for one hundred and seventy-eight days.

Unauthorized attempts had occurred in nine places.

Iona had audited four. Receiver workers caught three before activation. Aru's pressure-names identified one through the channel. The ninth reached a local coupling gate and failed because Bex's old central certificates no longer opened any root.

The Compact did not treat those attempts as proof that ratification had failed. It published them in a common exception register. Each entry named who had tried, which local need they claimed, what route they chose, and which safeguard stopped them. People could see that most violations began with a defensible emergency. A hospital wanted two minutes of stable load. A mining spindle lacked enough propellant to turn before an ore swarm arrived. A governor feared an evacuation announcement would trigger panic. Familiar need still knew how to become secret harm.

Var's entry drew more readers than the others because a local court had refused extradition. The engineer who opened the coupling gate remained under Var jurisdiction. Her testimony described twenty-seven thousand residents in a failing ring and a receiver that could have held them for another month by sending one pulse to Eidos. She had known the Olun would receive it. She had proceeded until the certificate failed.

Iona understood the decision and wanted the conviction.

That combination had unsettled commentators who preferred clean factions. Compact hearings preserved it anyway. The engineer received a public trial under Var law, while the ring received evacuation fuel from three signatory systems. Aid did not wait for agreement about guilt. Guilt did not disappear because aid arrived.

Olun observers maintained their own exception register through the translation cradles. Aru had not recovered its former coherence after holding Nacre's channel. Some days it spoke in one translated voice. On others, two pressure-names shared the signal and disagreed over human intent. The hard faction no longer returned pulses automatically, but it marked every attempted opening as evidence that cessation could fail.

That judgment also belonged in the settlement. Humans supplied public routes and local payment. Olun pressure-names kept debt channels closed to involuntary transfer while allowing low-band observation under explicit consent. Neither side granted the other permanent trust.

At Perihelion, the first calibration channel had been physically removed. At Nacre, engineers filled the old export conduit with inspection glass so visitors could see that no live path hid inside it. Across the seven outside systems, Olun seals used pressure intervals rather than Republic certificates. Any attempt to wake a receiver

had to break a pattern someone on Eidos could feel.

The victory consisted partly of making betrayal harder. It also required witnesses who survived long enough to report the attempt.

Nemi joined her beside the dock display. "Nacre sent the tribunal recess notice."

"Verdict?"

"Not yet. Bex asked to add testimony on the Var standby receivers."

"He no longer receives engineering reports."

"He reads public ledgers."

"Of course he does."

Their transport to Nacre departed under conventional thrust. The acceleration couch pressed Iona's shoulders with an honest discomfort. No compensator softened the burn. A display above each row showed propellant use, exhaust corridor, beam support, and the tug that would restore docking momentum at arrival. Most passengers ignored it after the first minute.

Iona did not mind. Public knowledge did not require permanent attention. It required availability before a decision and accountability afterward.

Nacre appeared two days later, its spindle orbit visibly eccentric against the shade array. Conventional tugs corrected the worst drift at every periapsis. Outer residential bands remained dark. Eighteen million displaced people lived in converted dock modules, temporary cylinders, and six systems that had accepted resettlement quotas.

Temporary had already become dishonest for many of them.

At the civil works yard, Tavin Serr stood beneath the drive cone of a tug that had never carried an RME compensator. His left shoulder remained stiff after the service-pod launch. He compensated by using his right hand for the inspection mirror and making Keir hold every heavy tool.

"This weld wanders," Tavin said.

Keir looked up from the scaffold. "By point four millimeters."

"Then it has wandered into another profession."

"The frame tolerance allows point six."

"Frame tolerance assumes crews enjoy vibration."

Pella Jor, supervising from the lower platform, tapped her three fingers against the hull. "Crews complain for free. Fixing point four costs twelve hours."

"Build it straight," Tavin said.

"Build it funded," Pella replied.

The yard contained six hulls in various stages. They lacked the Morrowline's long cargo spine and impossible

flexibility. Each tug had a thick conventional drive frame, detachable ballast arms, and service pods large enough for the full crew. Their design assumed no invisible load relief. They would carry less, burn more fuel, and require longer transfer windows.

Iona walked under the first hull. A small plate near the keel read MORROW CLASS.

Tavin saw her reading it. "Pella named the series while I was sedated."

"It won the yard vote," Pella said.

"The yard contains extortionists."

Keir climbed down. "He voted against and then corrected the drive calculations."

Iona touched the unpainted plate. "It is not the same name."

"No," Tavin said.

No new ship would become the Morrowline. The fragments remained on a disposal trajectory beyond Nacre, tracked by Haru until even their smallest risks passed out of the inhabited corridor. Tavin had refused salvage tugs. Recovering commemorative metal would have spent fuel needed for displaced residents.

"How many hulls?" Iona asked.

"Six here. Eleven contracted across four systems."

"Enough?"

"Never."

"When do you launch?"

"First tug in eight months if Pella stops finding places where the plans insult her."

Pella held up the inspection mirror. "The plans started it."

A shift horn sounded. Yard workers moved toward the hull with lunch packets, continuing arguments about coolant access and pod capacity. Keir stayed beside Tavin.

"The outer-band bearing report came back," he said to Iona. "Tower three cannot be stabilized."

"Can the band reopen around it?"

"No. We can recover archives and sealed belongings from four decks. Nothing more."

Eighteen million people had not been displaced for a season. The correction had removed their homes as permanently as a fire or impact, though the spindle continued around the star.

Iona entered the new assessment into Nacre's ledger from the yard. Outer residential band: closed without restoration path. Keir countersigned. A relocation office received the change and began converting temporary claims into permanent ones.

The tribunal convened that afternoon in the same Assembly chamber where Bex had surrendered his key. Its proceedings had remained public from the first charge. Survivors from Meridian spoke beside receiver

workers, Olun translations, ministry clerks, and habitat governors whose populations would have died under abrupt shutdown.

Before Iona testified, the court called three witnesses whose accounts resisted the story each side wanted.

The first governed a hospital spindle during Bex's emergency authority. She described how a continuity order kept surgical gravity stable through a drive repair and saved nine patients who could not have tolerated changing load. Then she displayed the same order's destination code. Eidos had received the debt. She refused to call the patients' survival false. She also refused to make it payment for an injury no patient had consented to send.

Next came a ministry routing clerk who had processed sealed responses for six years. He had never seen an Olun image or heard Aru speak. His task had consisted of moving files from scientific review into continuity archive and applying a retention code. Twice he asked why prime-ratio replies counted as noise. Twice supervisors showed him shutdown models. He continued processing.

"Were you threatened?" the judge asked.

"No."

"Did you believe the files might describe life?"

"Yes."

"Why did you remain?"

"I had a daughter on a receiver-dependent ring."

The clerk's daughter sat behind him. Her presence explained him without acquitting him.

Aru supplied the third account through a translation cradle bolted to the witness floor. The diminished speaker could not identify which exact human order struck which pressure lineage. Olun histories recorded acceleration, fracture, and interruption across different scales. Aru therefore gave ranges and refused the prosecutor's invitation to convert them into a single death total.

"Does uncertainty reduce the offense?" the prosecutor asked.

"Uncertainty limits count," Aru answered. "It does not create vacancy."

The hard pressure-names had submitted a separate statement demanding permanent closure of every human dark-lattice channel. Aru read that demand into the same record as its own support for observed, consenting contact. The tribunal heard disagreement instead of receiving an alien verdict convenient to either sentence.

Finally, a Meridian archivist placed the nursery ring's last structural trace on the chamber display. The ring broke in three stages after Bex's diversion returned. Evacuation craft saved most occupied decks. One section rotated beyond docking tolerance before its air gates closed. The archivist named the dead and then left the wreckage image on screen.

"No repair plan exists," she said. "No replacement ring inherits its identity. Meridian is permanently lost."

Bex lowered his head only after she finished, not while she spoke.

Iona took the witness rail.

The presiding judge asked, “Did Elian Bex’s engineering key make the Nacre localization possible?”

“Yes.”

“Does that act change your account of his concealment?”

“It changes the account by one act.”

“Does it excuse the prior conduct?”

“No.”

Bex watched her without appeal.

The judge turned to him. “You have requested a final statement before findings.”

Bex rose. “I concealed structured replies for six years. I authorized continued exports after I had reason to believe they caused organized harm. At Meridian I diverted debt to Eidos against the agreed staging plan. People died when that debt came back.”

He paused while the court record confirmed each admission.

“I believed continuity required a command that could act before consent. That belief saved people on some days. It also taught me to classify every life beyond my command boundary as uncertainty. I do not ask the tribunal to subtract Nacre from Meridian.”

The Meridian survivors’ register remained lit above the chamber. No habitat camera joined it. No nursery ring sent local testimony. Meridian stayed dead.

The findings took forty-three minutes to read. Bex was convicted of unlawful concealment, unauthorized debt export, obstruction of public audit, and command diversion at Meridian. The tribunal permanently barred him from public authority and custodial engineering access. Sentencing on loss of life would continue after individual claims, but no future government could return him to command.

When officers led him out, he stopped at the witness rail.

“Var’s inactive receiver,” he said to Iona. “Its ceramic coupler can hold a latent root after physical separation.”

“I know. We inspected it this morning.”

“Good.”

He continued toward custody.

Outside the chamber, no crowd celebrated. Some had wanted a death sentence. Others considered any conviction a betrayal of the minister who kept their habitats alive. Most returned to ration boards, relocation claims, or jobs on machines that no longer promised effortless weight.

Iona took a field ferry to the Meridian exclusion boundary before her next audit rotation.

The gas giant occupied half the forward port. Wrecked cylinders followed separate orbits around it, too damaged to join again. The nursery ring had broken into three principal arcs and thousands of tracked fragments. Recovery craft had removed bodies, records, and hazardous fuel where they could. They had not

rebuilt homes. No plan proposed that they should.

A memorial transmitter read paired names from the two archives. Human dead entered in maintenance time. Olun pressure lineages entered in translated acceleration intervals. Where no defensible count existed, the record said absence and did not invent precision.

Sella Rhyll's name arrived beside the sixteen-year-old incident marker. Iona listened without answering.

Afterward, she opened a training channel to twelve new field auditors scattered across the thirty-six signatory systems. Their faces appeared over the port, superimposed on Meridian's debris.

One trainee asked, "What do we certify when every available destination causes harm?"

"You certify the harms you can establish," Iona said. "You mark uncertainty where you cannot. You do not convert uncertainty into emptiness."

"Then who chooses?"

"The people who bear the local cost, under rules visible to those who may receive the distant one. Your work does not make the choice innocent."

Another trainee lifted a thrust schedule. "The new form rejects unnamed exhaust corridors."

"It should."

"Even below collision threshold?"

"Every force needs a named destination. Small forces accumulate. Thresholds require owners too."

She assigned them the Kalliope finding. They traced the hidden nozzle through reserve water, ferry risk, evacuation time, and Var law. No single column contained the right answer. Together, the columns prevented an easy lie.

The training ferry turned away from Meridian on conventional thrust. Its exhaust crossed a registered empty corridor surveyed by instruments built to find more than human signatures. Fuel declined. The return schedule lengthened by eleven minutes. Each consequence appeared on the public route. Nothing entered an unnamed margin or disappeared behind authority.

Through the port, Meridian's arcs continued around the gas giant without drawing closer. Nacre held its coarse orbit far away. Thirty-six systems worked inside the Compact. Seven worked outside it under their own cessation laws. None sent hidden weight into Eidos.

Iona set the next field audit on her calendar and watched the ferry's real path replace its promised one.

Epilogue — The Apple and the Moon

Lena Serr missed the apple because she expected it to fall straight.

It struck the garden mesh beside her shoe, bounced once, and rolled toward the moonward wall. She trapped it

with her heel before it reached the irrigation channel.

“You leaned spindleward,” Uncle Keir said.

“The tree leans spindleward.”

“Trees cheat.”

He knelt beside the school garden’s release frame and checked the little claw. A pale scar crossed his cheek where a thermal shield had failed years before Lena could remember. He had promised not to repair anything during Family Mechanics Day, which meant he had brought only two tools.

Around them, Nacre’s youngest orchard occupied a sealed crescent of the rebuilt inner band. The spindle provided most of its downward pull. Rail ballast under the floor trimmed the rest. Through the garden roof, Lena could see one of the conventional tugs holding station near a shade vane, its fusion plume reduced to a blue needle.

Her teacher, Mx. Soren, lifted the bruised fruit. “What changed its path?”

“The floor moved,” Lena said.

“The floor always moves.”

“The rail car moved under us.”

“Good. Why?”

Lena looked at the public strip running along the garden wall. It showed a diagram of Nacre, the Moon, three tugs, two ballast trains, and the afternoon shade correction. One train had shifted six tonnes toward the spindle. A tug had burned propellant to keep the moonward docks inside their corridor.

“Shade vane seven pulled the hub,” she said. “The train paid the small part. Tug Morrow Four paid the rest.”

Keir folded his arms. “Paid where?”

“Exhaust corridor blue-nine. The water recovery takes the heat tomorrow.”

Mx. Soren gave Lena the apple. “Then the garden can call its gravity ordinary.”

The class gathered around the Moon Ledger display. Its name came from the first page, which tracked every correction between Nacre’s spindle and its small industrial moon. The name had stayed after the ledger grew to include tugs, shades, ballast, docks, farms, and all the places force could go.

Five years earlier, Keir told them, some of those destinations had remained hidden.

Lena knew the story in the way children knew disasters that happened before their own memory. Meridian appeared in lessons as wreckage that still circled a gas giant. The Morrowline appeared as a ship with a long spine and too much work. Minister Bex appeared in tribunal records without the clean outline of a monster. Aru spoke sometimes through the school cradle, several pressure-names sharing a lesson about ice and acceleration.

None of those histories made the apple heavier in her hand.

Mx. Soren opened the afternoon entry. "Today's shade correction costs seventeen kilograms of tug propellant, adds nine minutes to a water barge, and moves the orchard rail ballast far enough to bend Lena's drop path. Who authorized it?"

"The garden council," said one student.

"The tug crew," said another.

"The moon dock," Lena added. "Because the exhaust crosses their approach."

"All three," Mx. Soren said. "Could they choose a different correction?"

Keir pointed toward the dark residential band beyond the garden roof. No lights had returned there. Its broken towers remained sealed inside a maintenance shell.

"They could close the orchard during shade turns," he said. "That saves propellant and keeps the apple path straight."

A boy beside Lena frowned. "Then the trees die."

"Some of them."

"Bad choice."

"Maybe. Put it in the ledger before you decide."

The class divided into groups. One counted water. Another counted fuel. Lena's group measured how much food the orchard produced and which dock approach would need to change. No column told them what to choose. The Moon Ledger kept the columns from vanishing when someone preferred an answer.

After the lesson, Lena carried the apple to the garden rail. Below the transparent floor, ballast carriages moved along bright tracks. She could see the machinery that adjusted the direction under her feet. It looked less impressive than the old receiver diagrams: steel wheels, cables, magnets, inspection lights, workers in orange suits.

"Did gravity feel different when you were little?" she asked Keir.

He considered. "Smoother."

"Better?"

"Until we learned where the roughness went."

The ballast train stopped. Lena's body noticed a faint sideways tug, then settled. No alarm sounded. No adult looked afraid. Overhead, the tug's narrow plume continued across its marked patch of dark.

She held the apple over the mesh and released it again.

This time she watched the wall strip as it fell. The fruit crossed the expected line, touched down beside her shoe, and remained there. A tiny destination mark appeared beside the orchard entry, naming the rail, the tug, the fuel, and the moon dock that had agreed.

Lena picked up the apple and bit through its bruised side.